Weakly Pointed Fusion Π-supercategories over Fields k with $\operatorname{char} k \neq 2$

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Abstract

We study a class of *weakly pointed* fusion Π-supercategories over a field k of characteristic not equal to 2, with unit $\mathbf{1}$ satisfying $\mathrm{End}(\mathbf{1}) = k$. Assuming the existence of a *Brauer–Wall reference category* over k, we classify these categories up to monoidal equivalence. We show that Brauer–Wall reference categories exist for $k = \mathbb{C}, \mathbb{R}$ but not for $k = \mathbb{F}_5$. For each finite group G, we also study a product of these categories equipped with a G-grading and define a group $\mathcal{T}_k(G)$, giving an explicit formula for the multiplication in terms of group cochains for $k = \mathbb{C}$ and $\mathbb{R}$.

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1 Introduction

Let k be a field of characteristic not equal to 2. All vector spaces, algebras, and categories are over k unless stated otherwise. All k-vector spaces are assumed to be finite-dimensional. Elements of groups are written g, h, ℓ, m; the letter k is reserved for the field.

The general theory of monoidal categories enriched in a braided monoidal category was developed in [MP19, MPP18]. Fusion categories over non-algebraically closed fields were studied in [San24]. The extension theory for non-enriched fusion categories over algebraically closed fields in [ENO10] was further developed for enriched fusion categories over algebraically closed fields in [JMPP21, DN21] and for non-enriched fusion categories over non-algebraically closed fields in [San26].

The present paper studies a narrow classification problem of enriched fusion categories over non-algebraically closed fields. Let sVec_k be the symmetric monoidal category of super vector spaces over k. A monoidal Π-supercategory is a monoidal sVec_k-enriched category equipped with a parity object π and an odd isomorphism $\pi \to \mathbf{1}$ as defined in [BE17]. We call such a category a fusion Π-supercategory if the category of even morphisms is a fusion category over k in the sense of [San24]. It is *weakly pointed* if every simple object X satisfies

$$X \otimes X^* \cong \mathbf{1}^{\oplus p} \oplus \pi^{\oplus q} \qquad \text{for some } p, q \geq 0 \text{ with } p + q > 0.$$

We classify weakly pointed fusion Π-supercategories $\mathcal{A}$ with split unit $\mathrm{End}_{\mathcal{A}}(\mathbf{1}) \cong k$ up to monoidal equivalence. Over $\mathbb{C}$, this classification problem is covered by the general theory of [JMPP21, DN21]. In the present paper, we give an explicit treatment over fields that need not be algebraically closed.

A physical motivation for studying this class comes from anomalies of finite group symmetries in $1+1$-dimensional systems. Let G be a finite group. In physics, bosonic anomalies in $1+1$-dimensions are described by $H^3(BG; U(1))$ [CGLW13, CGLW12]. On the mathematical side, this agrees with the classification of pointed fusion categories with Picard group G. For fermionic anomalies in $1+1$-dimensions, the approach initiated in [GW14] was followed by the development of a general connection with algebraic topology [Kap14, FH21]. Based on this connection, a general description was obtained in [BM18]. On the other hand, the classification of pointed fusion supercategories over $\mathbb{C}$ gives a group that differs in general from $H^3(BG; U(1))$, but can still be a proper subgroup of the group described in [BM18]. This motivates the study of weakly pointed fusion Π-supercategories over $\mathbb{C}$. Time-reversal symmetry also motivates considering the real case [GJF19].

In Section 2, we collect basic definitions and results of central simple superalgebras and their bimodules.

In Section 3, we fix conventions for supercategories, and define and classify weakly pointed fusion Π-supercategories over k with split unit. As described in [BE17], there exist several competing notions of monoidal supercategories. In this paper, we stick to monoidal Π-supercategories as in Definition 3.3.

We first show that the homogeneous isomorphism classes of simple objects form a finite group $G_{\mathcal{A}}$, with multiplication $[X][Y] = [S]$, where S is any simple summand of $X \otimes Y$ (Proposition 3.14). This gives a $G_{\mathcal{A}}$-grading of $\mathcal{A}$, and the endomorphism superalgebras of simple objects determine a homomorphism $\rho_{\mathcal{A}} : G_{\mathcal{A}} \to \mathrm{BW}(k)$.

In Theorem 3.29, assuming the existence of a Brauer–Wall reference category over k (Definition 3.12), we classify these categories up to monoidal superequivalence by data (G, ρ, ν, ω) modulo the gauge transformations of Definition 3.19. Here G is a finite group, $\rho : G \to \mathrm{BW}(k)$ is a homomorphism, and $\nu \in Z^2(G, \mathbb{Z}/2\mathbb{Z})$ and $\omega \in C^3(G, k^\times)$ are normalized cochains satisfying

$$\delta\omega = (-1)^{\nu \smile \nu}.$$

For $k = \mathbb{C}$, equivalent equations appear in [GW14, Eq. (A51)] and [BM18, Definition 3.15]. The existence of a reference category is addressed in Section 5.

In Section 4, we study the product obtained by taking the additive and even idempotent completion of the tensor product of supercategories. For a fixed finite group G, we consider categories $\mathcal{A}$ equipped with an isomorphism $G \cong G_{\mathcal{A}}$, and take the diagonal part of their product, which we call *stacking*. In Theorem 4.18, we show that this product makes the grade-preserving monoidal equivalence classes $\mathcal{W}_k(G)/\simeq_G$ into an abelian group, with unit represented by the category of G-graded super vector spaces and inverse represented by the opposite supercategory.

In order to describe the multiplication law of this group explicitly in terms of cochain data of Theorem 3.29, we need some braiding data for a Brauer–Wall reference category.

These data do not in general define an ordinary braiding. A more structural description of these data can be found in [SS21, Sections 3.3 and 3.6]. Proposition 4.23 states that every Brauer–Wall reference category can be equipped with such braiding data.

Let $\mathcal{T}_k(G)$ denote the set of classification data (G, ρ, ν, ω) modulo gauge transformations (ϕ, λ, β) with $\phi = \mathrm{id}_G$. Once a Brauer–Wall reference category and braiding data are fixed, Definition 4.27 gives an explicit multiplication on the cochain data. Theorem 4.30 shows that this product makes $\mathcal{T}_k(G)$ an abelian group and that the classification induces an isomorphism

$$\mathcal{T}_k(G) \cong \mathcal{W}_k(G)/\simeq_G .$$

Theorem 4.32 shows that the map $d : [(G, \rho, \nu, \omega)] \mapsto \rho$ gives an exact sequence

$$0 \longrightarrow \mathrm{SH}_k(G) \longrightarrow \mathcal{T}_k(G) \xrightarrow{d} \mathrm{Hom}(G, \mathrm{BW}(k)) \longrightarrow 0.$$

Here $\mathrm{SH}_k(G)$ is the subgroup with $\rho = 0$. It is the analogue with coefficients in $k^\times$ of the degree-three Gu–Wen supercohomology group [GW14, Appendices C and D].

In Sections 5 and 6, we give a criterion for the existence of Brauer–Wall reference categories and construct them when the criterion is satisfied. In particular, we show that such categories exist over $\mathbb{C}$ and $\mathbb{R}$, but not over $\mathbb{F}_5$. Our method is essentially the same as that used in the extension theory of [ENO10, Section 8]. More abstractly, the construction should admit a formulation in an appropriate super version of the theory of symmetric monoidal bicategories as described in [SP11, Chapter 2 and Appendix C]. Here we give an elementary treatment in terms of superalgebras and bimodules.

In Section 7, we give explicit cochain formulas for the product on $\mathcal{T}_k(G)$ for $k = \mathbb{C}$ and $\mathbb{R}$, without reference to auxiliary categorical data. We show that the exact sequence above need not split for $k = \mathbb{C}$, whereas it always splits for $k = \mathbb{R}$.

Lean formalization. A Lean 4 formalization of the real case is available in [Kaw26]. It includes the construction of a Brauer–Wall reference category, classification up to monoidal superequivalence, and stacking for a fixed finite grading group, together with a splitting of the degree map. The proof library was successfully rebuilt from a fresh clone using Lean 4.31.0. Build instructions and verification records are included in the repository.

2 Central Simple Superalgebras

Definition 2.1. A *super vector space* is a $\mathbb{Z}/2\mathbb{Z}$-graded k-vector space $X = X_0 \oplus X_1$. A vector $x \in X_0 \cup X_1$ is *homogeneous*, of *degree* $|x| \in \mathbb{Z}/2\mathbb{Z}$; the vectors in X_0 are *even* and those in X_1 are *odd*. A k-linear map $f : X \to Y$ between super vector spaces has *degree* $d \in \mathbb{Z}/2\mathbb{Z}$ if $f(X_i) \subseteq Y_{i+d}$ for $i \in \mathbb{Z}/2\mathbb{Z}$. The tensor product $X \otimes_k Y$ of super vector spaces is graded by $(X \otimes_k Y)_a = \bigoplus_{i+j=a} X_i \otimes_k Y_j$. For a super vector space X, ΠX denotes X with the reversed grading, $(\Pi X)_a = X_{a+1}$. The identity map $\Pi : X \to \Pi X, x \mapsto \Pi x = x$ is an odd isomorphism.

A *superalgebra* is a super vector space A with an associative k-bilinear multiplication $A \times A \to A$ and a unit $1 \in A_0$ such that $A_i A_j \subseteq A_{i+j}$ for $i, j \in \mathbb{Z}/2\mathbb{Z}$. A *homomorphism* of superalgebras is a k-linear map of degree 0 which preserves the multiplication and the unit. We write $A \cong B$ if superalgebras A and B are isomorpshc. The *supercenter* of A is $Z_{\mathrm{s}}(A) = Z_{\mathrm{s}}(A)_0 \oplus Z_{\mathrm{s}}(A)_1$, where

$$Z_{\mathrm{s}}(A)_d = \{z \in A_d : za = (-1)^{d|a|} az \text{ for every homogeneous } a \in A\}. \tag{2.1}$$

A superalgebra A is *central* if $Z_{\mathrm{s}}(A) = k$, *simple* if it is nonzero and has no nonzero proper graded two-sided ideals, and a *division superalgebra* if it is nonzero and every nonzero

homogeneous element is invertible. The *opposite superalgebra* A^{sop} of A is the supervector space A with the same grading and the multiplication

$$a \cdot_{\mathrm{sop}} b = (-1)^{|a||b|} ba. \tag{2.2}$$

For a superalgebra A and $p, q \geq 0$ with $p + q > 0$, the *matrix superalgebra* $M_{p|q}(A)$ is $\mathrm{Mat}_{p+q}(A)$ with ordinary matrix multiplication and grading

$$M_{p|q}(A)_d = \{(a_{ij}) : a_{ij} \in A_{d+\epsilon_i+\epsilon_j}\}, \qquad \epsilon_i = \begin{cases} 0 & i \leq p, \\ 1 & i > p. \end{cases} \tag{2.3}$$

The *graded tensor product* $A \mathbin{\widehat{\otimes}} B$ of superalgebras A and B is the super vector space $A \otimes_k B$ with the unit $1 \mathbin{\widehat{\otimes}} 1$ and the multiplication

$$(a \mathbin{\widehat{\otimes}} b)(a' \mathbin{\widehat{\otimes}} b') = (-1)^{|b||a'|} aa' \mathbin{\widehat{\otimes}} bb' \tag{2.4}$$

for homogeneous $a, a' \in A$ and $b, b' \in B$.

Lemma 2.2. *Let D be a division superalgebra. Then D_0 is a division algebra, and the only even idempotents of D are 0 and 1.*

Proof. The first assertion follows from the definition; see also [Haz16, Section 1.1.4, p. 22]. A nonzero idempotent in a division algebra is 1, which gives the second assertion. □

Lemma 2.3. *Let D be a division superalgebra and let e be a nonzero even idempotent of $M_{p|q}(D)$. If $eM_{p|q}(D)e$ is a division superalgebra, then it is isomorphic to D.*

Proof. This follows from the homogeneous-idempotent calculus and graded freeness over a division superalgebra; see [Haz16, Section 1.3.2 and Proposition 1.4.1]. The corner is the endomorphism superalgebra of the image of e in the defining graded free right D-module. The division assumption forces this image to have graded rank one, so its endomorphism superalgebra is isomorphic to D. □

Definition 2.4. Let A be a superalgebra. A *graded right A-module* is a super vector space V with an even k-linear right A-action. For graded right A-modules V and W, a k-linear map $f : V \to W$ of degree d is *right A-linear* if $f(vb) = f(v)b$ for $v \in V$ and $b \in A$. We write $\mathrm{End}_A(V)$ for the superalgebra whose degree-d part consists of the right A-linear maps $V \to V$ of degree d, with the composition as the multiplication.

Definition 2.5. Let A, B, and C be finite-dimensional superalgebras. A *graded (A, B)-bimodule* is a finite-dimensional super vector space M with even k-linear left A-action and right B-action satisfying $(am)b = a(mb)$. It is *simple* if it is nonzero and has no nonzero proper graded subbimodules. We write $M : A \to B$ for a graded (A, B)-bimodule M. For graded (A, B)-bimodules M, M', let $\mathrm{Hom}_{A-B}(M, M')$ be the super vector space whose degree-d part consists of the k-linear maps $u : M \to M'$ of degree d such that

$$u(amb) = (-1)^{|u||a|} a\, u(m) b \tag{2.5}$$

for homogeneous $a \in A$, $m \in M$, and $b \in B$. For $M : A \to B$ and $N : B \to C$, the balanced tensor product $M \otimes_B N$ is a graded (A, C)-bimodule, and for homogeneous $u \in \mathrm{Hom}_{A-B}(M, M')$ and $v \in \mathrm{Hom}_{B-C}(N, N')$,

$$u \otimes_B v \in \mathrm{Hom}_{A-C}(M \otimes_B N, M' \otimes_B N'), \qquad (u \otimes_B v)(m \otimes_B n) = (-1)^{|v||m|} u(m) \otimes_B v(n). \tag{2.6}$$

For $M : A \to B$, $N : B \to C$, and $P : C \to C'$, we use the even bimodule isomorphisms

$$\mathsf{a}_{M,N,P} : (M \otimes_B N) \otimes_C P \longrightarrow M \otimes_B (N \otimes_C P), \qquad (m \otimes_B n) \otimes_C p \longmapsto m \otimes_B (n \otimes_C p), \tag{2.7}$$

$$\mathsf{l}_M : A \otimes_A M \longrightarrow M, \qquad a \otimes_A m \longmapsto am, \tag{2.8}$$

$$\mathsf{r}_M : M \otimes_B B \longrightarrow M, \qquad m \otimes_B b \longmapsto mb. \tag{2.9}$$

For $M : A \to B$ and $N : A' \to B'$, we define the graded $(A \mathbin{\widehat{\otimes}} A', B \mathbin{\widehat{\otimes}} B')$-bimodule $M \mathbin{\widehat{\otimes}} N$ to be the super vector space $M \otimes_k N$ with the actions

$$(a \mathbin{\widehat{\otimes}} a')(m \mathbin{\widehat{\otimes}} n)(b \mathbin{\widehat{\otimes}} b') = (-1)^{|a'||m|+(|a'|+|n|)|b|} amb \mathbin{\widehat{\otimes}} a'nb' \tag{2.10}$$

for homogeneous $a \in A$, $a' \in A'$, $m \in M$, $n \in N$, $b \in B$, and $b' \in B'$. For homogeneous $u \in \mathrm{Hom}_{A-B}(M, M')$ and $v \in \mathrm{Hom}_{A'-B'}(N, N')$, we define $u \mathbin{\widehat{\otimes}} v \in \mathrm{Hom}_{A\widehat{\otimes}A'-B\widehat{\otimes}B'}(M \mathbin{\widehat{\otimes}} N, M' \mathbin{\widehat{\otimes}} N')$ by

$$(u \mathbin{\widehat{\otimes}} v)(m \mathbin{\widehat{\otimes}} n) = (-1)^{|v||m|} u(m) \mathbin{\widehat{\otimes}} v(n). \tag{2.11}$$

For $M : A \to B$, $M' : B \to C$, $N : A' \to B'$, and $N' : B' \to C'$, the *interchange isomorphism* is the even isomorphism of graded $(A \mathbin{\widehat{\otimes}} A', C \mathbin{\widehat{\otimes}} C')$-bimodules

$$\begin{gathered}\chi_{M,N,M',N'} : (M \mathbin{\widehat{\otimes}} N) \otimes_{B\widehat{\otimes}B'} (M' \mathbin{\widehat{\otimes}} N') \longrightarrow (M \otimes_B M') \mathbin{\widehat{\otimes}} (N \otimes_{B'} N'), \\ (m \mathbin{\widehat{\otimes}} n) \otimes (m' \mathbin{\widehat{\otimes}} n') \longmapsto (-1)^{|n||m'|} (m \otimes_B m') \mathbin{\widehat{\otimes}} (n \otimes_{B'} n').\end{gathered} \tag{2.12}$$

An isomorphism of superalgebras $f : A \to B$ is regarded as the graded (A, B)-bimodule ${}_fB_B$, the bimodule B with left A-action through f; in particular the swap $\tau_{A,B} : A \mathbin{\widehat{\otimes}} B \to B \mathbin{\widehat{\otimes}} A$, $a \mathbin{\widehat{\otimes}} b \mapsto (-1)^{|a||b|} b \mathbin{\widehat{\otimes}} a$, is a graded $(A \mathbin{\widehat{\otimes}} B, B \mathbin{\widehat{\otimes}} A)$-bimodule. A graded (A, B)-bimodule M is a *Morita equivalence* if there is a graded (B, A)-bimodule N with even isomorphisms $M \otimes_B N \cong A$ and $N \otimes_A M \cong B$. Superalgebras A and B are *Morita equivalent* if there is a Morita equivalence.

Proposition 2.6. *Let A and B be central simple superalgebras.*

(1) $A \cong M_{p|q}(D)$ for a central division superalgebra D, and D is unique up to isomorphism.

(2) A and B are Morita equivalent if and only if $A \cong M_{p|q}(D)$ and $B \cong M_{u|v}(D)$ for a central divisoin superalgebra D.

(3) Morita equivalence is an equivalence relation.

(4) A^{sop} is central simple.

(5) $A \mathbin{\widehat{\otimes}} B$ is central simple, and

$$A \mathbin{\widehat{\otimes}} A^{\mathrm{sop}} \xrightarrow{\sim} \mathrm{End}_k(A), \qquad a \mathbin{\widehat{\otimes}} b \longmapsto (x \mapsto (-1)^{|b||x|} axb), \tag{2.13}$$

is an isomorphism of superalgebras.

(6) The even part A_0 is a separable k-algebra.

(7) $Z(A) = k$ or $Z(A) = k \oplus kz$ for an odd element z with $z^2 \in k^\times$, where $Z(A)$ is the ordinary center. If we define $\epsilon(A) \in \mathbb{Z}/2\mathbb{Z}$ by

$$\epsilon(A) = \begin{cases} 0 & \dim_k Z(A) = 1, \\ 1 & \dim_k Z(A) = 2, \end{cases} \tag{2.14}$$

then $\epsilon(A)$ is invariant under Morita equivalence, and $\epsilon(A \mathbin{\widehat{\otimes}} B) = \epsilon(A) + \epsilon(B)$.

Proof. For (1), the graded Artin–Wedderburn theorem gives the matrix description, and the graded division algebra is unique by [Haz16, Remark 1.4.8 and Proposition 1.4.4]. Its centrality follows because passage to a matrix superalgebra preserves the supercenter.

Part (2) follows from graded Morita theory in [Haz16, Theorems 2.3.7–2.3.8] and the fact that every graded module over a graded division algebra is graded free [Haz16, Proposition 1.4.1]. Part (3) follows using identity bimodules, inverse bimodules, and balanced tensor products.

For (4), see [Lam05, Chapter IV, Section 4, p. 99]. The two assertions in (5) are [Lam05, Chapter IV, Theorem 2.3(3) and Proposition 4.1], respectively; the latter uses exactly the sign in (2.13).

For (6), if $A_1 = 0$, then $A_0 = A$ is an ordinary central simple algebra and hence is separable. If $A_1 \neq 0$, then A_0 is central simple in the odd-type case by [Lam05, Chapter IV, Theorem 3.4(2)], and its separability in the even-type case is stated in [Lam05, Chapter IV, Theorem 3.8].

The description of the center in (7) is [Lam05, Chapter IV, Theorems 3.4 and 3.6]. Our $\epsilon(A)$ is the type in [Lam05, Chapter IV, Definition 3.5]; its invariance and additivity follow from [Lam05, Chapter IV, Product Theorem 3.12 and Theorem 4.3]. By (1)–(2), the graded similarity used there agrees with our Morita equivalence. □

Definition 2.7. The *Brauer–Wall group* $\mathrm{BW}(k)$ is the abelian group of graded Morita equivalence classes of central simple superalgebras given by

$$[A] + [B] = [A \mathbin{\widehat{\otimes}} B], \qquad 0 = [k], \qquad -[A] = [A^{\mathrm{sop}}]. \tag{2.15}$$

For each $r \in \mathrm{BW}(k)$ fix a central division superalgebra D_r of class r, with $D_0 = k$. The *parity* $\epsilon : \mathrm{BW}(k) \to \mathbb{Z}/2\mathbb{Z}$ is the homomorphism given by (2.14).

Proposition 2.8. *Let A and B be Morita equivalent central simple superalgebras. A graded (A, B)-bimodule is a Morita equivalence if and only if it is simple. For Morita equivalences $M, N : A \xrightarrow{\sim} B$, the super vector space $\mathrm{Hom}_{A-B}(M, N)$ is one-dimensional, and its every nonzero element is invertible. Consequently, N is evenly isomorphic to one of M and ΠM, where ΠM has the actions*

$$a \cdot (\Pi m) \cdot b = (-1)^{|a|} \Pi(amb) \tag{2.16}$$

for homogeneous $a \in A$, $m \in M$, and $b \in B$.

Proof. These assertions follow, after transport along a Morita equivalence, from the equivalence between graded (B, B)-bimodules and super vector spaces in [Kwo13, Theorem 4.3.1]. We give a direct proof using the sign convention (2.5).

Let $P : A \to B$ be a Morita equivalence and let $Q : B \to A$ be an inverse as in Definition 2.5. The functors $Q \otimes_A (-)$ and $P \otimes_B (-)$ are mutually inverse equivalences between the categories of graded (A, B)-bimodules and graded (B, B)-bimodules with even maps. The first sends P to B. Since B is graded simple, B is a simple graded (B, B)-bimodule, and hence P is simple. This proves that every Morita equivalence is simple.

Put $E = A \mathbin{\widehat{\otimes}} B^{\mathrm{sop}}$, which is central simple by Proposition 2.6(4)–(5). It acts on every graded (A, B)-bimodule V by

$$(a \mathbin{\widehat{\otimes}} b)v = (-1)^{|b||v|} avb \qquad (a \in A,\ b \in B^{\mathrm{sop}},\ v \in V \text{ homogeneous}), \tag{2.17}$$

and the graded E-submodules of V are exactly its graded (A, B)-subbimodules. Since A and B are Morita equivalent, $[E] = [A] - [B] = 0$ by (2.15); by Proposition 2.6(1)–(2), E is therefore a full matrix superalgebra over k:

$$E \cong M_{r|s}(k) = \mathrm{End}_k(k^{r|s}) \qquad \text{for some } r, s \geq 0,\ r + s > 0. \tag{2.18}$$

Let e_{ij}, $1 \leq i, j \leq r + s$, be the matrix units of E under this isomorphism; they are homogeneous, and $\sum_i e_{i1} e_{1i} = 1$.

Let V be a nonzero simple graded E-module. The identity $\sum_i e_{i1}e_{1i} = 1$ implies $e_{11}V \neq 0$. A nonzero homogeneous $v \in e_{11}V$ generates V by simplicity, and the matrix-unit relations show that the vectors $e_{i1}v$, $1 \leq i \leq r+s$, form a homogeneous basis of V; in particular $e_{11}V = kv$ is one-dimensional. Now let M and N be simple graded (A,B)-bimodules, regarded as simple graded E-modules through (2.17). Choose nonzero homogeneous $v \in e_{11}M$ and $w \in e_{11}N$, and put $d = |w| - |v|$. Define the k-linear map $u : M \to N$ of degree d on the basis $(e_{i1}v)_i$ of M by

$$u(e_{i1}v) = (-1)^{d|e_{i1}|}e_{i1}w \qquad (1 \leq i \leq r+s). \tag{2.19}$$

The matrix-unit relations give $u(zm) = (-1)^{d|z|}z\,u(m)$ for homogeneous $z \in E$ and $m \in M$, and u is invertible because $(e_{i1}w)_i$ is a basis of N. Call a homogeneous k-linear map $u' : M \to N$ of degree d' *signed E-linear* if $u'(zm) = (-1)^{d'|z|}z\,u'(m)$ for homogeneous $z \in E$ and $m \in M$. Such a map is determined by $u'(v)$, since $(e_{i1}v)_i$ is a basis, and $u'(v) = u'(e_{11}v) = e_{11}u'(v)$ lies in the one-dimensional space $e_{11}N$; hence the signed E-linear maps form the line ku. Under (2.17), signed E-linearity of u' is equivalent to

$$u'(am) = (-1)^{d'|a|}a\,u'(m), \qquad u'(mb) = u'(m)b \qquad (a \in A,\ b \in B,\ m \in M \text{ homogeneous}), \tag{2.20}$$

which are the conditions (2.5). Hence $\mathrm{Hom}_{A-B}(M,N) = ku$, and its nonzero elements are invertible.

If $d = 0$, then u is an even isomorphism $M \to N$. If $d = 1$, then the map $\Pi M \to N$, $\Pi m \mapsto u(m)$, is even, and it is an isomorphism of graded (A,B)-bimodules for the actions (2.16), since $u(amb) = (-1)^{|a|}a\,u(m)b$ for homogeneous $a \in A$, $m \in M$, and $b \in B$ by (2.5). So N is evenly isomorphic to M or to ΠM. It is not evenly isomorphic to both: otherwise there is an even bimodule isomorphism $\phi : M \to \Pi M$, and the deletion map $\delta : \Pi M \to M$, $\Pi m \mapsto m$, is an odd element of $\mathrm{Hom}_{A-B}(\Pi M, M)$ by (2.16), so that $\delta\phi$ is a nonzero odd element of $\mathrm{Hom}_{A-B}(M,M)$; but $\mathrm{Hom}_{A-B}(M,M)$ is a line containing id_M, hence equals $k\,\mathrm{id}_M$ and is even.

Finally, let N be a simple graded (A,B)-bimodule and choose a Morita equivalence $P : A \to B$ with inverse $Q : B \to A$, which exists by the hypothesis. The bimodule ΠP is a Morita equivalence with inverse ΠQ: by (2.16), the map $\Pi P \otimes_B \Pi Q \to P \otimes_B Q$, $\Pi x \otimes_B \Pi y \mapsto (-1)^{|x|}x \otimes_B y$ for homogeneous $x \in P$ and $y \in Q$, is an even isomorphism of graded (A,A)-bimodules, and in the same way $\Pi Q \otimes_A \Pi P \cong Q \otimes_A P$. Since P is simple, N is evenly isomorphic to P or to ΠP, and a graded bimodule evenly isomorphic to a Morita equivalence is a Morita equivalence. Hence N is a Morita equivalence. □

Proposition 2.9. *Let A be a central simple superalgebra and let $\tau : A \mathbin{\widehat{\otimes}} A \to A \mathbin{\widehat{\otimes}} A$, $\tau(x \mathbin{\widehat{\otimes}} y) = (-1)^{|x||y|}y \mathbin{\widehat{\otimes}} x$, be the swap. The homogeneous elements $v \in A \mathbin{\widehat{\otimes}} A$ such that*

$$vz = (-1)^{|v||z|}\tau(z)v \qquad (z \in A \mathbin{\widehat{\otimes}} A \text{ homogeneous}) \tag{2.21}$$

span a one-dimensional super vector space concentrated in degree $\epsilon([A])$, and its nonzero elements are invertible.

Proof. Put $B = A \mathbin{\widehat{\otimes}} A$, a central simple superalgebra by Proposition 2.6(5), and let τ^*B be the graded (B,B)-bimodule B with the left action $z \cdot m = \tau(z)m$. For homogeneous $v \in B$, the map $u_v : B \to \tau^*B$, $m \mapsto vm$, is right B-linear, and (2.21) says precisely that u_v is a homogeneous bimodule map of degree $|v|$ in the sense of (2.5); conversely, a homogeneous bimodule map $u : B \to \tau^*B$ satisfies $u(m) = u(1)m$, so $u = u_{u(1)}$ and $u(1)$ satisfies (2.21). The bimodules B and τ^*B are simple graded (B,B)-bimodules, because B is graded simple, and B is graded Morita equivalent to itself. Proposition 2.8 shows that the solutions of

(2.21) form a one-dimensional super vector space whose nonzero elements are invertible. The existence of a homogeneous invertible solution also follows from the graded Skolem–Noether theorem [EV07, Proposition 8], applied to τ.

The equations defining the solution space and the ordinary center are linear over k, so their formation commutes with extension of scalars. Thus passage to a separable closure k^s preserves the dimensions of the graded pieces of the solution space and the parity $\epsilon([A])$. Over k^s every central simple superalgebra is isomorphic to a matrix superalgebra $E = \mathrm{End}_{k^s}(V)$, for a finite-dimensional super vector space V over k^s, or to $E \mathbin{\widehat{\otimes}} C$ with $C = k^s[e]/(e^2 - 1)$, e odd, according as $\epsilon([A]) = 0$ or 1, by [Wal64, Theorem 3] and Proposition 2.6(1)–(2). For E, let $\Phi : E \mathbin{\widehat{\otimes}} E \to \mathrm{End}_{k^s}(V \otimes V)$ be the isomorphism of superalgebras with $\Phi(f \mathbin{\widehat{\otimes}} g)(x \otimes y) = (-1)^{|g||x|} f(x) \otimes g(y)$ for homogeneous $f, g \in E$ and $x, y \in V$, and let $\sigma : V \otimes V \to V \otimes V$ be the even map $\sigma(x \otimes y) = (-1)^{|x||y|} y \otimes x$. A direct calculation gives $\sigma\Phi(f \mathbin{\widehat{\otimes}} g) = (-1)^{|f||g|}\Phi(g \mathbin{\widehat{\otimes}} f)\sigma = \Phi(\tau(f \mathbin{\widehat{\otimes}} g))\sigma$, so $\Phi^{-1}(\sigma)$ is an even solution of (2.21). For C, put $e_1 = e \mathbin{\widehat{\otimes}} 1$, $e_2 = 1 \mathbin{\widehat{\otimes}} e$, and $v_1 = e_1 - e_2$. Then $e_1 e_2 = -e_2 e_1$, $e_i^2 = 1$, and

$$v_1 e_1 = 1 + e_1 e_2 = -e_2 v_1 = -\tau(e_1) v_1, \qquad v_1 e_2 = e_1 e_2 - 1 = -e_1 v_1 = -\tau(e_2) v_1, \qquad v_1^2 = 2, \tag{2.22}$$

so v_1 satisfies (2.21) on the generators e_1, e_2. Applying this relation successively to two homogeneous elements shows that it also holds for their product; hence it holds on all of $C \mathbin{\widehat{\otimes}} C$. The element v_1 is odd and invertible since $\mathrm{char}\, k \neq 2$. Compare [GJF19, Section 5.5] for the Clifford self-braiding calculation over $\mathbb{C}$; the calculation here uses the signed bimodule-map convention (2.5). For $E \mathbin{\widehat{\otimes}} C$, write τ_E and τ_C for the swaps of $E \mathbin{\widehat{\otimes}} E$ and $C \mathbin{\widehat{\otimes}} C$, and let $\psi : (E \mathbin{\widehat{\otimes}} C) \mathbin{\widehat{\otimes}} (E \mathbin{\widehat{\otimes}} C) \to (E \mathbin{\widehat{\otimes}} E) \mathbin{\widehat{\otimes}} (C \mathbin{\widehat{\otimes}} C)$ be the even isomorphism of superalgebras $\psi(f \mathbin{\widehat{\otimes}} c \mathbin{\widehat{\otimes}} f' \mathbin{\widehat{\otimes}} c') = (-1)^{|c||f'|} f \mathbin{\widehat{\otimes}} f' \mathbin{\widehat{\otimes}} c \mathbin{\widehat{\otimes}} c'$ for homogeneous $f, f' \in E$ and $c, c' \in C$. Then $\psi \circ \tau = (\tau_E \mathbin{\widehat{\otimes}} \tau_C) \circ \psi$, and $\Phi^{-1}(\sigma) \mathbin{\widehat{\otimes}} v_1$ satisfies (2.21) with $\tau_E \mathbin{\widehat{\otimes}} \tau_C$ in place of τ by (2.4); hence $\psi^{-1}(\Phi^{-1}(\sigma) \mathbin{\widehat{\otimes}} v_1)$ is an odd solution of (2.21) for $E \mathbin{\widehat{\otimes}} C$. Since the solution space is a line in one degree, that degree is $\epsilon([A])$. □

3 Supercategories

Definition 3.1. The category sVec_k is the symmetric monoidal category of super vector spaces (Definition 2.1) and degree-preserving linear maps, with the tensor product

$$\begin{aligned} (X \otimes Y)_a &= \bigoplus_{i+j=a} X_i \otimes_k Y_j, \\ (f \otimes g)(x \otimes y) &= f(x) \otimes g(y) \qquad (f : X \to X',\ g : Y \to Y'), \end{aligned} \tag{3.1}$$

the unit k concentrated in degree zero, and the usual associativity and unit constraints. Its braiding is given by

$$\beta_{X,Y}(x \otimes y) = (-1)^{|x||y|} y \otimes x, \tag{3.2}$$

for homogeneous vectors $x \in X$ and $y \in Y$.

Definition 3.2. A *supercategory* is an sVec_k-enriched category $\mathcal{C}$. Composition has the order of [Kel05, §1.2]:

$$\mathcal{C}(Y, Z) \otimes_k \mathcal{C}(X, Y) \longrightarrow \mathcal{C}(X, Z), \qquad g \otimes f \longmapsto g \circ f. \tag{3.3}$$

The *endomorphism superalgebra* $\mathrm{End}_{\mathcal{C}}(X)$ is $\mathcal{C}(X, X)$, with its given grading and multiplication

$$fg = f \circ g, \qquad 1 = \mathrm{id}_X. \tag{3.4}$$

A *superfunctor* is an sVec_k-enriched functor. A *supernatural transformation* of degree $d \in \mathbb{Z}/2\mathbb{Z}$ between superfunctors $F, F' : \mathcal{C} \to \mathcal{D}$ is a family of morphisms $\theta_X : F(X) \to F'(X)$ of degree d satisfying

$$\theta_Y \circ F(f) = (-1)^{d|f|} F'(f) \circ \theta_X \tag{3.5}$$

for every homogeneous morphism $f : X \to Y$. We write $\mathrm{Nat}(F, F')$ for the super vector space whose degree-d part consists of supernatural transformations $F \Rightarrow F'$ of degree d. A superfunctor $F : \mathcal{C} \to \mathcal{D}$ is a *superequivalence* if we have a superfunctor $G : \mathcal{D} \to \mathcal{C}$ and even supernatural isomorphisms $GF \cong \mathrm{Id}_{\mathcal{C}}$ and $FG \cong \mathrm{Id}_{\mathcal{D}}$. The *underlying category* $\underline{\mathcal{C}}$ of a supercategory $\mathcal{C}$ is obtained by retaining all objects and only the even morphisms:

$$\underline{\mathcal{C}}(X, Y) = \mathcal{C}(X, Y)_0. \tag{3.6}$$

A supercategory $\mathcal{C}$ is *additive* if its underlying category $\underline{\mathcal{C}}$ is additive. An object of $\mathcal{C}$ is *simple* if it is simple in $\underline{\mathcal{C}}$. *Direct sums* in $\mathcal{C}$ are taken in $\underline{\mathcal{C}}$. An object X is an *even retract* of Y if there exist even morphisms $i : X \to Y$ and $p : Y \to X$ such that $p \circ i = \mathrm{id}_X$.

Definition 3.3. For supercategories $\mathcal{C}$ and $\mathcal{D}$, the *product* $\mathcal{C} \boxtimes \mathcal{D}$ is a supercategory with objects $X \boxtimes Y = (X, Y)$, morphism spaces

$$(\mathcal{C} \boxtimes \mathcal{D})(X \boxtimes Y, X' \boxtimes Y') = \mathcal{C}(X, X') \otimes_k \mathcal{D}(Y, Y'), \tag{3.7}$$

and composition

$$(f' \boxtimes g') \circ (f \boxtimes g) = (-1)^{|g'||f|} (f' \circ f) \boxtimes (g' \circ g) \tag{3.8}$$

for homogeneous morphisms, as in [BE17, (1.1), Definition 1.4]. Here $f \boxtimes g$ denotes $f \otimes_k g$. A *monoidal supercategory* is a supercategory $\mathcal{C}$ equipped with a superfunctor $\otimes : \mathcal{C} \boxtimes \mathcal{C} \to \mathcal{C}$, a unit object $\mathbf{1}$, and even supernatural associativity and unit isomorphisms satisfying the pentagon and triangle axioms. Its underlying category $\underline{\mathcal{C}}$ inherits the monoidal structure of $\mathcal{C}$. A *monoidal* Π*-supercategory* is a monoidal supercategory equipped with an object π and an odd isomorphism $\zeta : \pi \xrightarrow{\sim} \mathbf{1}$.

Definition 3.4. Let $\mathcal{C}$ be a monoidal Π-supercategory. For $a \in \mathbb{Z}/2\mathbb{Z}$, put

$$P_a = \begin{cases} \mathbf{1} & a = 0, \\ \pi & a = 1. \end{cases} \tag{3.9}$$

Define a superfunctor $\Pi : \mathcal{C} \to \mathcal{C}$ by

$$\Pi X = \pi \otimes X, \qquad \Pi(f) = \mathrm{id}_\pi \otimes f. \tag{3.10}$$

The morphisms

$$z_X = l_X \circ (\zeta \otimes \mathrm{id}_X) : \Pi X \longrightarrow X \tag{3.11}$$

form an odd supernatural isomorphism $z : \Pi \Rightarrow \mathrm{id}_{\mathcal{C}}$.

Definition 3.5. A *monoidal superfunctor* between monoidal supercategories $\mathcal{C}$ and $\mathcal{D}$ is a triple (F, J, φ), where $F : \mathcal{C} \to \mathcal{D}$ is a superfunctor, the *tensorator*

$$J_{X,Y} : F(X) \otimes F(Y) \xrightarrow{\sim} F(X \otimes Y) \tag{3.12}$$

is an even supernatural isomorphism, and the *unit map* $\varphi : \mathbf{1}_{\mathcal{D}} \xrightarrow{\sim} F(\mathbf{1}_{\mathcal{C}})$ is an even isomorphism, satisfying the usual monoidal functor axioms. It is *strict monoidal* if φ and all $J_{X,Y}$ are identity morphisms, and a *monoidal superequivalence* if F is a superequivalence; we write $\mathcal{C} \simeq_{\otimes} \mathcal{D}$ when such a superequivalence exists.

Proposition 3.6. *Let $(F, J, \varphi) : \mathcal{C} \to \mathcal{D}$ be a monoidal superfunctor between monoidal Π-supercategories with parity data $(\pi_{\mathcal{C}}, \zeta_{\mathcal{C}})$ and $(\pi_{\mathcal{D}}, \zeta_{\mathcal{D}})$. Then there is a unique even isomorphism $c_F : F(\pi_{\mathcal{C}}) \to \pi_{\mathcal{D}}$ such that*

$$\zeta_{\mathcal{D}} \circ c_F = \varphi^{-1} \circ F(\zeta_{\mathcal{C}}), \tag{3.13}$$

namely $c_F = \zeta_{\mathcal{D}}^{-1} \circ \varphi^{-1} \circ F(\zeta_{\mathcal{C}})$.

Proof. A superfunctor preserves degrees, so $F(\zeta_{\mathcal{C}}) : F(\pi_{\mathcal{C}}) \to F(\mathbf{1}_{\mathcal{C}})$ is an odd isomorphism, and $\varphi^{-1} \circ F(\zeta_{\mathcal{C}}) : F(\pi_{\mathcal{C}}) \to \mathbf{1}_{\mathcal{D}}$ is an odd isomorphism. Since $\zeta_{\mathcal{D}} : \pi_{\mathcal{D}} \to \mathbf{1}_{\mathcal{D}}$ is an odd isomorphism, $c_F = \zeta_{\mathcal{D}}^{-1} \circ \varphi^{-1} \circ F(\zeta_{\mathcal{C}})$ is an even isomorphism satisfying (3.13), and it is the only morphism satisfying it because $\zeta_{\mathcal{D}}$ is invertible. □

Definition 3.7. A *semisimple tensor Π-supercategory* is a monoidal Π-supercategory $\mathcal{C}$ whose underlying category $\underline{\mathcal{C}}$ is an essentially small semisimple k-linear abelian category with finite-dimensional Hom spaces, is rigid, and has simple unit $\mathbf{1}$. A semisimple tensor Π-supercategory $\mathcal{C}$ is *fusion* if it has finitely many isomorphism classes of simple objects. An object X of a semisimple tensor Π-supercategory $\mathcal{C}$ is *weakly invertible* if

$$X \otimes X^* \cong \mathbf{1}^{\oplus p} \oplus \pi^{\oplus q} \qquad \text{for some } p, q \geq 0 \text{ with } p + q > 0, \tag{3.14}$$

where X^* is a left dual of X in $\underline{\mathcal{C}}$. A semisimple tensor Π-supercategory is *weakly pointed* if every simple object is weakly invertible, and has *split unit* if $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) \cong k$. We write $\mathcal{W}$ for the class of weakly pointed fusion Π-supercategories with split unit.

Remark 3.8. If k is perfect, the underlying category of a fusion Π-supercategory is a fusion category over k in the sense of [San24, Definition 2.9], since a finite-dimensional semisimple algebra over a perfect field is separable.

Lemma 3.9. *Let $\mathcal{A}$ be a supercategory such that the underlying category $\underline{\mathcal{A}}$ is abelian, let $\Pi : \mathcal{A} \to \mathcal{A}$ be a superfunctor, and let $z : \Pi \Rightarrow \mathrm{id}_{\mathcal{A}}$ be an odd supernatural isomorphism.*

(1) Every nonzero homogeneous morphism between simple objects of $\mathcal{A}$ is invertible. If two simple objects of $\mathcal{A}$ are isomorphic then there is a homogeneous isomorphism between them.

(2) Suppose moreover that $\underline{\mathcal{A}}$ is semisimple with finite-dimensional Hom spaces, let S be a simple object of $\mathcal{A}$, let S' be an object of $\mathcal{A}$, and let $u : S' \to S$ be an odd isomorphism. An object isomorphic in $\mathcal{A}$ to a finite direct sum of copies of S and S' is evenly isomorphic to $S^{\oplus p} \oplus S'^{\oplus q}$ for some $p, q \geq 0$.

Proof. (1) Since Π is a superfunctor, it preserves degrees and restricts to a functor $\underline{\mathcal{A}} \to \underline{\mathcal{A}}$. For every object X of $\mathcal{A}$, the composite $z_X z_{\Pi X} : \Pi^2 X \to X$ is an even isomorphism, and for every even morphism $f : X \to Y$ of $\mathcal{A}$, (3.5) gives $z_Y \Pi(f) = f z_X$ and hence $z_Y z_{\Pi Y} \Pi^2(f) = f z_X z_{\Pi X}$; so Π is an autoequivalence of $\underline{\mathcal{A}}$ with quasi-inverse Π, and ΠX is simple for every simple object X. For an odd morphism $f : X \to Y$ between objects of $\mathcal{A}$,

$$f z_X : \Pi X \longrightarrow Y \text{ is even}, \qquad f = (f z_X) z_X^{-1}. \tag{3.15}$$

Let X and Y be simple objects of $\mathcal{A}$. For nonzero odd $f : X \to Y$, Schur's lemma in the abelian category $\underline{\mathcal{A}}$, applied to the nonzero even morphism $f z_X$ between the simple objects ΠX and Y, shows that $f z_X$ is invertible, so f is invertible by (3.15); for nonzero even f, Schur's lemma applies to f directly. So nonzero homogeneous morphisms between simple objects are invertible. An arbitrary isomorphism between simple objects has a nonzero homogeneous component, which is therefore invertible. In particular, for simple X, Y,

$$X \ncong Y \implies \mathcal{A}(X, Y) = 0, \qquad \mathrm{End}(X) \text{ is a division superalgebra.} \tag{3.16}$$

(2) All Hom spaces of $\mathcal{A}$ are finite-dimensional, since $f \mapsto fz_X$ in (3.15) is an isomorphism $\mathcal{A}(X,Y)_1 \cong \underline{\mathcal{A}}(\Pi X, Y)$. Since Π is an autoequivalence of $\underline{\mathcal{A}}$, as shown in (1), and $z_S^{-1}u : S' \to \Pi S$ is an even isomorphism, S' is simple. Let V be an object of $\mathcal{A}$ with an isomorphism $f : V \to S^{\oplus p'} \oplus S'^{\oplus q'}$ in $\mathcal{A}$ for some $p', q' \geq 0$. Since $\underline{\mathcal{A}}$ is semisimple with finite-dimensional Hom spaces, V is evenly isomorphic to a finite direct sum of simple objects. Let T be a simple summand of V, with even inclusion $i : T \to V$. The morphism fi is nonzero, so its composite with the projection onto one of the summands S or S' is nonzero, and a nonzero homogeneous component of this composite is a nonzero homogeneous morphism from T to S or to S'. It is invertible by (1), and composing with u^{-1} or u if it is odd makes it even with target S' or S. Thus V is evenly isomorphic to $S^{\oplus p} \oplus S'^{\oplus q}$ for some $p, q \geq 0$. □

Proposition 3.10. *Let $\mathcal{A}$ be a weakly pointed semisimple tensor Π-supercategory with split unit. For simple objects X, Y, the map*

$$\operatorname{End}(X) \mathbin{\widehat{\otimes}} \operatorname{End}(Y) \xrightarrow{\sim} \operatorname{End}(X \otimes Y), \qquad f \mathbin{\widehat{\otimes}} g \longmapsto f \otimes g \tag{3.17}$$

is an isomorphism of superalgebras.

Proof. For every object V, composition gives a perfect pairing

$$\mathcal{A}(V, \mathbf{1}) \times \mathcal{A}(\mathbf{1}, V) \longrightarrow k, \qquad (v, u) \longmapsto vu, \tag{3.18}$$

where V is an object of $\mathcal{A}$ and $\operatorname{End}(\mathbf{1}) = k$ by the split unit condition. To see this, decompose V into simple objects in $\underline{\mathcal{A}}$. By Lemma 3.9, only summands homogeneously isomorphic to $\mathbf{1}$ contribute. Such a summand is evenly isomorphic to $\mathbf{1}$ or π, using ζ in the odd case. On these two objects the pairing is the ordinary scalar pairing, using ζ and ζ^{-1} for π.

We first prove that for arbitrary objects V and W of $\mathcal{A}$ the map $\operatorname{End}(V) \mathbin{\widehat{\otimes}} \operatorname{End}(W) \to \operatorname{End}(V \otimes W)$, $f \mathbin{\widehat{\otimes}} g \mapsto f \otimes g$, is injective. Choose a right dual $V^\vee$ of V with even coevaluation $u : \mathbf{1} \to V^\vee \otimes V$. The duality adjunction is the isomorphism

$$\operatorname{End}(V) \xrightarrow{\sim} \mathcal{A}(\mathbf{1}, V^\vee \otimes V), \qquad f \longmapsto (\mathrm{id}_{V^\vee} \otimes f)u. \tag{3.19}$$

By (3.18), for each linear functional φ on $\operatorname{End}(V)$ there is a morphism $v_\varphi : V^\vee \otimes V \to \mathbf{1}$ such that

$$v_\varphi(\mathrm{id}_{V^\vee} \otimes f)u = \varphi(f)\mathrm{id}_{\mathbf{1}} \qquad (f \in \operatorname{End}(V)). \tag{3.20}$$

Define $\Sigma_\varphi : \operatorname{End}(V \otimes W) \to \operatorname{End}(W)$ by

$$\begin{aligned} \Sigma_\varphi(T) = {} & l_W(v_\varphi \otimes \mathrm{id}_W)\alpha^{-1}_{V^\vee,V,W} \\ & (\mathrm{id}_{V^\vee} \otimes T)\alpha_{V^\vee,V,W}(u \otimes \mathrm{id}_W)l_W^{-1} \qquad (T \in \operatorname{End}(V \otimes W)). \end{aligned} \tag{3.21}$$

For homogeneous $f \in \operatorname{End}(V)$ and $g \in \operatorname{End}(W)$, the interchange law (3.8) gives

$$\Sigma_\varphi(f \otimes g) = \varphi(f)g; \tag{3.22}$$

there is no sign, because in the two interchange steps the morphisms moved past f are the even coevaluation u and id_W. Let (f_i) be a homogeneous basis of $\operatorname{End}(V)$, let (φ_i) be the dual basis of the dual space, and let $g_i \in \operatorname{End}(W)$. Then

$$\sum_i f_i \otimes g_i = 0 \quad \Longrightarrow \quad g_j = \Sigma_{\varphi_j}\Big(\sum_i f_i \otimes g_i\Big) = 0 \quad \text{for every } j. \tag{3.23}$$

This proves injectivity. In particular, $\mathrm{id}_V \otimes \mathrm{id}_W \neq 0$ whenever $V, W \neq 0$, so $V \otimes W \neq 0$. The map preserves multiplication by (3.8) and (2.4).

By Lemma 3.9, an object isomorphic to a sum of copies of $\mathbf{1}$ and π admits an even decomposition into such summands. For simple Y, weak invertibility consequently gives an even isomorphism $Y \otimes Y^* \cong \mathbf{1}^{\oplus p} \oplus \pi^{\oplus q}$. Duality and this decomposition give, with total dimensions,

$$\begin{aligned} \dim_k \operatorname{End}(Y) &= \dim_k \mathcal{A}(\mathbf{1}, Y \otimes Y^*) = p + q, \\ \dim_k \operatorname{End}(X \otimes Y) &= \dim_k \mathcal{A}(X, X \otimes (Y \otimes Y^*)) \\ &= (p+q) \dim_k \operatorname{End}(X). \end{aligned} \tag{3.24}$$

Here $X \otimes \pi$ is oddly isomorphic to X via $r_X(\mathrm{id}_X \otimes \zeta)$, so the two kinds of summands have the same total Hom dimension. The injective map (3.17) is therefore an isomorphism. □

Definition 3.11. Let $\mathcal{A}$ be a weakly pointed semisimple tensor Π-supercategory with split unit, and let $G_{\mathcal{A}}$ be the set of classes of simple objects of $\mathcal{A}$ under homogeneous isomorphism in $\mathcal{A}$; by Lemma 3.9, two simple objects have the same class if and only if they are isomorphic in $\mathcal{A}$. For $g \in G_{\mathcal{A}}$, the *component* $\mathcal{A}_g$ is the full subsupercategory of objects whose simple summands have class g, including the zero object.

Definition 3.12. Let $\mathcal{C}$ be a weakly pointed semisimple tensor Π-supercategory with split unit. A simple object $X \in \mathcal{C}$ has *Brauer–Wall type* $r \in \mathrm{BW}(k)$ if there is an isomorphism of superalgebras

$$D_r \cong \operatorname{End}_{\mathcal{C}}(X). \tag{3.25}$$

We call $\mathcal{C}$ a *Brauer–Wall reference category* if every $r \in \mathrm{BW}(k)$ occurs as the type of a simple object, and two simple objects are isomorphic in $\mathcal{C}$, that is, have the same class in $G_{\mathcal{C}}$, if and only if their types agree. Section 5 decides when such a category exists.

Proposition 3.13. *Every component of a weakly pointed semisimple tensor* Π*-supercategory with split unit has a unique Brauer–Wall type, given by the type of any of its simple objects.*

Proof. Let $\mathcal{A}$ be a weakly pointed semisimple tensor Π-supercategory with split unit and let X be a simple object of $\mathcal{A}$. By (3.16), $D = \operatorname{End}(X)$ is a finite-dimensional division superalgebra. Let X^* be a left dual of X in $\underline{\mathcal{A}}$. By (3.14) and Lemma 3.9, there is an even isomorphism $X \otimes X^* \cong \mathbf{1}^{\oplus p} \oplus \pi^{\oplus q}$ for some $p, q \geq 0$. Duality in $\underline{\mathcal{A}}$ makes X^* simple. Let

$$\Phi : D \mathbin{\widehat{\otimes}} \operatorname{End}(X^*) \xrightarrow{\sim} \operatorname{End}(X \otimes X^*) \xrightarrow{\sim} M_{p|q}(k) \tag{3.26}$$

be the composite isomorphism of superalgebras whose first map is (3.17) and whose second map is conjugation by the even isomorphism above followed by the identification of $\operatorname{End}(\mathbf{1}^{\oplus p} \oplus \pi^{\oplus q})$ with $M_{p|q}(k)$ through $\operatorname{End}(\mathbf{1}) = k$ and the odd isomorphism $\zeta : \pi \to \mathbf{1}$. For homogeneous $z \in Z_{\mathrm{s}}(D)$, the element $z \mathbin{\widehat{\otimes}} \mathrm{id}_{X^*}$ is supercentral in $D \mathbin{\widehat{\otimes}} \operatorname{End}(X^*)$, so $\Phi(z \mathbin{\widehat{\otimes}} \mathrm{id}_{X^*})$ is supercentral in $M_{p|q}(k)$. Commutation with the even diagonal matrix units forces it to be diagonal. An odd diagonal matrix is zero, and an even diagonal matrix commuting with all matrix units is scalar; thus $\Phi(z \mathbin{\widehat{\otimes}} \mathrm{id}_{X^*}) = \lambda$ for some $\lambda \in k$. Then $(z - \lambda) \mathbin{\widehat{\otimes}} \mathrm{id}_{X^*} = \Phi^{-1}(0) = 0$, and since $\mathrm{id}_{X^*} \neq 0$ this forces $z = \lambda$. Thus D is central. A division superalgebra is simple, since a nonzero graded ideal contains a homogeneous invertible element. By Proposition 2.6(1), D is isomorphic to a unique representative D_r, which gives a unique type for X. By Lemma 3.9, isomorphic simple objects admit a homogeneous isomorphism, and conjugation by it is an isomorphism of endomorphism superalgebras; so the type is constant on each component. □

Proposition 3.14. *Let* $\mathcal{A}$ *be a weakly pointed semisimple tensor* Π*-supercategory with split unit. The rule*

$$[X][Y] = [S], \qquad S \text{ any simple summand of } X \otimes Y, \tag{3.27}$$

makes $G_{\mathcal{A}}$ a group, finite when $\mathcal{A} \in \mathcal{W}$, with identity $[\mathbf{1}]$ and inverse $[X]^{-1} = [X^*]$. Brauer–Wall type defines a homomorphism

$$\rho_{\mathcal{A}} : G_{\mathcal{A}} \longrightarrow \mathrm{BW}(k), \qquad \rho_{\mathcal{A}}([X]) = r \quad \textit{if } \mathrm{End}(X) \cong D_r. \tag{3.28}$$

The category $\mathcal{A}$ decomposes as

$$\mathcal{A} = \bigoplus_{g \in G_{\mathcal{A}}} \mathcal{A}_g, \qquad \mathcal{A}_g \otimes \mathcal{A}_h \subset \mathcal{A}_{gh}. \tag{3.29}$$

For a Brauer–Wall reference category $\mathcal{C}$, $\rho_{\mathcal{C}}$ is an isomorphism onto $\mathrm{BW}(k)$*; we index its components by $r \in \mathrm{BW}(k)$ and write them as $\mathcal{C}_r$.*

Proof. If $\mathcal{A} \in \mathcal{W}$, the set $G_{\mathcal{A}}$ is finite, being a quotient of the finite set of simple isomorphism classes in $\underline{\mathcal{A}}$. Taking finite direct sums in (3.16) gives

$$\mathcal{A}(X, Y) = 0 \quad (X \in \mathcal{A}_g,\ Y \in \mathcal{A}_h,\ g \neq h). \tag{3.30}$$

Grouping an even simple decomposition by isomorphism class then gives the first equality in (3.29).

For simple X, Y of types r, s, (3.17) and Proposition 2.6 give

$$E := \mathrm{End}(X \otimes Y) \cong D_r \mathbin{\widehat{\otimes}} D_s \cong M_{u|v}(D_{r+s}) \tag{3.31}$$

for some $u, v \geq 0$ with $u + v > 0$, and E is central simple. In particular, $X \otimes Y \neq 0$. Decompose $X \otimes Y$ into simple objects in $\underline{\mathcal{A}}$. Let e_g project onto the summands of class g. Equation (3.30) gives

$$e_g \in Z_{\mathrm{s}}(E)_0 = k\mathrm{id}, \qquad e_g^2 = e_g, \qquad \sum_g e_g = \mathrm{id} \neq 0. \tag{3.32}$$

Thus exactly one e_g is id and all others are zero. For a simple summand S with even projection idempotent e, $\mathrm{End}(S) \cong eEe$ is a division superalgebra. By (3.31) and Lemma 2.3, $\mathrm{End}(S) \cong D_{r+s}$.

Isomorphisms between simple objects can be chosen homogeneous. Tensoring such an isomorphism with an identity preserves invertibility by (3.8). Thus replacing either X or Y by an isomorphic simple object gives isomorphic nonzero tensor products, which belong to the same component by orthogonality. This proves that (3.27) defines a product on $G_{\mathcal{A}}$. Distributing tensor products over even simple decompositions now gives $\mathcal{A}_g \otimes \mathcal{A}_h \subset \mathcal{A}_{gh}$.

Choose simple $X \in \mathcal{A}_g$, $Y \in \mathcal{A}_h$ and $Z \in \mathcal{A}_\ell$. The objects $(X \otimes Y) \otimes Z$ and $X \otimes (Y \otimes Z)$ belong to $\mathcal{A}_{(gh)\ell}$ and $\mathcal{A}_{g(h\ell)}$, respectively. The first is nonzero: if S is a simple summand of $X \otimes Y$, then $S \otimes Z$ is a nonzero summand of $(X \otimes Y) \otimes Z$. The associator is an even isomorphism between these objects, so orthogonality gives $(gh)\ell = g(h\ell)$. The unitors give $[\mathbf{1}][X] = [X] = [X][\mathbf{1}]$.

Duality in $\underline{\mathcal{A}}$ makes X^* simple. The even morphisms

$$\mathrm{ev}_X : X^* \otimes X \longrightarrow \mathbf{1}, \qquad \mathrm{coev}_X : \mathbf{1} \longrightarrow X \otimes X^* \tag{3.33}$$

are nonzero by the snake identities. Since $\mathbf{1}$ is simple and $\underline{\mathcal{A}}$ is semisimple, ev_X is a split epimorphism and coev_X is a split monomorphism. Hence $\mathbf{1}$ is a simple summand of both products, giving $[X^*][X] = [\mathbf{1}] = [X][X^*]$. This proves the group assertion.

Proposition 3.13 makes $\rho_{\mathcal{A}}$ well-defined, and the tensor-product calculation above gives

$$\rho_{\mathcal{A}}(gh) = \rho_{\mathcal{A}}(g) + \rho_{\mathcal{A}}(h). \tag{3.34}$$

Also $\rho_{\mathcal{A}}([\mathbf{1}]) = 0$, since $\mathrm{End}(\mathbf{1}) \cong k = D_0$. Thus $\rho_{\mathcal{A}}$ is a homomorphism. For a reference category, every type occurs and two simple objects are isomorphic exactly when their types agree. These conditions make $\rho_{\mathcal{C}}$ surjective and injective, respectively. □

Proposition 3.15. *(1) Components of weakly pointed semisimple tensor* Π*-supercategories with split unit and the same Brauer–Wall type are superequivalent.*

(2) Let $n \geq 1$ *and let* $\mathcal{B}_1, \dots, \mathcal{B}_n, \mathcal{D}$ *be components of weakly pointed semisimple tensor* Π*-supercategories with split unit. Choose a simple object* $S_i \in \mathcal{B}_i$ *for each* i*, and put* $\mathcal{B} = \mathcal{B}_1 \boxtimes \cdots \boxtimes \mathcal{B}_n$ *and* $U = S_1 \boxtimes \cdots \boxtimes S_n \in \mathcal{B}$*, with both products parenthesized to the left. Suppose that the superfunctors*

$$F, F' : \mathcal{B} \longrightarrow \mathcal{D} \tag{3.35}$$

induce superalgebra isomorphisms

$$\begin{aligned} F_{U,U} &: \mathrm{End}_{\mathcal{B}}(U) \xrightarrow{\sim} \mathrm{End}_{\mathcal{D}}(F(U)), \quad a \longmapsto F(a), \\ F'_{U,U} &: \mathrm{End}_{\mathcal{B}}(U) \xrightarrow{\sim} \mathrm{End}_{\mathcal{D}}(F'(U)), \quad a \longmapsto F'(a). \end{aligned} \tag{3.36}$$

Then $\mathrm{Nat}(F, F')$ *is one-dimensional, every nonzero element is a supernatural isomorphism, and a transformation is determined by its value at* U*.*

Proof. (1) Let $\mathcal{E}$ be a component of a weakly pointed tensor Π-supercategory with split unit, let S be a simple object of $\mathcal{E}$, and put $D = \mathrm{End}_{\mathcal{E}}(S)$. Let $\mathsf{Mod}_{\mathrm{fd}}^{\mathrm{gr}}(D)$ be the supercategory whose objects are the finite-dimensional graded right D-modules and whose morphisms of degree d are the right D-linear maps of degree d. For an object X of $\mathcal{E}$, the super vector space $\mathcal{E}(S, X)$ is a graded right D-module by composition, $u \cdot a = ua$ for $u \in \mathcal{E}(S, X)$ and $a \in D$; and for a homogeneous morphism $f : X \to Y$ of $\mathcal{E}$, the map $u \mapsto fu$ is right D-linear of degree $|f|$. Hence

$$\mathrm{Hom}(S, -) : \mathcal{E} \longrightarrow \mathsf{Mod}_{\mathrm{fd}}^{\mathrm{gr}}(D), \qquad X \longmapsto \mathcal{E}(S, X), \qquad f \longmapsto (u \mapsto fu), \tag{3.37}$$

is a superfunctor. We show that it is a superequivalence. By Lemma 3.9, every simple object of $\mathcal{E}$ has a homogeneous isomorphism to S, hence an even isomorphism to S or to ΠS; so every object of $\mathcal{E}$ is evenly isomorphic to a finite direct sum of copies of S and ΠS. Let X and Y be two such sums, with p, respectively u, copies of S and q, respectively v, copies of ΠS. The inclusions $S \to X$ of the summands S, and the composites $S \xrightarrow{z_S^{-1}} \Pi S \to X$ of z_S^{-1} with the inclusions of the summands ΠS, form a homogeneous D-basis of $\mathcal{E}(S, X)$ with p even and q odd elements, and similarly for $\mathcal{E}(S, Y)$; in these bases both $\mathcal{E}(X, Y)$ and $\mathrm{Hom}_D(\mathcal{E}(S, X), \mathcal{E}(S, Y))$ are the space of $(u + v) \times (p + q)$ matrices over D with the grading of (2.3), the matrix of $f \in \mathcal{E}(X, Y)$ having as columns the coordinates of the images fb of the basis elements b of $\mathcal{E}(S, X)$ in the basis of $\mathcal{E}(S, Y)$, and (3.37) is the identity map between them. So the functor is fully faithful. Every finite-dimensional graded right D-module has a homogeneous basis, with p even and q odd elements say, and is therefore isomorphic to $\mathcal{E}(S, X)$ for X the sum of p copies of S and q copies of ΠS. So the functor is essentially surjective, and (3.37) is a superequivalence. Now let $\mathcal{E}'$ be a second component of a weakly pointed tensor Π-supercategory with split unit, of the same Brauer–Wall type r as $\mathcal{E}$, and let S' be a simple object of $\mathcal{E}'$. By Definition 3.12 and Proposition 3.13, there are superalgebra isomorphisms $D \cong D_r \cong \mathrm{End}_{\mathcal{E}'}(S')$, which give an isomorphism of supercategories $\mathsf{Mod}_{\mathrm{fd}}^{\mathrm{gr}}(D) \cong \mathsf{Mod}_{\mathrm{fd}}^{\mathrm{gr}}(\mathrm{End}_{\mathcal{E}'}(S'))$. Composing (3.37) for $\mathcal{E}$ with this isomorphism and with a quasi-inverse of (3.37) for $\mathcal{E}'$ gives a superequivalence $\mathcal{E} \simeq \mathcal{E}'$.

(2) Put $A = \mathrm{End}_{\mathcal{B}}(U)$. By (3.8), A is the left-parenthesized graded tensor product $\mathrm{End}(S_1) \mathbin{\widehat{\otimes}} \cdots \mathbin{\widehat{\otimes}} \mathrm{End}(S_n)$ of the central division superalgebras $\mathrm{End}(S_i)$, so A is central simple by Proposition 2.6(5). Choose a simple object T of $\mathcal{D}$ and put $D = \mathrm{End}_{\mathcal{D}}(T)$, $V_0 = \mathcal{D}(T, F(U))$, and $W_0 = \mathcal{D}(T, F'(U))$, graded right D-modules as in (1). They are graded (A, D)-bimodules with the left actions

$$a \cdot v = F(a)v, \qquad a \cdot w = F'(a)w \qquad (a \in A,\ v \in V_0,\ w \in W_0), \tag{3.38}$$

and by (3.36) and the full faithfulness of (3.37) for $\mathcal{D}$, the maps $a \mapsto (v \mapsto F(a)v)$ and $a \mapsto (w \mapsto F'(a)w)$ are superalgebra isomorphisms $A \to \mathrm{End}_D(V_0)$ and $A \to \mathrm{End}_D(W_0)$. A homogeneous D-basis of V_0 with p even and q odd elements gives $A \cong \mathrm{End}_D(V_0) \cong M_{p|q}(D)$, so A and D are graded Morita equivalent by Proposition 2.6(2). The bimodule V_0 is simple: a nonzero graded subbimodule contains a homogeneous vector with a nonzero, hence invertible, coordinate in this basis, and the matrix units in $\mathrm{End}_D(V_0) \cong A$ together with the right action of D then produce every basis vector. The same holds for W_0. Hence $\mathrm{Hom}_{A-D}(V_0, W_0)$ is one-dimensional and its nonzero elements are invertible, by Proposition 2.8.

We now compare $\mathrm{Hom}_{A-D}(V_0, W_0)$ with $\mathrm{Nat}(F, F')$. By the full faithfulness of (3.37) for $\mathcal{D}$, a homogeneous right D-linear map $V_0 \to W_0$ of degree d is of the form $v \mapsto uv$ for a unique homogeneous morphism $u : F(U) \to F'(U)$ of degree d in $\mathcal{D}$, and by (3.38) this map is left A-linear in the sense of (2.5) if and only if

$$u\,F(a) = (-1)^{d|a|}F'(a)\,u \qquad (a \in A \text{ homogeneous}). \tag{3.39}$$

Let V be an object of $\mathcal{B}$. By (1), each factor of V is a finite direct sum of copies of S_i and ΠS_i, so there are finitely many pairs of homogeneous morphisms $i_\alpha : U \to V$ and $p_\alpha : V \to U$ in $\mathcal{B}$, of a common degree e_α, such that $\sum_\alpha i_\alpha p_\alpha = \mathrm{id}_V$: take the tensor products of the inclusions and projections of the summands in the n factors, with the signs of (3.8) absorbed into the projections. For a homogeneous morphism $u : F(U) \to F'(U)$ of degree d satisfying (3.39), define

$$\theta_V = \sum_\alpha (-1)^{de_\alpha} F'(i_\alpha)\,u\,F(p_\alpha) : F(V) \longrightarrow F'(V). \tag{3.40}$$

Let $f : V \to W$ be a homogeneous morphism of $\mathcal{B}$, and let (i_α^V, p_α^V) and (i_β^W, p_β^W) be the chosen pairs for V and W. The morphism $p_\beta^W f i_\alpha^V$ lies in A and has degree $e_\beta + |f| + e_\alpha$, so (3.39) applied to it, together with $de_\beta + d(e_\beta + |f| + e_\alpha) = d|f| + de_\alpha$, gives

$$\begin{aligned} \theta_W F(f) &= \sum_{\alpha,\beta} (-1)^{d|f|+de_\alpha} F'(i_\beta^W p_\beta^W f i_\alpha^V)\,u\,F(p_\alpha^V) \\ &= (-1)^{d|f|} F'(f)\theta_V. \end{aligned} \tag{3.41}$$

Thus $\theta = (\theta_V)_V$ is a supernatural transformation $F \Rightarrow F'$ of degree d with $\theta_U = u$. Conversely, for every supernatural transformation $\theta : F \Rightarrow F'$ of degree d, (3.5) applied to i_α and p_α and $\sum_\alpha i_\alpha p_\alpha = \mathrm{id}_V$ give (3.40) with $u = \theta_U$, and θ_U satisfies (3.39). Hence evaluation at U,

$$\mathrm{Nat}(F, F') \xrightarrow{\sim} \mathrm{Hom}_{A-D}(V_0, W_0), \qquad \theta \longmapsto (v \mapsto \theta_U v), \tag{3.42}$$

is an isomorphism of super vector spaces. In particular $\mathrm{Nat}(F, F')$ is one-dimensional, a transformation is determined by its value at U, and (3.40) does not depend on the chosen pairs (i_α, p_α). Finally, let $\theta \in \mathrm{Nat}(F, F')$ be nonzero, so that $u = \theta_U$ is invertible, and let $\theta' : F' \Rightarrow F$ be the transformation with $\theta'_U = u^{-1}$. The composites $\theta'\theta$ and $\theta\theta'$ are even supernatural transformations with value id at U, so they are the identity transformations by (3.42) applied to the pairs (F, F) and (F', F'); hence θ is a supernatural isomorphism. □

Lemma 3.16. *Let $\mathcal{A}$ be a weakly pointed semisimple tensor Π-supercategory with split unit, and let $X, X' \in \mathcal{A}_g$ and $Y, Y' \in \mathcal{A}_h$. Then tensoring gives an isomorphism of super vector spaces*

$$\mathcal{A}(X, X') \otimes_k \mathcal{A}(Y, Y') \xrightarrow{\sim} \mathcal{A}(X \otimes Y, X' \otimes Y'). \tag{3.43}$$

Proof. Let $S \in \mathcal{A}_g$ and $T \in \mathcal{A}_h$ be simple objects. For $X = X' = S$ and $Y = Y' = T$, the map (3.43) is (3.17). Let $X, X' \in \mathcal{A}_g$ and $Y, Y' \in \mathcal{A}_h$ be simple, and choose homogeneous isomorphisms $\phi : X \to S$, $\phi' : X' \to S$, $\psi : Y \to T$, and $\psi' : Y' \to T$ (Lemma 3.9). The

isomorphisms $\mathcal{A}(X,X')\otimes_k\mathcal{A}(Y,Y')\to\operatorname{End}(S)\otimes_k\operatorname{End}(T)$, $f\otimes g\mapsto\phi' f\phi^{-1}\otimes\psi' g\psi^{-1}$, and $\mathcal{A}(X\otimes Y,X'\otimes Y')\to\operatorname{End}(S\otimes T)$, $u\mapsto(\phi'\otimes\psi')u(\phi\otimes\psi)^{-1}$, carry (3.43) for (X,X',Y,Y') to (3.17) for (S,T) up to the invertible signs of (3.8); so (3.43) is an isomorphism for simple X,X',Y,Y'. In general, let $X=\bigoplus_i X_i$, $X'=\bigoplus_j X'_j$, $Y=\bigoplus_{i'}Y_{i'}$, and $Y'=\bigoplus_{j'}Y'_{j'}$ be even decompositions into simple objects, with inclusions $\iota_i^X: X_i\to X$ and projections $p_i^X: X\to X_i$, and similarly for X', Y, Y'. The isomorphism $\mathcal{A}(X,X')\to\bigoplus_{i,j}\mathcal{A}(X_i,X'_j)$, $f\mapsto(p_j^{X'}f\iota_i^X)_{i,j}$, and the analogous isomorphisms for $\mathcal{A}(Y,Y')$ and for $\mathcal{A}(X\otimes Y,X'\otimes Y')$, with the inclusions $\iota_i^X\otimes\iota_{i'}^Y$ and projections $p_j^{X'}\otimes p_{j'}^{Y'}$, carry (3.43) to the direct sum

$$\bigoplus_{i,j,i',j'}\mathcal{A}(X_i,X'_j)\otimes_k\mathcal{A}(Y_{i'},Y'_{j'})\xrightarrow{\sim}\bigoplus_{i,j,i',j'}\mathcal{A}(X_i\otimes Y_{i'},X'_j\otimes Y'_{j'}) \tag{3.44}$$

of the maps (3.43) for the simple objects $X_i,X'_j,Y_{i'},Y'_{j'}$, which are isomorphisms. □

Proposition 3.17. *Let $\mathcal{C}$ be a Brauer–Wall reference category and let $H\le\mathrm{BW}(k)$ be a finite subgroup. The full subsupercategory*

$$\mathcal{C}_H=\bigoplus_{r\in H}\mathcal{C}_r \tag{3.45}$$

is a monoidal Π-subsupercategory of $\mathcal{C}$ with $\mathcal{C}_H\in\mathcal{W}$, and $\rho_{\mathcal{C}_H}$ is an isomorphism $G_{\mathcal{C}_H}\to H$. In particular $\mathcal{C}\in\mathcal{W}$ when $\mathrm{BW}(k)$ is finite.

Proof. By Proposition 3.14, $\rho_{\mathcal{C}}$ is an isomorphism $G_{\mathcal{C}}\to\mathrm{BW}(k)$, $\mathcal{C}=\bigoplus_{r\in\mathrm{BW}(k)}\mathcal{C}_r$, and $\mathcal{C}_r\otimes\mathcal{C}_s\subset\mathcal{C}_{r+s}$ for $r,s\in\mathrm{BW}(k)$. Since H is a subgroup, $\mathcal{C}_H$ is closed under $\otimes$ and under finite direct sums, and it contains $\mathbf{1}$ and π, which lie in $\mathcal{C}_0$: $\rho_{\mathcal{C}}([\mathbf{1}])=0$ since $\operatorname{End}(\mathbf{1})\cong k=D_0$, and $[\pi]=[\mathbf{1}]$ by the odd isomorphism ζ. The associator, the unitors, and ζ of $\mathcal{C}$ restrict to $\mathcal{C}_H$, so $\mathcal{C}_H$ is a monoidal Π-subsupercategory of $\mathcal{C}$.

The underlying category $\underline{\mathcal{C}_H}$ is the full subcategory of $\underline{\mathcal{C}}$ on the objects of $\mathcal{C}_H$. An object of $\underline{\mathcal{C}}$ lies in $\mathcal{C}_H$ if and only if all its simple summands do, so $\underline{\mathcal{C}_H}$ is closed under direct sums and direct summands in the semisimple abelian category $\underline{\mathcal{C}}$; hence it is a semisimple abelian k-linear category with finite-dimensional Hom spaces, in which $\mathbf{1}$ is simple, and $\operatorname{End}_{\mathcal{C}_H}(\mathbf{1})=\operatorname{End}_{\mathcal{C}}(\mathbf{1})\cong k$. The tensor product is k-linear in each variable. Let X be a simple object of $\mathcal{C}_H$, with left dual X^* and right dual ${}^*X$ in $\underline{\mathcal{C}}$. Then $[X^*]=[X]^{-1}$ by Proposition 3.14, and $[{}^*X]=[X]^{-1}$ by (3.27), since the evaluation $X\otimes{}^*X\to\mathbf{1}$ is nonzero and $\underline{\mathcal{C}}$ is semisimple, so that $\mathbf{1}$ is a summand of $X\otimes{}^*X$. Hence X^* and ${}^*X$ lie in $\mathcal{C}_H$, and $\underline{\mathcal{C}_H}$ is rigid with the duals of $\underline{\mathcal{C}}$, duals of direct sums being direct sums of duals.

Let $S_1,\dots,S_n$ be representatives of the isomorphism classes of simple objects of $\underline{\mathcal{C}_H}$. Their number is finite: for $r\in H$ choose a simple object $S_r\in\mathcal{C}_r$; by Lemma 3.9, every simple object of $\mathcal{C}_r$ has a homogeneous isomorphism to S_r, hence an even isomorphism to S_r or to ΠS_r, so $n\le 2|H|$. Put $P=S_1\oplus\cdots\oplus S_n$ and $E=\operatorname{End}_{\underline{\mathcal{C}}}(P)$. Since $\operatorname{Hom}_{\underline{\mathcal{C}}}(S_i,S_j)=0$ for $i\ne j$, $E=\prod_i E_i$ with $E_i=\operatorname{End}_{\underline{\mathcal{C}}}(S_i)$, a division algebra. Let $\mathsf{mod}_{\mathrm{fd}}(E)$ be the category of finite-dimensional right E-modules, and define

$$\operatorname{Hom}_{\underline{\mathcal{C}}}(P,-):\underline{\mathcal{C}_H}\longrightarrow\mathsf{mod}_{\mathrm{fd}}(E),\qquad X\longmapsto\operatorname{Hom}_{\underline{\mathcal{C}}}(P,X),\qquad f\longmapsto(u\mapsto fu), \tag{3.46}$$

where $\operatorname{Hom}_{\underline{\mathcal{C}}}(P,X)$ is a right E-module by precomposition. For objects $X\cong\bigoplus_i S_i^{\oplus m_i}$ and $Y\cong\bigoplus_i S_i^{\oplus n_i}$ of $\underline{\mathcal{C}_H}$, both $\operatorname{Hom}_{\underline{\mathcal{C}}}(X,Y)$ and $\operatorname{Hom}_E(\operatorname{Hom}_{\underline{\mathcal{C}}}(P,X),\operatorname{Hom}_{\underline{\mathcal{C}}}(P,Y))$ are the product over i of the spaces of $n_i\times m_i$ matrices over E_i, and (3.46) sends a matrix to left multiplication by the same matrix; so the functor is fully faithful. Every finite-dimensional right E-module is a direct sum of the modules $E_i=\operatorname{Hom}_{\underline{\mathcal{C}}}(P,S_i)$, so the functor is essentially surjective, and it is an equivalence. Since each E_i is a division algebra, E is semisimple, and $\underline{\mathcal{C}_H}\simeq\mathsf{mod}_{\mathrm{fd}}(E)$

has the n simple objects S_i up to isomorphism. Hence $\mathcal{C}_H$ is a fusion Π-supercategory with split unit by Definition 3.7.

For a simple object X of $\mathcal{C}_H$, (3.14) is an isomorphism in $\underline{\mathcal{C}}$ between objects of $\mathcal{C}_H$, so X is weakly invertible in $\mathcal{C}_H$, and $\mathcal{C}_H \in \mathcal{W}$. Finally, the simple objects of $\mathcal{C}_H$ are the simple objects X of $\mathcal{C}$ such that $\rho_{\mathcal{C}}([X]) \in H$, and the product (3.27) of $\mathcal{C}_H$ is computed by the same tensor products as that of $\mathcal{C}$; so $G_{\mathcal{C}_H}$ is the subgroup $\rho_{\mathcal{C}}^{-1}(H)$ of $G_{\mathcal{C}}$, and $\rho_{\mathcal{C}_H}$ is the restriction of $\rho_{\mathcal{C}}$ to it, an isomorphism onto H. If $\mathrm{BW}(k)$ is finite, then $H = \mathrm{BW}(k)$ gives $\mathcal{C}_H = \mathcal{C}$, so $\mathcal{C} \in \mathcal{W}$. □

Definition 3.18. Cochains, coboundaries, and cup products are as in Appendix A. A *twist* is a quadruple $t = (G, \rho, \nu, \omega)$, where G is a finite group, $\rho : G \to \mathrm{BW}(k)$ is a group homomorphism, $\nu \in C^2(G, \mathbb{Z}/2\mathbb{Z})$, and $\omega \in C^3(G, k^\times)$, satisfying

$$\delta\nu = 0 \quad \text{and} \quad \delta\omega = (-1)^{\nu \smile \nu}. \tag{3.47}$$

We write Tw for the class of all twists.

Definition 3.19. Let $s = (G, \rho, \nu, \omega)$ and $t = (H, \rho', \nu', \omega')$ be twists. A *gauge* from s to t is a triple (ϕ, λ, β), where $\phi : G \to H$ is a group isomorphism, $\lambda \in C^1(G, \mathbb{Z}/2\mathbb{Z})$, and $\beta \in C^2(G, k^\times)$, satisfying

$$\phi^*\rho' = \rho, \qquad \phi^*\nu' = \nu + \delta\lambda, \tag{3.48}$$

$$\phi^*\omega' = \omega(\delta\beta)^{-1}(-1)^{(\phi^*\nu') \smile \lambda + \lambda \smile \nu}. \tag{3.49}$$

The twists s and t are *gauge equivalent*, written $s \sim t$, if there exists a gauge from s to t.

Proposition 3.20. *Gauge equivalence is an equivalence relation on* Tw*:* $(\mathrm{id}_G, 0, 1)$ *is a gauge from s to s, the composite of gauges (ϕ, λ, β) from s to t and (ψ, μ, η) from t to u is a gauge from s to u with group isomorphism $\psi\phi$, and the inverse of (ϕ, λ, β) is a gauge from t to s with group isomorphism ϕ^{-1}.*

Proof. By Lemma A.2(1)–(2) and (A.8), the gauge formulas (3.48)–(3.49) preserve the twist equations and normalization.

Write $s = (G, \rho_s, \nu_s, \omega_s)$, $t = (G_t, \rho_t, \nu_t, \omega_t)$, and $u = (G_u, \rho_u, \nu_u, \omega_u)$. The identity gauge is $(\mathrm{id}_G, 0, 1)$. For gauges $(\phi, \lambda, \beta) : s \to t$ and $(\psi, \mu, \eta) : t \to u$, their composite is

$$(\psi\phi, \lambda + \phi^*\mu, \beta\, \phi^*\eta\, (-1)^{(\phi^*\mu) \smile \lambda}). \tag{3.50}$$

To verify its scalar equation, pull all cochains back to G and write $b = \phi^*\mu$. Put $\bar{\nu} = \phi^*\nu_t$ and $\tilde{\nu} = (\psi\phi)^*\nu_u$. Since $\bar{\nu} = \nu_s + \delta\lambda$ and $\tilde{\nu} = \bar{\nu} + \delta b$,

$$\begin{aligned} &(\bar{\nu} \smile \lambda + \lambda \smile \nu_s) + (\tilde{\nu} \smile b + b \smile \bar{\nu}) \\ &\quad + \tilde{\nu} \smile (\lambda + b) + (\lambda + b) \smile \nu_s \\ &= \delta b \smile \lambda + b \smile \delta\lambda = \delta(b \smile \lambda). \end{aligned} \tag{3.51}$$

This is the coboundary of the sign in (3.50). The degree equation follows from linearity of δ. Solving the gauge equations in the reverse direction, using (A.6), gives the inverse gauge

$$(\phi^{-1}, (\phi^{-1})^*\lambda, (\phi^{-1})^*(\beta^{-1}(-1)^{\lambda \smile \lambda})). \tag{3.52}$$

Substitution in (3.50) gives the identity in both orders. The cochains in (3.50) and (3.52) are normalized. Hence gauge equivalence is reflexive, transitive, and symmetric. □

Definition 3.21. Let $\mathcal{C}$ be a monoidal Π-supercategory. Put $\zeta_0 = \mathrm{id}_{\mathbf{1}}$ and $\zeta_1 = \zeta$. For $a, b \in \mathbb{Z}/2\mathbb{Z}$ and an object X, define even isomorphisms $c_{a,X} : P_a \otimes X \to X \otimes P_a$ and $m_{a,b} : P_a \otimes P_b \to P_{a+b}$ by

$$\begin{aligned} c_{a,X} &= (\mathrm{id}_X \otimes \zeta_a^{-1}) r_X^{-1} l_X (\zeta_a \otimes \mathrm{id}_X), \\ m_{a,b} &= \zeta_{a+b}^{-1} l_{\mathbf{1}} (\zeta_a \otimes \zeta_b). \end{aligned} \tag{3.53}$$

Let G be a group and $\nu \in C^2(G, \mathbb{Z}/2\mathbb{Z})$ with $\delta\nu = 0$. For $g, h, \ell \in G$ and objects X, Y, Z, define an even isomorphism

$$\begin{aligned} A^{\nu}_{g,h,\ell}(X,Y,Z) :\ & (((X \otimes Y) \otimes P_{\nu(g,h)}) \otimes Z) \otimes P_{\nu(gh,\ell)} \\ & \longrightarrow (X \otimes ((Y \otimes Z) \otimes P_{\nu(h,\ell)})) \otimes P_{\nu(g,h\ell)} \end{aligned} \tag{3.54}$$

by

$$\begin{aligned} A^{\nu}_{g,h,\ell}(X,Y,Z) = {} & \big(\mathrm{id}_{X\otimes(Y\otimes Z)} \otimes m^{-1}_{\nu(h,\ell),\nu(g,h\ell)} m_{\nu(g,h),\nu(gh,\ell)}\big) \\ & \circ \big(((\mathrm{id}_X \otimes \mathrm{id}_Y) \otimes c_{\nu(g,h),Z}) \otimes \mathrm{id}_{P_{\nu(gh,\ell)}}\big), \end{aligned} \tag{3.55}$$

with associators suppressed. Finally, define even isomorphisms

$$\begin{aligned} \tilde{l}_X &= l_X r_{\mathbf{1}\otimes X} : (\mathbf{1} \otimes X) \otimes \mathbf{1} \longrightarrow X, \\ \tilde{r}_X &= r_X r_{X\otimes\mathbf{1}} : (X \otimes \mathbf{1}) \otimes \mathbf{1} \longrightarrow X. \end{aligned} \tag{3.56}$$

Definition 3.22. Let $\mathcal{C}$ be a Brauer–Wall reference category. For each finite family $(X_i)_{i\in I}$ of objects of $\mathcal{C}$, fix a direct sum $\bigoplus_{i\in I} X_i$ with even inclusions ι_i and projections p_i. If $X_i = 0$ for all $i \neq j$, we choose X_j itself as the direct sum, with $\iota_j = p_j = \mathrm{id}_{X_j}$. In the formulas below, we identify $\bigoplus_i (\bigoplus_j X_{i,j})$ with $\bigoplus_{(i,j)} X_{i,j}$, and $(\bigoplus_i X_i) \otimes (\bigoplus_j Y_j)$ with $\bigoplus_{(i,j)} (X_i \otimes Y_j)$, using the isomorphisms induced by the inclusions and projections.

For $t = (G, \rho, \nu, \omega) \in \mathrm{Tw}$, we define a monoidal Π-supercategory $\mathcal{P}_{\mathcal{C}}(t)$ as follows. Its objects are families $X = (X_g)_{g\in G}$ with $X_g \in \mathcal{C}_{\rho(g)}$. Only the components $\mathcal{C}_r$ with $r \in \rho(G)$ enter the construction, so $\mathcal{C}$ may be replaced by $\mathcal{C}_{\rho(G)} \in \mathcal{W}$ of (3.45) in the statements below. Its morphism spaces are

$$\mathcal{P}_{\mathcal{C}}(t)(X,Y) = \bigoplus_{g\in G} \mathcal{C}(X_g, Y_g). \tag{3.57}$$

Identities and composition are defined componentwise. The tensor product is

$$\begin{aligned} (X \otimes_t Y)_\ell &= \bigoplus_{gh=\ell} \big((X_g \otimes Y_h) \otimes P_{\nu(g,h)}\big), \\ (f \otimes_t f')_\ell &= \bigoplus_{gh=\ell} \big((f_g \otimes f'_h) \otimes \mathrm{id}_{P_{\nu(g,h)}}\big). \end{aligned} \tag{3.58}$$

The unit object $\mathbf{1}_t$, parity object π_t, and odd isomorphism $\zeta_t : \pi_t \to \mathbf{1}_t$ are supported at $1 \in G$, with components $\mathbf{1}, \pi, \zeta$, respectively. The associator $\alpha^t_{X,Y,Z} : (X \otimes_t Y) \otimes_t Z \to X \otimes_t (Y \otimes_t Z)$ is

$$(\alpha^t_{X,Y,Z})_r = \bigoplus_{gh\ell=r} \omega(g,h,\ell)\, A^{\nu}_{g,h,\ell}(X_g, Y_h, Z_\ell). \tag{3.59}$$

The unitors $l^t_X : \mathbf{1}_t \otimes_t X \to X$ and $r^t_X : X \otimes_t \mathbf{1}_t \to X$ are

$$(l^t_X)_g = \tilde{l}_{X_g}, \qquad (r^t_X)_g = \tilde{r}_{X_g}. \tag{3.60}$$

The pentagon and triangle axioms follow from Proposition 3.23 below.

Proposition 3.23. *Let $\mathcal{C}$ be a Brauer–Wall reference category and let $t \in \mathrm{Tw}$. The associator α^t and unitors l^t, r^t defined in Definition 3.22 are even supernatural isomorphisms satisfying the pentagon and triangle identities.*

Proof. We first record the identities used in the calculation. Let $a, b, a' \in \mathbb{Z}/2\mathbb{Z}$ and let X, Y be objects of $\mathcal{C}$. Substitution in (3.53) gives

$$c_{0,X} = r_X^{-1} l_X, \qquad m_{0,b} = l_{P_b}, \qquad m_{b,0} = r_{P_b}, \tag{3.61}$$

and, with associators and unitors suppressed,

$$c_{a,X\otimes Y} = (\mathrm{id}_X \otimes c_{a,Y})(c_{a,X} \otimes \mathrm{id}_Y), \tag{3.62}$$

$$m_{a+b,a'}(m_{a,b} \otimes \mathrm{id}_{P_{a'}}) = m_{a,b+a'}(\mathrm{id}_{P_a} \otimes m_{b,a'}), \tag{3.63}$$

$$c_{a+b,X}(m_{a,b} \otimes \mathrm{id}_X) = (\mathrm{id}_X \otimes m_{a,b})(c_{a,X} \otimes \mathrm{id}_{P_b})(\mathrm{id}_{P_a} \otimes c_{b,X}), \tag{3.64}$$

$$c_{a,P_b} = (-1)^{ab} m_{b,a}^{-1} m_{a,b}. \tag{3.65}$$

For (3.63), postcompose with $\zeta_{a+b+a'}$; with associators and unitors suppressed,

$$\begin{aligned} \zeta_{a+b+a'} m_{a+b,a'}(m_{a,b} \otimes \mathrm{id}_{P_{a'}}) &= (\zeta_a \otimes \zeta_b) \otimes \zeta_{a'} \\ &= \zeta_{a+b+a'} m_{a,b+a'}(\mathrm{id}_{P_a} \otimes m_{b,a'}). \end{aligned} \tag{3.66}$$

For (3.62) and (3.64), substitute the two factors of c and cancel adjacent $\zeta_a^{-1}\zeta_a$. The same cancellation in (3.65) leaves one interchange of ζ_a and ζ_b, giving $(-1)^{ab}$. These verifications use only (3.8) and monoidal coherence, so apply to arbitrary objects of $\mathcal{C}$. For a homogeneous $f : X \to Y$, moving f through the two degree-a maps in c_a contributes $(-1)^{a|f|}$ twice. Thus

$$c_{a,Y}(\mathrm{id}_{P_a} \otimes f) = (-1)^{2a|f|}(f \otimes \mathrm{id}_{P_a}) c_{a,X} = (f \otimes \mathrm{id}_{P_a}) c_{a,X}. \tag{3.67}$$

Consequently (3.55) is even supernatural.

Let $g, h, \ell, m \in G$ and let X, Y, Z, W be objects of $\mathcal{C}$. We prove the following identity for the maps A^ν of (3.55):

$$\begin{aligned} &\big((\mathrm{id}_X \otimes A^\nu_{h,\ell,m}(Y,Z,W)) \otimes \mathrm{id}_{P_{\nu(g,h\ell m)}}\big) \\ &\quad \circ A^\nu_{g,h\ell,m}(X, (Y \otimes Z) \otimes P_{\nu(h,\ell)}, W) \\ &\quad \circ \big((A^\nu_{g,h,\ell}(X,Y,Z) \otimes \mathrm{id}_W) \otimes \mathrm{id}_{P_{\nu(gh\ell,m)}}\big) \\ &= (-1)^{\nu(g,h)\nu(\ell,m)} A^\nu_{g,h,\ell m}\big(X, Y, (Z \otimes W) \otimes P_{\nu(\ell,m)}\big) \\ &\quad \circ A^\nu_{gh,\ell,m}((X \otimes Y) \otimes P_{\nu(g,h)}, Z, W). \end{aligned} \tag{3.68}$$

For $g, h \in G$ and objects X, Y of $\mathcal{C}$, we have the triangle identity

$$\big((\mathrm{id}_X \otimes \tilde{l}_Y) \otimes \mathrm{id}_{P_{\nu(g,h)}}\big) A^\nu_{g,1,h}(X, \mathbf{1}, Y) = (\tilde{r}_X \otimes \mathrm{id}_Y) \otimes \mathrm{id}_{P_{\nu(g,h)}}. \tag{3.69}$$

Let T_3 be the composite of the three maps on the left of (3.68) and let T_2 be the composite of the two maps on the right, without the factor $(-1)^{\nu(g,h)\nu(\ell,m)}$. Put

$$\begin{aligned} (a_1, a_2, a_3) &= (\nu(g,h), \nu(gh,\ell), \nu(gh\ell, m)), \\ (b_1, b_2, b_3) &= (\nu(\ell,m), \nu(h,\ell m), \nu(g, h\ell m)). \end{aligned} \tag{3.70}$$

The cocycle equation gives $a_1 + a_2 + a_3 = b_1 + b_2 + b_3$. Put $Q = ((X \otimes Y) \otimes Z) \otimes W$ and postcompose both routes with the even isomorphism

$$\begin{aligned} \kappa &= \mathrm{id}_Q \otimes \big(m_{b_1+b_2,b_3}(m_{b_1,b_2} \otimes \mathrm{id}_{P_{b_3}})\big) \\ &\quad : (X \otimes ((Y \otimes ((Z \otimes W) \otimes P_{b_1})) \otimes P_{b_2})) \otimes P_{b_3} \longrightarrow Q \otimes P_{a_1+a_2+a_3}, \end{aligned} \tag{3.71}$$

with associators suppressed. Define, with the same convention,

$$\begin{aligned}\Psi = {} & (\mathrm{id}_Q \otimes m_{a_1+a_2,a_3}) \\ & \circ ((\mathrm{id}_{(X\otimes Y)\otimes Z} \otimes c_{a_1+a_2,W}) \otimes \mathrm{id}_{P_{a_3}}) \\ & \circ (((\mathrm{id}_{(X\otimes Y)\otimes Z} \otimes m_{a_1,a_2}) \otimes \mathrm{id}_W) \otimes \mathrm{id}_{P_{a_3}}) \\ & \circ ((((\mathrm{id}_{X\otimes Y} \otimes c_{a_1,Z}) \otimes \mathrm{id}_{P_{a_2}}) \otimes \mathrm{id}_W) \otimes \mathrm{id}_{P_{a_3}}).\end{aligned} \tag{3.72}$$

Its source is the common source of T_3, T_2, and its target is $Q \otimes P_{a_1+a_2+a_3}$. We have

$$\kappa T_3 = \Psi, \qquad \kappa T_2 = (-1)^{a_1 b_1}\Psi. \tag{3.73}$$

For the first identity, expand the three associators using (3.55). Equation (3.64) combines the crossings past W into $c_{a_1+a_2,W}$ after contracting $P_{a_1} \otimes P_{a_2}$. Equation (3.63) then cancels the intermediate contractions and their inverses, leaving the four factors of (3.72). For the second identity, (3.62) separates the crossing in $A^{\nu}_{g,h,\ell m}$ into crossings past Z, W, and P_{b_1}. Replacing the last by $c_{a_1,P_{b_1}} = (-1)^{a_1 b_1} m^{-1}_{b_1,a_1} m_{a_1,b_1}$ leaves the same contractions and crossings, by (3.64) and (3.63). This proves (3.73). Cancelling κ gives (3.68).

For (3.69), normalization gives $\nu(g,1) = \nu(1,h) = 0$. In (3.55) the two contractions are then $m_{0,\nu(g,h)}$ and $m^{-1}_{0,\nu(g,h)}$. They cancel, and $c_{0,Y}$ is the unitor comparison in (3.61). Both sides reduce to the unitor removing the two inserted copies of $\mathbf{1}$, with the same parentheses; monoidal coherence gives the asserted equality.

On a summand indexed by (g,h,ℓ,m), the ratio of the scalar on the three-edge route for α^t to that on the two-edge route is

$$\frac{\omega(h,\ell,m)\omega(g,h\ell,m)\omega(g,h,\ell)}{\omega(gh,\ell,m)\omega(g,h,\ell m)} = \delta\omega(g,h,\ell,m) = (-1)^{\nu(g,h)\nu(\ell,m)}. \tag{3.74}$$

This cancels the sign in (3.68). The triangle follows from (3.69) and $\omega(g,1,h) = 1$. Equality on all summand inclusions proves the identities for arbitrary families. □

Proposition 3.24. *Let $\mathcal{C}$ be a Brauer–Wall reference category and $t = (G,\rho,\nu,\omega)$ a twist. Then $\mathcal{P}_{\mathcal{C}}(t) \in \mathcal{W}$. Choose simple $S_r \in \mathcal{C}_r$ with $S_0 = \mathbf{1}$, and let X_g be $S_{\rho(g)}$ supported at g. Every simple object is isomorphic to a unique X_g. The map $g \mapsto [X_g]$ is a group isomorphism*

$$G \xrightarrow{\sim} G_{\mathcal{P}_{\mathcal{C}}(t)}, \qquad \rho_{\mathcal{P}_{\mathcal{C}}(t)}([X_g]) = \rho(g). \tag{3.75}$$

Proof. Put $\mathcal{P} = \mathcal{P}_{\mathcal{C}}(t)$, and write $S[g]$ for the family supported at g with value $S \in \mathcal{C}_{\rho(g)}$. Equation (3.57) gives

$$\underline{\mathcal{P}} = \prod_{g\in G} \underline{\mathcal{C}_{\rho(g)}}, \qquad V \cong \bigoplus_{g\in G} V_g[g] \quad \text{in } \underline{\mathcal{P}} \tag{3.76}$$

for every object $V = (V_g)_{g\in G}$ of $\mathcal{P}$. Each factor is a full semisimple abelian subcategory closed under direct sums and retracts. Since G is finite, $\underline{\mathcal{P}}$ is semisimple abelian, its Hom spaces are finite-dimensional, and it has finitely many simple objects up to even isomorphism. The tensor product is bilinear, its unit is simple, and $\mathrm{End}_{\mathcal{P}}(\mathbf{1}_t) \cong \mathrm{End}_{\mathcal{C}}(\mathbf{1}) \cong k$.

By (3.76), a simple family has exactly one nonzero component, and that component is simple. It is therefore evenly isomorphic to $S[g]$ for a simple $S \in \mathcal{C}_{\rho(g)}$. The reference condition makes S isomorphic to $S_{\rho(g)}$. Distinct supports have zero Hom spaces, so the X_g represent the simple isomorphism classes without repetition, and

$$\mathrm{End}_{\mathcal{P}}(X_g) \cong \mathrm{End}_{\mathcal{C}}(S_{\rho(g)}) \cong D_{\rho(g)}. \tag{3.77}$$

We construct duals before applying any result about $\mathcal{W}$ to $\mathcal{P}$. Let $g \in G$, let $S \in \mathcal{C}_{\rho(g)}$ be simple, and put $X = S[g]$. Choose a left dual S^* of S in $\underline{\mathcal{C}}$, with evaluation ev_S and coevaluation coev_S. Put

$$a = \nu(g, g^{-1}) = \nu(g^{-1}, g), \qquad T = S^* \otimes P_a, \qquad Y = T[g^{-1}]. \tag{3.78}$$

The cocycle identity at (g, g^{-1}, g) gives $\nu(g, g^{-1}) = \nu(g^{-1}, g)$. Duality in $\underline{\mathcal{C}}$ preserves simplicity. Right tensoring with P_a is an equivalence, since $m_{a,a}$ identifies its square with the identity functor. Thus T is simple. Since $\rho_{\mathcal{C}}([P_a]) = 0$, equations (3.29) and (3.34) give $T \in \mathcal{C}_{-\rho(g)}$; hence Y is an object of $\mathcal{P}$.

Define even morphisms $\eta : \mathbf{1}_t \to X \otimes_t Y$ and $\vartheta : Y \otimes_t X \to \mathbf{1}_t$, supported at 1, by

$$\begin{aligned} \eta_1 &= \iota_{(g,g^{-1})}(\mathrm{id}_{S\otimes S^*} \otimes m_{a,a}^{-1}) r_{S\otimes S^*}^{-1} \,\mathrm{coev}_S, \\ \vartheta_1 &= \omega(g, g^{-1}, g)^{-1} r_{\mathbf{1}}(\mathrm{ev}_S \otimes m_{a,a}) \\ &\qquad \circ ((\mathrm{id}_{S^*} \otimes c_{a,S}) \otimes \mathrm{id}_{P_a}) p_{(g^{-1},g)}. \end{aligned} \tag{3.79}$$

The inclusion and projection select the indicated convolution summands; associators in $\mathcal{C}$ are suppressed. Consider the first snake composite

$$s = r_X^t(\mathrm{id}_X \otimes_t \vartheta)\alpha_{X,Y,X}^t(\eta \otimes_t \mathrm{id}_X)(l_X^t)^{-1} = \mathrm{id}_X. \tag{3.80}$$

Only the (g, g^{-1}, g) summand contributes, and the four indices of A^ν are $(a, 0, a, 0)$. The crossing in that associator and the crossing in ϑ_1 combine with the inserted pair of parity factors to give

$$\begin{aligned} &r_S(\mathrm{id}_S \otimes m_{a,a})(c_{a,S} \otimes \mathrm{id}_{P_a}) \\ &\quad \circ (\mathrm{id}_{P_a} \otimes c_{a,S})(m_{a,a}^{-1} \otimes \mathrm{id}_S) l_S^{-1} = \mathrm{id}_S, \end{aligned} \tag{3.81}$$

with associators suppressed. Equation (3.64) with both indices a reduces its left side to $c_{0,S}$ between the two unitors, and (3.61) gives the identity. All structural maps here are even, so their interchange with ev_S contributes no sign. After this cancellation, the composite is the first snake for S, multiplied by $\omega(g, g^{-1}, g)$ and its reciprocal. This proves (3.80).

Let

$$e = l_Y^t(\vartheta \otimes_t \mathrm{id}_Y)(\alpha_{Y,X,Y}^t)^{-1}(\mathrm{id}_Y \otimes_t \eta)(r_Y^t)^{-1}. \tag{3.82}$$

To check the second snake, expand e^2 and bring the middle evaluation and coevaluation together. The even interchange law gives, with associators and unitors suppressed,

$$\begin{aligned} e^2 &= (\vartheta \otimes_t \mathrm{id}_Y)((\mathrm{id}_Y \otimes_t s) \otimes_t \mathrm{id}_Y)(\mathrm{id}_Y \otimes_t \eta) = e, \\ (\mathrm{id}_X \otimes_t e)\eta &= (s \otimes_t \mathrm{id}_Y)\eta = \eta. \end{aligned} \tag{3.83}$$

The first snake and $X \neq 0$ imply $\eta \neq 0$; the second equality then gives $e \neq 0$. Since T is simple in $\underline{\mathcal{C}}$, equation (3.16) makes $\mathrm{End}_{\mathcal{P}}(Y) = \mathrm{End}_{\mathcal{C}}(T)$ a division superalgebra. The nonzero even idempotent e is therefore the identity by Lemma 2.2. This proves both snakes, with no assumption that $\mathrm{End}_{\mathcal{C}}(T)_0 = k$.

The left duals just constructed are simple. Their assignment is injective on even simple isomorphism classes: two objects with evenly isomorphic left duals are evenly isomorphic by uniqueness of a right dual of that common object. Finiteness makes this assignment surjective, so every simple also has a right dual. Let $V = \bigoplus_i V_i$ be an even direct sum of simple objects V_i of $\mathcal{P}$, with inclusions i_i and projections p_i; let V_i^* be the left dual of V_i constructed above, with evaluation ϑ_i and coevaluation η_i, and let $V^* = \bigoplus_i V_i^*$ with inclusions j_i and projections q_i. Set

$$\mathrm{ev}_V = \sum_i \vartheta_i(q_i \otimes_t p_i), \qquad \mathrm{coev}_V = \sum_i (i_i \otimes_t j_i)\eta_i. \tag{3.84}$$

The biproduct identities reduce both snakes to those for the V_i. The same formulas give right duals, proving rigidity of $\underline{\mathcal{P}}$.

Every simple in $\underline{\mathcal{C}}$ of Brauer–Wall type zero is homogeneously isomorphic to $\mathbf{1}$. An odd such isomorphism becomes even after composition with ζ^{-1}, with target π. Consequently every object of the neutral component of $\mathcal{C}$ is an even direct sum of $\mathbf{1}$ and π. The product of $S[g]$ with its constructed left dual is supported at 1, and is therefore an even direct sum of $\mathbf{1}_t$ and π_t. Together with the split unit and the fusion properties already proved, this gives $\mathcal{P} \in \mathcal{W}$.

The nonzero tensor product $X_g \otimes_t X_h$ is supported at gh: nonvanishing follows from (3.17) in $\mathcal{C}$ and invertibility of right tensoring with $P_{\nu(g,h)}$. Hence all its simple summands are isomorphic to X_{gh}. The established bijection $g \mapsto [X_g]$ preserves multiplication and the unit, giving the group isomorphism in (3.75). Finally (3.77) gives its compatibility with the type homomorphisms. □

Definition 3.25. Let $\mathcal{A} \in \mathcal{W}$ and let $\mathcal{C}$ be a Brauer–Wall reference category. Put $G = G_{\mathcal{A}}$ and $\rho = \rho_{\mathcal{A}}$. *Normalized comparison data* for $\mathcal{A}$ consist of a pair $N = (F, \theta)$ of a family $F = (F_g)_{g \in G}$ of superequivalences and a family $\theta = (\theta_{g,h})_{g,h \in G}$ of homogeneous supernatural isomorphisms,

$$\begin{aligned} F_g &: \mathcal{A}_g \longrightarrow \mathcal{C}_{\rho(g)} && (g \in G), \\ \theta_{g,h}(X,Y) &: F_{gh}(X \otimes Y) \longrightarrow F_g X \otimes F_h Y && (g,h \in G,\ X \in \mathcal{A}_g,\ Y \in \mathcal{A}_h), \end{aligned} \tag{3.85}$$

where $\theta_{g,h}$ is supernatural between the superfunctors $(X,Y) \mapsto F_{gh}(X \otimes Y)$ and $(X,Y) \mapsto F_g X \otimes F_h Y$ from $\mathcal{A}_g \boxtimes \mathcal{A}_h$ to $\mathcal{C}_{\rho(gh)}$, such that $F_1 \mathbf{1} = \mathbf{1}$, $F_1 \pi = \pi$, $F_1 \zeta = \zeta$, and

$$\theta_{1,h}(\mathbf{1}, Y) = l^{-1}_{F_h Y} F_h(l_Y), \qquad \theta_{g,1}(X, \mathbf{1}) = r^{-1}_{F_g X} F_g(r_X). \tag{3.86}$$

Proposition 3.26. *Normalized comparison data exist for every $\mathcal{A} \in \mathcal{W}$ and every Brauer–Wall reference category $\mathcal{C}$. Let $N = (F, \theta)$ be normalized comparison data for $\mathcal{A}$. Then $\nu(g,h) = |\theta_{g,h}|$ defines a normalized cocycle $\nu \in C^2(G, \mathbb{Z}/2\mathbb{Z})$. For $V \in \mathcal{C}$, put $e_{a,V} = r_V(\mathrm{id}_V \otimes \zeta_a)$, a homogeneous isomorphism of degree a. For $X \in \mathcal{A}_g$ and $Y \in \mathcal{A}_h$, define the even isomorphism*

$$\begin{aligned} K_{g,h}(X,Y) &= e^{-1}_{\nu(g,h),\, F_g X \otimes F_h Y} \theta_{g,h}(X,Y) \\ &\quad : F_{gh}(X \otimes Y) \longrightarrow (F_g X \otimes F_h Y) \otimes P_{\nu(g,h)}. \end{aligned} \tag{3.87}$$

For $X \in \mathcal{A}_g$, $Y \in \mathcal{A}_h$, $Z \in \mathcal{A}_\ell$, define even isomorphisms

$$\begin{aligned} \mathcal{L}_{X,Y,Z} &= A^{\nu}_{g,h,\ell}(F_g X, F_h Y, F_\ell Z) \\ &\quad \circ \big((K_{g,h}(X,Y) \otimes \mathrm{id}_{F_\ell Z}) \otimes \mathrm{id}_{P_{\nu(gh,\ell)}}\big) K_{gh,\ell}(X \otimes Y, Z), \\ \mathcal{R}_{X,Y,Z} &= \big((\mathrm{id}_{F_g X} \otimes K_{h,\ell}(Y,Z)) \otimes \mathrm{id}_{P_{\nu(g,h\ell)}}\big) \\ &\quad \circ K_{g,h\ell}(X, Y \otimes Z) F_{gh\ell}(\alpha^{\mathcal{A}}_{X,Y,Z}) \\ &\quad : F_{gh\ell}((X \otimes Y) \otimes Z) \\ &\quad \longrightarrow (F_g X \otimes ((F_h Y \otimes F_\ell Z) \otimes P_{\nu(h,\ell)})) \otimes P_{\nu(g,h\ell)}. \end{aligned} \tag{3.88}$$

There is a unique map $\omega : G^3 \to k^\times$ such that

$$\mathcal{R}_{X,Y,Z} = \omega(g,h,\ell) \mathcal{L}_{X,Y,Z} \tag{3.89}$$

for all X, Y, Z in these components. Then $t = (G, \rho, \nu, \omega)$ is a twist.

Proof. Equation (3.37) supplies the component superequivalences F_g. In the neutral component the matrix construction in its proof can be made with $\mathbf{1}$ and π as the chosen representatives, using ζ as the odd coordinate map. It then gives $F_1\mathbf{1} = \mathbf{1}$, $F_1\pi = \pi$, and $F_1\zeta = \zeta$. For each g, h, the two binary functors in (3.85) satisfy (3.36), by (3.43) and full faithfulness of the F_g. Choose a nonzero element of their comparison line. When $g = 1$ or $h = 1$, choose its value at the simple object containing $\mathbf{1}$ to be $l^{-1}_{F_hY}F_h(l_Y)$, respectively $r^{-1}_{F_gX}F_g(r_X)$, as in (3.86). These are even intertwiners and hence extend uniquely by (3.40). For $g = h = 1$ the two prescriptions agree since $l_{\mathbf{1}} = r_{\mathbf{1}}$. This proves existence of normalized comparison data, and also $\nu(1, h) = \nu(g, 1) = 0$.

Consider the two homogeneous transformations

$$\begin{aligned} B_L &= \alpha^{\mathcal{C}}(\theta_{g,h} \otimes \mathrm{id})\theta_{gh,\ell}, \\ B_R &= (\mathrm{id} \otimes \theta_{h,\ell})\theta_{g,h\ell}F_{gh\ell}(\alpha^{\mathcal{A}}), \end{aligned} \tag{3.90}$$

where the components are evaluated at X, Y, Z in that order. Both compare $F_{gh\ell}((X \otimes Y) \otimes Z)$ with $F_gX \otimes (F_hY \otimes F_\ell Z)$: they are homogeneous supernatural transformations between the superfunctors $T_L, T_R : \mathcal{A}_g \boxtimes \mathcal{A}_h \boxtimes \mathcal{A}_\ell \to \mathcal{C}_{\rho(gh\ell)}$ with $T_L(X, Y, Z) = F_{gh\ell}((X \otimes Y) \otimes Z)$ and $T_R(X, Y, Z) = F_gX \otimes (F_hY \otimes F_\ell Z)$. Repeated use of (3.43) shows that T_L and T_R satisfy (3.36) at simple objects, so $\mathrm{Nat}(T_L, T_R)$ is a line by Proposition 3.15, in a single degree; hence

$$\nu(g, h) + \nu(gh, \ell) = \nu(h, \ell) + \nu(g, h\ell). \tag{3.91}$$

Thus $\delta\nu = 0$, and A^ν in (3.88) is defined. The maps $e_{a,V}$ are supernatural of degree a by (3.8); consequently each $K_{g,h}$ is even supernatural.

Let $T'_L, T'_R : \mathcal{A}_g \boxtimes \mathcal{A}_h \boxtimes \mathcal{A}_\ell \to \mathcal{C}$ be the superfunctors with $T'_L(X, Y, Z) = F_{gh\ell}((X \otimes Y) \otimes Z)$ and $T'_R(X, Y, Z) = (F_gX \otimes ((F_hY \otimes F_\ell Z) \otimes P_{\nu(h,\ell)})) \otimes P_{\nu(g,h\ell)}$, the source and target of $\mathcal{L}_{X,Y,Z}$ and $\mathcal{R}_{X,Y,Z}$ in (3.88). Since tensoring with P_a is a superequivalence, T'_L and T'_R satisfy (3.36) at simple objects as above, so by (3.42)

$$\mathrm{Nat}(T'_L, T'_R) = k\mathcal{L}, \qquad \mathcal{R} = \omega(g, h, \ell)\mathcal{L}, \quad \omega(g, h, \ell) \neq 0, \tag{3.92}$$

which defines $\omega : G^3 \to k^\times$ and proves (3.89). To check the cocycle equation, define for $X \in \mathcal{A}_g$, $Y \in \mathcal{A}_h$, and $Z \in \mathcal{A}_\ell$ the transported associator

$$\begin{aligned} \tilde{\alpha}_{X,Y,Z} = {} & ((\mathrm{id}_{F_gX} \otimes K_{h,\ell}(Y, Z)) \otimes \mathrm{id})K_{g,h\ell}(X, Y \otimes Z)F_{gh\ell}(\alpha^{\mathcal{A}}_{X,Y,Z}) \\ & \circ K_{gh,\ell}(X \otimes Y, Z)^{-1}((K_{g,h}(X, Y)^{-1} \otimes \mathrm{id}) \otimes \mathrm{id}), \end{aligned} \tag{3.93}$$

with the source and target of $A^\nu_{g,h,\ell}(F_gX, F_hY, F_\ell Z)$; by (3.88) and (3.89), $\tilde{\alpha}_{X,Y,Z} = \mathcal{R}_{X,Y,Z}\mathcal{L}^{-1}_{X,Y,Z}A^\nu_{g,h,\ell}(F_gX, F_hY, F_\ell Z) = \omega(g, h, \ell)A^\nu_{g,h,\ell}(F_gX, F_hY, F_\ell Z)$. Let $m \in G$ and $W \in \mathcal{A}_m$. The pentagon for $\tilde{\alpha}$ at (X, Y, Z, W) is formed with $\tilde{\alpha}$ at the compound arguments $X \otimes Y$, $Y \otimes Z$, and $Z \otimes W$ transported along the even isomorphisms K; for instance, $\tilde{\alpha}_{X\otimes Y,Z,W}$ conjugated by $K_{g,h}(X, Y)$ is $\omega(gh, \ell, m)A^\nu_{gh,\ell,m}((F_gX \otimes F_hY) \otimes P_{\nu(g,h)}, F_\ell Z, F_mW)$, by the even supernaturality of A^ν in each of its object arguments. In the two fourfold composites of this pentagon, the internal pairs $K^{-1}K$ cancel, leaving the two sides of the pentagon of $\mathcal{A}$ at (X, Y, Z, W), applied by $F_{gh\ell m}$ and conjugated by K; so $\tilde{\alpha}$ satisfies the pentagon without any assumption on ω. The identity (3.68) for the maps A^ν, whose proof used only $\delta\nu = 0$, reduces that equality to

$$\omega(h, \ell, m)\omega(g, h\ell, m)\omega(g, h, \ell) = (-1)^{\nu(g,h)\nu(\ell,m)}\omega(gh, \ell, m)\omega(g, h, \ell m) \tag{3.94}$$

for all $g, h, \ell, m \in G$; the common isomorphism multiplying both sides is cancelled by evaluating at nonzero simple objects of the four components, whose iterated tensor product is nonzero by (3.43). This is $\delta\omega = (-1)^{\nu\smile\nu}$.

The unit comparisons (3.86) transport the unitors to (3.56). Comparing the triangle with (3.69), and then evaluating (3.94) at $(1,1,g,h)$ and $(g,h,1,1)$, gives

$$\begin{aligned} &\omega(g,1,h) = 1, \\ &\omega(1,g,h)^2 = \omega(1,g,h), \quad \omega(g,h,1)^2 = \omega(g,h,1). \end{aligned} \tag{3.95}$$

All values are nonzero, so ω is normalized. Together with (3.91) and (3.94), this proves (3.47). □

Proposition 3.27. *Let $\mathcal{A} \in \mathcal{W}$ and let $\mathcal{C}$ be a Brauer–Wall reference category. Let $N = (F, \theta)$ be normalized comparison data for $\mathcal{A}$, and let $K_{g,h}$ and t be as in Proposition 3.26. The following data define a monoidal superequivalence $(F, J, \varphi) : \mathcal{A} \to \mathcal{P}_{\mathcal{C}}(t)$. For every object X of $\mathcal{A}$ choose a decomposition $X \cong \bigoplus_g X_g$ into components, with even inclusions i_g^X and projections p_g^X. The superfunctor F is*

$$(FX)_g = F_g X_g, \qquad (Ff)_g = F_g(p_g^Y f i_g^X) \quad (f : X \to Y), \tag{3.96}$$

and the tensorator $J_{X,Y} : FX \otimes_t FY \to F(X \otimes Y)$ and the unit map $\varphi : \mathbf{1}_t \to F\mathbf{1}$ are

$$\begin{aligned} (J_{X,Y})_\ell &= \sum_{gh=\ell} F_\ell(p_\ell^{X\otimes Y}(i_g^X \otimes i_h^Y))\, K_{g,h}(X_g, Y_h)^{-1}\, p_{(g,h)}, \\ \varphi_1 &= F_1(p_1^{\mathbf{1}}), \qquad \varphi_g = 0 \quad (g \neq 1), \end{aligned} \tag{3.97}$$

where $p_{(g,h)}$ is the projection of the direct sum $(FX \otimes_t FY)_\ell = \bigoplus_{gh=\ell}(F_g X_g \otimes F_h Y_h) \otimes P_{\nu(g,h)}$ onto its (g,h) summand, as fixed in Definition 3.22.

Proof. Let X, Y, Z be objects of $\mathcal{A}$. For $f : X \to Y$, equation (3.30) gives $p_h^Y f i_g^X = 0$ for $g \neq h$. For $f : X \to Y$ and $f' : Y \to Z$,

$$\begin{aligned} p_g^Z f' f i_g^X &= \sum_h p_g^Z f' i_h^Y p_h^Y f i_g^X = (p_g^Z f' i_g^Y)(p_g^Y f i_g^X), \\ (F(f'f))_g &= (Ff')_g (Ff)_g, \qquad (F\mathrm{id}_X)_g = \mathrm{id}_{F_g X_g}. \end{aligned} \tag{3.98}$$

The map $f \mapsto (p_g^Y f i_g^X)_g$ and full faithfulness of each F_g give isomorphisms

$$\mathcal{A}(X,Y) \xrightarrow{\sim} \bigoplus_g \mathcal{A}_g(X_g, Y_g) \xrightarrow{\bigoplus_g F_g} \bigoplus_g \mathcal{C}(F_g X_g, F_g Y_g) = \mathcal{P}_{\mathcal{C}}(t)(FX, FY). \tag{3.99}$$

For a family in the pullback, choose an even preimage in each component under F_g and take their direct sum in $\mathcal{A}$. This proves essential surjectivity in the underlying categories, and hence F is a superequivalence.

The morphisms $p_\ell^{X\otimes Y}(i_g^X \otimes i_h^Y)$, for $gh = \ell$, exhibit $(X \otimes Y)_\ell$ as the direct sum of the $X_g \otimes Y_h$. Indeed, their projections are $(p_g^X \otimes p_h^Y) i_\ell^{X\otimes Y}$, and the biproduct identities follow from those for X, Y, since all inclusions and projections are even. Consequently the inverse of the tensorator $J_{X,Y}$ is

$$(J_{X,Y}^{-1})_\ell = \sum_{gh=\ell} \iota_{(g,h)} K_{g,h}(X_g, Y_h) F_\ell((p_g^X \otimes p_h^Y) i_\ell^{X\otimes Y}). \tag{3.100}$$

Even supernaturality of K gives that of J. The map φ is an even isomorphism because $\mathbf{1}$ is concentrated in the neutral component.

For objects X, Y, Z of $\mathcal{A}$, the required coherence equations are

$$\begin{aligned} F(\alpha^{\mathcal{A}}_{X,Y,Z}) J_{X\otimes Y,Z}(J_{X,Y}\otimes_t \mathrm{id}_{FZ}) &= J_{X,Y\otimes Z}(\mathrm{id}_{FX}\otimes_t J_{Y,Z})\alpha^{t}_{FX,FY,FZ}, \\ F(l_X) J_{\mathbf{1},X}(\varphi\otimes_t \mathrm{id}_{FX}) &= l^{t}_{FX}, \\ F(r_X) J_{X,\mathbf{1}}(\mathrm{id}_{FX}\otimes_t \varphi) &= r^{t}_{FX}. \end{aligned} \tag{3.101}$$

On the (g,h,ℓ) summand, the first equality is (3.89) with tensorators K^{-1} and associator (3.59). The other two are (3.86), using $e_{0,V} = r_V$. The summand inclusions are jointly epimorphic, so these equalities hold on the whole direct sums. □

Proposition 3.28. *Let $s = (G,\rho,\nu,\omega)$ and $t = (H,\rho',\nu',\omega')$ be twists, and let $\mathcal{C}$ be a Brauer–Wall reference category. Then*

$$\mathcal{P}_{\mathcal{C}}(s) \simeq_{\otimes} \mathcal{P}_{\mathcal{C}}(t) \quad\Longleftrightarrow\quad s \sim t. \tag{3.102}$$

More precisely, a gauge (ϕ,λ,β) from s to t defines a monoidal superequivalence $(F,J,\varphi) : \mathcal{P}_{\mathcal{C}}(s) \to \mathcal{P}_{\mathcal{C}}(t)$ by

$$(FX)_{\phi(g)} = X_g \otimes P_{\lambda(g)}, \qquad (Ff)_{\phi(g)} = f_g \otimes \mathrm{id}_{P_{\lambda(g)}}, \tag{3.103}$$

$$\begin{aligned} (J_{X,Y})_{\phi(\ell)} &= \sum_{gh=\ell} \beta(g,h)\,(\iota_{(g,h)}\otimes \mathrm{id}_{P_{\lambda(\ell)}})\,\Gamma_{g,h}(X_g,Y_h)\,p_{(\phi(g),\phi(h))}, \\ \varphi_1 &= r_{\mathbf{1}}^{-1}, \qquad \varphi_h = 0 \quad (h\neq 1). \end{aligned} \tag{3.104}$$

Here $p_{(\phi(g),\phi(h))}$ is the projection of the fixed direct sum $(FX\otimes_t FY)_{\phi(\ell)}$ onto its $(\phi(g),\phi(h))$ summand, $\iota_{(g,h)}$ is the inclusion of the (g,h) summand of $(X\otimes_s Y)_\ell$, and $\Gamma_{g,h}(X_g,Y_h)$ is the even isomorphism

$$\begin{aligned} \Gamma_{g,h}(X_g,Y_h) = {} & \big(\mathrm{id}_{X_g\otimes Y_h}\otimes m^{-1}_{\nu(g,h),\lambda(gh)}\, m_{\lambda(g)+\lambda(h),\nu'(\phi(g),\phi(h))}\,(m_{\lambda(g),\lambda(h)}\otimes \mathrm{id}_{P_{\nu'(\phi(g),\phi(h))}})\big) \\ & \circ \big(((\mathrm{id}_{X_g}\otimes c_{\lambda(g),Y_h})\otimes \mathrm{id}_{P_{\lambda(h)}})\otimes \mathrm{id}_{P_{\nu'(\phi(g),\phi(h))}}\big) \\ & : ((X_g\otimes P_{\lambda(g)})\otimes(Y_h\otimes P_{\lambda(h)}))\otimes P_{\nu'(\phi(g),\phi(h))} \\ & \longrightarrow ((X_g\otimes Y_h)\otimes P_{\nu(g,h)})\otimes P_{\lambda(gh)} \end{aligned} \tag{3.105}$$

with associators suppressed. The contractions and the splitting are composable by (3.48)*. Conversely, every monoidal superequivalence E from $\mathcal{P}_{\mathcal{C}}(s)$ to $\mathcal{P}_{\mathcal{C}}(t)$ induces a gauge (ϕ,λ,β) from s to t, where $\phi : G\to H$ is the isomorphism induced by E on the groups* (3.75) *of simple objects. In particular, if $G = H$ and $E(X_g)\cong X_g$ for every $g\in G$, with X_g as in Proposition 3.24, then $(\mathrm{id}_G,\lambda,\beta)$ is a gauge from s to t.*

Proof. The construction of (F,J,φ) from a gauge proves the implication from right to left in (3.102), and the converse construction proves the implication from left to right. Write $\bar\nu = \phi^*\nu'$ and $\bar\omega = \phi^*\omega'$. The condition $\phi^*\rho' = \rho$ makes F well defined. For $a\in\mathbb{Z}/2\mathbb{Z}$, right tensoring with P_a is a superequivalence of $\mathcal{C}$: applying it twice is evenly naturally isomorphic to the identity by $m_{a,a}$ and the right unitor. Reindexing by ϕ therefore makes F a superequivalence. The formula for Γ is a composite of even supernatural isomorphisms, so J has the same properties.

We verify the associativity of J, including the scalar sign. Fix $g,h,\ell\in G$ and put

$$\begin{aligned} (a_1,a_2,a_3,a_{12},a_{23},a_{123}) &= (\lambda(g),\lambda(h),\lambda(\ell),\lambda(gh),\lambda(h\ell),\lambda(gh\ell)), \\ (n_1,n_2,n_3,n_4) &= (\nu(g,h),\nu(gh,\ell),\nu(h,\ell),\nu(g,h\ell)), \\ (n_1',n_2',n_3',n_4') &= (\bar\nu(g,h),\bar\nu(gh,\ell),\bar\nu(h,\ell),\bar\nu(g,h\ell)). \end{aligned} \tag{3.106}$$

Thus $n_1+n_2=n_3+n_4$, $n_1'+n_2'=n_3'+n_4'$, and, by (3.48),

$$\begin{aligned} a_1+a_2+n_1'&=n_1+a_{12}, & a_{12}+a_3+n_2'&=n_2+a_{123},\\ a_2+a_3+n_3'&=n_3+a_{23}, & a_1+a_{23}+n_4'&=n_4+a_{123}. \end{aligned} \tag{3.107}$$

Use the maps $e_{a,V}=r_V(\mathrm{id}_V\otimes\zeta_a)$ of Proposition 3.26 to delete parity factors. For objects U_1,U_2,U_3 of $\mathcal{C}$ and $u,v\in\mathbb{Z}/2\mathbb{Z}$, define the homogeneous isomorphisms of degree $u+v$

$$\begin{aligned} D_L^{U_1,U_2,U_3}(u,v)&=\alpha_{U_1,U_2,U_3}\,e_{v,(U_1\otimes U_2)\otimes U_3}\big((e_{u,U_1\otimes U_2}\otimes\mathrm{id}_{U_3})\otimes\mathrm{id}_{P_v}\big)\\ &\qquad:(((U_1\otimes U_2)\otimes P_u)\otimes U_3)\otimes P_v\longrightarrow U_1\otimes(U_2\otimes U_3),\\ D_R^{U_1,U_2,U_3}(u,v)&=e_{v,U_1\otimes(U_2\otimes U_3)}\big((\mathrm{id}_{U_1}\otimes e_{u,U_2\otimes U_3})\otimes\mathrm{id}_{P_v}\big)\\ &\qquad:(U_1\otimes((U_2\otimes U_3)\otimes P_u))\otimes P_v\longrightarrow U_1\otimes(U_2\otimes U_3). \end{aligned} \tag{3.108}$$

Now let $U_1\in\mathcal{C}_{\rho(g)}$, $U_2\in\mathcal{C}_{\rho(h)}$, and $U_3\in\mathcal{C}_{\rho(\ell)}$. Direct substitution of the maps $c_{-,-}$ and $m_{-,-}$ of (3.53) into (3.55) gives

$$D_R^{U_1,U_2,U_3}(n_3,n_4)\,A^{\nu}_{g,h,\ell}(U_1,U_2,U_3)=(-1)^{n_1n_2+n_3n_4}D_L^{U_1,U_2,U_3}(n_1,n_2). \tag{3.109}$$

For the binary comparison, with $V=U_1\otimes U_2$, the same substitution gives

$$\begin{aligned} &e_{a_{12},V}(e_{n_1,V}\otimes\mathrm{id}_{P_{a_{12}}})\Gamma_{g,h}(U_1,U_2)\\ &\quad=(-1)^{(a_1+a_2)n_1'+n_1a_{12}}e_{n_1',V}\big((e_{a_1,U_1}\otimes e_{a_2,U_2})\otimes\mathrm{id}_{P_{n_1'}}\big). \end{aligned} \tag{3.110}$$

For these calculations, a contraction $m_{u,v}$ followed by a deletion differs from two successive deletions by $(-1)^{uv}$, for $u,v\in\mathbb{Z}/2\mathbb{Z}$ the two contracted degrees. This follows from $\zeta_{u+v}m_{u,v}=l_{\mathbf{1}}(\zeta_u\otimes\zeta_v)$ and (3.8). It gives $n_1n_2+n_3n_4$ in (3.109). In (3.110), cancel the crossing c_{a_1,U_2} against the two component deletions; the two remaining contractions give $(a_1+a_2)n_1'+n_1a_{12}$.

Let $\mathbf{X}=U_1[g]$, $\mathbf{Y}=U_2[h]$, and $\mathbf{Z}=U_3[\ell]$ be the families supported at g, h, and ℓ with values U_1, U_2, and U_3. The monoidal functor equation compares

$$\begin{aligned} &F(\alpha^{s}_{\mathbf{X},\mathbf{Y},\mathbf{Z}})J_{\mathbf{X}\otimes_s\mathbf{Y},\mathbf{Z}}(J_{\mathbf{X},\mathbf{Y}}\otimes_t\mathrm{id}_{F\mathbf{Z}}),\\ &J_{\mathbf{X},\mathbf{Y}\otimes_s\mathbf{Z}}(\mathrm{id}_{F\mathbf{X}}\otimes_t J_{\mathbf{Y},\mathbf{Z}})\alpha^{t}_{F\mathbf{X},F\mathbf{Y},F\mathbf{Z}}. \end{aligned} \tag{3.111}$$

On their component at $\phi(gh\ell)$, postcompose both with $e_{a_{123},U_1\otimes(U_2\otimes U_3)}(D_R^{U_1,U_2,U_3}(n_3,n_4)\otimes\mathrm{id}_{P_{a_{123}}})$. Apply (3.110) twice and (3.109) once to each route. Both become scalar multiples of the same homogeneous isomorphism

$$D_L^{U_1,U_2,U_3}(n_1',n_2')\Big(\big(((e_{a_1,U_1}\otimes e_{a_2,U_2})\otimes\mathrm{id}_{P_{n_1'}})\otimes e_{a_3,U_3}\big)\otimes\mathrm{id}_{P_{n_2'}}\Big). \tag{3.112}$$

The coefficients are respectively

$$\omega(g,h,\ell)\beta(g,h)\beta(gh,\ell)(-1)^{E_L},\qquad \bar{\omega}(g,h,\ell)\beta(h,\ell)\beta(g,h\ell)(-1)^{E_R}, \tag{3.113}$$

where

$$\begin{aligned} E_L&=(a_1+a_2)n_1'+n_1a_{12}+(a_{12}+a_3)n_2'+n_2a_{123}+a_3n_1+n_1n_2+n_3n_4,\\ E_R&=(a_2+a_3)n_3'+n_3a_{23}+(a_1+a_{23})n_4'+n_4a_{123}+a_1n_3'+n_1'n_2'+n_3'n_4'. \end{aligned} \tag{3.114}$$

Here a_3n_1 comes from interchanging the deletion of P_{n_1} with the degree-a_3 deletion on U_3 on the first route. The term a_1n_3' comes from interchanging the degree-a_1 deletion on U_1 with the deletion of the inner $P_{n_3'}$ on the second route. The other terms are exactly those

in the two deletion identities. Substitute $a_{12} = a_1 + a_2 + n_1 + n_1'$, $a_{23} = a_2 + a_3 + n_3 + n_3'$, $a_{123} = a_{12} + a_3 + n_2 + n_2'$, $n_4 = n_1 + n_2 + n_3$, and $n_4' = n_1' + n_2' + n_3'$ to obtain

$$E_L + E_R = n_1' a_3 + a_1 n_3 = \bar{\nu}(g,h)\lambda(\ell) + \lambda(g)\nu(h,\ell). \tag{3.115}$$

By the coboundary formula (A.1), equality of the coefficients is exactly (3.49). Cancelling the common deletion proves associativity on each summand, and hence on the direct sum. For the unit diagrams, set $g = 1$ or $h = 1$ in Γ. Since λ, ν, $\bar{\nu}$, and β are normalized, their values at arguments containing 1 are 0, respectively 1. Equation (3.61), together with $c_{a,\mathbf{1}} = l_{P_a}^{-1} r_{P_a}$, which follows from (3.53) by the naturality of l and r and $l_{\mathbf{1}} = r_{\mathbf{1}}$, then reduces both diagrams to the unitors, with unit map $r_{\mathbf{1}}^{-1}$. This proves the forward assertion.

Conversely, let $E : \mathcal{P}_{\mathcal{C}}(s) \to \mathcal{P}_{\mathcal{C}}(t)$ be a monoidal superequivalence. It preserves simple objects, their endomorphism superalgebras, and the simple summands of tensor products. Equation (3.75) therefore gives a group isomorphism $\phi : G \to H$ with $\phi^* \rho' = \rho$, namely the map induced by E on isomorphism classes of simple objects; if $G = H$ and $E(X_g) \cong X_g$ for all g, then $\phi = \mathrm{id}_G$. For $g \in G$ and an object U of $\mathcal{C}_{\rho(g)}$, write $U[g]$ for the family supported at g with value U; since E preserves components, $E(U[g])$ is supported at $\phi(g)$, and $\rho'(\phi(g)) = \rho(g)$. Define the superequivalence $E_g : \mathcal{C}_{\rho(g)} \to \mathcal{C}_{\rho(g)}$ by $E_g(U) = (E(U[g]))_{\phi(g)}$ on objects and $E_g(f) = (E(f[g]))_{\phi(g)}$ on morphisms. Equation (3.42) gives a homogeneous supernatural isomorphism $\eta_g : E_g \Rightarrow \mathrm{Id}$ of some degree $\lambda(g)$. At $g = 1$, the unit map of E supplies an even comparison at $\mathbf{1}$, so uniqueness of the comparison degree gives $\lambda(1) = 0$. The comparisons $e^{-1}_{\lambda(g),U} \eta_{g,U} : E_g(U) \to U \otimes P_{\lambda(g)}$ are even; let $\eta : E \Rightarrow F$ be the even supernatural isomorphism with these components on the families $U[g]$, extended to direct sums, where F is the superfunctor (3.103). Transporting the tensorator and the unit map of E along η gives a monoidal structure $(\widehat{J}, \widehat{\varphi})$ on F. For $U \in \mathcal{C}_{\rho(g)}$ and $V \in \mathcal{C}_{\rho(h)}$, let $\widehat{J}_{g,h}(U,V)$ be the component of $\widehat{J}_{U[g],V[h]}$ from the $(\phi(g),\phi(h))$ summand of $FU[g] \otimes_t FV[h]$ to $F(U[g] \otimes_s V[h])$; it is even and supernatural in (U,V).

Delete the parity factors from the source and target of $\widehat{J}_{g,h}$, as in (3.110). This gives a nonzero supernatural endomorphism of the binary tensor superfunctor $(U,V) \mapsto U \otimes V$ on $\mathcal{C}_{\rho(g)} \boxtimes \mathcal{C}_{\rho(h)}$ of degree $n_1 + a_{12} + a_1 + a_2 + n_1'$. Equations (3.42) and (3.43) give $\mathrm{Nat}(\otimes,\otimes) = k\mathrm{id}$. Consequently $n_1 + a_{12} + a_1 + a_2 + n_1' = 0$, that is, $\bar{\nu} = \nu + \delta\lambda$, and there is a unique scalar $\beta(g,h) \in k^\times$ such that

$$\widehat{J}_{g,h} = \beta(g,h)\Gamma_{g,h}; \tag{3.116}$$

this defines a function $\beta : G \times G \to k^\times$. The associativity calculation (3.109)–(3.115) applies to these coefficients as well. On nonzero simple inputs the common deletion is a nonzero isomorphism, so the coherence of $\widehat{J}$ forces (3.49).

The function β need not be normalized. Apply its scalar equation at $(1,1,g)$ and $(g,1,1)$. Normalization of the twists and $\lambda(1) = 0$ give $\beta(1,g) = \beta(1,1) = \beta(g,1)$. Put $\beta_0 = \beta/\beta(1,1) \in C^2(G,k^\times)$; then β_0 is normalized and $\delta\beta_0 = \delta\beta$. Thus (ϕ, λ, β_0) satisfies (3.48)–(3.49). □

Theorem 3.29. *Let $\mathcal{C}$ be a Brauer–Wall reference category. The map $t \mapsto \mathcal{P}_{\mathcal{C}}(t)$ induces a bijection*

$$\mathrm{Tw}/\!\sim \xrightarrow{\sim} \mathcal{W}/\!\simeq_{\otimes}. \tag{3.117}$$

Proof. By Proposition 3.20, $\sim$ is an equivalence relation on Tw; and $\simeq_{\otimes}$ is an equivalence relation on $\mathcal{W}$, since a quasi-inverse E' of a monoidal superequivalence E, with the monoidal structure transported along the isomorphisms $EE' \cong \mathrm{id}$ and $E'E \cong \mathrm{id}$, is a monoidal superequivalence. By Proposition 3.24, $\mathcal{P}_{\mathcal{C}}(t) \in \mathcal{W}$ for every $t \in \mathrm{Tw}$. By (3.102), for $s,t \in \mathrm{Tw}$ we have $\mathcal{P}_{\mathcal{C}}(s) \simeq_{\otimes} \mathcal{P}_{\mathcal{C}}(t)$ if and only if $s \sim t$, so $t \mapsto \mathcal{P}_{\mathcal{C}}(t)$ induces a well-defined injective map $\mathrm{Tw}/\!\sim \to \mathcal{W}/\!\simeq_{\otimes}$. For $\mathcal{A} \in \mathcal{W}$, Proposition 3.26 provides normalized comparison data

and a twist t, and Proposition 3.27 a monoidal superequivalence $\mathcal{A} \to \mathcal{P}_{\mathcal{C}}(t)$; so the map is surjective. □

4 Stacking

Definition 4.1. Let $\mathcal{E}$ be a supercategory. An even idempotent $e : X \to X$ is *split* if there are an object Y and even morphisms $i : Y \to X$ and $p : X \to Y$ such that $i \circ p = e$ and $p \circ i = \mathrm{id}_Y$. The supercategory $\mathcal{E}$ has *split even idempotents* if every even idempotent in $\mathcal{E}$ is split. The *additive envelope* $\mathrm{Add}(\mathcal{E})$ is the supercategory whose objects are finite formal direct sums of objects of $\mathcal{E}$, and whose morphisms are matrices of morphisms in $\mathcal{E}$. A matrix has degree d if all its entries have degree d. The *even idempotent completion* $\mathrm{Kar}_0(\mathcal{E})$ is the supercategory whose objects are the pairs (X, e) such that $e \in \mathrm{End}_{\mathcal{E}}(X)_0$ and $e^2 = e$, and whose morphism spaces are

$$\mathrm{Kar}_0(\mathcal{E})((X, e), (Y, f)) = f\mathcal{E}(X, Y)e, \tag{4.1}$$

with the grading inherited from $\mathcal{E}(X, Y)$.

Definition 4.2. The *completion* of a supercategory $\mathcal{E}$ is

$$\overline{\mathcal{E}} = \mathrm{Kar}_0(\mathrm{Add}(\mathcal{E})). \tag{4.2}$$

Define a superfunctor

$$\iota : \mathcal{E} \longrightarrow \overline{\mathcal{E}}, \qquad X \longmapsto ((X), \mathrm{id}_X), \quad f \longmapsto (f). \tag{4.3}$$

In $\overline{\mathcal{E}}$, we write X for $\iota(X)$ and f for $\iota(f)$.

Proposition 4.3. *For every supercategory $\mathcal{E}$, its completion $\overline{\mathcal{E}}$ is additive and has split even idempotents. The superfunctor ι is fully faithful, and every object of $\overline{\mathcal{E}}$ is an even retract of a finite direct sum of objects of $\mathcal{E}$.*

Proof. For objects $X = (X_1, \ldots, X_n)$ and $Y = (Y_1, \ldots, Y_m)$ of $\mathrm{Add}(\mathcal{E})$, the concatenation $X \oplus Y = (X_1, \ldots, X_n, Y_1, \ldots, Y_m)$ with the block inclusions and projections is a direct sum in $\mathrm{Add}(\mathcal{E})$. For objects (X, e) and (Y, f) of $\overline{\mathcal{E}}$, with X, Y objects of $\mathrm{Add}(\mathcal{E})$ and e, f even idempotents,

$$(X, e) \oplus (Y, f) = \left(X \oplus Y, \begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix} \right) \tag{4.4}$$

is a direct sum in $\overline{\mathcal{E}}$, with the block inclusions and projections composed with the idempotents. Let $u : (X, e) \to (X, e)$ be even with $u^2 = u$; then $eu = u = ue$. Define $i : (X, u) \to (X, e)$ and $p : (X, e) \to (X, u)$ to be the morphism u; then

$$ip = u, \qquad pi = u = \mathrm{id}_{(X,u)}, \tag{4.5}$$

so i and p split u. For objects X, Y of $\mathcal{E}$, (4.1) gives

$$\overline{\mathcal{E}}(\iota X, \iota Y) = \mathcal{E}(X, Y), \tag{4.6}$$

with the original grading and composition, so ι is fully faithful. Finally, let (X, e) be an object of $\overline{\mathcal{E}}$ with $X = (X_1, \ldots, X_n)$. The morphisms $e : (X, e) \to (X, \mathrm{id}_X)$ and $e : (X, \mathrm{id}_X) \to (X, e)$ exhibit (X, e) as an even retract of (X, id_X), and $(X, \mathrm{id}_X) = \bigoplus_i \iota X_i$ by (4.4). □

Proposition 4.4. *Let $\mathcal{E}$ be a supercategory and let $\mathcal{D}$ be an additive supercategory with split even idempotents. Every superfunctor $F : \mathcal{E} \to \mathcal{D}$ admits an extension*

$$\overline{F} : \overline{\mathcal{E}} \longrightarrow \mathcal{D}, \qquad \overline{F} \circ \iota = F. \tag{4.7}$$

For any two such extensions $\overline{F}, \overline{F}'$, there is a unique even supernatural isomorphism

$$\theta : \overline{F} \xRightarrow{\sim} \overline{F}', \qquad \theta_X = \mathrm{id}_{FX} \quad (X \in \mathcal{E}). \tag{4.8}$$

For superfunctors $H, H' : \mathcal{E} \to \mathcal{D}$ and chosen extensions $\overline{H}, \overline{H}' : \overline{\mathcal{E}} \to \mathcal{D}$, extension induces an isomorphism of super vector spaces

$$\begin{gathered} \mathrm{Nat}(H, H') \xrightarrow{\sim} \mathrm{Nat}(\overline{H}, \overline{H}'), \qquad \theta \longmapsto \overline{\theta}, \\ \overline{\theta}_X = \theta_X \quad (X \in \mathcal{E}). \end{gathered} \tag{4.9}$$

These statements also hold for finitely many source variables, with each source completed separately.

Proof. Choose finite direct sums in $\mathcal{D}$, with inclusions $\iota_j^{\mathcal{D}}$ and projections $p_j^{\mathcal{D}}$. Define a superfunctor $F^+ : \mathrm{Add}(\mathcal{E}) \to \mathcal{D}$ on an object $X = (X_1, \dots, X_n)$ and a morphism $(f_{ji}) : (X_1, \dots, X_n) \to (Y_1, \dots, Y_m)$ by

$$F^+(X) = \bigoplus_i FX_i, \qquad F^+((f_{ji})) = \sum_{i,j} \iota_j^{\mathcal{D}} F(f_{ji}) p_i^{\mathcal{D}}, \tag{4.10}$$

taking $F^+(X_1) = FX_1$ for one-term sums. For every even idempotent e on an object X of $\mathrm{Add}(\mathcal{E})$, choose an object Q_e of $\mathcal{D}$ and even maps $I_e : Q_e \to F^+X$ and $P_e : F^+X \to Q_e$ with $I_e P_e = F^+(e)$ and $P_e I_e = \mathrm{id}_{Q_e}$, taking $Q_e = F^+X$ and $I_e = P_e = \mathrm{id}$ when $e = \mathrm{id}_X$. Define $\overline{F} : \overline{\mathcal{E}} \to \mathcal{D}$ by

$$\overline{F}(X, e) = Q_e, \qquad \overline{F}(u) = P_f F^+(u) I_e \quad (u : (X, e) \to (Y, f)). \tag{4.11}$$

For $v : (Y, f) \to (Z, \ell)$, the identities $fu = u = ue$ and $\ell v = v = vf$ give

$$\begin{aligned} \overline{F}(v)\overline{F}(u) &= P_\ell F^+(v) I_f P_f F^+(u) I_e = P_\ell F^+(vfu) I_e = \overline{F}(vu), \\ \overline{F}(e) &= P_e F^+(e) I_e = \mathrm{id}_{Q_e}. \end{aligned} \tag{4.12}$$

All splitting maps are even, so $\overline{F}$ is a superfunctor, and $\overline{F} \circ \iota = F$ by the choices for one-term sums and identity idempotents; this proves (4.7).

Let $\overline{F}'$ be any extension. For an object X of $\mathrm{Add}(\mathcal{E})$, the images under $\overline{F}'$ of the block inclusions and projections of $(X, \mathrm{id}_X) = \bigoplus_i \iota X_i$ give an even isomorphism $\kappa_X : F^+X \to \overline{F}'(X, \mathrm{id}_X)$, and for an even idempotent e on X the maps $I'_e = \kappa_X^{-1} \overline{F}'(e) : \overline{F}'(X, e) \to F^+X$ and $P'_e = \overline{F}'(e) \kappa_X : F^+X \to \overline{F}'(X, e)$, with e read as the morphisms $(X, e) \to (X, \mathrm{id}_X)$ and $(X, \mathrm{id}_X) \to (X, e)$, satisfy $I'_e P'_e = F^+(e)$ and $P'_e I'_e = \mathrm{id}$, and $\overline{F}'(u) = P'_f F^+(u) I'_e$ for $u : (X, e) \to (Y, f)$. Thus $\overline{F}$ and $\overline{F}'$ use two splittings (I_e, P_e) and (I'_e, P'_e) of the same idempotents, and

$$\theta_{(X,e)} = P'_e I_e : \overline{F}(X, e) \to \overline{F}'(X, e), \qquad \theta_{(X,e)}^{-1} = P_e I'_e, \qquad \theta_{\iota X} = \mathrm{id}_{FX}, \tag{4.13}$$

is an even supernatural isomorphism. Naturality with respect to the block inclusions and projections and the two morphisms e forces this formula, proving (4.8).

Let $H, H' : \mathcal{E} \to \mathcal{D}$ be superfunctors with chosen extensions $\overline{H}$ and $\overline{H}'$. By (4.13) we may assume that $\overline{H}$ and $\overline{H}'$ are constructed as in (4.11), from H^+ and H'^+ as in (4.10) and

splittings (I_e^H, P_e^H) and $(I_e^{H'}, P_e^{H'})$. For a supernatural transformation $\theta : H \Rightarrow H'$ of degree d and an object $X = (X_1, \dots, X_n)$ of $\mathrm{Add}(\mathcal{E})$, put $\theta_X^+ = \bigoplus_i \theta_{X_i} : H^+X \to H'^+X$ and

$$\overline{\theta}_{(X,e)} = P_e^{H'} \theta_X^+ I_e^H : \overline{H}(X,e) \to \overline{H}'(X,e). \tag{4.14}$$

For homogeneous $u : (X,e) \to (Y,f)$, signed naturality (3.5) of θ gives

$$\begin{aligned} \overline{\theta}_{(Y,f)} \overline{H}(u) &= P_f^{H'} \theta_Y^+ H^+(u) I_e^H \\ &= (-1)^{d|u|} P_f^{H'} H'^+(u) \theta_X^+ I_e^H \\ &= (-1)^{d|u|} \overline{H}'(u) \overline{\theta}_{(X,e)}, \end{aligned} \tag{4.15}$$

so $\overline{\theta}$ is supernatural of degree d with $\overline{\theta}_{\iota X} = \theta_X$. Conversely, naturality of a transformation $\overline{H} \Rightarrow \overline{H}'$ with respect to the block inclusions and projections and the retraction maps forces (4.14). Thus extension and restriction are inverse in each degree, proving (4.9).

For n source variables, let $F : \mathcal{E}_1 \boxtimes \cdots \boxtimes \mathcal{E}_n \to \mathcal{D}$ be a superfunctor. Define $F^+ : \mathrm{Add}(\mathcal{E}_1) \boxtimes \cdots \boxtimes \mathrm{Add}(\mathcal{E}_n) \to \mathcal{D}$ by (4.10) in each variable separately. For even idempotents e_i on objects X_i of $\mathrm{Add}(\mathcal{E}_i)$ put

$$\begin{aligned} E_i &= F^+(\mathrm{id}_{X_1} \boxtimes \cdots \boxtimes e_i \boxtimes \cdots \boxtimes \mathrm{id}_{X_n}), \\ E_i E_j = E_j E_i, \qquad E &= \prod_{i=1}^n E_i = F^+(e_1 \boxtimes \cdots \boxtimes e_n), \qquad E^2 = E. \end{aligned} \tag{4.16}$$

Split E and use (4.11) and (4.14) with E in place of $F^+(e)$. Equations (4.12) and (4.15) remain valid, the only interchanges being those of even idempotents, which carry no sign. This proves the finite-variable assertions. □

Proposition 4.5. *Let $\mathcal{E}$ be a monoidal supercategory. Choose an extension of $\iota \circ \otimes : \mathcal{E} \boxtimes \mathcal{E} \to \overline{\mathcal{E}}$ in both variables:*

$$\begin{aligned} \overline{\otimes} : \overline{\mathcal{E}} \boxtimes \overline{\mathcal{E}} &\longrightarrow \overline{\mathcal{E}}, \\ \overline{\otimes} \circ (\iota \boxtimes \iota) &= \iota \circ \otimes, \\ \mathbf{1}_{\overline{\mathcal{E}}} &= \iota(\mathbf{1}_{\mathcal{E}}). \end{aligned} \tag{4.17}$$

The associator and unitors are the unique extensions of $\iota(\alpha)$, $\iota(l)$, and $\iota(r)$ by (4.9) *and its finite-variable version. These define a monoidal structure on $\overline{\mathcal{E}}$ for which ι is strict monoidal. If $\mathcal{E}$ is a monoidal* Π*-supercategory, then $\overline{\mathcal{E}}$ is a monoidal* Π*-supercategory with*

$$\pi_{\overline{\mathcal{E}}} = \iota(\pi_{\mathcal{E}}), \qquad \zeta_{\overline{\mathcal{E}}} = \iota(\zeta_{\mathcal{E}}). \tag{4.18}$$

Let $\mathcal{D}$ be an additive monoidal supercategory with split even idempotents. For a monoidal superfunctor $(F, J, \varphi) : \mathcal{E} \to \mathcal{D}$, every extension $\overline{F}$ becomes monoidal with tensorator $\overline{J}$, the unique extension of J by (4.9) *in two variables, and unit map $\overline{\varphi} = \varphi$. Then $\overline{F} \circ \iota = F$ as monoidal superfunctors.*

Proof. Apply (4.7) in two variables to $\iota \circ \otimes$. Equation (4.9) extends the coherence maps and their inverses. For composable supernatural transformations η and θ between superfunctors $\mathcal{E}^{\boxtimes n} \to \overline{\mathcal{E}}$ with chosen extensions, uniqueness in (4.9) gives

$$\overline{\eta\theta} = \overline{\eta}\,\overline{\theta}, \qquad \overline{\mathrm{id}} = \mathrm{id}, \qquad \overline{\theta^{-1}} = (\overline{\theta})^{-1}. \tag{4.19}$$

In particular, the extended associator and unitors are even supernatural isomorphisms. If L, R are the two pentagon composites,

$$L|_{\mathcal{E}^{\boxtimes 4}} = R|_{\mathcal{E}^{\boxtimes 4}} \quad \Longrightarrow \quad L = R \tag{4.20}$$

by injectivity in the four-variable version of (4.9). The same implication in two variables proves the triangle. The restrictions are $\iota(\alpha), \iota(l), \iota(r)$ and the unit is $\iota(\mathbf{1})$, so ι is strict monoidal. Equation (4.18) gives an odd isomorphism with inverse $\iota(\zeta^{-1})$.

For (F, J, φ), the two functors $\overline{F}(-) \otimes \overline{F}(-)$ and $\overline{F}(-\overline{\otimes}-)$ restrict to the source and target of J. Their comparison is

$$\overline{J} : \overline{F}(-) \otimes \overline{F}(-) \xrightarrow{\sim} \overline{F}(-\overline{\otimes}-), \qquad \overline{\varphi} = \varphi. \tag{4.21}$$

Invertibility follows from (4.19). The associativity and unit equations have the form (3.101). They hold on the original objects because (F, J, φ) is monoidal, and hence on the completions by (4.9) in three and one variables. □

Proposition 4.6. *Let $\mathcal{E}$ be a supercategory and let $\mathcal{D}$ be an additive supercategory with split even idempotents. Let $F : \mathcal{E} \to \mathcal{D}$ be a fully faithful superfunctor. Suppose that every object of $\mathcal{D}$ is an even retract of a finite direct sum of objects FX. Every extension $\overline{F}$ in* (4.7) *is a superequivalence. If $\mathcal{E}$ and $\mathcal{D}$ are monoidal and F is monoidal, then $\overline{F}$, with the structure of Proposition 4.5, is a monoidal superequivalence.*

Proof. By (4.8) it suffices to treat the extension $\overline{F}$ constructed in (4.10)–(4.11), with F^+, Q_e, I_e, and P_e as there. For objects (X, e) and (Y, f) of $\overline{\mathcal{E}}$, full faithfulness of F gives isomorphisms on matrices and corners:

$$\begin{aligned} f \operatorname{Add}(\mathcal{E})(X, Y) e &\xrightarrow{\sim} F^+(f)\mathcal{D}(F^+X, F^+Y)F^+(e) \\ &\xrightarrow{\sim} \mathcal{D}(Q_e, Q_f), \qquad u \longmapsto P_f F^+(u) I_e. \end{aligned} \tag{4.22}$$

Thus $\overline{F}$ is fully faithful. Let V be an object of $\mathcal{D}$. By hypothesis there are an object X of $\operatorname{Add}(\mathcal{E})$ and even maps $i : V \to F^+X$ and $p : F^+X \to V$ with $pi = \mathrm{id}_V$, the chosen sum F^+X being isomorphic to any direct sum of the objects FX_i. By full faithfulness of F^+, the even idempotent ip on F^+X is $F^+(e)$ for a unique even idempotent e on X, and

$$V \xrightarrow{P_e i} Q_e, \qquad (P_e i)^{-1} = pI_e, \qquad (pI_e)(P_e i) = \mathrm{id}_V, \quad (P_e i)(pI_e) = \mathrm{id}_{Q_e}. \tag{4.23}$$

So $V \cong \overline{F}(X, e)$ evenly, which proves even essential surjectivity, hence that $\overline{F}$ is a superequivalence. In the monoidal case, $\overline{F}$ with the tensorator (4.21) is a monoidal superfunctor by Proposition 4.5, hence a monoidal superequivalence. □

Definition 4.7. Let $\mathcal{A}$ and $\mathcal{B}$ be monoidal supercategories. The monoidal structure on $\mathcal{A} \boxtimes \mathcal{B}$ is defined by

$$\begin{aligned} (X \boxtimes Y) \otimes (X' \boxtimes Y') &= (X \otimes X') \boxtimes (Y \otimes Y'), \\ (f \boxtimes g) \otimes (f' \boxtimes g') &= (-1)^{|g||f'|} (f \otimes f') \boxtimes (g \otimes g') \end{aligned} \tag{4.24}$$

for homogeneous morphisms. The unit object, associator, and unitors are

$$\begin{aligned} \mathbf{1}_{\mathcal{A}\boxtimes\mathcal{B}} &= \mathbf{1}_{\mathcal{A}} \boxtimes \mathbf{1}_{\mathcal{B}}, \\ \alpha_{X\boxtimes Y,\, X'\boxtimes Y',\, X''\boxtimes Y''} &= \alpha^{\mathcal{A}}_{X,X',X''} \boxtimes \alpha^{\mathcal{B}}_{Y,Y',Y''}, \\ l_{X\boxtimes Y} &= l^{\mathcal{A}}_X \boxtimes l^{\mathcal{B}}_Y, \qquad r_{X\boxtimes Y} = r^{\mathcal{A}}_X \boxtimes r^{\mathcal{B}}_Y. \end{aligned} \tag{4.25}$$

The *completed product* is

$$\mathcal{A} \,\overline{\boxtimes}\, \mathcal{B} = \overline{\mathcal{A} \boxtimes \mathcal{B}}, \tag{4.26}$$

with monoidal structure chosen as in Proposition 4.5. If $\mathcal{A}$ and $\mathcal{B}$ are monoidal Π-supercategories, the Π-structure on $\mathcal{A} \boxtimes \mathcal{B}$ is defined by

$$\begin{aligned} \pi_{\mathcal{A}\boxtimes\mathcal{B}} &= \pi_{\mathcal{A}} \boxtimes \mathbf{1}_{\mathcal{B}}, \\ \zeta_{\mathcal{A}\boxtimes\mathcal{B}} &= \zeta_{\mathcal{A}} \boxtimes \mathrm{id}_{\mathbf{1}_{\mathcal{B}}}. \end{aligned} \tag{4.27}$$

The Π-structure on $\mathcal{A} \,\overline{\boxtimes}\, \mathcal{B}$ is then defined by Proposition 4.5.

Proposition 4.8. *If $\mathcal{A}, \mathcal{B} \in \mathcal{W}$, then*

$$\mathcal{A} \mathbin{\overline{\boxtimes}} \mathcal{B} \in \mathcal{W}, \tag{4.28}$$

and there is a group isomorphism

$$G_{\mathcal{A}} \times G_{\mathcal{B}} \xrightarrow{\sim} G_{\mathcal{A}\overline{\boxtimes}\mathcal{B}}, \qquad ([X],[Y]) \longmapsto [S], \tag{4.29}$$

where S is any simple summand of $X \boxtimes Y$. Under this isomorphism,

$$\rho_{\mathcal{A}\overline{\boxtimes}\mathcal{B}}(g,h) = \rho_{\mathcal{A}}(g) + \rho_{\mathcal{B}}(h). \tag{4.30}$$

Proof. Write $\mathcal{T} = \mathcal{A} \mathbin{\overline{\boxtimes}} \mathcal{B}$. For an object W of a supercategory write $\mathrm{End}_0(W) = \mathrm{End}(W)_0$ for the algebra of even endomorphisms, and for a finite-dimensional k-algebra E write $\mathsf{mod}_{\mathrm{fd}}(E)$ for the category of finite-dimensional right E-modules and $\mathsf{proj}_{\mathrm{fd}}(E)$ for its full subcategory of projective modules. For $g \in G_{\mathcal{A}}$ and $h \in G_{\mathcal{B}}$ choose simple objects $X_g \in \mathcal{A}_g$ and $Y_h \in \mathcal{B}_h$, and put $Z_{g,h} = X_g \boxtimes Y_h$. Morphisms between distinct pairs of components vanish. The endomorphism superalgebra of $Z_{g,h}$ is

$$E_{g,h} = \mathrm{End}(X_g \boxtimes Y_h) \cong D_{\rho_{\mathcal{A}}(g)} \mathbin{\widehat{\otimes}} D_{\rho_{\mathcal{B}}(h)}. \tag{4.31}$$

It is central simple of type $\rho_{\mathcal{A}}(g) + \rho_{\mathcal{B}}(h)$ by Proposition 2.6(5) and (2.15).

By Definition 3.11 and Lemma 3.9(1), every simple object of $\mathcal{A}_g$ has a homogeneous isomorphism to X_g, hence an even isomorphism to X_g or to ΠX_g; since $\underline{\mathcal{A}}$ is semisimple, every object of $\mathcal{A}$ is evenly isomorphic to a finite direct sum of the objects X_g and ΠX_g, and likewise for $\mathcal{B}$. By (4.24), (4.27), and the odd isomorphisms z, the objects $\Pi X \boxtimes Y$, $X \boxtimes \Pi Y$, and $\Pi X \boxtimes \Pi Y$ are evenly isomorphic to $\Pi(X \boxtimes Y)$, $\Pi(X \boxtimes Y)$, and $X \boxtimes Y$. Hence every object of $\mathcal{T}$ is an even retract of a finite direct sum of copies of $P = \bigoplus_{g,h}(Z_{g,h} \oplus \Pi Z_{g,h})$. For an object Z, conjugation by $\mathrm{id}_Z \oplus z_Z : Z \oplus \Pi Z \to Z \oplus Z$ identifies $\mathrm{End}(Z \oplus \Pi Z)$ with $M_{1|1}(\mathrm{End}(Z))$, graded as in (2.3), and $e_{ij} \mathbin{\widehat{\otimes}} c \mapsto (-1)^{\epsilon_j|c|} e_{ij} c$, for the matrix units e_{ij} and homogeneous c, is an isomorphism of superalgebras $M_{1|1}(k) \mathbin{\widehat{\otimes}} \mathrm{End}(Z) \to M_{1|1}(\mathrm{End}(Z))$. The superalgebra $M_{1|1}(k)$ is simple as an ordinary algebra, and commutation with the matrix units shows that its supercenter is k. Hence $\mathrm{End}(P) \cong \prod_{g,h} M_{1|1}(k) \mathbin{\widehat{\otimes}} E_{g,h}$, whose factors are central simple by Proposition 2.6(5); so $E = \mathrm{End}_0(P)$ is a finite product of separable, hence semisimple, k-algebras by Proposition 2.6(6). Let $\mathrm{Hom}_0(P, -)$ be the functor sending an object W of $\underline{\mathcal{T}}$ to the even part of $\mathcal{T}(P, W)$, a right E-module under precomposition, and a morphism to postcomposition with it. Since every object is an even retract of a finite sum of copies of P, full faithfulness on sums and corners gives an equivalence

$$\mathrm{Hom}_0(P, -) : \underline{\mathcal{T}} \xrightarrow{\simeq} \mathsf{proj}_{\mathrm{fd}}(E) = \mathsf{mod}_{\mathrm{fd}}(E). \tag{4.32}$$

The last equality uses semisimplicity of E. It proves that $\underline{\mathcal{T}}$ is semisimple with finite-dimensional Hom spaces and finitely many isomorphism classes of simple objects, those of the simple E-modules.

For objects X of $\mathcal{A}$ and Y of $\mathcal{B}$ with left duals X^* and Y^*, take $(X \boxtimes Y)^* = X^* \boxtimes Y^*$ and

$$\mathrm{ev}_{X\boxtimes Y} = \mathrm{ev}_X \boxtimes \mathrm{ev}_Y, \qquad \mathrm{coev}_{X\boxtimes Y} = \mathrm{coev}_X \boxtimes \mathrm{coev}_Y\,. \tag{4.33}$$

The maps are even, so (4.24) gives both snakes without extra signs. Extend to sums by (3.84), with the present tensor product. For an object (Z, e) of the completion, with Z such a sum and e an even idempotent on Z, let Z^* be the dual sum, let $e^* = (\mathrm{ev}_Z \otimes \mathrm{id}_{Z^*})(\mathrm{id}_{Z^*} \otimes e \otimes \mathrm{id}_{Z^*})(\mathrm{id}_{Z^*} \otimes \mathrm{coev}_Z)$ be the dual idempotent, with associators and unitors suppressed, and take

$$\begin{aligned} &(Z,e)^* = (Z^*, e^*), \\ &\mathrm{ev}_{(Z,e)} = \mathrm{ev}_Z(e^* \otimes e), \quad \mathrm{coev}_{(Z,e)} = (e \otimes e^*)\,\mathrm{coev}_Z\,. \end{aligned} \tag{4.34}$$

The duality identity $\mathrm{ev}_Z(e^*\otimes \mathrm{id}_Z)=\mathrm{ev}_Z(\mathrm{id}_{Z^*}\otimes e)$ reduces the two snakes to $e=\mathrm{id}_{(Z,e)}$ and $e^*=\mathrm{id}_{(Z^*,e^*)}$. The same formulas give right duals. Finally,

$$\mathrm{End}(\mathbf{1}_{\mathcal{A}\overline{\boxtimes}\mathcal{B}})=\mathrm{End}(\mathbf{1}_{\mathcal{A}})\,\widehat{\otimes}\,\mathrm{End}(\mathbf{1}_{\mathcal{B}})\cong k. \tag{4.35}$$

In particular $\mathbf{1}_{\mathcal{T}}$ is simple: it is nonzero, and a decomposition into two or more nonzero summands in $\underline{\mathcal{T}}$ would give $\dim_k \mathrm{End}(\mathbf{1}_{\mathcal{T}})_0\geq 2$. Together with (4.18) and (4.27), this shows that $\mathcal{T}$ is a fusion Π-supercategory with split unit.

Let e and e' be even idempotents of $E_{g,h}$ such that the retracts $(X_g\boxtimes Y_h,e)$ and $(X_g\boxtimes Y_h,e')$ are simple objects. The corner $eE_{g,h}e$ is the endomorphism superalgebra of $(X_g\boxtimes Y_h,e)$, a division superalgebra, hence isomorphic to $D_{\rho_{\mathcal{A}}(g)+\rho_{\mathcal{B}}(h)}$ by Proposition 2.6(1) and Lemma 2.3; so both retracts have Brauer–Wall type $\rho_{\mathcal{A}}(g)+\rho_{\mathcal{B}}(h)$. Since $E_{g,h}$ is graded simple, $E_{g,h}eE_{g,h}=E_{g,h}$, so $e'E_{g,h}e\neq 0$; a nonzero homogeneous element of $e'E_{g,h}e$ is a homogeneous morphism between the two simple objects, and it is an isomorphism, since a nonzero even morphism between simple objects of the semisimple category $\underline{\mathcal{T}}$ is invertible and an odd one becomes even after composition with the odd isomorphism z of (3.11). Hence all simple retracts of $X_g\boxtimes Y_h$ and of its parity shift are isomorphic. Let $\mathcal{T}_{g,h}$ be the full subsupercategory of $\mathcal{T}$ whose objects are the even retracts of finite direct sums of $X_g\boxtimes Y_h$ and its parity shift. We obtain

$$\mathcal{T}=\bigoplus_{(g,h)}\mathcal{T}_{g,h},\qquad \mathcal{T}_{g,h}\otimes\mathcal{T}_{g',h'}\subset\mathcal{T}_{gg',hh'},\qquad (\mathcal{T}_{g,h})^*\subset\mathcal{T}_{g^{-1},h^{-1}}. \tag{4.36}$$

The tensor inclusion is (4.24); the dual inclusion follows from (4.33) and (4.34). Since $\mathcal{T}_{1,1}$ consists of sums of $\mathbf{1}$ and π,

$$S\in\mathcal{T}_{g,h}\text{ simple}\quad\Longrightarrow\quad S\otimes S^*\in\mathcal{T}_{1,1},\qquad S\otimes S^*\cong\mathbf{1}^{\oplus p}\oplus\pi^{\oplus q}\text{ for some }p,q\geq 0. \tag{4.37}$$

Thus $\mathcal{T}\in\mathcal{W}$ by (3.14). Equations (4.36) and (4.31) give (4.29) and (4.30). □

Definition 4.9. Let G be a finite group. A category $\mathcal{A}\in\mathcal{W}$ is *G-labeled* if it is equipped with a group isomorphism $\lambda_{\mathcal{A}}:G\to G_{\mathcal{A}}$; write $\mathcal{W}(G)$ for the class of G-labeled categories and $\mathcal{A}_{(g)}=\mathcal{A}_{\lambda_{\mathcal{A}}(g)}$ for the components. For $\mathcal{A},\mathcal{B}\in\mathcal{W}(G)$, write $\mathcal{A}\simeq_G\mathcal{B}$ if there is a monoidal superequivalence $E:\mathcal{A}\to\mathcal{B}$ such that $E(\mathcal{A}_{(g)})\subset\mathcal{B}_{(g)}$ for all $g\in G$; such an E is called *label-preserving*. The *stack* $\mathcal{S}(\mathcal{A},\mathcal{B})$ is the full subsupercategory of $\mathcal{A}\,\overline{\boxtimes}\,\mathcal{B}$ whose objects are the even retracts of finite direct sums of the objects

$$X\boxtimes Y,\qquad X\in\mathcal{A}_{(g)},\quad Y\in\mathcal{B}_{(g)},\quad g\in G; \tag{4.38}$$

$\mathcal{D}(\mathcal{A},\mathcal{B})$ denotes the full subsupercategory on the objects (4.38), with inclusion $j:\mathcal{D}(\mathcal{A},\mathcal{B})\hookrightarrow\mathcal{S}(\mathcal{A},\mathcal{B})$.

Proposition 4.10. *If $\mathcal{A},\mathcal{B}\in\mathcal{W}(G)$, then $\mathcal{D}(\mathcal{A},\mathcal{B})$ and $\mathcal{S}(\mathcal{A},\mathcal{B})$ are monoidal Π-subsupercategories of $\mathcal{A}\,\overline{\boxtimes}\,\mathcal{B}$, and $\mathcal{S}(\mathcal{A},\mathcal{B})\in\mathcal{W}(G)$ such that*

$$\lambda_{\mathcal{S}(\mathcal{A},\mathcal{B})}(g)=[S],\qquad S\text{ any simple summand of }X\boxtimes Y,\quad X\in\mathcal{A}_{(g)},\ Y\in\mathcal{B}_{(g)}\text{ simple.} \tag{4.39}$$

Proof. Write $\mathcal{T}=\mathcal{A}\,\overline{\boxtimes}\,\mathcal{B}$. By (4.28), $\mathcal{T}\in\mathcal{W}$, and (4.29) indexes its components by $G_{\mathcal{A}}\times G_{\mathcal{B}}$; for $(g',h')\in G_{\mathcal{A}}\times G_{\mathcal{B}}$, write $\mathcal{T}_{(g',h')}$ for the component of $\mathcal{T}$ indexed by (g',h'). The objects (4.38) of $\mathcal{D}(\mathcal{A},\mathcal{B})$ lie in the components $(\lambda_{\mathcal{A}}(g),\lambda_{\mathcal{B}}(g))$. Conversely, fix g and choose simple objects $X\in\mathcal{A}_{(g)}$ and $Y\in\mathcal{B}_{(g)}$, and let S be a simple summand of $X\boxtimes Y$. By Definition 3.11 and Lemma 3.9, every object of the component of S is evenly isomorphic to a finite direct sum of copies of S and ΠS, hence is an even retract of a finite sum of $X\boxtimes Y$ and its parity

shift. The parity shift of $X \boxtimes Y$ is $\pi_{\mathcal{T}} \otimes (X \boxtimes Y) \cong (\Pi X) \boxtimes Y$ evenly by (4.27) and (4.24), with $\Pi X \in \mathcal{A}_{(g)}$; so both lie in $\mathcal{D}(\mathcal{A}, \mathcal{B})$ up to even isomorphism. Hence, for $g, h \in G$,

$$\mathcal{S}(\mathcal{A}, \mathcal{B}) = \bigoplus_{g \in G} \mathcal{T}_{(\lambda_{\mathcal{A}}(g), \lambda_{\mathcal{B}}(g))}, \qquad \mathcal{T}_{(\lambda_{\mathcal{A}}(g), \lambda_{\mathcal{B}}(g))} \otimes \mathcal{T}_{(\lambda_{\mathcal{A}}(h), \lambda_{\mathcal{B}}(h))} \subset \mathcal{T}_{(\lambda_{\mathcal{A}}(gh), \lambda_{\mathcal{B}}(gh))}, \tag{4.40}$$

and the dual of an object of $\mathcal{T}_{(\lambda_{\mathcal{A}}(g), \lambda_{\mathcal{B}}(g))}$ lies in $\mathcal{T}_{(\lambda_{\mathcal{A}}(g^{-1}), \lambda_{\mathcal{B}}(g^{-1}))}$; the inclusions follow from (4.36) and the fact that $\lambda_{\mathcal{A}}, \lambda_{\mathcal{B}}$ are homomorphisms. In particular $\mathcal{S}(\mathcal{A}, \mathcal{B})$ is closed under $\otimes$ and contains $\mathbf{1} = \mathbf{1} \boxtimes \mathbf{1}$ and $\pi = \pi \boxtimes \mathbf{1}$, which are objects of $\mathcal{D}(\mathcal{A}, \mathcal{B})$; and $\mathcal{D}(\mathcal{A}, \mathcal{B})$ is closed under $\otimes$ by (4.24) and (3.29), since $X \otimes X' \in \mathcal{A}_{(gh)}$ and $Y \otimes Y' \in \mathcal{B}_{(gh)}$ for $X \in \mathcal{A}_{(g)}$, $X' \in \mathcal{A}_{(h)}$, $Y \in \mathcal{B}_{(g)}$, and $Y' \in \mathcal{B}_{(h)}$. The constraints of $\mathcal{A} \mathbin{\overline{\boxtimes}} \mathcal{B}$ restrict to both. The neutral component contains $\mathbf{1}$ and π; finite semisimplicity, rigidity, split unit, and (4.37) therefore restrict to the stack. Thus $\mathcal{S}(\mathcal{A}, \mathcal{B}) \in \mathcal{W}$. The map on component groups is $g \mapsto (\lambda_{\mathcal{A}}(g), \lambda_{\mathcal{B}}(g))$, which is (4.39) under (4.29); it is a group isomorphism by (4.40). □

Lemma 4.11. *If $\mathcal{A} \simeq_G \mathcal{A}'$ and $\mathcal{B} \simeq_G \mathcal{B}'$ in $\mathcal{W}(G)$, then $\mathcal{S}(\mathcal{A}, \mathcal{B}) \simeq_G \mathcal{S}(\mathcal{A}', \mathcal{B}')$.*

Proof. Let $E : \mathcal{A} \to \mathcal{A}'$ be a label-preserving monoidal superequivalence with tensorator J and unit map φ, and let $\mathcal{B}$ be G-labeled. The superfunctor $E \boxtimes \mathrm{id} : \mathcal{A} \boxtimes \mathcal{B} \to \mathcal{A}' \boxtimes \mathcal{B}$ is a superequivalence, and it is monoidal for the structures (4.24)–(4.25) with tensorator $J \boxtimes \mathrm{id}$ and unit map $\varphi \boxtimes \mathrm{id}$, the axioms holding factorwise. Composing with ι gives a fully faithful monoidal superfunctor $\mathcal{A} \boxtimes \mathcal{B} \to \mathcal{A}' \mathbin{\overline{\boxtimes}} \mathcal{B}$ whose image generates the target under finite direct sums and even retracts, because E is essentially surjective; by Propositions 4.5 and 4.6 it extends to a monoidal superequivalence $\overline{E} : \mathcal{A} \mathbin{\overline{\boxtimes}} \mathcal{B} \to \mathcal{A}' \mathbin{\overline{\boxtimes}} \mathcal{B}$. Since E preserves labels, $\overline{E}$ sends the objects (4.38) of $\mathcal{D}(\mathcal{A}, \mathcal{B})$ to objects isomorphic to objects of $\mathcal{D}(\mathcal{A}', \mathcal{B})$, and finite sums and even retracts to finite sums and even retracts; as $\mathcal{S}(\mathcal{A}', \mathcal{B})$ is generated by these objects, $\overline{E}$ restricts to a monoidal superequivalence $\mathcal{S}(\mathcal{A}, \mathcal{B}) \to \mathcal{S}(\mathcal{A}', \mathcal{B})$, which preserves the labelings (4.39) because it preserves the components $(\lambda_{\mathcal{A}}(g), \lambda_{\mathcal{B}}(g))$. Let $E' : \mathcal{B} \to \mathcal{B}'$ be a label-preserving monoidal superequivalence; in the same way, $\mathrm{id} \boxtimes E'$ composed with ι extends to a monoidal superequivalence $\overline{E'} : \mathcal{A}' \mathbin{\overline{\boxtimes}} \mathcal{B} \to \mathcal{A}' \mathbin{\overline{\boxtimes}} \mathcal{B}'$, which restricts to a label-preserving monoidal superequivalence $\mathcal{S}(\mathcal{A}', \mathcal{B}) \to \mathcal{S}(\mathcal{A}', \mathcal{B}')$. Composing the two restrictions gives $\mathcal{S}(\mathcal{A}, \mathcal{B}) \simeq_G \mathcal{S}(\mathcal{A}', \mathcal{B}) \simeq_G \mathcal{S}(\mathcal{A}', \mathcal{B}')$. □

Definition 4.12. For a finite group G, let sVec_G be the supercategory of families $V = (V_g)_{g \in G}$ of finite-dimensional super vector spaces, with morphism spaces $\bigoplus_{g \in G} \mathrm{Hom}_k(V_g, W_g)$ graded by the parity of linear maps. For a family indexed by G, $X[g]$ denotes the family with X at g and 0 elsewhere. It is a monoidal Π-supercategory with tensor product

$$(V \otimes W)_\ell = \bigoplus_{gh = \ell} V_g \otimes_k W_h, \qquad (f \otimes f')(v \otimes w) = (-1)^{|f'||v|} f(v) \otimes f'(w), \tag{4.41}$$

for homogeneous f, f' and v, unit $k[1]$, associator and unitors acting on each summand by

$$(v \otimes w) \otimes u \longmapsto v \otimes (w \otimes u), \qquad c \otimes v \longmapsto cv, \qquad v \otimes c \longmapsto cv \qquad (c \in k), \tag{4.42}$$

parity object $\pi = k^{0|1}[1]$, where $k^{0|1}$ is k placed in odd degree, and $\zeta : k^{0|1}[1] \to k[1]$ the identity map of k, which is odd.

Lemma 4.13. *Let G be a finite group. We have $\mathsf{sVec}_G \in \mathcal{W}(G)$ with $\lambda_{\mathsf{sVec}_G}(g) = [k[g]]$, and $\mathcal{A} \simeq_G \mathcal{S}(\mathcal{A}, \mathsf{sVec}_G)$ for every $\mathcal{A} \in \mathcal{W}(G)$.*

Proof. The underlying category of sVec_G is the category of finite-dimensional $(G \times \mathbb{Z}/2\mathbb{Z})$-graded vector spaces with degree-preserving maps, that is, of finite-dimensional modules over the semisimple algebra $k^{G \times \mathbb{Z}/2\mathbb{Z}}$; so it is semisimple with finite-dimensional Hom spaces, its

simple objects are the $k[g]$ and $k^{0|1}[g]$, and $\mathrm{End}(k[g]) = k$. The tensor product (4.41) is a superfunctor by the Koszul sign and k-linear in each variable, and the constraints (4.42) are even, supernatural, and satisfy the pentagon and triangle identities on each summand; so sVec_G is a monoidal supercategory with unit $k[1]$, which is simple with $\mathrm{End}(k[1]) = k$. It is rigid, with $V[g]^* = V^*[g^{-1}]$ for a finite-dimensional super vector space V, with the evaluation and coevaluation of sVec_k, and ζ is an odd isomorphism; so sVec_G is a fusion Π-supercategory with split unit. Since $k[g] \otimes k[g^{-1}] = k[1]$ and $k^{0|1}[g] \otimes k^{0|1}[g^{-1}] = (k^{0|1} \otimes_k k^{0|1})[1] \cong k[1]$ evenly, every simple object is weakly invertible, and $\mathsf{sVec}_G \in \mathcal{W}$. The odd map ζ shows $k^{0|1}[g] \cong k[g]$, so the classes of simple objects are the $[k[g]]$, and (3.27) with $k[g] \otimes k[h] = k[gh]$ shows that $g \mapsto [k[g]]$ is a group isomorphism $G \to G_{\mathsf{sVec}_G}$.

By (3.29), $\mathcal{A} = \bigoplus_{g \in G} \mathcal{A}_{(g)}$. Let $\mathcal{A}'$ be the full subsupercategory of $\mathcal{A}$ whose objects are the objects of the components $\mathcal{A}_{(g)}$, $g \in G$. By (3.29), $\mathcal{A}'$ is closed under $\otimes$ and contains $\mathbf{1}$ and π; the constraints of $\mathcal{A}$ restrict to it, so $\mathcal{A}'$ is a monoidal Π-subsupercategory of $\mathcal{A}$, and every object of $\mathcal{A}$ is a finite direct sum of objects of $\mathcal{A}'$. Define the superfunctor $K : \mathcal{A}' \to \mathcal{D}(\mathcal{A}, \mathsf{sVec}_G)$ by $K(X) = X \boxtimes k[g]$ and $K(f) = f \boxtimes \mathrm{id}_{k[g]}$ for a nonzero object X of $\mathcal{A}_{(g)}$ and a morphism f of $\mathcal{A}_{(g)}$, and $K(0) = 0 \boxtimes k[1]$; a morphism between objects of different components is zero and is sent to zero. It is fully faithful, since $\mathcal{A}(X, X') \otimes_k \mathsf{sVec}_G(k[g], k[g]) = \mathcal{A}(X, X')$ for $X, X' \in \mathcal{A}_{(g)}$ and both sides vanish for different labels. It is monoidal: for $X \in \mathcal{A}_{(g)}$ and $X' \in \mathcal{A}_{(h)}$, $K(X) \otimes K(X') = (X \boxtimes k[g]) \otimes (X' \boxtimes k[h]) = (X \otimes X') \boxtimes (k \otimes_k k)[gh]$ by (4.24), the tensorator is $\mathrm{id} \boxtimes (k \otimes_k k \to k)$, and the associator $\alpha \boxtimes \mathrm{id}$ of (4.25) is compatible with it by coherence in sVec_k. Every object of $\mathcal{D}(\mathcal{A}, \mathsf{sVec}_G)$ is $X \boxtimes V[g]$ with V a finite-dimensional super vector space, hence a finite direct sum of objects $X \boxtimes k[g]$ and $X \boxtimes k^{0|1}[g]$, and $X \boxtimes k^{0|1}[g] \cong (X \otimes \pi) \boxtimes k[g]$ evenly, through the even isomorphism $\mathbf{1} \boxtimes k^{0|1} \cong \pi \boxtimes \mathbf{1}$ of (4.27), $\zeta^{-1} \boxtimes \zeta$, and (4.24). Hence the image of K generates $\mathcal{S}(\mathcal{A}, \mathsf{sVec}_G)$ under finite direct sums and even retracts. Let $I : \mathcal{A}' \to \mathcal{A}$ be the inclusion, a fully faithful strict monoidal superfunctor whose image generates $\mathcal{A}$ under finite direct sums. Both $\mathcal{A}$ and $\mathcal{S}(\mathcal{A}, \mathsf{sVec}_G)$ are additive with split even idempotents, so Propositions 4.5 and 4.6, applied to I and to jK with j the inclusion of Definition 4.9, give monoidal superequivalences $\overline{I} : \overline{\mathcal{A}'} \to \mathcal{A}$ and $\overline{K} : \overline{\mathcal{A}'} \to \mathcal{S}(\mathcal{A}, \mathsf{sVec}_G)$ with $\overline{I}\iota = I$ and $\overline{K}\iota = jK$. Let I' be a monoidal quasi-inverse of $\overline{I}$, with the monoidal structure transported along the isomorphisms $\overline{I}I' \cong \mathrm{id}$ and $I'\overline{I} \cong \mathrm{id}$. Then $\overline{K}I'$ is a monoidal superequivalence $\mathcal{A} \to \mathcal{S}(\mathcal{A}, \mathsf{sVec}_G)$. It preserves labels: for $X \in \mathcal{A}_{(g)}$, $I'(X) \cong \iota(X)$ evenly, so $\overline{K}I'(X) \cong \overline{K}\iota(X) = X \boxtimes k[g]$, which lies in the component $(\lambda_{\mathcal{A}}(g), g)$. □

Definition 4.14. For a monoidal Π-supercategory $\mathcal{A}$, the *opposite* $\mathcal{A}^{\mathrm{op}}$ has the same objects, morphism spaces $\mathcal{A}^{\mathrm{op}}(X, Y) = \mathcal{A}(Y, X)$, composition and tensor product

$$g \circ^{\mathrm{op}} f = (-1)^{|f||g|} f \circ g, \qquad X \otimes^{\mathrm{op}} Y = X \otimes Y, \qquad f \otimes^{\mathrm{op}} g = f \otimes g, \tag{4.43}$$

associator α^{-1}, unitors l^{-1} and r^{-1}, unit $\mathbf{1}$, parity object π, and $\zeta^{\mathrm{op}} = \zeta^{-1}$.

Lemma 4.15. *$\mathcal{A}^{\mathrm{op}}$ is a monoidal Π-supercategory. If $\mathcal{A} \in \mathcal{W}(G)$, then $\mathcal{A}^{\mathrm{op}} \in \mathcal{W}(G)$ with the same simple objects and labeling.*

Proof. Composition in $\mathcal{A}^{\mathrm{op}}$ is associative, since for composable homogeneous morphisms f, f', f'' of $\mathcal{A}$ both $(f'' \circ^{\mathrm{op}} f') \circ^{\mathrm{op}} f$ and $f'' \circ^{\mathrm{op}} (f' \circ^{\mathrm{op}} f)$ equal $(-1)^{|f||f'|+|f||f''|+|f'||f''|} f \circ f' \circ f''$, and identities are identities. The tensor product is a superfunctor $\mathcal{A}^{\mathrm{op}} \boxtimes \mathcal{A}^{\mathrm{op}} \to \mathcal{A}^{\mathrm{op}}$: for homogeneous morphisms f, f', u, u' of $\mathcal{A}$ such that $f' \boxtimes u'$ and $f \boxtimes u$ are composable, both $(f' \otimes u') \circ^{\mathrm{op}} (f \otimes u)$ and $(-1)^{|u'||f|}(f' \circ^{\mathrm{op}} f) \otimes (u' \circ^{\mathrm{op}} u)$, where $(-1)^{|u'||f|}$ is the interchange sign of (3.8), equal $(-1)^{|f||f'|+|f||u'|+|u||u'|}(f \circ f') \otimes (u \circ u')$. The inverses of the constraints of $\mathcal{A}$ are even supernatural isomorphisms satisfying the pentagon and triangle identities, and ζ^{-1} is an odd isomorphism; so $\mathcal{A}^{\mathrm{op}}$ is a monoidal Π-supercategory. Its underlying category is $(\underline{\mathcal{A}})^{\mathrm{op}}$, which is semisimple with finite-dimensional Hom spaces, rigid with the roles of left and right

duals exchanged, and has simple unit. The simple objects are those of $\mathcal{A}$, and for an object X, $\mathrm{End}_{\mathcal{A}^{\mathrm{op}}}(X)$ has the multiplication $(f,g)\mapsto(-1)^{|f||g|}g\circ f$, that is, $\mathrm{End}_{\mathcal{A}^{\mathrm{op}}}(X)=\mathrm{End}_{\mathcal{A}}(X)^{\mathrm{sop}}$ in the notation of (2.2). Weak invertibility holds in $\mathcal{A}^{\mathrm{op}}$: a left dual of a simple object X in $\underline{\mathcal{A}^{\mathrm{op}}}$ is a right dual ${}^*X$ of X in $\underline{\mathcal{A}}$; the object $\mathbf{1}$ is a summand of $X\otimes X^*$ and of $X\otimes {}^*X$, since the coevaluation $\mathbf{1}\to X\otimes X^*$ and the evaluation $X\otimes {}^*X\to\mathbf{1}$ are nonzero and $\underline{\mathcal{A}}$ is semisimple, so $[X^*]=[X]^{-1}=[{}^*X]$ by (3.27), hence ${}^*X$ and X^* are isomorphic, by a homogeneous isomorphism by Lemma 3.9; so $X\otimes {}^*X$ is homogeneously isomorphic to $X\otimes X^*$ and decomposes as in (3.14), with p and q exchanged if the isomorphism is odd. Thus $\mathcal{A}^{\mathrm{op}}\in\mathcal{W}$; the product (3.27) is read off from $X\otimes Y$, so $G_{\mathcal{A}^{\mathrm{op}}}=G_{\mathcal{A}}$. A G-labeling of $\mathcal{A}$ is therefore a G-labeling of $\mathcal{A}^{\mathrm{op}}$. □

Lemma 4.16. *For $\mathcal{A}\in\mathcal{W}(G)$, $\mathcal{S}(\mathcal{A},\mathcal{A}^{\mathrm{op}})\simeq_G \mathsf{sVec}_G$.*

Proof. Let $\mathcal{V}$ be the supercategory $\mathsf{sVec}_{\{1\}}$ of Definition 4.12 for the trivial group: finite-dimensional super vector spaces and all linear maps, graded by parity, with the tensor product (4.41). Define $H:\mathcal{A}\boxtimes\mathcal{A}^{\mathrm{op}}\to\mathcal{V}$ by

$$H(X\boxtimes Y)=\mathcal{A}(Y,X),\qquad H(a\boxtimes b)(f)=(-1)^{|b||f|}\,a\circ f\circ b \tag{4.44}$$

for $a\in\mathcal{A}(X,X')$, $b\in\mathcal{A}^{\mathrm{op}}(Y,Y')=\mathcal{A}(Y',Y)$, and $f\in\mathcal{A}(Y,X)$. For $a\in\mathcal{A}(X,X')$, $a'\in\mathcal{A}(X',X'')$, $b\in\mathcal{A}^{\mathrm{op}}(Y,Y')$, $b'\in\mathcal{A}^{\mathrm{op}}(Y',Y'')$, and $f\in\mathcal{A}(Y,X)$, (3.8) and (4.43) give $(a'\boxtimes b')\circ(a\boxtimes b)=(-1)^{|b'||a|+|b||b'|}(a'a)\boxtimes(bb')$, and both $H((a'\boxtimes b')\circ(a\boxtimes b))(f)$ and $H(a'\boxtimes b')(H(a\boxtimes b)(f))$ equal $(-1)^{|b||f|+|b'||f|+|b'||a|+|b'||b|}\,a'afbb'$; also $H(\mathrm{id}\boxtimes\mathrm{id})=\mathrm{id}$. So H is a superfunctor. The tensorator is the map (3.43), $f\otimes f'\mapsto f\otimes f'$, which is an isomorphism when X,Y lie in one component and X',Y' lie in one component, by Lemma 3.16. It is supernatural: for $a\in\mathcal{A}(X,X_1)$, $b\in\mathcal{A}^{\mathrm{op}}(Y,Y_1)$, $a_1\in\mathcal{A}(X',X_1')$, $b_1\in\mathcal{A}^{\mathrm{op}}(Y',Y_1')$, $f\in\mathcal{A}(Y,X)$, and $f'\in\mathcal{A}(Y',X')$, we have $(a\boxtimes b)\otimes(a_1\boxtimes b_1)=(-1)^{|b||a_1|}(a\otimes a_1)\boxtimes(b\otimes b_1)$ by (4.24), and two applications of (3.8) give $(a\otimes a_1)(f\otimes f')(b\otimes b_1)=(-1)^{|a_1||f|+(|a_1|+|f'|)|b|}(afb)\otimes(a_1f'b_1)$, so that $H\big((a\boxtimes b)\otimes(a_1\boxtimes b_1)\big)(f\otimes f')$ and $\big(H(a\boxtimes b)\otimes H(a_1\boxtimes b_1)\big)(f\otimes f')=(-1)^{(|a_1|+|b_1|)|f|}H(a\boxtimes b)(f)\otimes H(a_1\boxtimes b_1)(f')$ both equal $(-1)^{|b||f|+|b_1||f|+|b_1||f'|+|a_1||f|}(afb)\otimes(a_1f'b_1)$. For the associativity axiom, the associator of $\mathcal{A}\boxtimes\mathcal{A}^{\mathrm{op}}$ is $\alpha\boxtimes\alpha^{-1}$ by (4.25) and (4.43), and $H(\alpha\boxtimes\alpha^{-1})(f)=\alpha\circ f\circ\alpha^{-1}$; by supernaturality of α, for $f\in\mathcal{A}(Y,X)$, $f'\in\mathcal{A}(Y',X')$, and $f''\in\mathcal{A}(Y'',X'')$, $\alpha\circ((f\otimes f')\otimes f'')\circ\alpha^{-1}=f\otimes(f'\otimes f'')$, which is the axiom, the associator of $\mathcal{V}$ being the identity on underlying spaces. The unit axioms follow from the naturality of the unitors and $l_{\mathbf{1}}=r_{\mathbf{1}}$. For $X,Y\in\mathcal{A}_{(g)}$ and $X',Y'\in\mathcal{A}_{(h)}$ the tensorator is an isomorphism onto $\mathcal{A}(Y\otimes Y',X\otimes X')$ with $X\otimes X',Y\otimes Y'\in\mathcal{A}_{(gh)}$; so the restriction of H to $\mathcal{D}(\mathcal{A},\mathcal{A}^{\mathrm{op}})$ is a monoidal superfunctor. Define the monoidal superfunctor $\widetilde{H}:\mathcal{D}(\mathcal{A},\mathcal{A}^{\mathrm{op}})\to\mathsf{sVec}_G$ by $\widetilde{H}(X\boxtimes Y)=H(X\boxtimes Y)[g]$ for $X,Y\in\mathcal{A}_{(g)}$, that is, $H(X\boxtimes Y)$ placed at g, with the same tensorator and with unit $\mathbf{1}\boxtimes\mathbf{1}\mapsto k[1]$; it sends $\pi\boxtimes\mathbf{1}$ to $\mathcal{A}(\mathbf{1},\pi)[1]\cong k^{0|1}[1]$.

The superfunctor $\widetilde{H}$ is fully faithful. For $X,X',Y,Y'\in\mathcal{A}_{(g)}$, choose a simple $S\in\mathcal{A}_{(g)}$ and put $D=\mathrm{End}(S)$; by (3.37) all four objects are finite direct sums of S and ΠS, and the map $\mathcal{A}(X,X')\otimes_k\mathcal{A}^{\mathrm{op}}(Y,Y')\to\mathrm{Hom}_k(\mathcal{A}(Y,X),\mathcal{A}(Y',X'))$, $a\otimes b\mapsto H(a\boxtimes b)$, is compatible with direct sums; for $X=X'=Y=Y'=S$ it is the map $D\mathbin{\widehat{\otimes}}D^{\mathrm{sop}}\to\mathrm{End}_k(D)$, $x\mathbin{\widehat{\otimes}}y\mapsto(f\mapsto(-1)^{|y||f|}xfy)$, where $\mathcal{A}^{\mathrm{op}}(S,S)=D^{\mathrm{sop}}$ by (4.43), and this is the isomorphism (2.13). For different labels both sides vanish. The image of $\widetilde{H}$ generates sVec_G: $H(S\boxtimes S)=D$ at g has $D_0\neq0$, so k at g is an even retract of it, and $H((\Pi S)\boxtimes S)=\mathcal{A}(S,\Pi S)\cong\Pi D$ at g has $k^{0|1}$ at g as an even retract. By Propositions 4.5 and 4.6, applied to the inclusion $j:\mathcal{D}(\mathcal{A},\mathcal{A}^{\mathrm{op}})\to\mathcal{S}(\mathcal{A},\mathcal{A}^{\mathrm{op}})$ of Definition 4.9 and to $\widetilde{H}$, there is a monoidal superequivalence $\overline{H}:\mathcal{S}(\mathcal{A},\mathcal{A}^{\mathrm{op}})\to\mathsf{sVec}_G$ with $\overline{H}j\cong\widetilde{H}$; it is label-preserving because $\widetilde{H}$ sends $\mathcal{D}(\mathcal{A},\mathcal{A}^{\mathrm{op}})_{(g)}$ to the objects placed at g. □

Lemma 4.17. *For $\mathcal{A},\mathcal{B},\mathcal{C}\in\mathcal{W}(G)$,*

$$\mathcal{S}(\mathcal{A},\mathcal{B})\simeq_G\mathcal{S}(\mathcal{B},\mathcal{A}),\qquad \mathcal{S}(\mathcal{S}(\mathcal{A},\mathcal{B}),\mathcal{C})\simeq_G\mathcal{S}(\mathcal{A},\mathcal{S}(\mathcal{B},\mathcal{C})). \tag{4.45}$$

Proof. Let $\mathcal{A}_1$ and $\mathcal{B}_1$ be monoidal supercategories. The superfunctor $\sigma:\mathcal{A}_1\boxtimes\mathcal{B}_1\to\mathcal{B}_1\boxtimes\mathcal{A}_1$, $X\boxtimes Y\mapsto Y\boxtimes X$ on objects and $f\boxtimes u\mapsto(-1)^{|f||u|}u\boxtimes f$ on homogeneous morphisms f of $\mathcal{A}_1$ and u of $\mathcal{B}_1$, is an isomorphism of supercategories by (3.8), and it is strict monoidal for the structures (4.24) and (4.25): for objects X,X' of $\mathcal{A}_1$ and Y,Y' of $\mathcal{B}_1$, $\sigma((X\boxtimes Y)\otimes(X'\boxtimes Y'))=(Y\otimes Y')\boxtimes(X\otimes X')=\sigma(X\boxtimes Y)\otimes\sigma(X'\boxtimes Y')$; for homogeneous morphisms f,f' of $\mathcal{A}_1$ and u,u' of $\mathcal{B}_1$, both $\sigma((f\boxtimes u)\otimes(f'\boxtimes u'))$ and $\sigma(f\boxtimes u)\otimes\sigma(f'\boxtimes u')$ equal $(-1)^{|f||u|+|f||u'|+|f'||u'|}(u\otimes u')\boxtimes(f\otimes f')$; and σ carries $\alpha^{\mathcal{A}_1}\boxtimes\alpha^{\mathcal{B}_1}$ to $\alpha^{\mathcal{B}_1}\boxtimes\alpha^{\mathcal{A}_1}$ and likewise for the unitors and the unit. By Propositions 4.3, 4.5, and 4.6, $\iota\sigma$ extends to a monoidal superequivalence $\overline{\sigma}:\mathcal{A}_1\mathbin{\overline{\boxtimes}}\mathcal{B}_1\to\mathcal{B}_1\mathbin{\overline{\boxtimes}}\mathcal{A}_1$. For $\mathcal{A}_1=\mathcal{A}$ and $\mathcal{B}_1=\mathcal{B}$ in $\mathcal{W}(G)$, $\overline{\sigma}$ sends the objects $X\boxtimes Y$ of $\mathcal{D}(\mathcal{A},\mathcal{B})$ to the objects $Y\boxtimes X$ of $\mathcal{D}(\mathcal{B},\mathcal{A})$ and finite sums and even retracts to finite sums and even retracts; since it is an equivalence and $\mathcal{S}(\mathcal{B},\mathcal{A})$ is generated by these objects, it restricts to a monoidal superequivalence $\mathcal{S}(\mathcal{A},\mathcal{B})\to\mathcal{S}(\mathcal{B},\mathcal{A})$, which preserves the labelings (4.39).

For the second equivalence, put $\mathcal{M}=(\mathcal{A}\boxtimes\mathcal{B})\boxtimes\mathcal{C}$. The isomorphism of supercategories $\mathcal{M}\to\mathcal{A}\boxtimes(\mathcal{B}\boxtimes\mathcal{C})$, $(X\boxtimes Y)\boxtimes W\mapsto X\boxtimes(Y\boxtimes W)$ on objects and $(f\boxtimes u)\boxtimes v\mapsto f\boxtimes(u\boxtimes v)$ on morphisms, is strict monoidal by (3.7) and (4.24); we use it to regard $\mathcal{A}\boxtimes(\mathcal{B}\boxtimes\mathcal{C})$ as $\mathcal{M}$. Let $j_1=\iota\circ(\iota\boxtimes\mathrm{id}):\mathcal{M}\to(\mathcal{A}\mathbin{\overline{\boxtimes}}\mathcal{B})\mathbin{\overline{\boxtimes}}\mathcal{C}$ and $j_2=\iota\circ(\mathrm{id}\boxtimes\iota):\mathcal{M}\to\mathcal{A}\mathbin{\overline{\boxtimes}}(\mathcal{B}\mathbin{\overline{\boxtimes}}\mathcal{C})$. These strict monoidal superfunctors are fully faithful and have generating images, because ι is k-linear in each variable and therefore carries finite direct sums and even retracts in one variable to finite direct sums and even retracts. By Proposition 4.6 there are monoidal superequivalences $\Theta_1:(\mathcal{A}\mathbin{\overline{\boxtimes}}\mathcal{B})\mathbin{\overline{\boxtimes}}\mathcal{C}\to\overline{\mathcal{M}}$ and $\Theta_2:\mathcal{A}\mathbin{\overline{\boxtimes}}(\mathcal{B}\mathbin{\overline{\boxtimes}}\mathcal{C})\to\overline{\mathcal{M}}$ with $\Theta_1 j_1\cong\iota\cong\Theta_2 j_2$. The stack $\mathcal{S}(\mathcal{S}(\mathcal{A},\mathcal{B}),\mathcal{C})$ is generated by the objects $Z\boxtimes W$ with $Z\in\mathcal{S}(\mathcal{A},\mathcal{B})_{(g)}$ and $W\in\mathcal{C}_{(g)}$; by Definition 4.9, Z is an even retract of a finite sum of objects $X_i\boxtimes Y_i$ with $X_i\in\mathcal{A}_{(g_i)}$, $Y_i\in\mathcal{B}_{(g_i)}$, and since morphisms between different components vanish by (3.29), Z is an even retract of the sum of those with $g_i=g$. Let $\Lambda_1:\mathcal{S}(\mathcal{A},\mathcal{B})\mathbin{\overline{\boxtimes}}\mathcal{C}\to(\mathcal{A}\mathbin{\overline{\boxtimes}}\mathcal{B})\mathbin{\overline{\boxtimes}}\mathcal{C}$ be the extension, by Propositions 4.4 and 4.5, of the composite of the inclusion $\mathcal{S}(\mathcal{A},\mathcal{B})\boxtimes\mathcal{C}\to(\mathcal{A}\mathbin{\overline{\boxtimes}}\mathcal{B})\boxtimes\mathcal{C}$ with ι; it is fully faithful and monoidal. Let $\mathcal{S}_3$ be the full subsupercategory of $\overline{\mathcal{M}}$ whose objects are the even retracts of finite direct sums of the objects $\iota((X\boxtimes Y)\boxtimes W)$ with $X\in\mathcal{A}_{(g)}$, $Y\in\mathcal{B}_{(g)}$, $W\in\mathcal{C}_{(g)}$, and $g\in G$. The composite $\Theta_1\Lambda_1$ is fully faithful, sends the generators $Z\boxtimes W$ of $\mathcal{S}(\mathcal{S}(\mathcal{A},\mathcal{B}),\mathcal{C})$ to even retracts of finite sums of generators of $\mathcal{S}_3$, and every generator of $\mathcal{S}_3$ is isomorphic to an object in its image; so $\Theta_1\Lambda_1$ restricts to a monoidal superequivalence $\mathcal{S}(\mathcal{S}(\mathcal{A},\mathcal{B}),\mathcal{C})\to\mathcal{S}_3$. In the same way, with $\Lambda_2:\mathcal{A}\mathbin{\overline{\boxtimes}}\mathcal{S}(\mathcal{B},\mathcal{C})\to\mathcal{A}\mathbin{\overline{\boxtimes}}(\mathcal{B}\mathbin{\overline{\boxtimes}}\mathcal{C})$, the composite $\Theta_2\Lambda_2$ restricts to a monoidal superequivalence $\mathcal{S}(\mathcal{A},\mathcal{S}(\mathcal{B},\mathcal{C}))\to\mathcal{S}_3$, since $\iota(X\boxtimes(Y\boxtimes W))=\iota((X\boxtimes Y)\boxtimes W)$ in $\overline{\mathcal{M}}$. All functors involved preserve the components indexed by g, so the composite $\mathcal{S}(\mathcal{S}(\mathcal{A},\mathcal{B}),\mathcal{C})\to\mathcal{S}_3\to\mathcal{S}(\mathcal{A},\mathcal{S}(\mathcal{B},\mathcal{C}))$ is label-preserving. □

Theorem 4.18. *Let G be a finite group. The set $\mathcal{W}(G)/\simeq_G$ is an abelian group under*

$$[\mathcal{A}]\cdot[\mathcal{B}]=[\mathcal{S}(\mathcal{A},\mathcal{B})], \tag{4.46}$$

with unit $[\mathsf{sVec}_G]$ and inverse $[\mathcal{A}]^{-1}=[\mathcal{A}^{\mathrm{op}}]$.

Proof. The relation $\simeq_G$ is an equivalence relation on $\mathcal{W}(G)$: identity functors are label-preserving monoidal superequivalences, composites of label-preserving monoidal superequivalences are such, and a quasi-inverse E' of a label-preserving monoidal superequivalence $E:\mathcal{A}\to\mathcal{B}$, with the monoidal structure transported along the isomorphisms $EE'\cong\mathrm{id}$ and $E'E\cong\mathrm{id}$, is a label-preserving monoidal superequivalence $\mathcal{B}\to\mathcal{A}$, since E' sends $\mathcal{B}_{(g)}$, the component containing $E(\mathcal{A}_{(g)})$, into $\mathcal{A}_{(g)}$. By Proposition 4.10, $\mathcal{S}(\mathcal{A},\mathcal{B})$ is a G-labeled

category in $\mathcal{W}$, and by Lemma 4.11 its class depends only on $[\mathcal{A}]$ and $[\mathcal{B}]$; so (4.46) is well defined. For $\mathcal{A},\mathcal{B},\mathcal{C}\in\mathcal{W}(G)$, Lemma 4.17 gives $[\mathcal{A}][\mathcal{B}]=[\mathcal{B}][\mathcal{A}]$ and

$$([\mathcal{A}][\mathcal{B}])[\mathcal{C}]=[\mathcal{S}(\mathcal{S}(\mathcal{A},\mathcal{B}),\mathcal{C})]=[\mathcal{S}(\mathcal{A},\mathcal{S}(\mathcal{B},\mathcal{C}))]=[\mathcal{A}]([\mathcal{B}][\mathcal{C}]). \tag{4.47}$$

Lemma 4.13 gives $[\mathcal{A}][\mathsf{sVec}_G]=[\mathcal{A}]$, and Lemmas 4.15 and 4.16 give $[\mathcal{A}][\mathcal{A}^{\mathrm{op}}]=[\mathsf{sVec}_G]$. Thus $\mathcal{W}(G)/{\simeq_G}$ is an abelian group. □

Definition 4.19. For a finite group G, let $\mathrm{Tw}(G)$ be the set of twists with group G; for $s,t\in\mathrm{Tw}(G)$, write $s\sim_G t$ if there is a gauge $(\mathrm{id}_G,\lambda,\beta)$ from s to t; by Proposition 3.20 this is an equivalence relation, and we put $\mathcal{T}_k(G)=\mathrm{Tw}(G)/{\sim_G}$. Put $e_G=(G,0,0,1)$ and $p_\rho=(G,\rho,0,1)$ for $\rho\in\mathrm{Hom}(G,\mathrm{BW}(k))$. For a Brauer–Wall reference category $\mathcal{C}$ and $s\in\mathrm{Tw}(G)$, we have $\mathcal{P}_{\mathcal{C}}(s)\in\mathcal{W}(G)$ with $\lambda(g)=[X_g]$ by (3.75).

Lemma 4.20. *Let $\mathcal{C}$ be a Brauer–Wall reference category and let G be a finite group. Then*

$$\mathcal{P}_{\mathcal{C}}(e_G)\simeq_G\mathsf{sVec}_G. \tag{4.48}$$

Proof. Let $\mathcal{N}$ be the component $\mathcal{C}_r$ of $\mathcal{C}$ at $r=0$. Every simple object of $\mathcal{N}$ is isomorphic to $\mathbf{1}$, hence evenly isomorphic to $\mathbf{1}$ or π by Lemma 3.9. Let $\mathcal{V}$ be the supercategory $\mathsf{sVec}_{\{1\}}$ of Definition 4.12 for the trivial group: finite-dimensional super vector spaces and all linear maps, graded by parity, with the tensor product (4.41). Then (3.37) with $S=\mathbf{1}$ and $D=\mathrm{End}(\mathbf{1})=k$ shows that $\mathcal{C}(\mathbf{1},-):\mathcal{N}\to\mathcal{V}$ is a superequivalence. For objects X,Y of $\mathcal{N}$, define the tensorator

$$J_{X,Y}:\mathcal{C}(\mathbf{1},X)\otimes_k\mathcal{C}(\mathbf{1},Y)\longrightarrow\mathcal{C}(\mathbf{1},X\otimes Y),\qquad f\otimes u\longmapsto(f\otimes u)\circ l_{\mathbf{1}}^{-1}, \tag{4.49}$$

for $f\in\mathcal{C}(\mathbf{1},X)$ and $u\in\mathcal{C}(\mathbf{1},Y)$, an even supernatural isomorphism by (3.43), and the unit map id_k under $\mathcal{C}(\mathbf{1},\mathbf{1})=k$. For objects X,Y,Z of $\mathcal{N}$ and $f\in\mathcal{C}(\mathbf{1},X)$, $u\in\mathcal{C}(\mathbf{1},Y)$, $v\in\mathcal{C}(\mathbf{1},Z)$, the two sides of the associativity axiom are $\alpha_{X,Y,Z}\circ((f\otimes u)l_{\mathbf{1}}^{-1}\otimes v)\circ l_{\mathbf{1}}^{-1}$ and $(f\otimes(u\otimes v)l_{\mathbf{1}}^{-1})\circ l_{\mathbf{1}}^{-1}$; by supernaturality of α and of the tensor product these are $(f\otimes(u\otimes v))\circ\alpha_{\mathbf{1},\mathbf{1},\mathbf{1}}(l_{\mathbf{1}}^{-1}\otimes\mathrm{id}_{\mathbf{1}})l_{\mathbf{1}}^{-1}$ and $(f\otimes(u\otimes v))\circ(\mathrm{id}_{\mathbf{1}}\otimes l_{\mathbf{1}}^{-1})l_{\mathbf{1}}^{-1}$, which agree by coherence for the unit constraints [ML98, Chapter VII, §2]. The unit axioms hold because $l_{\mathbf{1}}=r_{\mathbf{1}}$. Thus $\mathcal{C}(\mathbf{1},-)$ is a monoidal superequivalence $\mathcal{N}\to\mathcal{V}$; it sends π to $\mathcal{C}(\mathbf{1},\pi)\cong k^{0|1}$ through ζ. Define $K:\mathcal{P}_{\mathcal{C}}(e_G)\to\mathsf{sVec}_G$ on a family $V=(V_g)_g$ and a morphism $f=(f_g)_g$ by $K(V)_g=\mathcal{C}(\mathbf{1},V_g)$ and $K(f)_g=\mathcal{C}(\mathbf{1},f_g)$, where $V_g\in\mathcal{N}$ since $\rho=0$ for e_G. By (3.58) with $\nu=0$, for families V,W the component $(V\otimes_{e_G}W)_\ell=\bigoplus_{gh=\ell}(V_g\otimes W_h)\otimes\mathbf{1}$ is sent to $\bigoplus_{gh=\ell}\mathcal{C}(\mathbf{1},V_g)\otimes\mathcal{C}(\mathbf{1},W_h)$ through the maps J_{V_g,W_h} and $r_{V_g\otimes W_h}$, which we take as the tensorator of K; the associator (3.59) with $\omega=1$ is A^0 of (3.55), built from unitors alone, which goes to the identity by the same coherence. Hence K is a monoidal superequivalence $\mathcal{P}_{\mathcal{C}}(e_G)\to\mathsf{sVec}_G$; it sends the simple object $\mathbf{1}[g]$, which is the object X_g of Proposition 3.24 for $S_0=\mathbf{1}$, to $k[g]$, so it is label-preserving. □

Lemma 4.21. *Let $\mathcal{C}$ be a Brauer–Wall reference category and let $a,b\in\mathrm{BW}(k)$. Define superfunctors $T,T^{\mathrm{op}}:\mathcal{C}_a\boxtimes\mathcal{C}_b\to\mathcal{C}_{a+b}$ by*

$$\begin{aligned}T(X\boxtimes Y)&=X\otimes Y,\qquad T(f\boxtimes g)=f\otimes g,\\ T^{\mathrm{op}}(X\boxtimes Y)&=Y\otimes X,\quad T^{\mathrm{op}}(f\boxtimes g)=(-1)^{|f||g|}g\otimes f\end{aligned} \tag{4.50}$$

for homogeneous morphisms f,g. Then $\mathrm{Nat}(T,T^{\mathrm{op}})$ is a one-dimensional super vector space, and its nonzero elements are supernatural isomorphisms.

Proof. The superfunctor property of T^{op} follows from (3.8): for composable homogeneous $f' \boxtimes g'$ and $f \boxtimes g$, both $T^{\mathrm{op}}((f' \boxtimes g') \circ (f \boxtimes g))$ and $T^{\mathrm{op}}(f' \boxtimes g') \circ T^{\mathrm{op}}(f \boxtimes g)$ equal $(-1)^{|f'||g'|+|f'||g|+|f||g|}(g'g) \otimes (f'f)$. Choose simple $S_a \in \mathcal{C}_a$, $S_b \in \mathcal{C}_b$ and put $U = S_a \boxtimes S_b$. By Proposition 3.10, T induces the superalgebra isomorphism (3.17) at U, and T^{op} induces the composite of the superalgebra isomorphism $f \mathbin{\widehat{\otimes}} g \mapsto (-1)^{|f||g|} g \mathbin{\widehat{\otimes}} f$ with (3.17) for (S_b, S_a). Hence Proposition 3.15 applies: $\mathrm{Nat}(T, T^{\mathrm{op}})$ is one-dimensional, its nonzero elements are supernatural isomorphisms, and they are determined by their values at U. An element R of $\mathrm{Nat}(T, T^{\mathrm{op}})$ of degree d is a family of morphisms $R_{X,Y} : X \otimes Y \to Y \otimes X$ of degree d, for $X \in \mathcal{C}_a$ and $Y \in \mathcal{C}_b$, and its supernaturality (3.5) is $R_{X',Y'} T(f \boxtimes g) = (-1)^{d(|f|+|g|)} T^{\mathrm{op}}(f \boxtimes g) R_{X,Y}$, that is,

$$R_{X',Y'}(f \otimes g) = (-1)^{d(|f|+|g|)+|f||g|}(g \otimes f) R_{X,Y} \tag{4.51}$$

for homogeneous $f : X \to X'$ in $\mathcal{C}_a$ and $g : Y \to Y'$ in $\mathcal{C}_b$. □

Definition 4.22. Let $\mathcal{C}$ be a Brauer–Wall reference category. The *exchange degree* of $\mathcal{C}$ is the map $b_{\mathcal{C}} : \mathrm{BW}(k)^2 \to \mathbb{Z}/2\mathbb{Z}$ sending (a, b) to the degree of $\mathrm{Nat}(T, T^{\mathrm{op}})$ in Lemma 4.21. An *exchange datum* of $\mathcal{C}$ is a triple (R, β_1, β_2) consisting of a nonzero element R of $\mathrm{Nat}(T, T^{\mathrm{op}})$ for every $a, b \in \mathrm{BW}(k)$, with components

$$R_{X,Y} : X \otimes Y \longrightarrow Y \otimes X \qquad (X \in \mathcal{C}_a,\ Y \in \mathcal{C}_b) \tag{4.52}$$

and the unit values

$$R_{\mathbf{1},X} = r_X^{-1} l_X, \qquad R_{X,\mathbf{1}} = l_X^{-1} r_X, \tag{4.53}$$

and maps $\beta_1, \beta_2 : \mathrm{BW}(k)^3 \to k^\times$, the *hexagon scalars*, such that for $X \in \mathcal{C}_a$, $Y \in \mathcal{C}_b$, $Z \in \mathcal{C}_c$

$$\alpha_{Y,Z,X} R_{X,Y\otimes Z} \alpha_{X,Y,Z} = \beta_1(a; b, c)\, (\mathrm{id}_Y \otimes R_{X,Z}) \alpha_{Y,X,Z} (R_{X,Y} \otimes \mathrm{id}_Z), \tag{4.54}$$

$$\begin{aligned} R_{X\otimes Y,Z} = \beta_2(a, b; c)\, &\alpha_{Z,X,Y} (R_{X,Z} \otimes \mathrm{id}_Y) \alpha_{X,Z,Y}^{-1} \\ &\circ (\mathrm{id}_X \otimes R_{Y,Z}) \alpha_{X,Y,Z}. \end{aligned} \tag{4.55}$$

Proposition 4.23. *Every Brauer–Wall reference category $\mathcal{C}$ has an exchange datum. For a family R as in Definition 4.22, the hexagon scalars exist, are unique, and equal* 1 *whenever an argument is* 0*. The exchange degree satisfies*

$$b_{\mathcal{C}}(a, b) = b_{\mathcal{C}}(b, a), \qquad b_{\mathcal{C}}(a + b, c) = b_{\mathcal{C}}(a, c) + b_{\mathcal{C}}(b, c), \qquad b_{\mathcal{C}}(a, a) = \epsilon(a), \tag{4.56}$$

and $b_{\mathcal{C}}(a, b) = \epsilon(a)\epsilon(b)$ for all a, b if $\mathrm{BW}(k)$ *is cyclic.*

Proof. Let R be a family of nonzero elements $R_{X,Y}$ of the lines $\mathrm{Nat}(T, T^{\mathrm{op}})$ of Lemma 4.21, one for each $(a, b) \in \mathrm{BW}(k)^2$, of degrees $b_{\mathcal{C}}(a, b)$. The composite $(X, Y) \mapsto R_{Y,X} R_{X,Y}$ is a nonzero element of $\mathrm{Nat}(T, T)$ of degree $b_{\mathcal{C}}(a, b) + b_{\mathcal{C}}(b, a)$. Since $\mathrm{Nat}(T, T)$ is a line containing id_T, this degree is 0. For $c \in \mathrm{BW}(k)$, choose simple objects $S_a \in \mathcal{C}_a$, $S_b \in \mathcal{C}_b$, and $S_c \in \mathcal{C}_c$, and let $T_1, T_2 : \mathcal{C}_a \boxtimes \mathcal{C}_b \boxtimes \mathcal{C}_c \to \mathcal{C}_{a+b+c}$ be the superfunctors with $T_1(X, Y, Z) = (X \otimes Y) \otimes Z$ and $T_2(X, Y, Z) = Y \otimes (Z \otimes X)$, where $T_2(f \boxtimes g \boxtimes h) = (-1)^{|f|(|g|+|h|)} g \otimes (h \otimes f)$ for homogeneous morphisms f, g, h carries the Koszul sign of the permutation. Both satisfy (3.36) at $S_a \boxtimes S_b \boxtimes S_c$ by Proposition 3.10 and Lemma 3.16, applied to $S_a \otimes S_b$ and S_c, respectively to S_b and $S_c \otimes S_a$; so their comparison space is a line by Proposition 3.15. The two sides of (4.54) are nonzero homogeneous elements of this line, of degrees $b_{\mathcal{C}}(a, b+c)$ and $b_{\mathcal{C}}(a, b) + b_{\mathcal{C}}(a, c)$; hence these agree, and the two sides differ by a unique scalar $\beta_1(a; b, c) \in k^\times$. The right hexagon (4.55) gives $b_{\mathcal{C}}(a + b, c) = b_{\mathcal{C}}(a, c) + b_{\mathcal{C}}(b, c)$ and $\beta_2(a, b; c)$ in the same way.

For $a = b$, let $\Phi : \mathrm{End}(S_a) \mathbin{\widehat{\otimes}} \mathrm{End}(S_a) \to \mathrm{End}(S_a \otimes S_a)$ be the isomorphism (3.17) and put $v = \Phi^{-1}(R_{S_a,S_a})$, a homogeneous element of degree $b_{\mathcal{C}}(a, a)$. For homogeneous $f, g \in \mathrm{End}(S_a)$ and $z = f \mathbin{\widehat{\otimes}} g$, (4.51) reads $\Phi(v)\Phi(z) = (-1)^{|v||z|}\Phi(\tau(z))\Phi(v)$, that is, $vz = (-1)^{|v||z|}\tau(z)v$,

where τ is the swap of $\mathrm{End}(S_a) \mathbin{\widehat{\otimes}} \mathrm{End}(S_a)$. By Proposition 2.9 applied to the central simple superalgebra $\mathrm{End}(S_a) \cong D_a$, $|v| = \epsilon(a)$.

We now choose R with the unit values (4.53). For $b \in \mathrm{BW}(k)$, let $R_{X,Y}$, for $X \in \mathcal{C}_0$ and $Y \in \mathcal{C}_b$, be the unique element of the line $\mathrm{Nat}(T, T^{\mathrm{op}})$ for the pair of degrees $(0, b)$ whose value at $\mathbf{1} \boxtimes S_b$ is $r_{S_b}^{-1} l_{S_b}$ (Lemma 4.21); it is even. For fixed b, the families $Y \mapsto R_{\mathbf{1},Y}$ and $Y \mapsto r_Y^{-1} l_Y$ are both supernatural isomorphisms between the superfunctors $Y \mapsto \mathbf{1} \otimes Y$ and $Y \mapsto Y \otimes \mathbf{1}$ on $\mathcal{C}_b$, which satisfy (3.36) at S_b, and they agree at $Y = S_b$; so $R_{\mathbf{1},Y} = r_Y^{-1} l_Y$ for all $Y \in \mathcal{C}_b$ by Proposition 3.15(2) with $n = 1$. Likewise, for $a \in \mathrm{BW}(k)$, let $R_{X,Y}$, for $X \in \mathcal{C}_a$ and $Y \in \mathcal{C}_0$, be the element of the line for $(a, 0)$ whose value at $S_a \boxtimes \mathbf{1}$ is $l_{S_a}^{-1} r_{S_a}$; then $R_{X,\mathbf{1}} = l_X^{-1} r_X$ for all $X \in \mathcal{C}_a$. The two prescriptions agree for $(a, b) = (0, 0)$ because $l_{\mathbf{1}} = r_{\mathbf{1}}$. For the remaining pairs (a, b) choose $R_{X,Y}$ as an arbitrary nonzero element of the line. When one of a, b, c is 0 and the object X, Y, or Z of that degree is $\mathbf{1}$, both sides of each hexagon agree: the exchange isomorphisms with $\mathbf{1}$ as an argument are the unit values $r^{-1}l$ and $l^{-1}r$, the remaining exchange isomorphisms, such as $R_{X,\mathbf{1}\otimes Z}$ and $R_{X,Z}$ in (4.54) for $b = 0$, are related by the supernaturality (4.51) applied to l_Z or r_Z, and the resulting composites of associativity and unit constraints agree by coherence [ML98, Chapter VII, §2]; evaluating at the simple generators gives $\beta_1 = \beta_2 = 1$ in these cases.

If $\mathrm{BW}(k)$ is cyclic with generator g_0, then for integers m, n and $a = mg_0$, $b = ng_0$, biadditivity and symmetry in (4.56) give $b_{\mathcal{C}}(a, b) = mn\, b_{\mathcal{C}}(g_0, g_0) = mn\, \epsilon(g_0)$, while $\epsilon(a)\epsilon(b) = mn\, \epsilon(g_0)^2 = mn\, \epsilon(g_0)$ in $\mathbb{Z}/2\mathbb{Z}$. □

Definition 4.24. Let (R, β_1, β_2) be an exchange datum of a Brauer–Wall reference category $\mathcal{C}$, and let $\lambda : \mathrm{BW}(k)^2 \to k^\times$ satisfy $\lambda(0, b) = \lambda(a, 0) = 1$ for all a, b. Put $(\lambda R)_{X,Y} = \lambda(a, b) R_{X,Y}$ for $X \in \mathcal{C}_a$, $Y \in \mathcal{C}_b$, and

$$\beta_1^\lambda(a; b, c) = \beta_1(a; b, c) \frac{\lambda(a, b + c)}{\lambda(a, b)\lambda(a, c)}, \qquad \beta_2^\lambda(a, b; c) = \beta_2(a, b; c) \frac{\lambda(a + b, c)}{\lambda(a, c)\lambda(b, c)}. \tag{4.57}$$

Then $(\lambda R, \beta_1^\lambda, \beta_2^\lambda)$ is the *rescaling* of the exchange datum by λ.

Proposition 4.25. *Let (R, β_1, β_2) be an exchange datum of a Brauer–Wall reference category $\mathcal{C}$. Every rescaling of it is an exchange datum of $\mathcal{C}$, and every exchange datum of $\mathcal{C}$ is its rescaling by a unique λ.*

Proof. Let λ be as in Definition 4.24. Then λR is again a family of nonzero elements of the lines of Lemma 4.21 with the unit values, so it is part of an exchange datum, and its hexagon scalars are obtained by substituting λR into the hexagons: the left hexagon for λR has $\lambda(a, b + c)$ on the left and $\lambda(a, b)\lambda(a, c)$ on the right, the right hexagon $\lambda(a + b, c)$ on the left and $\lambda(a, c)\lambda(b, c)$ on the right; so they are (4.57), and the rescaling is an exchange datum. Conversely, let (R', β_1', β_2') be another exchange datum of $\mathcal{C}$. For each (a, b), the exchange isomorphisms R' and R are nonzero elements of the line $\mathrm{Nat}(T, T^{\mathrm{op}})$, so $R' = \lambda R$ for a unique $\lambda : \mathrm{BW}(k)^2 \to k^\times$, and the common unit values force $\lambda(0, b) = \lambda(a, 0) = 1$. Since the hexagon scalars of λR are unique (Proposition 4.23), they are $(\beta_1^\lambda, \beta_2^\lambda)$; so (R', β_1', β_2') is the rescaling of (R, β_1, β_2) by λ. □

Proposition 4.26. *Let (R, β_1, β_2) be an exchange datum of a Brauer–Wall reference category $\mathcal{C}$. For $a, b, c, d \in \mathrm{BW}(k)$,*

$$\begin{aligned}
\beta_1(a; b, c)\beta_1(a; b + c, d) &= \beta_1(a; b, c + d)\beta_1(a; c, d), \\
\beta_2(a + b, c; d)\beta_2(a, b; d) &= \beta_2(b, c; d)\beta_2(a, b + c; d), \\
\frac{\beta_1(b; c, d)\beta_1(a; c, d)}{\beta_1(a + b; c, d)} &= (-1)^{b_{\mathcal{C}}(a,c)\, b_{\mathcal{C}}(b,d)} \frac{\beta_2(a, b; c)\beta_2(a, b; d)}{\beta_2(a, b; c + d)}, \\
\frac{\beta_1(a; b, c)}{\beta_1(a; c, b)} &= (-1)^{b_{\mathcal{C}}(a+b,c)\, b_{\mathcal{C}}(a,b) + b_{\mathcal{C}}(a,b+c)\, b_{\mathcal{C}}(b,c)} \frac{\beta_2(b, a; c)}{\beta_2(a, b; c)}.
\end{aligned} \tag{4.58}$$

Proof. For objects X, Y, Z, W of components of types a, b, c, d, each identity compares two nonzero homogeneous supernatural isomorphisms between the same two superfunctors on a product of components, and such isomorphisms differ by a scalar by Proposition 3.15, as in the proof of Proposition 4.23. First, expand $R_{X,(Y\otimes Z)\otimes W}$ by (4.54) for the pair $(Y\otimes Z, W)$ and then for (Y, Z), or, after transporting along the even associator by (4.51), for the pair $(Y, Z\otimes W)$ and then for (Z, W). Both expansions are the same composite of $R_{X,Y}, R_{X,Z}, R_{X,W}$ and associators, by the pentagon, with scalars $\beta_1(a; b+c, d)\beta_1(a; b, c)$ and $\beta_1(a; b, c+d)\beta_1(a; c, d)$. This is the first identity; the second follows in the same way from (4.55) applied to $R_{(X\otimes Y)\otimes Z,W}$. Third, expand $R_{X\otimes Y,Z\otimes W}$ by the right hexagon for $(a, b; c+d)$ and then the left hexagons for $(a; c, d)$ and $(b; c, d)$, or by the left hexagon for $(a+b; c, d)$ and then the right hexagons for $(a, b; c)$ and $(a, b; d)$. The elementary crossings occur in the orders $R_{Y,Z}, R_{Y,W}, R_{X,Z}, R_{X,W}$ and $R_{Y,Z}, R_{X,Z}, R_{Y,W}, R_{X,W}$; the two words differ by interchanging $R_{Y,W}$ and $R_{X,Z}$, which act on disjoint tensor factors and hence commute up to $(-1)^{|R_{Y,W}||R_{X,Z}|} = (-1)^{b_{\mathcal{C}}(b,d)\,b_{\mathcal{C}}(a,c)}$ by (3.8). This is the third identity. Fourth, (4.51) applied to $f = R_{X,Y}$ and $g = \mathrm{id}_Z$, and to $f = \mathrm{id}_X$ and $g = R_{Y,Z}$, gives

$$\begin{aligned} R_{Y\otimes X,Z}(R_{X,Y}\otimes \mathrm{id}_Z) &= (-1)^{b_{\mathcal{C}}(a+b,c)\,b_{\mathcal{C}}(a,b)}(\mathrm{id}_Z\otimes R_{X,Y})R_{X\otimes Y,Z},\\ R_{X,Z\otimes Y}(\mathrm{id}_X\otimes R_{Y,Z}) &= (-1)^{b_{\mathcal{C}}(a,b+c)\,b_{\mathcal{C}}(b,c)}(R_{Y,Z}\otimes \mathrm{id}_X)R_{X,Y\otimes Z}. \end{aligned} \tag{4.59}$$

Expand the four block exchanges by the hexagons. All four sides become composites $(X\otimes Y)\otimes Z \to Z\otimes(Y\otimes X)$ of $R_{X,Y}$, $R_{X,Z}$, $R_{Y,Z}$, and associators; let Ω_1 be the composite in which $R_{X,Y}$ is applied first, then $R_{X,Z}$, then $R_{Y,Z}$, and Ω_2 the composite in which $R_{Y,Z}$ is applied first, then $R_{X,Z}$, then $R_{X,Y}$, with the associators required by the sources and targets. Then the first identity reads $\beta_2(b, a; c)\Omega_1 = (-1)^{b_{\mathcal{C}}(a+b,c)b_{\mathcal{C}}(a,b)}\beta_2(a, b; c)\Omega_2$ and the second $\beta_1(a; c, b)\Omega_2 = (-1)^{b_{\mathcal{C}}(a,b+c)b_{\mathcal{C}}(b,c)}\beta_1(a; b, c)\Omega_1$; eliminating Ω_1, Ω_2 gives the fourth identity. □

Definition 4.27. Let $\mathcal{C}$ be a Brauer–Wall reference category with an exchange datum (R, β_1, β_2). For $s, t \in \mathrm{Tw}(G)$ define quadruples $s\star t$ and $s^\vee$ by

$$s\star t = \big(G,\ \rho_s+\rho_t,\ \nu_s+\nu_t+c_{s,t},\ \omega_s\omega_t\,K_{s,t}\,(-1)^{\Xi_{s,t}}\big), \tag{4.60}$$

$$s^\vee = \big(G,\ -\rho_s,\ \nu_s+c_{s,s},\ (\omega_s\,K_s^\vee\,(-1)^{\Xi_s^\vee})^{-1}\big), \tag{4.61}$$

where

$$\begin{aligned} c_{s,t}(g,h) &= b_{\mathcal{C}}(\rho_t(g),\rho_s(h)), \qquad K_{s,t}(g,h,\ell) = \frac{\beta_2(\rho_t(g),\rho_t(h);\rho_s(\ell))}{\beta_1(\rho_t(g);\rho_s(h),\rho_s(\ell))},\\ \Xi_{s,t}(g,h,\ell) &= \big(\nu_s \smile_1 \nu_t + c_{s,t} \smile_1 (\nu_s+\nu_t)\big)(g,h,\ell) + c_{s,t}(g,h)\,c_{s,t}(h,\ell),\\ K_s^\vee(g,h,\ell) &= \frac{\beta_2(\rho_s(g),\rho_s(h);-\rho_s(\ell))}{\beta_1(\rho_s(g);-\rho_s(h),-\rho_s(\ell))},\\ \Xi_s^\vee(g,h,\ell) &= \big(\nu_s \smile_1 \nu_s + c_{s,s} \smile_1 (\nu_s+c_{s,s})\big)(g,h,\ell) + c_{s,s}(g,h)\,c_{s,s}(h,\ell), \end{aligned} \tag{4.62}$$

with $\smile_1$ as in (A.4). By construction, $c_{s^\vee,s} = c_{s,s}$, $K_{s^\vee,s} = K_s^\vee$, and $\Xi_{s^\vee,s} = \Xi_s^\vee$.

Lemma 4.28. *For $s, t \in \mathrm{Tw}(G)$ we have $s\star t \in \mathrm{Tw}(G)$ and $s^\vee \in \mathrm{Tw}(G)$, and*

$$s^\vee \star s = e_G. \tag{4.63}$$

Proof. All calculations with $\mathbb{Z}/2\mathbb{Z}$-valued cochains take place over $\mathbb{Z}/2\mathbb{Z}$. Let $T \in C^3(G, \mathbb{Z}/2\mathbb{Z})$ be the last term of $\Xi_{s,t}$, $T(g,h,\ell) = c_{s,t}(g,h)c_{s,t}(h,\ell)$, and define $\hat{c} \in C^4(G, \mathbb{Z}/2\mathbb{Z})$ by $\hat{c}(g_1,g_2,g_3,g_4) = c_{s,t}(g_1,g_3)c_{s,t}(g_2,g_4)$. Since ρ_s, ρ_t are homomorphisms and $b_{\mathcal{C}}$ is biadditive by (4.56),

$$\begin{aligned} c_{s,t}(gg',h) = c_{s,t}(g,h) + c_{s,t}(g',h), \qquad c_{s,t}(g,hh') = c_{s,t}(g,h) + c_{s,t}(g,h'),\\ c_{s,t}(1,h) = c_{s,t}(g,1) = 0 \qquad (g,g',h,h' \in G), \end{aligned} \tag{4.64}$$

so $\delta c_{s,t}(g,h,\ell) = c_{s,t}(h,\ell) + c_{s,t}(gh,\ell) + c_{s,t}(g,h\ell) + c_{s,t}(g,h) = 0$, and $c_{s,t}$ is a normalized 2-cocycle. For the coboundary of $K_{s,t}$, let $g_1,\dots,g_4 \in G$ and write $a_i = \rho_s(g_i)$ and $b_i = \rho_t(g_i)$. The five factors of $(\delta K_{s,t})(g_1,g_2,g_3,g_4)$ are

$$\frac{\beta_2(b_2,b_3;a_4)}{\beta_1(b_2;a_3,a_4)}\,\frac{\beta_1(b_1+b_2;a_3,a_4)}{\beta_2(b_1+b_2,b_3;a_4)}\,\frac{\beta_2(b_1,b_2+b_3;a_4)}{\beta_1(b_1;a_2+a_3,a_4)}\,\frac{\beta_1(b_1;a_2,a_3+a_4)}{\beta_2(b_1,b_2;a_3+a_4)}\,\frac{\beta_2(b_1,b_2;a_3)}{\beta_1(b_1;a_2,a_3)}. \tag{4.65}$$

The second identity in (4.58) with $(a,b,c,d) = (b_1,b_2,b_3,a_4)$ reduces the three β_2 factors involving b_3 to $\beta_2(b_1,b_2;a_4)$, and the first identity with $(a;b,c,d) = (b_1;a_2,a_3,a_4)$ reduces the three β_1 factors involving a_2 to $\beta_1(b_1;a_3,a_4)^{-1}$. What remains is

$$\frac{\beta_2(b_1,b_2;a_3)\beta_2(b_1,b_2;a_4)}{\beta_2(b_1,b_2;a_3+a_4)} \cdot \frac{\beta_1(b_1+b_2;a_3,a_4)}{\beta_1(b_1;a_3,a_4)\beta_1(b_2;a_3,a_4)} = (-1)^{b_{\mathcal{C}}(b_1,a_3)\,b_{\mathcal{C}}(b_2,a_4)}, \tag{4.66}$$

by the third identity with $(a,b,c,d) = (b_1,b_2,a_3,a_4)$; and

$$b_{\mathcal{C}}(b_1,a_3)\,b_{\mathcal{C}}(b_2,a_4) = c_{s,t}(g_1,g_3)\,c_{s,t}(g_2,g_4) = \hat{c}(g_1,g_2,g_3,g_4). \tag{4.67}$$

This proves

$$\delta K_{s,t} = (-1)^{\hat{c}}. \tag{4.68}$$

Next, by (4.64),

$$\begin{aligned}
\delta T(g_1,g_2,g_3,g_4) &= c_{s,t}(g_2,g_3)c_{s,t}(g_3,g_4) + \big(c_{s,t}(g_1,g_3) + c_{s,t}(g_2,g_3)\big)c_{s,t}(g_3,g_4) \\
&\quad + \big(c_{s,t}(g_1,g_2) + c_{s,t}(g_1,g_3)\big)\big(c_{s,t}(g_2,g_4) + c_{s,t}(g_3,g_4)\big) \\
&\quad + c_{s,t}(g_1,g_2)\big(c_{s,t}(g_2,g_3) + c_{s,t}(g_2,g_4)\big) + c_{s,t}(g_1,g_2)c_{s,t}(g_2,g_3) \\
&= c_{s,t}(g_1,g_2)c_{s,t}(g_3,g_4) + c_{s,t}(g_1,g_3)c_{s,t}(g_2,g_4) \\
&= (c_{s,t} \smile c_{s,t} + \hat{c})(g_1,g_2,g_3,g_4),
\end{aligned} \tag{4.69}$$

all other terms occurring twice. Finally, since ν_s, ν_t, and $c_{s,t}$ are cocycles, (A.7) and (4.69) give

$$\begin{aligned}
\delta\Xi_{s,t} &= \nu_s \smile \nu_t + \nu_t \smile \nu_s + c_{s,t} \smile (\nu_s+\nu_t) + (\nu_s+\nu_t) \smile c_{s,t} + c_{s,t} \smile c_{s,t} + \hat{c} \\
&= \nu_{s\star t} \smile \nu_{s\star t} + \nu_s \smile \nu_s + \nu_t \smile \nu_t + \hat{c},
\end{aligned} \tag{4.70}$$

and combining with (4.68) gives

$$\delta\big(K_{s,t}(-1)^{\Xi_{s,t}}\big) = (-1)^{\nu_{s\star t}\smile\nu_{s\star t} + \nu_s\smile\nu_s + \nu_t\smile\nu_t}. \tag{4.71}$$

The maps $\rho_s + \rho_t$ and $-\rho_s$ are homomorphisms. The cochain $K_{s,t}$ is normalized because β_1 and β_2 equal 1 whenever an argument is 0, and $c_{s,t}$ is normalized as shown above; so all cochains in (4.60) and (4.61) are normalized. Since $c_{s,t}$ is a cocycle, $\delta\nu_{s\star t} = 0$. By (3.47) and (4.71),

$$\delta\omega_{s\star t} = (-1)^{\nu_s\smile\nu_s + \nu_t\smile\nu_t}\delta\big(K_{s,t}(-1)^{\Xi_{s,t}}\big) = (-1)^{\nu_{s\star t}\smile\nu_{s\star t}}, \tag{4.72}$$

For $s^\vee$, the cochain $c_{s^\vee,s}(g,h) = b_{\mathcal{C}}(\rho_s(g),\rho_s(h))$ satisfies (4.64), hence is a normalized cocycle, and $\nu_{s^\vee}$ is a normalized cocycle. The pair $(s^\vee,s)$ has $\nu_{s^\vee} + \nu_s + c_{s^\vee,s} = 0$, and (4.71) for this pair, whose proof uses only that $\rho_{s^\vee}$ and ρ_s are homomorphisms, that $\nu_{s^\vee}$ and ν_s are normalized cocycles, and that $c_{s^\vee,s}$ is a normalized cocycle satisfying (4.64), and not the cocycle condition on $\omega_{s^\vee}$, gives $\delta(K_{s^\vee,s}(-1)^{\Xi_{s^\vee,s}}) = (-1)^{\nu_{s^\vee}\smile\nu_{s^\vee} + \nu_s\smile\nu_s}$; hence $\delta\omega_{s^\vee} = (\delta\omega_s)^{-1}(-1)^{\nu_{s^\vee}\smile\nu_{s^\vee}+\nu_s\smile\nu_s} = (-1)^{\nu_{s^\vee}\smile\nu_{s^\vee}}$. Thus $s\star t$ and $s^\vee$ satisfy (3.47).

For the pair $(s^\vee,s)$, $\nu_{s^\vee\star s} = \nu_{s^\vee} + \nu_s + c_{s^\vee,s} = \nu_s + c_{s^\vee,s} + \nu_s + c_{s^\vee,s} = 0$, and $\omega_{s^\vee\star s} = 1$ by (4.61). This proves (4.63). □

Proposition 4.29. *Let $\mathcal{C}$ and (R, β_1, β_2) be as in Definition 4.27 and let $s, t \in \mathrm{Tw}(G)$. There is a fully faithful monoidal superfunctor*

$$(F_0, J, \varphi) : \mathcal{D}(\mathcal{P}_{\mathcal{C}}(s), \mathcal{P}_{\mathcal{C}}(t)) \longrightarrow \mathcal{P}_{\mathcal{C}}(s \star t) \tag{4.73}$$

whose image generates $\mathcal{P}_{\mathcal{C}}(s \star t)$ under finite direct sums and even retracts. Its data are as follows. For morphisms $f : X \to X'$ and $f' : Y \to Y'$,

$$\begin{aligned} F_0(X[g] \boxtimes Y[g]) &= (X \otimes Y)[g], \\ F_0(f[g] \boxtimes f'[g]) &= (f \otimes f')[g], \end{aligned} \tag{4.74}$$

extended linearly, where $f[g]$ denotes the morphism supported at g with component f. For objects X, Y, Z, W of components of $\mathcal{C}$, let $\chi_{X,Y,Z,W}$ be the middle exchange

$$\begin{aligned} \chi_{X,Y,Z,W} = {} & \alpha^{-1}_{X,Z,Y\otimes W}(\mathrm{id}_X \otimes \alpha_{Z,Y,W})(\mathrm{id}_X \otimes (R_{Y,Z} \otimes \mathrm{id}_W)) \\ & \circ (\mathrm{id}_X \otimes \alpha^{-1}_{Y,Z,W})\alpha_{X,Y,Z\otimes W} \\ & : (X \otimes Y) \otimes (Z \otimes W) \longrightarrow (X \otimes Z) \otimes (Y \otimes W). \end{aligned} \tag{4.75}$$

Write $e_{d,Z} = r_Z(\mathrm{id}_Z \otimes \zeta_d)$ as in Proposition 3.26. For $X \in \mathcal{C}_{\rho_s(g)}$, $Y \in \mathcal{C}_{\rho_t(g)}$, $X' \in \mathcal{C}_{\rho_s(h)}$, $Y' \in \mathcal{C}_{\rho_t(h)}$, put $a = \nu_s(g,h)$, $b = \nu_t(g,h)$, $n = \nu_{s\star t}(g,h)$, and

$$\begin{aligned} J_{g,h}(X, Y; X', Y') = {} & (e_{a,X\otimes X'} \otimes e_{b,Y\otimes Y'})^{-1} \\ & \circ \chi_{X,Y,X',Y'}\, e_{n,(X\otimes Y)\otimes(X'\otimes Y')} \\ & : ((X \otimes Y) \otimes (X' \otimes Y')) \otimes P_n \\ & \longrightarrow ((X \otimes X') \otimes P_a) \otimes ((Y \otimes Y') \otimes P_b). \end{aligned} \tag{4.76}$$

The tensorator and unit map are

$$\begin{aligned} J_{X[g]\boxtimes Y[g],\, X'[h]\boxtimes Y'[h]} &= J_{g,h}(X, Y; X', Y')[gh], \\ \varphi &= (l_{\mathbf{1}}^{-1})[1]. \end{aligned} \tag{4.77}$$

Proof. The tensor superfunctor of $\mathcal{C}$ gives F_0 on each support, and distinct supports have zero Hom spaces. Write $v = \nu_s$, $w = \nu_t$, $c = c_{s,t}$, and $\hat{n} = \nu_{s\star t} = v + w + c$, cochains on G with values in $\mathbb{Z}/2\mathbb{Z}$. Let X, Y, X', Y' and a, b, n be as in the statement. The middle exchange $R_{Y,X'}$ in (4.76) has degree $b_{\mathcal{C}}(\rho_t(g), \rho_s(h)) = c(g,h)$ by Lemma 4.21, so $J_{g,h}(X, Y; X', Y')$ has degree $n + a + b + c(g,h) = 0$. All its factors are homogeneous isomorphisms, and

$$(e_{a,X\otimes X'} \otimes e_{b,Y\otimes Y'})J_{g,h}(X, Y; X', Y') = \chi_{X,Y,X',Y'} e_{n,(X\otimes Y)\otimes(X'\otimes Y')}. \tag{4.78}$$

The inverse on the left side of (4.76) is the inverse of the entire tensor product:

$$(e_{a,X\otimes X'} \otimes e_{b,Y\otimes Y'})^{-1} = (-1)^{ab}(e^{-1}_{a,X\otimes X'} \otimes e^{-1}_{b,Y\otimes Y'}). \tag{4.79}$$

This sign follows by composing the two tensors and using (3.8).

For $d \in \mathbb{Z}/2\mathbb{Z}$, objects V, V' of $\mathcal{C}$, and a homogeneous morphism $u : V \to V'$,

$$e_{d,V'}(u \otimes \mathrm{id}_{P_d}) = (-1)^{d|u|} u\, e_{d,V}. \tag{4.80}$$

Let $f : X \to X_1$ and $f' : Y \to Y_1$ be homogeneous morphisms of $\mathcal{C}$ with $X_1 \in \mathcal{C}_{\rho_s(g)}$ and $Y_1 \in \mathcal{C}_{\rho_t(g)}$; morphisms in the variables X' and Y' are treated in the same way. Apply (4.80) to the three deletion factors in $J_{g,h}$ and (4.51) to its middle exchange. The Koszul sign for interchanging the two middle morphisms is the sign in the tensor product of the source category. The signs from the three deletion factors and from the exchange have total

exponent $(n+a+b+c(g,h))(|f|+|f'|)=0$. Thus $J_{g,h}$ is supernatural in $(X,Y;X',Y')$, and for homogeneous morphisms $u:Z\to Z'$ and $u':W\to W'$ of $\mathcal{D}(\mathcal{P}_{\mathcal{C}}(s),\mathcal{P}_{\mathcal{C}}(t))$,

$$J_{Z',W'}(F_0u\otimes_{s\star t}F_0u')=F_0(u\otimes u')J_{Z,W}. \tag{4.81}$$

The unit map is also even and invertible.

We compute the associativity equation. First record the six-object shuffle identity. For objects A,B,C,D,E,H of components of $\mathcal{C}$, define

$$\begin{aligned} U_L&=(\alpha_{A,C,E}\otimes\alpha_{B,D,H})\chi_{A\otimes C,B\otimes D,E,H}(\chi_{A,B,C,D}\otimes\mathrm{id}_{E\otimes H}),\\ U_R&=\chi_{A,B,C\otimes E,D\otimes H}(\mathrm{id}_{A\otimes B}\otimes\chi_{C,D,E,H})\alpha_{A\otimes B,C\otimes D,E\otimes H}. \end{aligned} \tag{4.82}$$

Both have source $((A\otimes B)\otimes(C\otimes D))\otimes(E\otimes H)$ and target $(A\otimes(C\otimes E))\otimes(B\otimes(D\otimes H))$. If B,C,D,E have types b_0,c_0,d_0,e_0, respectively, then

$$U_L=\frac{\beta_2(b_0,d_0;e_0)}{\beta_1(b_0;c_0,e_0)}(-1)^{b_{\mathcal{C}}(b_0,c_0)\,b_{\mathcal{C}}(d_0,e_0)}U_R. \tag{4.83}$$

Indeed, expand the block exchange $R_{B\otimes D,E}$ in U_L by (4.55), which contributes $\beta_2(b_0,d_0;e_0)$, and the block exchange $R_{B,C\otimes E}$ in U_R by (4.54), which contributes $\beta_1(b_0;c_0,e_0)$ on the right side. The elementary crossings are then $R_{B,C},R_{D,E},R_{B,E}$ in U_L and $R_{D,E},R_{B,C},R_{B,E}$ in U_R. The crossings of B with C and of D with E occur in opposite orders on disjoint tensor factors, and interchanging them contributes the product $b_{\mathcal{C}}(b_0,c_0)b_{\mathcal{C}}(d_0,e_0)$ of their degrees by (3.8), which is the sign in (4.83). The associators in U_L and in U_R contribute equally, by the pentagon in $\mathcal{C}$. This proves (4.83).

Fix labels $(g_1,g_2,g_3)=(g,h,\ell)$ and, for $i=1,2,3$, objects $X_i\in\mathcal{C}_{\rho_s(g_i)}$ and $Y_i\in\mathcal{C}_{\rho_t(g_i)}$; put $Z_i=X_i[g_i]\boxtimes Y_i[g_i]$, an object of $\mathcal{D}(\mathcal{P}_{\mathcal{C}}(s),\mathcal{P}_{\mathcal{C}}(t))$, and $V_i=X_i\otimes Y_i$. Write $\alpha^{\mathcal{D}}$ for the associator of $\mathcal{D}(\mathcal{P}_{\mathcal{C}}(s),\mathcal{P}_{\mathcal{C}}(t))$ and $\alpha^{s\star t}$ for that of $\mathcal{P}_{\mathcal{C}}(s\star t)$. For a 2-cochain $z\in C^2(G,\mathbb{Z}/2\mathbb{Z})$, use subscripts $12,L,23,R$ for its values at $(g,h),(gh,\ell),(h,\ell),(g,h\ell)$; thus $\hat{n}_i=v_i+w_i+c_i$ for $i\in\{12,L,23,R\}$. Each of the cocycles $v,w,c,\hat{n}$ satisfies $z_{12}+z_L=z_{23}+z_R$. Put $Q(z)=z_{12}z_L+z_{23}z_R$. For $(A,B,C,D,E,H)=(X_1,Y_1,X_2,Y_2,X_3,Y_3)$ the types in (4.83) are $(b_0,c_0,d_0,e_0)=(\rho_t(g),\rho_s(h),\rho_t(h),\rho_s(\ell))$, so the scalar in (4.83) is $K_{s,t}(g,h,\ell)$ and the sign is $(-1)^{c(g,h)c(h,\ell)}$, by (4.62).

Use $D_L^{U_1,U_2,U_3}$ and $D_R^{U_1,U_2,U_3}$ from (3.108). Define the homogeneous isomorphisms

$$\begin{aligned} \Delta&=D_R^{X_1,X_2,X_3}(v_{23},v_R)\otimes D_R^{Y_1,Y_2,Y_3}(w_{23},w_R),\\ M&=\chi_{X_1,Y_1,X_2\otimes X_3,Y_2\otimes Y_3}(\mathrm{id}_{V_1}\otimes\chi_{X_2,Y_2,X_3,Y_3})D_L^{V_1,V_2,V_3}(\hat{n}_{12},\hat{n}_L), \end{aligned} \tag{4.84}$$

whose common target is $(X_1\otimes(X_2\otimes X_3))\otimes(Y_1\otimes(Y_2\otimes Y_3))$. Both have degree $v_{23}+v_R+w_{23}+w_R$: Δ by (3.108), and M because its two middle exchanges have degrees $c(h,\ell)$ and $c(g,h\ell)$ by Lemma 4.21, so that M has degree $\hat{n}_{12}+\hat{n}_L+c(h,\ell)+c(g,h\ell)$, which equals $v_{23}+v_R+w_{23}+w_R$ since $\hat{n}=v+w+c$ and $\delta v=\delta w=\delta c=0$. Let L^0,R^0 be the two monoidal-functor routes

$$\begin{aligned} L^0&=F_0(\alpha^{\mathcal{D}}_{Z_1,Z_2,Z_3})J_{Z_1\otimes Z_2,Z_3}(J_{Z_1,Z_2}\otimes_{s\star t}\mathrm{id}),\\ R^0&=J_{Z_1,Z_2\otimes Z_3}(\mathrm{id}\otimes_{s\star t}J_{Z_2,Z_3})\alpha^{s\star t}_{F_0Z_1,F_0Z_2,F_0Z_3}. \end{aligned} \tag{4.85}$$

Successive use of (4.78) gives

$$\Delta L^0=\omega_s(g,h,\ell)\omega_t(g,h,\ell)K_{s,t}(g,h,\ell)(-1)^{E_L}M,\qquad \Delta R^0=\omega_{s\star t}(g,h,\ell)(-1)^{E_R}M, \tag{4.86}$$

where

$$E_L=w_Lv_{12}+\hat{n}_Lc_{12}+Q(w)+Q(v)+c(g,h)c(h,\ell),\qquad E_R=w_Rv_{23}+\hat{n}_Rc_{23}+Q(\hat{n}). \tag{4.87}$$

Here the sign calculation can be read from the following individual operations. In the left nesting, exchanging the outer w_L deletion with the inner v_{12} deletion gives $w_L v_{12}$; moving the outer degree-$\hat{n}_L$ deletion past the first shuffle gives $\hat{n}_L c_{12}$ by (4.80). Deleting the two input associators contributes $Q(w)+Q(v)$ by (3.109). The shuffle of (4.83) contributes $K_{s,t}(g,h,\ell)$ and $c(g,h)c(h,\ell)$ by (4.83). In the right nesting the same operations give $w_R v_{23}$ and $\hat{n}_R c_{23}$; deleting the output associator contributes $Q(\hat{n})$. The binary deletion identity has no further sign, because it uses the inverse in (4.79). These operations leave exactly the morphism M in both cases, which proves (4.86).

Expanding $Q(\hat{n})$ using $\hat{n} = w + v + c$ now gives

$$E_L + E_R = (v \smile_1 w + c \smile_1 (w+v))(g,h,\ell) + c(g,h)c(h,\ell) = \Xi_{s,t}(g,h,\ell). \tag{4.88}$$

Indeed, all terms below are evaluated at (g,h,ℓ):

$$\begin{aligned} w_L v_{12} + w_R v_{23} &= w \smile_1 v, \\ \hat{n}_L c_{12} + \hat{n}_R c_{23} &= (v+w+c) \smile_1 c, \\ Q(\hat{n}) + Q(v) + Q(w) &= v \smile_1 w + w \smile_1 v \\ &\quad + c \smile_1 (v+w) + (v+w) \smile_1 c + c \smile_1 c. \end{aligned} \tag{4.89}$$

Substitution in (4.86) gives (4.88). Thus the two routes are equal if and only if

$$\omega_{s\star t}(g,h,\ell) = \omega_s(g,h,\ell)\omega_t(g,h,\ell)K_{s,t}(g,h,\ell)(-1)^{\Xi_{s,t}(g,h,\ell)}, \tag{4.90}$$

which is (4.60). Cancelling the invertible Δ proves associativity.

If $g = 1$ or $h = 1$, that is, if one of the two inputs of $J_{g,h}$ is the unit, then $a = b = n = 0$, and the exchange is given by (4.53). Inserting $e_{0,V} = r_V$ in (4.76), the two unit equations reduce to the triangle for $\mathcal{C}$ and $l_{\mathbf{1}} = r_{\mathbf{1}}$, with unit map $(l_{\mathbf{1}}^{-1})[1]$. This proves the monoidal-functor axioms.

For equal supports, the Hom map in (4.74) is (3.43); for distinct supports, both spaces vanish by (3.57). Thus F_0 is fully faithful. For each g, take simple $X \in \mathcal{C}_{\rho_s(g)}$ and $Y \in \mathcal{C}_{\rho_t(g)}$. Equations (3.29) and (3.34) give $0 \neq X \otimes Y \in \mathcal{C}_{\rho_s(g)+\rho_t(g)}$. Every simple object of that component is an even retract of a finite sum of $X \otimes Y$ and its parity shift. The parity shift $\Pi(X \otimes Y)$ is evenly isomorphic to $\pi \otimes (X \otimes Y)$, hence to the image of $(\Pi X)[g] \boxtimes Y[g]$. Taking finite sums over g proves the generation assertion. □

Theorem 4.30. *Let $\mathcal{C}$ be a Brauer–Wall reference category with an exchange datum and let G be a finite group. For $s,t \in \mathrm{Tw}(G)$ there is a label-preserving monoidal superequivalence*

$$F : \mathcal{S}(\mathcal{P}_{\mathcal{C}}(s), \mathcal{P}_{\mathcal{C}}(t)) \xrightarrow{\simeq_{\otimes}} \mathcal{P}_{\mathcal{C}}(s \star t). \tag{4.91}$$

The set $\mathcal{T}_k(G)$ is an abelian group under

$$[s]\cdot[t] = [s\star t], \qquad 1 = [e_G], \qquad [s]^{-1} = [s^{\vee}], \tag{4.92}$$

the map

$$\mathcal{T}_k(G) \longrightarrow \mathcal{W}(G)/\simeq_G, \qquad [s] \longmapsto [\mathcal{P}_{\mathcal{C}}(s)], \tag{4.93}$$

is a group isomorphism onto the stacking group of Theorem 4.18.

Proof. Write $\mathcal{D} = \mathcal{D}(\mathcal{P}_{\mathcal{C}}(s), \mathcal{P}_{\mathcal{C}}(t))$, $\mathcal{S} = \mathcal{S}(\mathcal{P}_{\mathcal{C}}(s), \mathcal{P}_{\mathcal{C}}(t))$. The inclusion $j : \mathcal{D} \to \mathcal{S}$ is fully faithful and monoidal, and its image generates $\mathcal{S}$ by finite sums and even retracts. Both $\mathcal{S}$ and $\mathcal{P}_{\mathcal{C}}(s\star t)$ are additive with split even idempotents. Equations (4.7) and (4.21), followed by Proposition 4.6, show that the extensions

$$\overline{j} : \overline{\mathcal{D}} \longrightarrow \mathcal{S}, \qquad \overline{F}_0 : \overline{\mathcal{D}} \longrightarrow \mathcal{P}_{\mathcal{C}}(s\star t) \tag{4.94}$$

are monoidal superequivalences; for F_0 the hypotheses are Proposition 4.29. Choose a superequivalence $\Upsilon : \mathcal{S} \to \overline{\mathcal{D}}$ quasi-inverse to $\overline{j}$ such that $\Upsilon j = \iota$ and the counit $\overline{j}\Upsilon \Rightarrow \mathrm{Id}_{\mathcal{S}}$ is the identity on the objects jX, $X \in \mathcal{D}$. Such a choice is possible because j is an inclusion and $\overline{j}\iota = j$: choose these preimages on $j\mathcal{D}$, choose preimages of the other objects arbitrarily, and use full faithfulness of $\overline{j}$ to define Υ on morphisms. Let Υ carry the monoidal structure transported from that of $\overline{j}$ along this counit. Define

$$F = \overline{F}_0\Upsilon : \mathcal{S} \longrightarrow \mathcal{P}_{\mathcal{C}}(s \star t), \qquad \text{so that} \qquad Fj = \overline{F}_0\Upsilon j = \overline{F}_0\iota = F_0 \tag{4.95}$$

as monoidal superfunctors. The functor F_0 sends $X[g] \boxtimes Y[g]$, which lies in the component $\mathcal{S}_{(g)}$ by (4.39), to $(X \otimes Y)[g]$, which lies in the component of $\mathcal{P}_{\mathcal{C}}(s \star t)$ labeled by g; since every object of $\mathcal{S}_{(g)}$ is an even retract of a finite sum of such objects with the same g, by Definition 4.9 and (3.29), F is label-preserving. This proves (4.91).

Let $(\mathcal{A}, \lambda_{\mathcal{A}})$ be G-labeled. By Theorem 3.29 and Proposition 3.27, there is a twist $t'_{\mathcal{A}} = (G_{\mathcal{A}}, \rho_{\mathcal{A}}, \nu', \omega')$ and a monoidal superequivalence $\mathcal{A} \to \mathcal{P}_{\mathcal{C}}(t'_{\mathcal{A}})$ sending $\mathcal{A}_{g'}$ into the families supported at g'. Put $t_{\mathcal{A}} = (G, \rho_{\mathcal{A}}\lambda_{\mathcal{A}}, \lambda_{\mathcal{A}}^*\nu', \lambda_{\mathcal{A}}^*\omega')$. Then $(\lambda_{\mathcal{A}}, 0, 1)$ is a gauge from $t_{\mathcal{A}}$ to $t'_{\mathcal{A}}$, and the functor F^{λ} of (3.103) in Proposition 3.28, $(F^{\lambda}V)_{\lambda_{\mathcal{A}}(g)} = V_g \otimes P_0 = V_g \otimes \mathbf{1}$ for a family V, is a monoidal superequivalence $\mathcal{P}_{\mathcal{C}}(t_{\mathcal{A}}) \to \mathcal{P}_{\mathcal{C}}(t'_{\mathcal{A}})$ carrying the families supported at g to the families supported at $\lambda_{\mathcal{A}}(g)$. Composing with a quasi-inverse of the first equivalence gives a label-preserving monoidal superequivalence $\mathcal{P}_{\mathcal{C}}(t_{\mathcal{A}}) \to \mathcal{A}$, and its quasi-inverse is label-preserving as well.

For $s, t \in \mathrm{Tw}(G)$, a gauge $(\mathrm{id}_G, \lambda, \beta)$ from s to t gives by (3.103) a monoidal superequivalence F^{λ} with $(F^{\lambda}V)_g = V_g \otimes P_{\lambda(g)}$ for a family V, which sends the component at g to the component at g; it is label-preserving. Conversely, if $E : \mathcal{P}_{\mathcal{C}}(s) \to \mathcal{P}_{\mathcal{C}}(t)$ is label-preserving, then $E(X_g)$ is a simple object of the component of X_g, hence isomorphic to X_g by Proposition 3.24, and the last assertion of Proposition 3.28 gives $s \sim_G t$. The map (4.93) is well defined and injective by the second assertion and surjective by the first.

By Theorem 4.18, there is a unique group structure on $\mathcal{T}_k(G)$ for which (4.93) is an isomorphism. By (4.91) and (4.46), $[\mathcal{P}_{\mathcal{C}}(s)][\mathcal{P}_{\mathcal{C}}(t)] = [\mathcal{S}(\mathcal{P}_{\mathcal{C}}(s), \mathcal{P}_{\mathcal{C}}(t))] = [\mathcal{P}_{\mathcal{C}}(s \star t)]$, so $[s][t] = [s \star t]$; the unit $[\mathsf{sVec}_G]$ corresponds to $[e_G]$ by (4.48); and (4.63) gives $s^{\vee} \star s = e_G$, so $[s]^{-1} = [s^{\vee}]$. □

Proposition 4.31. *Let $\mathcal{C}$ be a Brauer–Wall reference category with an exchange datum and let G be a finite group. The subset*

$$\mathrm{SH}_k(G) = \{t \in \mathrm{Tw}(G) : \rho_t = 0\}/\!\sim_G \tag{4.96}$$

of $\mathcal{T}_k(G)$ is a subgroup, on which the product (4.92) *is induced by*

$$(\nu, \omega) \star (\nu', \omega') = \big(\nu + \nu',\, \omega\omega'(-1)^{\nu \smile_1 \nu'}\big), \tag{4.97}$$

for every exchange datum, with $[(\nu, \omega)]^{-1} = [(\nu, \omega^{-1}(-1)^{\nu \smile_1 \nu})]$. There is an exact sequence

$$\begin{gathered} 0 \longrightarrow H^3(G, k^{\times}) \longrightarrow \mathrm{SH}_k(G) \longrightarrow H^2(G, \mathbb{Z}/2\mathbb{Z})^+ \longrightarrow 0, \\ H^2(G, \mathbb{Z}/2\mathbb{Z})^+ = \{[\nu] \in H^2(G, \mathbb{Z}/2\mathbb{Z}) : (-1)^{\nu \smile \nu} \in B^4(G, k^{\times})\}, \end{gathered} \tag{4.98}$$

in which the first map is $[\omega] \mapsto [(0, \omega)]$ and the second is $[(\nu, \omega)] \mapsto [\nu]$.

Proof. Let $\mathcal{C}$ be a Brauer–Wall reference category with an exchange datum (R, β_1, β_2), as in Definition 4.27, and let $s, t \in \mathrm{Tw}(G)$ with $\rho_s = \rho_t = 0$. Then $c_{s,t} = 0$ and $K_{s,t} = 1$, since $b_{\mathcal{C}}(0, 0) = 0$ and $\beta_2(0, 0; 0) = \beta_1(0; 0, 0) = 1$; whence $\Xi_{s,t} = \nu_s \smile_1 \nu_t$ and (4.60) is (4.97), whatever the exchange datum is. A gauge $(\mathrm{id}_G, \lambda, \beta)$ preserves ρ by (3.48), so (4.96) is a subset of $\mathcal{T}_k(G)$; it is a subgroup because $\rho_{s\star t} = \rho_s + \rho_t$, $\rho_{s^{\vee}} = -\rho_s$, and $\rho_{e_G} = 0$, with the

product $[s][t] = [s \star t]$ of Theorem 4.30; and $(\nu, \omega) \star (\nu, \omega^{-1}(-1)^{\nu \smile_1 \nu}) = (0, 1) = e_G$ gives the inverse.

The map $[(\nu, \omega)] \mapsto [\nu]$ is well defined, since a gauge $(\mathrm{id}_G, \lambda, \beta)$ changes ν by $\delta\lambda$, and it is a homomorphism by (4.97). Its image consists of the classes $[\nu]$ with $(-1)^{\nu \smile \nu} = \delta\omega$ for some normalized ω, that is, of $H^2(G, \mathbb{Z}/2\mathbb{Z})^+$; this set does not depend on the representative ν, because $(-1)^{(\nu+\delta\lambda) \smile (\nu+\delta\lambda)} = (-1)^{\nu \smile \nu}\, \delta((-1)^{\lambda \smile \nu + \nu \smile \lambda + \lambda \smile \delta\lambda})$ by (A.8) and (A.5). If $\nu = \delta\lambda$, then $(\mathrm{id}_G, \lambda, 1)$ is a gauge from (ν, ω) to $(0, \omega(-1)^{\lambda \smile \nu})$ by (3.49), so the kernel consists of classes of pairs $(0, \omega)$ with $\delta\omega = 1$. Two such pairs $(0, \omega)$ and $(0, \omega')$ are gauge equivalent if and only if $\omega' = \omega(\delta\beta)^{-1}(-1)^{0 \smile \lambda + \lambda \smile 0} = \omega(\delta\beta)^{-1}$ for some β, that is, if and only if $[\omega] = [\omega']$ in $H^3(G, k^\times)$; and $(0, \omega) \star (0, \omega') = (0, \omega\omega')$. This identifies the kernel with $H^3(G, k^\times)$ and proves exactness of (4.98). □

Theorem 4.32. *Let $\mathcal{C}$ be a Brauer–Wall reference category with an exchange datum and let G be a finite group. The degree map*

$$d : \mathcal{T}_k(G) \longrightarrow \mathrm{Hom}(G, \mathrm{BW}(k)), \qquad [s] \longmapsto \rho_s, \tag{4.99}$$

is a surjective group homomorphism with $d[p_\rho] = \rho$ and kernel $\mathrm{SH}_k(G)$, so that there is an exact sequence

$$0 \longrightarrow \mathrm{SH}_k(G) \longrightarrow \mathcal{T}_k(G) \xrightarrow{d} \mathrm{Hom}(G, \mathrm{BW}(k)) \longrightarrow 0. \tag{4.100}$$

Proof. Let $s, s', t \in \mathrm{Tw}(G)$. The degree map is well defined, since a gauge $(\mathrm{id}_G, \lambda, \beta)$ from s to s' has $\rho_{s'} = \rho_s$ by (3.48); it is a homomorphism because $\rho_{s \star t} = \rho_s + \rho_t$; $d[p_\rho] = \rho$ shows that it is surjective; and its kernel is $\mathrm{SH}_k(G)$ by definition. So (4.100) is exact. □

5 Reference categories over k

Definition 5.1. A *braided categorical group* is a braided monoidal category in which every morphism is invertible and every object is invertible with respect to the tensor product [JS93, §3]. A map $q : \Gamma \to \Lambda$ between abelian groups is *quadratic* if $q(-a) = q(a)$ for all $a \in \Gamma$ and the map $\Gamma \times \Gamma \to \Lambda$, $(a, a') \mapsto q(a + a') - q(a) - q(a')$, is additive in each variable.

Proposition 5.2. *Let $\mathcal{G}$ be a braided categorical group with tensor product $\otimes$, unit $\mathbf{1}$, and braiding c. The set $\pi_0\mathcal{G}$ of isomorphism classes of objects of $\mathcal{G}$ is an abelian group with the multiplication induced by $\otimes$, and $\mathrm{Aut}(\mathbf{1})$ is abelian. For every object Y of $\mathcal{G}$, the map $f \mapsto \mathrm{id}_Y \otimes f$ is an isomorphism $\mathrm{Aut}(\mathbf{1}) \to \mathrm{Aut}(Y \otimes \mathbf{1}) \cong \mathrm{Aut}(Y)$. The map*

$$\begin{gathered} q : \pi_0\mathcal{G} \longrightarrow \mathrm{Aut}(\mathbf{1}), \\ q([X]) = \textit{the element of } \mathrm{Aut}(\mathbf{1}) \textit{ corresponding to } c_{X,X} \in \mathrm{Aut}(X \otimes X), \end{gathered} \tag{5.1}$$

is well defined and quadratic; it is called the quadratic function *of $\mathcal{G}$.*

Proof. Every object X has an object X' with $X \otimes X' \cong \mathbf{1} \cong X' \otimes X$, so $[X]$ has the inverse $[X']$, and $[X][Y] = [X \otimes Y] = [Y \otimes X] = [Y][X]$ because $c_{X,Y}$ is an isomorphism; thus $\pi_0\mathcal{G}$ is an abelian group. Let l and r be the unitors of $\mathcal{G}$ and let $u = l_{\mathbf{1}} = r_{\mathbf{1}} : \mathbf{1} \otimes \mathbf{1} \to \mathbf{1}$ [ML98, Chapter VII, §2]. For $f, f' \in \mathrm{Aut}(\mathbf{1})$, naturality of r and l gives $u \circ (f \otimes \mathrm{id}) = f \circ u$ and $u \circ (\mathrm{id} \otimes f') = f' \circ u$, so

$$f \circ f' \circ u = u \circ (f \otimes \mathrm{id}) \circ (\mathrm{id} \otimes f') = u \circ (f \otimes f') = u \circ (\mathrm{id} \otimes f') \circ (f \otimes \mathrm{id}) = f' \circ f \circ u, \tag{5.2}$$

and $f \circ f' = f' \circ f$ since u is invertible.

Let Y be an object and Y' an inverse. The functor $Y \otimes -$ has the quasi-inverse $Y' \otimes -$, so it is an equivalence and induces an isomorphism $\mathrm{Aut}(\mathbf{1}) \to \mathrm{Aut}(Y \otimes \mathbf{1})$, $f \mapsto \mathrm{id}_Y \otimes f$;

composing with conjugation by $r_Y : Y \otimes \mathbf{1} \to Y$ gives the isomorphism $\mathrm{Aut}(\mathbf{1}) \to \mathrm{Aut}(Y)$. For objects Z, Z' and an isomorphism $\psi : Z \to Z'$, these isomorphisms are compatible with conjugation by ψ, because $(\psi \otimes \mathrm{id}) \circ (\mathrm{id}_Z \otimes f) = \psi \otimes f = (\mathrm{id}_{Z'} \otimes f) \circ (\psi \otimes \mathrm{id})$ for $f \in \mathrm{Aut}(\mathbf{1})$ and r is natural.

If $\phi : X \to X'$ is an isomorphism, naturality of the braiding gives $(\phi \otimes \phi) \circ c_{X,X} = c_{X',X'} \circ (\phi \otimes \phi)$, so $c_{X,X}$ and $c_{X',X'}$ correspond to the same element of $\mathrm{Aut}(\mathbf{1})$ by the compatibility just shown; hence q is well defined. That q is quadratic, with $q([X] + [Y]) - q([X]) - q([Y])$ the element corresponding to $c_{Y,X} \circ c_{X,Y} \in \mathrm{Aut}(X \otimes Y)$, is [JS93, §3]. □

Definition 5.3. Let $\mathcal{G}$ be a braided categorical group with tensor product $\otimes$, unit $\mathbf{1}$, associator α, and unitors l and r, and let $(X_a)_{a \in \pi_0\mathcal{G}}$ be objects of $\mathcal{G}$ such that X_a is of class a and $X_0 = \mathbf{1}$. Automorphism groups of objects are identified with $\mathrm{Aut}(\mathbf{1})$ by Proposition 5.2, and cochains with values in $\mathrm{Aut}(\mathbf{1})$ are written multiplicatively. A *section* of $\mathcal{G}$ on $(X_a)_{a \in \pi_0\mathcal{G}}$ is a family $t = (t_{a,b})_{a,b \in \pi_0\mathcal{G}}$ of isomorphisms $t_{a,b} : X_a \otimes X_b \to X_{a+b}$ with $t_{0,a} = l_{X_a}$ and $t_{a,0} = r_{X_a}$. Its *associativity cocycle* $h_t \in C^3(\pi_0\mathcal{G}, \mathrm{Aut}(\mathbf{1}))$ is defined by

$$t_{a,b+c} \circ (\mathrm{id}_{X_a} \otimes t_{b,c}) \circ \alpha_{X_a,X_b,X_c} = h_t(a,b,c) \circ t_{a+b,c} \circ (t_{a,b} \otimes \mathrm{id}_{X_c}), \tag{5.3}$$

with $h_t(a,b,c)$ acting on X_{a+b+c}. For a normalized $n \in C^2(\pi_0\mathcal{G}, \mathrm{Aut}(\mathbf{1}))$, the section $n \cdot t$ is $(n(a,b) \circ t_{a,b})_{a,b}$, with $n(a,b)$ acting on X_{a+b}. The section t is *monoidal* if $h_t = 1$.

Proposition 5.4. *Let $\mathcal{G}$ and $(X_a)_{a \in \pi_0\mathcal{G}}$ be as in Definition 5.3, and let t be a section of $\mathcal{G}$ on $(X_a)_{a \in \pi_0\mathcal{G}}$.*

(1) The cochain h_t is a normalized 3-cocycle, every section on $(X_a)_{a \in \pi_0\mathcal{G}}$ is $n \cdot t$ for a unique normalized $n \in C^2(\pi_0\mathcal{G}, \mathrm{Aut}(\mathbf{1}))$, and $h_{n \cdot t} = h_t\, \delta n$. The class

$$[h_t] \in H^3(\pi_0\mathcal{G}, \mathrm{Aut}(\mathbf{1})) \tag{5.4}$$

is independent of t and of $(X_a)_{a \in \pi_0\mathcal{G}}$; it is called the associator class *of $\mathcal{G}$, and braided equivalent braided categorical groups have the same associator class [JS93, §3].*

(2) The section t is monoidal if and only if $a \mapsto X_a$ with the structure isomorphisms $t_{a,b}$ is a monoidal functor $(\pi_0\mathcal{G})_{\mathrm{disc}} \to \mathcal{G}$, where $(\pi_0\mathcal{G})_{\mathrm{disc}}$ is the strict monoidal category with object set $\pi_0\mathcal{G}$ and only identity morphisms, whose tensor product and unit are the multiplication and the unit of the group $\pi_0\mathcal{G}$. Two monoidal sections t, t' are monoidally isomorphic as such functors if and only if $t' = (\delta\mu) \cdot t$ for some normalized $\mu \in C^1(\pi_0\mathcal{G}, \mathrm{Aut}(\mathbf{1}))$.

(3) A monoidal section on $(X_a)_{a \in \pi_0\mathcal{G}}$ exists if and only if the associator class of $\mathcal{G}$ is trivial. In that case $[n] \cdot t = n \cdot t$ makes the set of monoidal sections on $(X_a)_{a \in \pi_0\mathcal{G}}$ modulo monoidal isomorphism a torsor under $H^2(\pi_0\mathcal{G}, \mathrm{Aut}(\mathbf{1}))$.

(4) Let q be the quadratic function of $\mathcal{G}$. The associator class of $\mathcal{G}$ is trivial if and only if there is a biadditive map $\beta : \pi_0\mathcal{G} \times \pi_0\mathcal{G} \to \mathrm{Aut}(\mathbf{1})$ such that $q(a) = \beta(a,a)$ for all $a \in \pi_0\mathcal{G}$.

(5) Let $(X'_a)_{a \in \pi_0\mathcal{G}}$ be another family of objects such that X'_a is of class a and $X'_0 = \mathbf{1}$, and let $\xi_a : X_a \to X'_a$ be isomorphisms with $\xi_0 = \mathrm{id}_{\mathbf{1}}$. Then

$$t \longmapsto t^\xi, \qquad t^\xi_{a,b} = \xi_{a+b} \circ t_{a,b} \circ (\xi_a \otimes \xi_b)^{-1}, \tag{5.5}$$

is a bijection from the sections on $(X_a)_{a \in \pi_0\mathcal{G}}$ to the sections on $(X'_a)_{a \in \pi_0\mathcal{G}}$ with $h_{t^\xi} = h_t$ and $(n \cdot t)^\xi = n \cdot t^\xi$; it induces a bijection between the torsors of (3) which does not depend on the ξ_a.

Proof. Throughout, automorphisms of objects are identified with elements of $\mathrm{Aut}(\mathbf{1})$ by Proposition 5.2; these identifications are compatible with $\otimes$ and with conjugation by isomorphisms, and $\mathrm{Aut}(\mathbf{1})$ is abelian, so an automorphism of the form $u \otimes \mathrm{id}$ or $\mathrm{id} \otimes u$ may be moved through any isomorphism and through any tensor product with identities.

(1) *Normalization.* For $a = 0$, both sides of (5.3) equal $t_{b,c} \circ (l_{X_b} \otimes \mathrm{id})$: the left side is $l_{X_{b+c}} \circ (\mathrm{id} \otimes t_{b,c}) \circ \alpha_{\mathbf{1},X_b,X_c} = t_{b,c} \circ l_{X_b \otimes X_c} \circ \alpha_{\mathbf{1},X_b,X_c}$ by naturality of l, and $l_{X_b \otimes X_c} \circ \alpha_{\mathbf{1},X_b,X_c} = l_{X_b} \otimes \mathrm{id}$ by coherence [ML98, Chapter VII, §2]. The cases $b = 0$ and $c = 0$ follow in the same way from the triangle axiom and from the naturality of r. Hence h_t is normalized.

Cocycle. For $a, b, c, d \in \pi_0\mathcal{G}$, consider the five parenthesizations of $X_a \otimes X_b \otimes X_c \otimes X_d$ and, for each, the isomorphism to $X_{a+b+c+d}$ obtained by applying the t's from the innermost bracket outwards. Each of the five associativity isomorphisms of $\mathcal{G}$ between adjacent parenthesizations, composed with these isomorphisms, is by (5.3) and naturality the automorphism $h_t(b, c, d)$, $h_t(a, b + c, d)$, $h_t(a, b, c)$, $h_t(a + b, c, d)$, or $h_t(a, b, c + d)$ of $X_{a+b+c+d}$, the first three along one path of the pentagon and the last two along the other. The pentagon axiom of $\mathcal{G}$ says that the composites along the two paths are equal, so $h_t(b, c, d)h_t(a, b + c, d)h_t(a, b, c) = h_t(a + b, c, d)h_t(a, b, c + d)$, that is, $\delta h_t = 1$.

Change of section. If t' is another section, then $t'_{a,b} \circ t_{a,b}^{-1}$ is an automorphism of X_{a+b}, so $t' = n \cdot t$ for a unique $n \in C^2(\pi_0\mathcal{G}, \mathrm{Aut}(\mathbf{1}))$, which is normalized because $t'_{0,a} = t_{0,a}$ and $t'_{a,0} = t_{a,0}$. Substituting $n \cdot t$ into (5.3) and moving the central automorphisms $n(b, c)$, $n(a, b+c)$ to the left side and $n(a, b)$, $n(a+b, c)$ to the right side gives $h_{n\cdot t} = h_t\, n(b, c)n(a, b + c)n(a, b)^{-1}n(a + b, c)^{-1} = h_t\, \delta n$. Hence $[h_t]$ does not depend on t. If (X'_a) is another family of objects with X'_a of class a and $X'_0 = \mathbf{1}$, choose isomorphisms $\xi_a : X_a \to X'_a$ with $\xi_0 = \mathrm{id}$ and put $t'_{a,b} = \xi_{a+b} \circ t_{a,b} \circ (\xi_a \otimes \xi_b)^{-1}$; this is a section on (X'_a), and naturality of α together with the compatibility of the identifications of automorphism groups with conjugation gives $h_{t'} = h_t$. Invariance under braided equivalences is [JS93, §3], where the associator class is shown to be the invariant of the underlying categorical group.

(2) A monoidal functor $(\pi_0\mathcal{G})_{\mathrm{disc}} \to \mathcal{G}$ with $a \mapsto X_a$ consists of the structure isomorphisms $t_{a,b} : X_a \otimes X_b \to X_{a+b}$ subject to the associativity axiom, which is (5.3) with $h_t = 1$, and to the unit axioms $t_{0,a} = l_{X_a}$, $t_{a,0} = r_{X_a}$, which are the normalization of a section. A monoidal natural isomorphism from t to t' is a family of automorphisms $\mu(a)$ of X_a with $\mu(0) = \mathrm{id}$ and $t'_{a,b} \circ (\mu(a) \otimes \mu(b)) = \mu(a + b) \circ t_{a,b}$; moving the central automorphisms, this reads $t'_{a,b} = \mu(a + b)\mu(a)^{-1}\mu(b)^{-1} \circ t_{a,b}$, that is, $t' = (\delta\mu^{-1}) \cdot t$, and every normalized μ arises.

(3) If t' is a monoidal section then $[h_t] = [h_{t'}] = 1$ by (1). Conversely, if $h_t = \delta n$ for a normalized n, then $h_{n^{-1}\cdot t} = h_t\, \delta n^{-1} = 1$. If the associator class is trivial, let t_0 be a monoidal section on (X_a); by (1) the monoidal sections are the $n \cdot t_0$ with $\delta n = 1$, and by (2) two of them are monoidally isomorphic if and only if the cocycles differ by a coboundary. Hence $[n] \cdot t_0 = n \cdot t_0$, for monoidal t_0, is a well-defined free and transitive action of $H^2(\pi_0\mathcal{G}, \mathrm{Aut}(\mathbf{1}))$.

(4) Let c be the braiding of $\mathcal{G}$. Suppose that $\beta : \pi_0\mathcal{G} \times \pi_0\mathcal{G} \to \mathrm{Aut}(\mathbf{1})$ is biadditive with $q(a) = \beta(a, a)$ for all a. Let $\mathcal{G}'$ be the braided categorical group whose objects are the elements of $\pi_0\mathcal{G}$, with $\mathrm{Aut}(a) = \mathrm{Aut}(\mathbf{1})$ and no other morphisms, with tensor product the multiplication of $\pi_0\mathcal{G}$ on objects and of $\mathrm{Aut}(\mathbf{1})$ on morphisms, with identity associator and unitors, and with braiding $c'_{a,b} = \beta(a, b)$; the hexagon axioms are the biadditivity of β, and the quadratic function of $\mathcal{G}'$ is $a \mapsto \beta(a, a) = q(a)$. By [JS93, §3], $\mathcal{G}$ and $\mathcal{G}'$ are braided equivalent, so by (1) the associator class of $\mathcal{G}$ is that of $\mathcal{G}'$, which is trivial. Conversely, suppose the associator class of $\mathcal{G}$ is trivial, let t_0 be a monoidal section on (X_a), which exists by (3), and define $\beta : \pi_0\mathcal{G} \times \pi_0\mathcal{G} \to \mathrm{Aut}(\mathbf{1})$ by $\beta(a, b) = (t_0)_{b,a} \circ c_{X_a,X_b} \circ (t_0)_{a,b}^{-1} \in \mathrm{Aut}(X_{a+b}) = \mathrm{Aut}(\mathbf{1})$. Let $\mathcal{G}''$ be the category whose objects are the elements of $\pi_0\mathcal{G}$, with $\mathrm{Aut}(a) = \mathrm{Aut}(\mathbf{1})$ and no other morphisms, with tensor product the multiplication of $\pi_0\mathcal{G}$ on objects and of $\mathrm{Aut}(\mathbf{1})$ on morphisms, with identity associator and unitors, and with braiding $c''_{a,b} = \beta(a, b)$. The functor $\mathcal{G}'' \to \mathcal{G}$, $a \mapsto X_a$, with structure isomorphisms $(t_0)_{a,b}$ is an equivalence of categories, by Proposition 5.2, and since $h_{t_0} = 1$ and β is defined by conjugation with t_0, it carries the tensor product, associator, unitors, and braiding of $\mathcal{G}''$ to those of $\mathcal{G}$. Hence $\mathcal{G}''$ is a braided categorical group braided equivalent to $\mathcal{G}$; its hexagon axioms read $\beta(a + b, c) = \beta(a, c)\beta(b, c)$ and $\beta(a, b + c) = \beta(a, b)\beta(a, c)$ [JS93, §3], and $\beta(a, a) = q(a)$ by (5.1).

(5) The map $t \mapsto t^\xi$ has the inverse $t' \mapsto t'^{\xi^{-1}}$, and t^ξ is a section on (X'_a) because $\xi_0 = \mathrm{id}$ and the unit isomorphisms are natural. Naturality of α and the compatibility of the identifications of automorphism groups with conjugation give $h_{t^\xi} = h_t$, as in the proof of (1), and $(n \cdot t)^\xi = n \cdot t^\xi$ because the automorphisms $n(a,b)$ are central. Hence $t \mapsto t^\xi$ sends monoidal sections to monoidal sections, $(\delta\mu) \cdot t$ to $(\delta\mu) \cdot t^\xi$, and $n \cdot t$ to $n \cdot t^\xi$, so it induces a bijection of the torsors of (3). Let $\xi'_a : X_a \to X'_a$ be other isomorphisms with $\xi'_0 = \mathrm{id}$, and write $\xi'_a = \xi_a \circ u_a$ with $u_a = \xi_a^{-1}\xi'_a \in \mathrm{Aut}(X_a)$, so that $u_0 = \mathrm{id}$; then, moving the central automorphisms, $t^{\xi'}_{a,b} = u_{a+b}u_a^{-1}u_b^{-1} \circ t^\xi_{a,b}$, so $t^{\xi'} = (\delta u^{-1}) \cdot t^\xi$ is monoidally isomorphic to t^ξ by (2); the induced bijection is therefore independent of the ξ_a. □

Definition 5.5. The *Morita groupoid* Mor_k is the category whose objects are the finite-dimensional central simple superalgebras and whose morphisms $A \to B$ are the even isomorphism classes $[M]$ of Morita equivalences $M : A \to B$, with the composite of $[M] : A \to B$ and $[N] : B \to C$ equal to $[M \otimes_B N]$ and with identities $[A]$. It is equipped with the product $\widehat{\otimes}$ on objects and on classes and with the unit k. An isomorphism of superalgebras is regarded as a Morita equivalence by Definition 2.5. The associator is the class of the isomorphism $(A \mathbin{\widehat{\otimes}} B) \mathbin{\widehat{\otimes}} C \to A \mathbin{\widehat{\otimes}} (B \mathbin{\widehat{\otimes}} C)$, $(a \mathbin{\widehat{\otimes}} b) \mathbin{\widehat{\otimes}} c \mapsto a \mathbin{\widehat{\otimes}} (b \mathbin{\widehat{\otimes}} c)$, the unitors are the classes of the isomorphisms $k \mathbin{\widehat{\otimes}} A \to A$, $\lambda \mathbin{\widehat{\otimes}} a \mapsto \lambda a$, and $A \mathbin{\widehat{\otimes}} k \to A$, $a \mathbin{\widehat{\otimes}} \lambda \mapsto a\lambda$, and the braiding is the class of the swap $\tau_{A,B} : A \mathbin{\widehat{\otimes}} B \to B \mathbin{\widehat{\otimes}} A$, $a \mathbin{\widehat{\otimes}} b \mapsto (-1)^{|a||b|} b \mathbin{\widehat{\otimes}} a$.

Lemma 5.6. *The category* Mor_k *with the structure of Definition 5.5 is a braided categorical group. By Definitions 2.5 and 2.7, the isomorphism classes of objects of* Mor_k *are the graded Morita classes, so* $\pi_0\mathsf{Mor}_k = \mathrm{BW}(k)$*, the multiplication induced by* $\widehat{\otimes}$ *being the addition of Definition 2.7. For every object* X*, the automorphism group* $\mathrm{Aut}(X)$ *of* X *in* Mor_k *consists of the classes of the two Morita equivalences* ${}_XX_X$ *and* $\Pi({}_XX_X)$*, where* $\Pi({}_XX_X)$ *has the actions* (2.16)*, and the isomorphism* $\mathrm{Aut}(k) \to \mathrm{Aut}(X)$ *of Proposition 5.2 sends the class of* Πk *to the class of* $\Pi({}_XX_X)$*. We identify* $\mathrm{Aut}(k)$ *with* $\mathbb{Z}/2\mathbb{Z}$ *by sending the class of* Πk *to* 1*. Its quadratic function is then*

$$q(a) = \epsilon(a), \tag{5.6}$$

where $\epsilon : \mathrm{BW}(k) \to \mathbb{Z}/2\mathbb{Z}$ *is the parity* (2.14)*.*

Proof. The composite of two Morita equivalences is a Morita equivalence, and the balanced tensor product of two even isomorphisms is an even isomorphism, so composition of classes is well defined; the associativity and unit isomorphisms of balanced tensor products are even, so Mor_k is a category, and a groupoid since every Morita equivalence has an inverse up to even isomorphism. The graded tensor product of Morita equivalences $M : A \to B$ and $M' : A' \to B'$ is a Morita equivalence $A \mathbin{\widehat{\otimes}} A' \to B \mathbin{\widehat{\otimes}} B'$, with inverse the graded tensor product of the inverses by (2.10), and $\widehat{\otimes}$ is functorial on classes because the interchange isomorphism is an even isomorphism natural in the four bimodules. An isomorphism of superalgebras $f : A \to B$, regarded as a 1-morphism, is a Morita equivalence with inverse f^{-1}, and $f \mapsto [f]$ is compatible with composition, with identities, and with $\widehat{\otimes}$. The associativity and unit isomorphisms of $\widehat{\otimes}$ and the swaps $\tau_{A,B}$ are isomorphisms of superalgebras satisfying the pentagon, triangle, and hexagon identities, because they are the structure isomorphisms of the symmetric monoidal category sVec_k [ML98, Chapter XI] applied to the underlying super vector spaces. Their classes are natural in Mor_k: naturality of the associativity and unit isomorphisms with respect to bimodules is the compatibility of balanced tensor products with $\widehat{\otimes}$, given by (2.12), and for Morita equivalences $M : A \to A'$ and $N : B \to B'$ the map

$$\begin{aligned} (M \mathbin{\widehat{\otimes}} N) \otimes_{A' \widehat{\otimes} B'} \tau_{A',B'} &\longrightarrow \tau_{A,B} \otimes_{B \widehat{\otimes} A} (N \mathbin{\widehat{\otimes}} M), \\ (m \mathbin{\widehat{\otimes}} n) \otimes (1 \mathbin{\widehat{\otimes}} 1) &\longmapsto (-1)^{|m||n|} (1 \mathbin{\widehat{\otimes}} 1) \otimes (n \mathbin{\widehat{\otimes}} m) \end{aligned} \tag{5.7}$$

for homogeneous $m \in M$ and $n \in N$ is an even isomorphism of $(A \mathbin{\widehat{\otimes}} B, B' \mathbin{\widehat{\otimes}} A')$-bimodules, by (2.10) and the definition of τ; every element of the source is a sum of elements $(m \mathbin{\widehat{\otimes}} n) \otimes (1 \mathbin{\widehat{\otimes}} 1)$, since $(m \mathbin{\widehat{\otimes}} n) \otimes (b' \mathbin{\widehat{\otimes}} a') = (-1)^{|a'||b'|+|n||a'|}(ma' \mathbin{\widehat{\otimes}} nb') \otimes (1 \mathbin{\widehat{\otimes}} 1)$ for homogeneous $a' \in A'$ and $b' \in B'$. Hence Mor_k is a braided monoidal groupoid. Every object is invertible, since $X \mathbin{\widehat{\otimes}} X^{\mathrm{sop}} \cong \mathrm{End}_k(X)$ by (2.13) is Morita equivalent to k. Thus Mor_k is a braided categorical group.

Two objects are isomorphic in Mor_k if and only if a Morita equivalence between them exists, that is, if and only if they are Morita equivalent in the sense of Definition 2.5; so the isomorphism classes of objects are the graded Morita classes, and the multiplication induced by $\widehat{\otimes}$ is the addition (2.15) of $\mathrm{BW}(k)$. By Proposition 2.8, applied to the Morita equivalence ${}_XX_X$, the automorphisms of an object X are $[{}_XX_X]$ and $[\Pi({}_XX_X)]$, which are distinct, and the (X,X)-bimodule ${}_XX_X \mathbin{\widehat{\otimes}} \Pi k$, with X acting on both sides through the unitor $X \mathbin{\widehat{\otimes}} k \to X$ of Definition 5.5, is evenly isomorphic to $\Pi({}_XX_X)$ via $x \mathbin{\widehat{\otimes}} \Pi\lambda \mapsto (-1)^{|x|}\Pi(x\lambda)$ for homogeneous $x \in X$ and $\lambda \in k$, by (2.10) and (2.16). Hence the isomorphism $\mathrm{Aut}(k) \to \mathrm{Aut}(X)$ of Proposition 5.2, which sends the class of a Morita equivalence $E : k \to k$ to the class of ${}_XX_X \mathbin{\widehat{\otimes}} E$ conjugated by the unitor, sends $[\Pi k]$ to $[\Pi({}_XX_X)]$. For $a \in \mathrm{BW}(k)$ put $B = D_a \mathbin{\widehat{\otimes}} D_a$. The self-braiding τ_{D_a,D_a} is the class of the (B,B)-bimodule τ^*B, that is, B with the left action twisted by τ. As in the proof of Proposition 2.9, the homogeneous bimodule maps $B \to \tau^*B$ are the left multiplications by the solutions of (2.21), which form a line of degree $\epsilon(a)$; hence $\tau^*B \cong \Pi^{\epsilon(a)}B$ evenly, and $q(a) = \epsilon(a)$. □

Lemma 5.7. *The associator class of* Mor_k*, read in* $H^3(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$ *through the identification* $\mathrm{Aut}(k) \cong \mathbb{Z}/2\mathbb{Z}$ *of Lemma 5.6 and denoted* $\kappa_3(k)$*, is trivial.*

Proof. By Lemma 5.6, the quadratic function of Mor_k, read through $\mathrm{Aut}(k) \cong \mathbb{Z}/2\mathbb{Z}$, is the parity ϵ. Define $\beta : \mathrm{BW}(k) \times \mathrm{BW}(k) \to \mathbb{Z}/2\mathbb{Z}$ by $\beta(a,b) = \epsilon(a)\epsilon(b)$. Since $\epsilon : \mathrm{BW}(k) \to \mathbb{Z}/2\mathbb{Z}$ is a homomorphism, β is biadditive, and $\beta(a,a) = \epsilon(a)^2 = \epsilon(a)$ for all $a \in \mathrm{BW}(k)$. Hence the associator class of Mor_k is trivial by Proposition 5.4(4). □

Definition 5.8. Let $D = (D_a)_{a \in \mathrm{BW}(k)}$ be the family of representatives of Definition 2.7. A *Morita system* with respect to D is a family $M = (M_{a,b})_{a,b \in \mathrm{BW}(k)}$ of Morita equivalences $M_{a,b} : D_a \mathbin{\widehat{\otimes}} D_b \to D_{a+b}$ such that $M_{0,a} = D_a = M_{a,0}$, where $k \mathbin{\widehat{\otimes}} D_a$ and $D_a \mathbin{\widehat{\otimes}} k$ are identified with D_a by the unit isomorphisms. For $a, b, c \in \mathrm{BW}(k)$ put

$$\begin{aligned} L_{a,b,c} &= (M_{a,b} \mathbin{\widehat{\otimes}} D_c) \otimes_{D_{a+b} \mathbin{\widehat{\otimes}} D_c} M_{a+b,c}, \\ U_{a,b,c} &= (D_a \mathbin{\widehat{\otimes}} M_{b,c}) \otimes_{D_a \mathbin{\widehat{\otimes}} D_{b+c}} M_{a,b+c}, \end{aligned} \tag{5.8}$$

both Morita equivalences $(D_a \mathbin{\widehat{\otimes}} D_b) \mathbin{\widehat{\otimes}} D_c \to D_{a+b+c}$, the left action on $U_{a,b,c}$ being transported along the associativity isomorphism of $\widehat{\otimes}$. By Proposition 2.8, the homogeneous bimodule maps $L_{a,b,c} \to U_{a,b,c}$ form a one-dimensional super vector space concentrated in a single degree. The *parity cochain* $p_M \in C^3(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$ of M is this degree,

$$p_M(a,b,c) = |A_{a,b,c}| \qquad \text{for any nonzero homogeneous bimodule map } A_{a,b,c} : L_{a,b,c} \to U_{a,b,c}. \tag{5.9}$$

The system M is *even* if $p_M = 0$. When one of a, b, c is 0, the normalization $M_{0,a} = D_a = M_{a,0}$ and the unit isomorphisms (2.8), (2.9), $k \mathbin{\widehat{\otimes}} N \cong N$, and $N \mathbin{\widehat{\otimes}} k \cong N$ identify both $L_{a,b,c}$ and $U_{a,b,c}$ with $M_{b,c}$, $M_{a,c}$, or $M_{a,b}$, respectively; the composite even isomorphism $L_{a,b,c} \to U_{a,b,c}$ is the *unit identification*. Explicitly, by the normalization, $L_{0,b,c} = (D_b \mathbin{\widehat{\otimes}} D_c) \otimes_{D_b \mathbin{\widehat{\otimes}} D_c} M_{b,c}$, $U_{0,b,c} = (k \mathbin{\widehat{\otimes}} M_{b,c}) \otimes_{k \mathbin{\widehat{\otimes}} D_{b+c}} D_{b+c}$, $L_{a,0,c} = U_{a,0,c} = (D_a \mathbin{\widehat{\otimes}} D_c) \otimes_{D_a \mathbin{\widehat{\otimes}} D_c} M_{a,c}$, $L_{a,b,0} = (M_{a,b} \mathbin{\widehat{\otimes}} k) \otimes_{D_{a+b} \mathbin{\widehat{\otimes}} k} D_{a+b}$, and $U_{a,b,0} = (D_a \mathbin{\widehat{\otimes}} D_b) \otimes_{D_a \mathbin{\widehat{\otimes}} D_b} M_{a,b}$, and for $x \in D_b \mathbin{\widehat{\otimes}} D_c$ and $m \in M_{b,c}$, for

$x \in D_a \mathbin{\widehat{\otimes}} D_c$ and $m \in M_{a,c}$, and for $m \in M_{a,b}$, $\lambda \in k$, and $y \in D_{a+b}$, the unit identifications are

$$\begin{aligned} L_{0,b,c} &\longrightarrow U_{0,b,c}, & x \otimes m &\longmapsto (1 \mathbin{\widehat{\otimes}} xm) \otimes 1, \\ L_{a,0,c} &\longrightarrow U_{a,0,c}, & x \otimes m &\longmapsto x \otimes m, \\ L_{a,b,0} &\longrightarrow U_{a,b,0}, & (m \mathbin{\widehat{\otimes}} \lambda) \otimes y &\longmapsto (1 \mathbin{\widehat{\otimes}} 1) \otimes \lambda my; \end{aligned} \tag{5.10}$$

no signs occur, since 1 and λ are even. An *even system* (M, A) is an even Morita system M together with even bimodule isomorphisms $A_{a,b,c} : L_{a,b,c} \to U_{a,b,c}$ for all $a, b, c \in \mathrm{BW}(k)$, the *comparisons* of (M, A), such that $A_{a,b,c}$ is the unit identification (5.10) whenever one of a, b, c is 0. For a normalized $n \in C^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$, write $\Pi^n M$ for the Morita system $(\Pi^{n(a,b)} M_{a,b})_{a,b}$, with the actions (2.16). An *isomorphism* of Morita systems $M \to M'$ is a family of even bimodule isomorphisms $\phi_{a,b} : M_{a,b} \to M'_{a,b}$, $a, b \in \mathrm{BW}(k)$; two Morita systems are *isomorphic* if an isomorphism between them exists, and M' is a *parity transport* of M if M' is isomorphic to $\Pi^{\delta\mu} M$ for some normalized $\mu \in C^1(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$.

Lemma 5.9. *Let $D = (D_a)_{a \in \mathrm{BW}(k)}$ be the family of representatives of Definition 2.7, regarded as objects of Mor_k, identify $\mathrm{Aut}(k)$ with $\mathbb{Z}/2\mathbb{Z}$ as in Lemma 5.6, and let $\mathrm{Sec}(D)$ be the set of even Morita systems with respect to D modulo isomorphism and parity transport. For a Morita system M with respect to D, the family $t_M = ([M_{a,b}])_{a,b \in \mathrm{BW}(k)}$ is a section of Mor_k on D, and $M \mapsto t_M$ is a bijection from the set of Morita systems modulo isomorphism onto the set of sections of Mor_k on D, such that*

$$h_{t_M} = p_M, \qquad t_{\Pi^n M} = n \cdot t_M \qquad (n \in C^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z}) \text{ normalized}). \tag{5.11}$$

Consequently M is even if and only if t_M is monoidal, parity transports of M correspond to sections monoidally isomorphic to t_M, even Morita systems exist, and $M \mapsto t_M$ identifies $\mathrm{Sec}(D)$ with the set of monoidal sections of Mor_k on D modulo monoidal isomorphism; in particular $\mathrm{Sec}(D)$ is a torsor under $H^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$, acting by $[n] \cdot M = \Pi^n M$.

Proof. By Definition 5.5, the morphisms $D_a \mathbin{\widehat{\otimes}} D_b \to D_{a+b}$ of Mor_k are the classes $[N]$ of Morita equivalences $N : D_a \mathbin{\widehat{\otimes}} D_b \to D_{a+b}$, and $[N] = [N']$ if and only if N and N' are evenly isomorphic. For a Morita system M, the classes $[M_{0,a}] = [D_a]$ and $[M_{a,0}] = [D_a]$ are the unitors of Mor_k, so t_M is a section on (D_a); conversely, a section t consists of morphisms $t_{a,b} = [M_{a,b}]$ for Morita equivalences $M_{a,b}$, which may be chosen equal to D_a when $a = 0$ or $b = 0$, and two such choices are isomorphic Morita systems. This proves the bijection.

Let $a, b, c \in \mathrm{BW}(k)$. In Mor_k, the composite of $[M_{a,b}] \widehat{\otimes} \mathrm{id}_{D_c}$ and $[M_{a+b,c}]$ is $[L_{a,b,c}]$, and the composite of the associator, $\mathrm{id}_{D_a} \mathbin{\widehat{\otimes}} [M_{b,c}]$, and $[M_{a,b+c}]$ is $[U_{a,b,c}]$, by (5.8) and Definition 5.5, the left action on $U_{a,b,c}$ having been transported along the associativity isomorphism of $\widehat{\otimes}$. Since $U_{a,b,c} \cong \Pi^{p_M(a,b,c)} L_{a,b,c}$ evenly by Proposition 2.8 and the definition of p_M, the two composites differ by the automorphism $[\Pi^{p_M(a,b,c)}({}_E E_E)]$ of the object $E = D_{a+b+c}$, which corresponds to $p_M(a, b, c) \in \mathbb{Z}/2\mathbb{Z}$ under the identification of Lemma 5.6; this is $h_{t_M} = p_M$. For normalized $n \in C^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$ and $E' = D_{a+b}$, the parity shift $\Pi^{n(a,b)} M_{a,b}$ is the composite of $M_{a,b}$ with $\Pi^{n(a,b)}({}_{E'} E'_{E'})$, so $[\Pi^{n(a,b)} M_{a,b}] = n(a, b) \circ [M_{a,b}]$, that is, $t_{\Pi^n M} = n \cdot t_M$.

The assertions on even systems, parity transports, existence, and $\mathrm{Sec}(D)$ follow: M is even if and only if $p_M = 0$, that is, $h_{t_M} = 1$; parity transports $M' \cong \Pi^{\delta\mu} M$ correspond to the sections $(\delta\mu) \cdot t_M$, which are the sections monoidally isomorphic to t_M by Proposition 5.4(2); since the associator class of Mor_k is trivial (Lemma 5.7), monoidal sections exist by Proposition 5.4(3), hence even Morita systems exist; and the bijection $M \mapsto t_M$ sends even systems modulo isomorphism and parity transport to monoidal sections modulo monoidal isomorphism and $[n] \cdot M = \Pi^n M$ to $[n] \cdot t_M = n \cdot t_M$, so $\mathrm{Sec}(D)$ inherits the torsor structure of Proposition 5.4(3). □

Definition 5.10. Let (M, A) be an even system. A *word* is either a letter $a \in \mathrm{BW}(k)$ or $(w_1 w_2)$ for words w_1, w_2. Define $|w| \in \mathrm{BW}(k)$, a superalgebra D_w, and a Morita equivalence $M_w : D_w \to D_{|w|}$ by recursion on w, starting from D:

$$\begin{aligned} |a| &= a, \qquad |(w_1 w_2)| = |w_1| + |w_2|, \qquad D_{(w_1 w_2)} = D_{w_1} \mathbin{\widehat{\otimes}} D_{w_2}, \\ M_a &= D_a, \qquad M_{(w_1 w_2)} = (M_{w_1} \mathbin{\widehat{\otimes}} M_{w_2}) \otimes_{D_{|w_1|} \widehat{\otimes} D_{|w_2|}} M_{|w_1|,|w_2|}. \end{aligned} \tag{5.12}$$

For words w_1, w_2, w_3, let

$$\begin{aligned} \iota^L_{w_1,w_2,w_3} &= \mathsf{a}_{(M_{w_1} \widehat{\otimes} M_{w_2}) \widehat{\otimes} M_{w_3},\, M_{|w_1|,|w_2|} \widehat{\otimes} D_{|w_3|},\, M_{|w_1|+|w_2|,|w_3|}} \\ &\qquad \circ \Big((\chi^{-1}_{M_{w_1} \widehat{\otimes} M_{w_2},\, M_{w_3},\, M_{|w_1|,|w_2|},\, D_{|w_3|}} \circ (\mathrm{id} \mathbin{\widehat{\otimes}} \mathsf{r}^{-1}_{M_{w_3}})) \otimes \mathrm{id} \Big) \\ &\qquad : M_{((w_1 w_2) w_3)} \longrightarrow \big((M_{w_1} \mathbin{\widehat{\otimes}} M_{w_2}) \mathbin{\widehat{\otimes}} M_{w_3} \big) \otimes_{D_{|w_1|} \widehat{\otimes} D_{|w_2|} \widehat{\otimes} D_{|w_3|}} L_{|w_1|,|w_2|,|w_3|}, \\ \iota^U_{w_1,w_2,w_3} &= \mathsf{a}_{M_{w_1} \widehat{\otimes} (M_{w_2} \widehat{\otimes} M_{w_3}),\, D_{|w_1|} \widehat{\otimes} M_{|w_2|,|w_3|},\, M_{|w_1|,|w_2|+|w_3|}} \\ &\qquad \circ \Big((\chi^{-1}_{M_{w_1},\, M_{w_2} \widehat{\otimes} M_{w_3},\, D_{|w_1|},\, M_{|w_2|,|w_3|}} \circ (\mathsf{r}^{-1}_{M_{w_1}} \mathbin{\widehat{\otimes}} \mathrm{id})) \otimes \mathrm{id} \Big) \\ &\qquad : M_{(w_1 (w_2 w_3))} \longrightarrow \big(M_{w_1} \mathbin{\widehat{\otimes}} (M_{w_2} \mathbin{\widehat{\otimes}} M_{w_3}) \big) \otimes_{D_{|w_1|} \widehat{\otimes} D_{|w_2|} \widehat{\otimes} D_{|w_3|}} U_{|w_1|,|w_2|,|w_3|}, \end{aligned} \tag{5.13}$$

with a, r, and χ the isomorphisms (2.7), (2.9), and (2.12), and with $M_{w_1} \mathbin{\widehat{\otimes}} (M_{w_2} \mathbin{\widehat{\otimes}} M_{w_3})$ identified with $(M_{w_1} \mathbin{\widehat{\otimes}} M_{w_2}) \mathbin{\widehat{\otimes}} M_{w_3}$ along the associativity isomorphism of $\mathbin{\widehat{\otimes}}$. Words w, w' *differ by one reassociation* if w' is obtained from w by replacing one subword of the form $((w_1 w_2) w_3)$ by $(w_1 (w_2 w_3))$. For such w, w', define the even isomorphism $c_{w,w'} : M_w \to M_{w'}$ by recursion on w, according to whether the replaced subword is w itself, lies in the left factor of w, or lies in the right factor of w:

$$\begin{aligned} c_{((w_1 w_2) w_3),(w_1 (w_2 w_3))} &= (\iota^U_{w_1,w_2,w_3})^{-1} \circ (\mathrm{id} \otimes A_{|w_1|,|w_2|,|w_3|}) \circ \iota^L_{w_1,w_2,w_3}, \\ c_{(w_1 w_2),(w_1' w_2)} &= (c_{w_1,w_1'} \mathbin{\widehat{\otimes}} \mathrm{id}_{M_{w_2}}) \otimes \mathrm{id}, \\ c_{(w_1 w_2),(w_1 w_2')} &= (\mathrm{id}_{M_{w_1}} \mathbin{\widehat{\otimes}} c_{w_2,w_2'}) \otimes \mathrm{id}. \end{aligned} \tag{5.14}$$

For $a, b, c, d \in \mathrm{BW}(k)$, the even isomorphisms

$$\begin{aligned} P_L &= c_{(a((bc)d)),(a(b(cd)))} \circ c_{((a(bc))d),(a((bc)d))} \circ c_{(((ab)c)d),((a(bc))d)}, \\ P_R &= c_{((ab)(cd)),(a(b(cd)))} \circ c_{(((ab)c)d),((ab)(cd))} \end{aligned} \tag{5.15}$$

from $M_{(((ab)c)d)}$ to $M_{(a(b(cd)))}$ differ by a unique scalar (Proposition 2.8),

$$P_L = O_{M,A}(a, b, c, d)\, P_R, \qquad O_{M,A}(a, b, c, d) \in k^\times. \tag{5.16}$$

The function $O_{M,A} : \mathrm{BW}(k)^4 \to k^\times$ is the *pentagon scalar* of (M, A), and an even system with $O_{M,A} = 1$ is called a *reference system.*

Lemma 5.11. *Let (M, A) be an even system with respect to D and let $n \in C^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$ be normalized.*

(1) The map

$$Q : H^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z}) \longrightarrow H^4(\mathrm{BW}(k), k^\times), \qquad Q([n]) = \big[(-1)^{n \smile n}\big], \tag{5.17}$$

is a well-defined group homomorphism.

(2) The Morita system $\Pi^n M$ is even if and only if $\delta n = 0$. In that case, for every even system $(\Pi^n M, A')$ there is a normalized $\lambda \in C^3(\mathrm{BW}(k), k^\times)$ such that

$$O_{\Pi^n M, A'} = O_{M,A}\, (-1)^{n \smile n}\, \delta\lambda. \tag{5.18}$$

(3) If (M', A') is an even system with M' isomorphic to M or a parity transport of M, then $O_{M',A'} = O_{M,A}\, \delta\lambda$ for a normalized $\lambda \in C^3(\mathrm{BW}(k), k^\times)$.

Proof. (1) The class of $(-1)^{n\smile n}$ depends only on $[n]$, because for normalized $\mu \in C^1(\mathrm{BW}(k),\mathbb{Z}/2\mathbb{Z})$ and $n' = n + \delta\mu$ we have $(-1)^{n'\smile n'} = (-1)^{n\smile n}\,\delta\big((-1)^{\mu\smile n+n\smile\mu+\mu\smile\delta\mu}\big)$ by (A.8) and (A.5); and Q is a homomorphism because $(n+n')\smile(n+n') = n\smile n+n'\smile n'+\delta(n\smile_1 n')$ for normalized $n,n'\in Z^2(\mathrm{BW}(k),\mathbb{Z}/2\mathbb{Z})$, by (A.7).

(2) By Lemma 5.9 and Proposition 5.4(1), $p_{\Pi^n M} = p_M + \delta n$, so $\Pi^n M$ is even if and only if $\delta n = 0$. Let $n \in Z^2(\mathrm{BW}(k),\mathbb{Z}/2\mathbb{Z})$ be normalized. For a graded bimodule V, let $d_V : \Pi V \to V$, $\Pi v \mapsto v$, be the deletion map; by (2.16) it is an odd bimodule isomorphism in the sense of (2.5). Write $d_V^0 = \mathrm{id}_V$ and $d_V^1 = d_V$. Let $L'_{a,b,c}$ and $U'_{a,b,c}$ be the bimodules (5.8) of $\Pi^n M$, and define the homogeneous bimodule isomorphisms

$$\begin{aligned}\Delta^L_{a,b,c} &= \big(\mathrm{id}\otimes d^{n(a+b,c)}_{M_{a+b,c}}\big)\circ\big((d^{n(a,b)}_{M_{a,b}}\mathbin{\widehat{\otimes}}\mathrm{id}_{D_c})\otimes\mathrm{id}\big) : L'_{a,b,c}\longrightarrow L_{a,b,c},\\ \Delta^U_{a,b,c} &= \big(\mathrm{id}\otimes d^{n(a,b+c)}_{M_{a,b+c}}\big)\circ\big((\mathrm{id}_{D_a}\mathbin{\widehat{\otimes}} d^{n(b,c)}_{M_{b,c}})\otimes\mathrm{id}\big) : U'_{a,b,c}\longrightarrow U_{a,b,c},\end{aligned} \tag{5.19}$$

of degrees $n(a,b)+n(a+b,c)$ and $n(b,c)+n(a,b+c)$, which agree since $\delta n = 0$; here $\widehat{\otimes}$ and $\otimes$ of maps are (2.11) and (2.6). Put

$$A'_{a,b,c} = (\Delta^U_{a,b,c})^{-1}\circ A_{a,b,c}\circ\Delta^L_{a,b,c} : L'_{a,b,c}\longrightarrow U'_{a,b,c}, \tag{5.20}$$

an even bimodule isomorphism; $A'_{a,b,c}$ is the unit identification when one of a,b,c is 0, since n is normalized. Thus $(\Pi^n M, A')$ is an even system. Let P'_L and P'_R be the composites (5.15) for $(\Pi^n M, A')$ and write $O = O_{M,A}$. For $a,b,c,d\in\mathrm{BW}(k)$, expanding P'_L and P'_R by (5.20), every deletion map occurs in the same order on both sides, except the deletions of the shifts $\Pi^{n(a,b)}$ and $\Pi^{n(c,d)}$, attached to the disjoint pairs (a,b) and (c,d), which occur in opposite orders. Interchanging two odd deletions in (2.11) and (2.6) produces the sign -1 of the Koszul braiding (3.2) on two odd lines, so

$$P'_L = O(a,b,c,d)\,(-1)^{n(a,b)n(c,d)}\,P'_R; \tag{5.21}$$

this is the computation of (3.68), with the deletion maps d_V in place of ζ and the maps (5.19) in place of the contractions $m_{a,b}$. Any other even system $(\Pi^n M, A'')$ has $A'' = \lambda A'$, that is, $A''_{a,b,c} = \lambda(a,b,c)A'_{a,b,c}$ for a normalized $\lambda\in C^3(\mathrm{BW}(k),k^\times)$, by Proposition 2.8 and the normalization of comparisons; replacing A' by $\lambda A'$ multiplies P'_L by $\lambda(b,c,d)\lambda(a,b+c,d)\lambda(a,b,c)$ and P'_R by $\lambda(a,b,c+d)\lambda(a+b,c,d)$, so $O_{\Pi^n M,A''} = O_{\Pi^n M,A'}\,\delta\lambda$; this is (5.18).

(3) Let $\phi = (\phi_{a,b})_{a,b} : M\to M'$ be an isomorphism of Morita systems. For $a\in\mathrm{BW}(k)$, $\phi_{0,a}$ and $\phi_{a,0}$ are even bimodule automorphisms of D_a, hence multiplications by nonzero elements of $Z(D_a)_0 = k$, by Proposition 2.6(7); since Definition 5.8 imposes no relation between the $\phi_{a,b}$ for different (a,b), replacing $\phi_{0,a}$ and $\phi_{a,0}$ by id_{D_a} gives again an isomorphism $M\to M'$, and we assume $\phi_{0,a} = \phi_{a,0} = \mathrm{id}_{D_a}$ for all a. For $a,b,c\in\mathrm{BW}(k)$, the even bimodule isomorphisms $\phi^L_{a,b,c} = (\phi_{a,b}\widehat{\otimes}\mathrm{id}_{D_c})\otimes\phi_{a+b,c} : L_{a,b,c}\to L'_{a,b,c}$ and $\phi^U_{a,b,c} = (\mathrm{id}_{D_a}\mathbin{\widehat{\otimes}}\phi_{b,c})\otimes\phi_{a,b+c} : U_{a,b,c}\to U'_{a,b,c}$, where L' and U' are the bimodules (5.8) of M', carry the unit identifications (5.10) of M to those of M': for $a=0$, $\phi^L_{0,b,c}$ sends $x\otimes m$ to $x\otimes\phi_{b,c}(m)$ and $\phi^U_{0,b,c}$ sends $(1\mathbin{\widehat{\otimes}} xm)\otimes 1$ to $(1\mathbin{\widehat{\otimes}} x\phi_{b,c}(m))\otimes 1$, because $\phi_{0,b} = \mathrm{id}_{D_b}$ and $\phi_{0,b+c} = \mathrm{id}_{D_{b+c}}$, and similarly for $b=0$ and $c=0$. So $A''_{a,b,c} = \phi^U_{a,b,c}A_{a,b,c}(\phi^L_{a,b,c})^{-1}$ are comparisons of an even system (M',A''). Define even isomorphisms $\phi_w : M_w\to M'_w$ between the iterated bimodules (5.12) by recursion, $\phi_a = \mathrm{id}_{D_a}$ and $\phi_{(w_1w_2)} = (\phi_{w_1}\mathbin{\widehat{\otimes}}\phi_{w_2})\otimes\phi_{|w_1|,|w_2|}$. By naturality of a, r, and χ, the comparisons (5.14) satisfy $c'_{w,w'} = \phi_{w'}c_{w,w'}\phi_w^{-1}$, hence $P'_L = \phi P_L\phi^{-1}$ and $P'_R = \phi P_R\phi^{-1}$ for the composites (5.15) of (M',A''), and $O_{M',A''} = O_{M,A}$. Any even system (M',A') has $A' = \lambda A''$ for a normalized $\lambda\in C^3(\mathrm{BW}(k),k^\times)$, as in (2), normalized because $A'_{a,b,c}$ and $A''_{a,b,c}$ are both the unit identification when one of a,b,c is 0; so $O_{M',A'} = O_{M,A}\,\delta\lambda$. If M' is a parity transport of M, then $M'\cong\Pi^{\delta\mu}M$ for a normalized $\mu\in C^1(\mathrm{BW}(k),\mathbb{Z}/2\mathbb{Z})$ (Definition 5.8), and by (2) and the isomorphism case, $O_{M',A'} = O_{M,A}\,(-1)^{\delta\mu\smile\delta\mu}\,\delta\lambda$ with $(-1)^{\delta\mu\smile\delta\mu} = \delta\big((-1)^{\mu\smile\delta\mu}\big)$ by (A.6), which is again of the required form. □

Lemma 5.12. *Let $D' = (D'_a)_{a\in \mathrm{BW}(k)}$ be another family of central division superalgebras with D'_a of class a and $D'_0 = k$, and let $\phi_a : D_a \to D'_a$, $a \in \mathrm{BW}(k)$, be isomorphisms of superalgebras with $\phi_0 = \mathrm{id}_k$. For a Morita system M with respect to D, let $M^{\phi}_{a,b}$ be the super vector space $M_{a,b}$ with the actions*

$$x \cdot m \cdot y = (\phi_a \mathbin{\widehat{\otimes}} \phi_b)^{-1}(x)\, m\, \phi^{-1}_{a+b}(y) \qquad (x \in D'_a \mathbin{\widehat{\otimes}} D'_b,\ m \in M_{a,b},\ y \in D'_{a+b}), \tag{5.22}$$

and define

$$M'_{0,a} = M'_{a,0} = D'_a, \qquad M'_{a,b} = M^{\phi}_{a,b} \quad (a, b \neq 0), \tag{5.23}$$

together with the maps $\psi_{a,b} : M_{a,b} \to M'_{a,b}$ given by $\psi_{0,a} = \psi_{a,0} = \phi_a$ and by the identity map of $M_{a,b}$ for $a, b \neq 0$. Then M' is a Morita system with respect to D' such that $t_{M'} = (t_M)^{[\phi]}$ in the notation of (5.5)*, where $[\phi_a] : D_a \to D'_a$ is the class of ϕ_a regarded as a Morita equivalence. For $a, b, c \in \mathrm{BW}(k)$, the maps*

$$\begin{aligned} \theta^L_{a,b,c} &: L_{a,b,c} \longrightarrow L'_{a,b,c}, \quad (m \mathbin{\widehat{\otimes}} d) \otimes n \longmapsto (\psi_{a,b}(m) \mathbin{\widehat{\otimes}} \phi_c(d)) \otimes \psi_{a+b,c}(n), \\ \theta^U_{a,b,c} &: U_{a,b,c} \longrightarrow U'_{a,b,c}, \quad (d \mathbin{\widehat{\otimes}} m) \otimes n \longmapsto (\phi_a(d) \mathbin{\widehat{\otimes}} \psi_{b,c}(m)) \otimes \psi_{a,b+c}(n), \end{aligned} \tag{5.24}$$

where L' and U' are the bimodules (5.8) *of M', are even isomorphisms such that $\theta(x\,u\,y) = \Phi(x)\,\theta(u)\,\phi_{a+b+c}(y)$ for $\Phi = (\phi_a \mathbin{\widehat{\otimes}} \phi_b) \mathbin{\widehat{\otimes}} \phi_c$, and for an even system (M, A) the maps*

$$A'_{a,b,c} = \theta^U_{a,b,c}\, A_{a,b,c}\, (\theta^L_{a,b,c})^{-1} \tag{5.25}$$

are the comparisons of an even system (M', A') with $O_{M',A'} = O_{M,A}$. Hence $M \mapsto M'$ induces the bijection $\mathrm{Sec}(D) \to \mathrm{Sec}(D')$ of Proposition 5.4(5), which does not depend on the ϕ_a.

Proof. Twisting the actions of a graded bimodule by isomorphisms of superalgebras preserves Morita equivalences, even isomorphisms, balanced tensor products, and the isomorphisms of Definition 2.5, since all of these are defined by formulas in the actions; so each $M^{\phi}_{a,b}$ is a Morita equivalence $D'_a \mathbin{\widehat{\otimes}} D'_b \to D'_{a+b}$. For $a \in \mathrm{BW}(k)$ and $x, y \in D'_a$, the map $\phi_a : M^{\phi}_{0,a} \to D'_a$ satisfies $\phi_a(x \cdot m \cdot y) = \phi_a(\phi_a^{-1}(x) m \phi_a^{-1}(y)) = x\phi_a(m)y$, so it is an even isomorphism of $(D'_0 \mathbin{\widehat{\otimes}} D'_a, D'_a)$-bimodules, and likewise for $M^{\phi}_{a,0}$. Hence M' is a Morita system with respect to D', normalized by (5.23) and $\phi_0 = \mathrm{id}_k$.

For an isomorphism of superalgebras $\psi : E \to E'$, the graded (E', E)-bimodule ${}_{\psi^{-1}}E_E$, which is E with the left E'-action $y \cdot x = \psi^{-1}(y)x$, is inverse to the Morita equivalence ${}_{\psi}E'_{E'} : E \to E'$ of Definition 2.5: the maps ${}_{\psi}E'_{E'} \otimes_{E'} {}_{\psi^{-1}}E_E \to E$, $y \otimes x \mapsto \psi^{-1}(y)x$, and ${}_{\psi^{-1}}E_E \otimes_E {}_{\psi}E'_{E'} \to E'$, $x \otimes y \mapsto \psi(x)y$, are even isomorphisms of bimodules. Put $\Phi_{a,b} = \phi_a \mathbin{\widehat{\otimes}} \phi_b : D_a \mathbin{\widehat{\otimes}} D_b \to D'_a \mathbin{\widehat{\otimes}} D'_b$. By (5.22), $m \mapsto 1 \otimes m \otimes 1$ is an even isomorphism $M^{\phi}_{a,b} \to {}_{\Phi^{-1}_{a,b}}(D_a \mathbin{\widehat{\otimes}} D_b)_{D_a \mathbin{\widehat{\otimes}} D_b} \otimes_{D_a \mathbin{\widehat{\otimes}} D_b} M_{a,b} \otimes_{D_{a+b}} {}_{\phi_{a+b}}(D'_{a+b})_{D'_{a+b}}$ of $(D'_a \mathbin{\widehat{\otimes}} D'_b, D'_{a+b})$-bimodules, so the class of $M'_{a,b}$ in Mor_k is $[\phi_{a+b}] \circ [M_{a,b}] \circ ([\phi_a] \mathbin{\widehat{\otimes}} [\phi_b])^{-1}$, also at the positions where a or b is 0; this is $t_{M'} = (t_M)^{[\phi]}$.

The maps (5.24) are well defined on the balanced tensor products and satisfy $\theta(x\,u\,y) = \Phi(x)\,\theta(u)\,\phi_{a+b+c}(y)$, because $\psi_{a,b}$, $\psi_{a+b,c}$, $\psi_{b,c}$, $\psi_{a,b+c}$, and the ϕ's are compatible with the actions through the ϕ's; they are bijective, with inverses given by the same formulas for ϕ^{-1} and ψ^{-1}. When one of a, b, c is 0, they carry the unit identification (5.10) of M to that of M'. For $a = 0$, $x \in D_b \mathbin{\widehat{\otimes}} D_c$, and $m \in M_{b,c}$, we have $\theta^L_{0,b,c}(x \otimes m) = \Phi_{b,c}(x) \otimes \psi_{b,c}(m)$, whose unit identification in M' is $(1 \mathbin{\widehat{\otimes}} \Phi_{b,c}(x) \cdot \psi_{b,c}(m)) \otimes 1 = (1 \mathbin{\widehat{\otimes}} \psi_{b,c}(xm)) \otimes 1 = \theta^U_{0,b,c}((1 \mathbin{\widehat{\otimes}} xm) \otimes 1)$, using $\phi_b(1) = 1$ and $\phi_{b+c}(1) = 1$; the cases $b = 0$ and $c = 0$ are the same. Hence (5.25) defines comparisons of an even system (M', A'). Define $\theta_w : M_w \to M'_w$ for the iterated bimodules (5.12) by recursion, $\theta_a = \phi_a$ and $\theta_{(w_1 w_2)} = (\theta_{w_1} \mathbin{\widehat{\otimes}} \theta_{w_2}) \otimes \psi_{|w_1|,|w_2|}$. By the naturality of a, r, and χ in their bimodule arguments, the comparisons (5.14) of (M', A') satisfy $c'_{w,w'} = \theta_{w'} c_{w,w'} \theta_w^{-1}$,

so $P'_L = \theta P_L \theta^{-1}$ and $P'_R = \theta P_R \theta^{-1}$ in (5.15), and $O_{M',A'} = O_{M,A}$. Under the bijections of Lemma 5.9 for D and for D', the map $M \mapsto M'$ is $t \mapsto t^{[\phi]}$, so it induces the bijection $\mathrm{Sec}(D) \to \mathrm{Sec}(D')$ of Proposition 5.4(5), which does not depend on the ϕ_a. □

Proposition 5.13. *Let (M, A) be an even system with respect to D.*

(1) $O_{M,A}$ is a normalized 4-cocycle.

(2) The class $o_4(M) = [O_{M,A}] \in H^4(\mathrm{BW}(k), k^\times)$ does not depend on A, and it depends only on the class of M in $\mathrm{Sec}(D)$.

(3) The class $o_4(D) = o_4(M) \operatorname{im} Q \in H^4(\mathrm{BW}(k), k^\times)/\operatorname{im} Q$ does not depend on M.

(4) For a family D' as in Lemma 5.12, $o_4(D) = o_4(D')$; we define $o_4(k) = o_4(D)$.

(5) A reference system exists if and only if $o_4(k) = 1$.

Proof. Write $O = O_{M,A}$.

(1) Let $a = 0$. In (5.15), the three comparisons $c_{(((0b)c)d),((0(bc))d)}$, $c_{((0(bc))d),(0((bc)d))}$, and $c_{((0b)(cd)),(0(b(cd)))}$ are built from the unit identifications $A_{0,b,c}$, $A_{0,b+c,d}$, and $A_{0,b,c+d}$ respectively. The comparison $c_{(0((bc)d)),(0(b(cd)))}$ equals $(\mathrm{id}_{D_0} \mathbin{\widehat{\otimes}} c_{((bc)d),(b(cd))}) \otimes \mathrm{id}$, and the comparison $c_{(((0b)c)d),((0b)(cd))}$ is built from $A_{b,c,d}$. By the coherence of the unit isomorphisms of $\widehat{\otimes}$ and the naturality of a, r, and χ, both P_L and P_R equal the composite of the unit identification $M_{(((0b)c)d)} \to M_{((bc)d)}$, the comparison $c_{((bc)d),(b(cd))}$, and the inverse of the unit identification $M_{(0(b(cd)))} \to M_{(b(cd))}$; hence $O(0, b, c, d) = 1$. The cases $b = 0$, $c = 0$, and $d = 0$ are proved in the same way, so O is normalized. For $a, b, c, d, e \in \mathrm{BW}(k)$, consider the fourteen iterated bimodules M_w of (5.12) for the parenthesizations w of $abcde$ and the comparisons $c_{w,w'}$ of (5.14) between adjacent ones. They form the 3-dimensional associahedron, whose faces are six pentagons and three squares. The pentagons for the subwords $bcde$ and $abcd$ are instances of (5.15) tensored with id_{M_a}, respectively id_{M_e}, and composed with a bimodule $M_{u,v}$, $u, v \in \mathrm{BW}(k)$; for the four pentagons in which two adjacent letters are grouped into a word w, such as $((ab), c, d, e)$, the five comparisons (5.14) are conjugate, by the naturality of a, r, and χ in their bimodule arguments, under the identifications (5.13) to the comparisons of (5.15) for the letters $(|w|, c, d, e)$ tensored with id_{M_w}; so each pentagon face commutes up to the pentagon scalar O of its four letters or words. The squares commute because even comparisons acting on disjoint positions commute by (2.5). Comparing the two composites of pentagon faces between the two paths around the associahedron, as in Mac Lane's coherence theorem [ML98, Chapter VII, §2], gives

$$O(b,c,d,e)\, O(a{+}b,c,d,e)^{-1}\, O(a,b{+}c,d,e)\, O(a,b,c{+}d,e)^{-1}\, O(a,b,c,d{+}e)\, O(a,b,c,d)^{-1} = 1, \tag{5.26}$$

which is $\delta O = 1$. For a normalized $\lambda \in C^3(\mathrm{BW}(k), k^\times)$, write $\lambda A = (\lambda(a,b,c) A_{a,b,c})_{a,b,c}$. Replacing A by λA multiplies P_L by $\lambda(b,c,d)\lambda(a,b+c,d)\lambda(a,b,c)$ and P_R by $\lambda(a,b,c+d)\lambda(a+b,c,d)$, so O becomes $O\,\delta\lambda$.

(2) For normalized $\lambda \in C^3(\mathrm{BW}(k), k^\times)$, replacing A by λA replaces $O_{M,A}$ by $O_{M,A}\,\delta\lambda$, which has the same class; and by Lemma 5.11(3) the class $[O_{M,A}]$ is unchanged when M is replaced by an isomorphic Morita system or a parity transport, so $o_4(M)$ depends only on the class of M in $\mathrm{Sec}(D)$.

(3) Let (M', A') be another even system with respect to D. By Lemma 5.9, M' is isomorphic to $\Pi^n M$ for a normalized n, and $n \in Z^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$ by Lemma 5.11(2) since $\Pi^n M$ is even. Hence $o_4(M') = o_4(M)\, Q([n])$ by Lemma 5.11(2) and (3), so $o_4(M') \operatorname{im} Q = o_4(M) \operatorname{im} Q$.

(4) By Proposition 2.6(1) there are isomorphisms of superalgebras $\phi_a : D_a \to D'_a$, $a \in \mathrm{BW}(k)$, with $\phi_0 = \mathrm{id}_k$; let M' be the Morita system (5.23) with respect to D', and let A' be the family of comparisons (5.25). By Lemma 5.12, (M', A') is an even system with respect to D' with $O_{M',A'} = O_{M,A}$, so $o_4(D') = [O_{M',A'}] \operatorname{im} Q = o_4(D)$.

(5) If (M, A) is an even system with $o_4(M) = 1$, then $O_{M,A} = \delta\lambda^{-1}$ for a normalized λ, and $(M, \lambda A)$ is a reference system; conversely a reference system (M, A) has $O_{M,A} = 1$

and $o_4(M) = 1$. If $o_4(k) = 1$, then $o_4(M) \in \operatorname{im} Q$, say $o_4(M) = Q([n])^{-1}$ with $n \in Z^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$, and $o_4(\Pi^n M) = o_4(M)Q([n]) = 1$ by Lemma 5.11(2), so a reference system with Morita system $\Pi^n M$ exists; conversely a reference system has $o_4(M) = 1$, hence $o_4(k) = 1$. □

Definition 5.14. Let (M, A) be an even system. Let $\mathcal{R} = \mathcal{R}(M, A)$ be the supercategory whose objects are the families $X = (X_a)_{a \in \mathrm{BW}(k)}$ of a finite-dimensional graded right D_a-module X_a such that $X_a = 0$ for all but finitely many a, and whose morphism spaces are

$$\mathcal{R}(X, Y)_d = \bigoplus_{a \in \mathrm{BW}(k)} \{f \in \mathrm{Hom}_k(X_a, Y_a)_d : f(vb) = f(v)b \text{ for } v \in X_a,\ b \in D_a\}, \tag{5.27}$$

with componentwise composition. For $a \in \mathrm{BW}(k)$, we say that X is *supported at* a if $X_b = 0$ for all $b \neq a$. For $a, b \in \mathrm{BW}(k)$, the super vector space $X_a \otimes_k Y_b$ is a graded right $D_a \mathbin{\widehat{\otimes}} D_b$-module with

$$(x \otimes y)(u \mathbin{\widehat{\otimes}} v) = (-1)^{|y||u|} xu \otimes yv \qquad (x \in X_a,\ y \in Y_b,\ u \in D_a,\ v \in D_b \text{ homogeneous}). \tag{5.28}$$

The tensor product of $X, Y \in \mathcal{R}$ is the family

$$(X \otimes Y)_c = \bigoplus_{a+b=c} (X_a \otimes_k Y_b) \otimes_{D_a \widehat{\otimes} D_b} M_{a,b} \qquad (c \in \mathrm{BW}(k)), \tag{5.29}$$

where the summand indexed by (a, b) is a graded right D_c-module through the right action of $D_{a+b} = D_c$ on $M_{a,b}$, and $\bigoplus$ is the direct sum of graded right D_c-modules. The tensor product of $f = (f_a)_a \in \mathcal{R}(X, X')_d$ and $g = (g_b)_b \in \mathcal{R}(Y, Y')_{d'}$ is the element $f \otimes g \in \mathcal{R}(X \otimes Y, X' \otimes Y')_{d+d'}$ with

$$(f \otimes g)\big((x \otimes y) \otimes_{D_a \widehat{\otimes} D_b} m\big) = (-1)^{d'|x|} \big(f_a(x) \otimes g_b(y)\big) \otimes_{D_a \widehat{\otimes} D_b} m \tag{5.30}$$

for homogeneous $x \in X_a$, $y \in Y_b$, and $m \in M_{a,b}$. The unit object $\mathbf{1}$ is the family supported at 0 with $\mathbf{1}_0 = k$, the regular right $D_0 = k$-module; the parity object π is the family supported at 0 with $\pi_0 = \Pi k$; and $\zeta \in \mathcal{R}(\pi, \mathbf{1})_1$ is the odd isomorphism with $\zeta_0(\Pi\lambda) = \lambda$ for $\lambda \in k$. For $a, b, c \in \mathrm{BW}(k)$ and homogeneous $m \in M_{a,b}$ and $n \in M_{a+b,c}$, write

$$A_{a,b,c}\big((m \mathbin{\widehat{\otimes}} 1) \otimes_{D_{a+b} \widehat{\otimes} D_c} n\big) = \sum_i (1 \mathbin{\widehat{\otimes}} m_i) \otimes_{D_a \widehat{\otimes} D_{b+c}} n_i \tag{5.31}$$

with homogeneous $m_i \in M_{b,c}$ and $n_i \in M_{a,b+c}$; every element of $U_{a,b,c}$ is of this form, since $(u \mathbin{\widehat{\otimes}} m') \otimes n' = (-1)^{|u||m'|} (1 \mathbin{\widehat{\otimes}} m') \otimes (u \mathbin{\widehat{\otimes}} 1) n'$. The associator $\alpha_{X,Y,Z} : (X \otimes Y) \otimes Z \to X \otimes (Y \otimes Z)$ is the even isomorphism with

$$\begin{aligned} &\alpha_{X,Y,Z}\Big(\big(((x \otimes y) \otimes_{D_a \widehat{\otimes} D_b} m) \otimes z\big) \otimes_{D_{a+b} \widehat{\otimes} D_c} n\Big) \\ &\qquad = (-1)^{|m||z|} \sum_i \big(x \otimes ((y \otimes z) \otimes_{D_b \widehat{\otimes} D_c} m_i)\big) \otimes_{D_a \widehat{\otimes} D_{b+c}} n_i \end{aligned} \tag{5.32}$$

for homogeneous $x \in X_a$, $y \in Y_b$, $z \in Z_c$, $m \in M_{a,b}$, and $n \in M_{a+b,c}$. The unitors $l_X : \mathbf{1} \otimes X \to X$ and $r_X : X \otimes \mathbf{1} \to X$ are the even isomorphisms with

$$l_X\big((\lambda \otimes x) \otimes_{k \widehat{\otimes} D_a} u\big) = \lambda x u, \qquad r_X\big((x \otimes \lambda) \otimes_{D_a \widehat{\otimes} k} u\big) = \lambda x u \tag{5.33}$$

for $\lambda \in k$, $x \in X_a$, and $u \in D_a = M_{0,a} = M_{a,0}$. For $a \in \mathrm{BW}(k)$, let $S_a \in \mathcal{R}$ be the family supported at a with $(S_a)_a = D_a$, the regular right D_a-module.

Proposition 5.15. *Let (M, A) be a reference system. Then Definition 5.14 makes $\mathcal{R}(M, A)$ a Brauer–Wall reference category. Its component $\mathcal{R}_a$ consists of the families supported at a, the object S_a is simple in $\mathcal{R}_a$, and left multiplication is an isomorphism of superalgebras*

$$D_a \xrightarrow{\sim} \mathrm{End}_{\mathcal{R}}(S_a), \qquad d \longmapsto (v \mapsto dv). \tag{5.34}$$

Proof. Let (M, A) be a reference system, that is, an even system with pentagon scalar $O_{M,A} = 1$, and put $\mathcal{R} = \mathcal{R}(M, A)$. The rule (5.28) makes $X_a \otimes_k Y_b$ a graded right $D_a \mathbin{\widehat{\otimes}} D_b$-module: for homogeneous elements both $((x \otimes y)(u \mathbin{\widehat{\otimes}} v))(u' \mathbin{\widehat{\otimes}} v')$ and $(x \otimes y)((u \mathbin{\widehat{\otimes}} v)(u' \mathbin{\widehat{\otimes}} v'))$ equal $(-1)^{|y||u|+|y||u'|+|v||u'|} xuu' \otimes yvv'$, by (2.4). For $f = (f_a)_a \in \mathcal{R}(X, X')_{d_1}$ and $g = (g_b)_b \in \mathcal{R}(Y, Y')_{d_2}$, the map $x \otimes y \mapsto (-1)^{d_2|x|} f_a(x) \otimes g_b(y)$ is right $D_a \mathbin{\widehat{\otimes}} D_b$-linear, since applied to $(x \otimes y)(u \mathbin{\widehat{\otimes}} v)$ and followed by $u \mathbin{\widehat{\otimes}} v$ it gives in both cases $(-1)^{|y||u|+d_2|x|+d_2|u|} f_a(x)u \otimes g_b(y)v$; hence (5.30) defines a right D_{a+b}-linear map on the balanced tensor product with $M_{a,b}$, and composing two such maps gives the sign of (3.8). Thus $\otimes$ is a superfunctor $\mathcal{R} \boxtimes \mathcal{R} \to \mathcal{R}$.

The elements on the left of (5.32) span $((X \otimes Y) \otimes Z)_e$ for $e = a+b+c$, and the right-hand side of (5.32) is the image of the left-hand side under the composite

$$\begin{aligned} ((X \otimes Y) \otimes Z)_e &\xrightarrow{j^L} \bigoplus_{a+b+c=e} ((X_a \otimes_k Y_b) \otimes_k Z_c) \otimes_{(D_a \widehat{\otimes} D_b) \widehat{\otimes} D_c} L_{a,b,c} \\ &\xrightarrow{\bigoplus \mathrm{id} \otimes A_{a,b,c}} \bigoplus_{a+b+c=e} ((X_a \otimes_k Y_b) \otimes_k Z_c) \otimes_{(D_a \widehat{\otimes} D_b) \widehat{\otimes} D_c} U_{a,b,c} \\ &\xrightarrow{(j^U)^{-1}} (X \otimes (Y \otimes Z))_e, \end{aligned} \tag{5.35}$$

where, for homogeneous $x \in X_a$, $y \in Y_b$, $z \in Z_c$, $m \in M_{a,b}$, $n \in M_{a+b,c}$, $m' \in M_{b,c}$, and $n' \in M_{a,b+c}$, j^L and j^U are the even isomorphisms

$$\begin{aligned} j^L((((x \otimes y) \otimes m) \otimes z) \otimes n) &= (-1)^{|m||z|}((x \otimes y) \otimes z) \otimes ((m \mathbin{\widehat{\otimes}} 1) \otimes n), \\ j^U((x \otimes ((y \otimes z) \otimes m')) \otimes n') &= (x \otimes (y \otimes z)) \otimes ((1 \mathbin{\widehat{\otimes}} m') \otimes n'), \end{aligned} \tag{5.36}$$

with $X_a \otimes_k (Y_b \otimes_k Z_c)$ identified with $(X_a \otimes_k Y_b) \otimes_k Z_c$: in the notation of (5.13), j^L is $\mathsf{a} \circ ((\chi^{-1} \circ (\mathrm{id} \mathbin{\widehat{\otimes}} \mathsf{r}^{-1})) \otimes \mathrm{id})$ and j^U is $\mathsf{a} \circ ((\chi^{-1} \circ (\mathsf{r}^{-1} \mathbin{\widehat{\otimes}} \mathrm{id})) \otimes \mathrm{id})$, with $X_a \otimes_k Y_b$ and Z_c in place of $M_{w_1} \mathbin{\widehat{\otimes}} M_{w_2}$ and M_{w_3}; the sign in j^L is the sign of (2.12), and the middle step uses that $((x \otimes y) \otimes z) \otimes (1 \mathbin{\widehat{\otimes}} m_i) \otimes n_i$ is the image of $((x \otimes y) \otimes z) \otimes A_{a,b,c}((m \mathbin{\widehat{\otimes}} 1) \otimes n)$ by (5.31). Hence $\alpha_{X,Y,Z}$ is a well-defined even isomorphism, independent of the expansion (5.31), and natural in X, Y, and Z. Similarly, the maps (5.33) are well-defined even isomorphisms of right D_a-modules, since $k \mathbin{\widehat{\otimes}} D_a$ and $D_a \mathbin{\widehat{\otimes}} k$ act on $M_{0,a} = D_a = M_{a,0}$ by multiplication. The tensor product and the associator respect the decompositions (5.29) by supports, so the pentagon and triangle identities need only be checked for families supported at single elements. Let X, Y, Z, W be supported at a, b, c, d, and for a word w in the letters a, b, c, d in this order let T_w be the tensor product of X, Y, Z, W parenthesized as w and V_w the tensor product of X_a, Y_b, Z_c, W_d over k parenthesized as w. Define even isomorphisms $j_w : (T_w)_{a+b+c+d} \to V_w \otimes_{D_w} M_w$ by recursion on w: for a letter, $j_a = \mathsf{r}_{X_a}^{-1} : X_a \to X_a \otimes_{D_a} D_a$ and similarly for b, c, d, and $j_{(w_1 w_2)} = \mathsf{a} \circ (\chi^{-1} \otimes \mathrm{id}) \circ ((j_{w_1} \otimes_k j_{w_2}) \otimes \mathrm{id})$, with a and χ as in (5.13). For $w = ((w_1 w_2) w_3)$ and $w' = (w_1 (w_2 w_3))$, j^L and j^U are the cases of j_w and $j_{w'}$ with $X_a \otimes_k Y_b$ and Z_c in place of $V_{w_1} \otimes_k V_{w_2}$ and V_{w_3}, and by the naturality of a and χ, $j_{w'} \circ \alpha \circ j_w^{-1} = \mathrm{id} \otimes c_{w,w'}$ for each instance of α in the pentagon, with $c_{w,w'}$ of (5.14); so the two sides of the pentagon, conjugated by $j_{(((ab)c)d)}$ and $j_{(a(b(cd)))}$, are $\mathrm{id} \otimes P_L$ and $\mathrm{id} \otimes P_R$, which are equal since $O_{M,A} = 1$. The triangle follows from the unit normalization of A and $M_{0,a} = D_a = M_{a,0}$. Thus $\mathcal{R}$ is a monoidal supercategory with unit $\mathbf{1}$, and π with the odd isomorphism ζ makes it a monoidal Π-supercategory.

Morphisms between families with disjoint supports vanish, so $\mathcal{R} = \bigoplus_a \mathcal{R}_a$ with $\mathcal{R}_a$ the families supported at a, and $\mathcal{R}_a$ is the category of finite-dimensional graded right D_a-modules with homogeneous right-linear maps. Since D_a is a division superalgebra, every such module has a homogeneous basis, so $\mathcal{R}_a$ is semisimple with simple objects S_a and ΠS_a, which are oddly isomorphic, and the right-linear endomorphisms of D_a are the left multiplications, which proves (5.34). Hence the underlying category of $\mathcal{R}$ is an essentially small semisimple k-linear abelian category with finite-dimensional Hom spaces, $\mathrm{End}(\mathbf{1}) = k$, every simple object is isomorphic to a unique S_a, and S_a has type a.

For $X, X' \in \mathcal{R}_a$ and $Y, Y' \in \mathcal{R}_b$, the Morita equivalence $M_{a,b}$ and exactness of $\otimes_k$ give

$$\mathcal{R}(X \otimes Y, X' \otimes Y') \cong \mathrm{Hom}_{D_a \widehat{\otimes} D_b}(X_a \otimes_k Y_b, X'_a \otimes_k Y'_b) \cong \mathrm{Hom}_{D_a}(X_a, X'_a) \otimes_k \mathrm{Hom}_{D_b}(Y_b, Y'_b), \tag{5.37}$$

the second isomorphism being $f \otimes g \mapsto (x \otimes y \mapsto (-1)^{|g||x|} f(x) \otimes g(y))$, which is right $D_a \mathbin{\widehat{\otimes}} D_b$-linear by (2.10) and bijective: both sides are additive in each of X_a, X'_a, Y_b, and Y'_b and compatible with the parity shifts of these modules; every finite-dimensional graded right module over D_a or D_b is a finite direct sum of copies of the regular module and of its parity shift; and for $X_a = X'_a = D_a$ and $Y_b = Y'_b = D_b$ both sides are $D_a \mathbin{\widehat{\otimes}} D_b$ acting by left multiplication, with the map the identity. In particular $S_a \otimes S_b \cong M_{a,b}$ as a right D_{a+b}-module is a nonzero sum of copies of S_{a+b} and ΠS_{a+b}, and $\mathrm{End}(S_a \otimes S_b) \cong D_a \mathbin{\widehat{\otimes}} D_b$.

Rigidity. Let $X \in \mathcal{R}_a$ and put $S = S_{-a}$. The graded right D_{-a}-module $\Phi = \mathcal{R}(X \otimes S, \mathbf{1})$, with D_{-a} acting through $\mathrm{End}(S)$ in the second tensor slot, defines an object $X^\vee \in \mathcal{R}_{-a}$ with $(X^\vee)_{-a} = \Phi$. For $Y \in \mathcal{R}_{-a}$, define $\Theta_Y : \mathcal{R}_{-a}(Y, X^\vee) \to \mathcal{R}(X \otimes Y, \mathbf{1})$ by $\Theta_Y(f)((x \otimes y) \otimes m) = (-1)^{|x||y|} f(y)((x \otimes 1) \otimes m)$ for homogeneous $x \in X_a$, $y \in Y_{-a}$, and $m \in M_{a,-a}$, where $f(y) \in \Phi$ is evaluated at $(x \otimes 1) \otimes m \in (X \otimes S)_0$; the sign makes the right side balanced over $D_a \mathbin{\widehat{\otimes}} D_{-a}$, by (2.10) and (5.30), and Θ natural in Y. For $Y = S$, $\Theta_S(f) = f(1)$, since $f(y) = f(1)y$ and $(f(1)y)((x \otimes 1) \otimes m) = (-1)^{|x||y|} f(1)((x \otimes y) \otimes m)$ by the action of D_{-a} on Φ through the second tensor slot; so Θ_S is an isomorphism, and Θ_Y is an isomorphism for every Y, both sides being additive in Y and every Y a finite sum of copies of S and ΠS. Let $\mathrm{ev} = \Theta_{X^\vee}(\mathrm{id}_{X^\vee}) : X \otimes X^\vee \to \mathbf{1}$. For $T \in \mathcal{R}_{b-a}$ and $V \in \mathcal{R}_b$ consider

$$\mathcal{R}(T, X^\vee \otimes V) \longrightarrow \mathcal{R}(X \otimes T, V), \qquad f \longmapsto l_V(\mathrm{ev} \otimes \mathrm{id}_V)\alpha^{-1}_{X,X^\vee,V}(\mathrm{id}_X \otimes f). \tag{5.38}$$

Every T is an even retract of a finite sum of objects $Y \otimes Z$ with $Y \in \mathcal{R}_{-a}$, $Z \in \mathcal{R}_b$, by the previous paragraph. For $T = Y \otimes Z$, the isomorphisms (5.37) for the pairs $(Y, X^\vee; Z, V)$ and $(X \otimes Y, \mathbf{1}; Z, V)$, together with conjugation by $\alpha_{X,Y,Z}$ and l_V, carry (5.38) to the isomorphism $\Theta_Y \otimes \mathrm{id} : \mathcal{R}(Y, X^\vee) \otimes_k \mathcal{R}(Z, V) \to \mathcal{R}(X \otimes Y, \mathbf{1}) \otimes_k \mathcal{R}(Z, V)$; by naturality in T it is an isomorphism for every T. Let $\mathrm{coev} : \mathbf{1} \to X^\vee \otimes X$ be the preimage of $r_X : X \otimes \mathbf{1} \to X$ under (5.38) with $T = \mathbf{1}$ and $V = X$. Naturality of (5.38) in T and V then gives both snake identities for ev and coev, so $X^\vee$ is a right dual of X. The same argument with $Y \mapsto \mathcal{R}(Y \otimes X, \mathbf{1})$ gives a left dual. Hence the underlying category of $\mathcal{R}$ is rigid, and $\mathcal{R}$ is a semisimple tensor Π-supercategory. The product of S_a with a dual lies in $\mathcal{R}_0$, whose simple objects are $\mathbf{1}$ and π, so every simple object is weakly invertible. Therefore $\mathcal{R}$ is a weakly pointed semisimple tensor Π-supercategory with split unit in which every type occurs exactly once: a Brauer–Wall reference category. □

Proposition 5.16. *Let $\mathcal{C}$ be a Brauer–Wall reference category. Choose simple objects $S_a \in \mathcal{C}_a$ with $S_0 = \mathbf{1}$ and superalgebra isomorphisms $D_a \cong \mathrm{End}(S_a)$. Put*

$$M_{a,b} = \mathcal{C}(S_{a+b}, S_a \otimes S_b) \quad (a, b \neq 0), \qquad M_{0,a} = M_{a,0} = D_a, \tag{5.39}$$

with the left action of $D_a \mathbin{\widehat{\otimes}} D_b \cong \mathrm{End}(S_a \otimes S_b)$ on $\mathcal{C}(S_{a+b}, S_a \otimes S_b)$ by postcomposition and the right action of D_{a+b} by precomposition; here D_a replaces the bimodules $\mathcal{C}(S_a, \mathbf{1} \otimes S_a)$ and

$\mathcal{C}(S_a, S_a \otimes \mathbf{1})$ *along the even isomorphisms*

$$\begin{aligned} \mathcal{C}(S_a, \mathbf{1} \otimes S_a) &\longrightarrow \mathrm{End}(S_a) \cong D_a, \quad f \longmapsto l_{S_a} \circ f, \\ \mathcal{C}(S_a, S_a \otimes \mathbf{1}) &\longrightarrow \mathrm{End}(S_a) \cong D_a, \quad f \longmapsto r_{S_a} \circ f. \end{aligned} \tag{5.40}$$

Then M is a Morita system; the associator of $\mathcal{C}$, transported along (5.40) *at the positions where an argument is* 0*, induces comparisons $A_{a,b,c}$, and (M, A) is a reference system.*

Proof. Let $\mathcal{C}$ be a Brauer–Wall reference category, with S_a and $D_a \cong \mathrm{End}(S_a)$ as in the statement. By (3.37), $\mathcal{C}(S_{a+b}, -)$ is an equivalence from $\mathcal{C}_{a+b}$ to graded right D_{a+b}-modules; it sends $S_a \otimes S_b$ to $\mathcal{C}(S_{a+b}, S_a \otimes S_b)$, a free module of positive rank, and induces $\mathrm{End}(S_a \otimes S_b) \cong \mathrm{End}_{D_{a+b}}(\mathcal{C}(S_{a+b}, S_a \otimes S_b))$. Since $D_a \mathbin{\widehat{\otimes}} D_b \cong \mathrm{End}(S_a \otimes S_b)$ by (3.17), $\mathcal{C}(S_{a+b}, S_a \otimes S_b)$ is an invertible $(D_a \mathbin{\widehat{\otimes}} D_b, D_{a+b})$-bimodule for all a, b. The maps (5.40) are even isomorphisms of $(D_0 \mathbin{\widehat{\otimes}} D_a, D_a)$-bimodules, respectively of $(D_a \mathbin{\widehat{\otimes}} D_0, D_a)$-bimodules, by the naturality of l and r, $l_{S_a} \circ (\mathrm{id}_{\mathbf{1}} \otimes d) = d \circ l_{S_a}$ and $r_{S_a} \circ (d \otimes \mathrm{id}_{\mathbf{1}}) = d \circ r_{S_a}$ for $d \in \mathrm{End}(S_a)$; so M, with $M_{0,a} = M_{a,0} = D_a$, is a Morita system. For $m \in M_{a,b}$ write $\tilde{m} \in \mathcal{C}(S_{a+b}, S_a \otimes S_b)$ for m itself if $a, b \neq 0$, for $l_{S_b}^{-1} \circ m$ if $a = 0$, and for $r_{S_a}^{-1} \circ m$ if $b = 0$, the preimages of m under (5.40). For $a, b, c \in \mathrm{BW}(k)$ define

$$\begin{aligned} \lambda^L_{a,b,c} &: L_{a,b,c} \longrightarrow \mathcal{C}(S_{a+b+c}, (S_a \otimes S_b) \otimes S_c), \qquad (m \mathbin{\widehat{\otimes}} d) \otimes n \longmapsto (\tilde{m} \otimes d) \circ \tilde{n}, \\ \lambda^U_{a,b,c} &: U_{a,b,c} \longrightarrow \mathcal{C}(S_{a+b+c}, S_a \otimes (S_b \otimes S_c)), \quad (d' \mathbin{\widehat{\otimes}} m') \otimes n' \longmapsto (d' \otimes \tilde{m}') \circ \tilde{n}', \end{aligned} \tag{5.41}$$

for $m \in M_{a,b}$, $d \in D_c$, $n \in M_{a+b,c}$, $d' \in D_a$, $m' \in M_{b,c}$, and $n' \in M_{a,b+c}$; by (3.43) and the equivalence $\mathcal{C}(S_{a+b+c}, -)$ these are even isomorphisms of graded $((D_a \mathbin{\widehat{\otimes}} D_b) \mathbin{\widehat{\otimes}} D_c, D_{a+b+c})$-bimodules. Put

$$A_{a,b,c} = (\lambda^U_{a,b,c})^{-1} \circ (\alpha_{S_a,S_b,S_c} \circ -) \circ \lambda^L_{a,b,c}, \tag{5.42}$$

an even bimodule isomorphism $L_{a,b,c} \to U_{a,b,c}$. When an argument is 0, it is the unit identification (5.10). For $a = 0$, $x \in D_b \mathbin{\widehat{\otimes}} D_c$ acting on $S_b \otimes S_c$ through (3.17), and $m \in M_{b,c}$, $\alpha_{\mathbf{1},S_b,S_c} \circ \lambda^L_{0,b,c}(x \otimes m) = l^{-1}_{S_b \otimes S_c} \circ x \circ \tilde{m} = \lambda^U_{0,b,c}((1 \mathbin{\widehat{\otimes}} xm) \otimes 1)$, by the identity $l_{Y \otimes Z} \circ \alpha_{\mathbf{1},Y,Z} = l_Y \otimes \mathrm{id}_Z$ and the naturality of l. For $b = 0$, the triangle identity $\alpha_{X,\mathbf{1},Z} \circ (r_X^{-1} \otimes \mathrm{id}_Z) = \mathrm{id}_X \otimes l_Z^{-1}$ and the naturality of α give $\alpha_{S_a,\mathbf{1},S_c} \circ \lambda^L_{a,0,c} = \lambda^U_{a,0,c}$. For $c = 0$, $m \in M_{a,b}$, $\lambda \in k$, and $y \in D_{a+b}$, the identity $r_{X \otimes Y} = (\mathrm{id}_X \otimes r_Y) \circ \alpha_{X,Y,\mathbf{1}}$ and the naturality of r give $\alpha_{S_a,S_b,\mathbf{1}} \circ \lambda^L_{a,b,0}((m \mathbin{\widehat{\otimes}} \lambda) \otimes y) = (\mathrm{id}_{S_a} \otimes r_{S_b}^{-1}) \circ \lambda \tilde{m} y = \lambda^U_{a,b,0}((1 \mathbin{\widehat{\otimes}} 1) \otimes \lambda m y)$. The two identities for l and r follow from the pentagon and triangle identities [ML98, Chapter VII, §1]. For a word w in the letters a, b, c, d let S_w be the tensor product of S_a, S_b, S_c, S_d parenthesized as w, and define even isomorphisms $\lambda_w : M_w \to \mathcal{C}(S_{|w|}, S_w)$ by recursion, $\lambda_a = \mathrm{id}_{D_a}$ and $\lambda_{(w_1 w_2)}((m_1 \mathbin{\widehat{\otimes}} m_2) \otimes n) = (\lambda_{w_1}(m_1) \otimes \lambda_{w_2}(m_2)) \circ \tilde{n}$ for $m_1 \in M_{w_1}$, $m_2 \in M_{w_2}$, and $n \in M_{|w_1|,|w_2|}$, where $\tilde{n} \in \mathcal{C}(S_{|w_1|+|w_2|}, S_{|w_1|} \otimes S_{|w_2|})$ is n itself if $|w_1|, |w_2| \neq 0$, is $l^{-1}_{S_{|w_2|}} \circ n$ if $|w_1| = 0$, and is $r^{-1}_{S_{|w_1|}} \circ n$ if $|w_2| = 0$. By naturality of a, r, and χ, $\lambda_{w'} \circ c_{w,w'} \circ \lambda_w^{-1}$ is composition with the associator $S_w \to S_{w'}$ of $\mathcal{C}$ for words w, w' differing by one reassociation. Hence the pentagon of $\mathcal{C}$ at (S_a, S_b, S_c, S_d) gives $P_L = P_R$, so $O_{M,A} = 1$ and (M, A) is a reference system. □

Theorem 5.17. *The following are equivalent:*

(1) A Brauer–Wall reference category over k exists,

(2) A reference system over k exists,

(3) $o_4(k) = 1 \in H^4(\mathrm{BW}(k), k^\times)/\operatorname{im} Q$.

Proof. The implications (1) ⇒ (2) and (2) ⇒ (1) follow from Proposition 5.16 and Proposition 5.15, respectively. The equivalence of (2) and (3) is Proposition 5.13(5). □

Proposition 5.18. *Suppose that* $\mathrm{BW}(k)$ *is finite and let* $\mathcal{C}$ *be a Brauer–Wall reference category. For normalized* $n \in Z^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$ *and* $\theta \in C^3(\mathrm{BW}(k), k^\times)$ *with* $\delta\theta = (-1)^{n\smile n}$, *the twisted pullback* $\mathcal{P}_\mathcal{C}(\mathrm{BW}(k), \mathrm{id}, n, \theta)$ *is a Brauer–Wall reference category, and*

$$[(n,\theta)] \longmapsto [\mathcal{P}_\mathcal{C}(\mathrm{BW}(k), \mathrm{id}, n, \theta)] \tag{5.43}$$

is a bijection from $\mathrm{SH}_k(\mathrm{BW}(k))$ *onto the set of Brauer–Wall reference categories up to monoidal superequivalence.*

Proof. Since $\mathrm{BW}(k)$ is finite, $t = (\mathrm{BW}(k), \mathrm{id}, n, \theta)$ is a twist by Definition 3.18, and $\mathcal{C} \in \mathcal{W}$ by Proposition 3.17. Put $\mathcal{P} = \mathcal{P}_\mathcal{C}(t)$. By Proposition 3.24, $\mathcal{P} \in \mathcal{W}$, every simple object of $\mathcal{P}$ is isomorphic to a unique X_r, $r \in \mathrm{BW}(k)$, and $\rho_\mathcal{P}([X_r]) = r$. Hence every $r \in \mathrm{BW}(k)$ is the type of a simple object of $\mathcal{P}$, and two simple objects of $\mathcal{P}$ are isomorphic if and only if their types agree; so $\mathcal{P}$ is a Brauer–Wall reference category.

A monoidal superequivalence $E : \mathcal{A} \to \mathcal{A}'$ between weakly pointed semisimple tensor Π-supercategories with split unit induces a bijection $G_\mathcal{A} \to G_{\mathcal{A}'}$, $[X] \mapsto [E(X)]$, and superalgebra isomorphisms $\mathrm{End}(X) \cong \mathrm{End}(E(X))$; so $\mathcal{A}'$ is a Brauer–Wall reference category if and only if $\mathcal{A}$ is. Let $t' = (\mathrm{BW}(k), \mathrm{id}, n', \theta')$ be a second twist of the same form as t. By (3.102), $\mathcal{P}_\mathcal{C}(t) \simeq_\otimes \mathcal{P}_\mathcal{C}(t')$ if and only if there is a gauge (ϕ, λ, β) from t to t'. For such a gauge, (3.48) gives $\phi^*\mathrm{id} = \mathrm{id}$, that is, $\phi = \mathrm{id}$; and since (3.48) and (3.49) do not involve ρ once $\phi = \mathrm{id}$, a gauge $(\mathrm{id}, \lambda, \beta)$ from t to t' is the same as a gauge $(\mathrm{id}, \lambda, \beta)$ from $(\mathrm{BW}(k), 0, n, \theta)$ to $(\mathrm{BW}(k), 0, n', \theta')$. Hence $\mathcal{P}_\mathcal{C}(t) \simeq_\otimes \mathcal{P}_\mathcal{C}(t')$ if and only if $(\mathrm{BW}(k), 0, n, \theta) \sim_{\mathrm{BW}(k)} (\mathrm{BW}(k), 0, n', \theta')$ in the sense of Definition 4.19, so (5.43) is well defined and injective on $\mathrm{SH}_k(\mathrm{BW}(k))$.

Finally, let $\mathcal{C}'$ be a Brauer–Wall reference category. Then $\mathcal{C}' \in \mathcal{W}$ by Proposition 3.17, and by Theorem 3.29 there is a twist $s = (G, \rho, \nu, \omega)$ with $\mathcal{C}' \simeq_\otimes \mathcal{P}_\mathcal{C}(s)$. Under the bijections $G \to G_{\mathcal{P}_\mathcal{C}(s)} \to G_{\mathcal{C}'}$ of (3.75) and of the preceding paragraph, ρ corresponds to $\rho_{\mathcal{C}'}$, which is an isomorphism by Proposition 3.14; so $\rho : G \to \mathrm{BW}(k)$ is a group isomorphism. Put $n = (\rho^{-1})^*\nu$ and $\theta = (\rho^{-1})^*\omega$. Then n is a normalized cocycle, θ is normalized with $\delta\theta = (-1)^{n\smile n}$, and $(\rho, 0, 1)$ is a gauge from s to $(\mathrm{BW}(k), \mathrm{id}, n, \theta)$ by Definition 3.19. Hence $\mathcal{C}' \simeq_\otimes \mathcal{P}_\mathcal{C}(\mathrm{BW}(k), \mathrm{id}, n, \theta)$ by (3.102), and (5.43) is surjective. □

6 The fields $\mathbb{C}$, $\mathbb{R}$, and $\mathbb{F}_5$

Lemma 6.1. *(1)* $\mathrm{BW}(\mathbb{C}) \cong \mathbb{Z}/2\mathbb{Z}$, *generated by the class of* $\mathbb{C}[\xi]/(\xi^2 - 1)$ *with* ξ *odd, and the parity* $\epsilon : \mathrm{BW}(\mathbb{C}) \to \mathbb{Z}/2\mathbb{Z}$ *of* (2.14) *is an isomorphism.*

(2) $\mathrm{BW}(\mathbb{R}) \cong \mathbb{Z}/8\mathbb{Z}$. *Let* $\mathrm{Cl}_{p,q}$ *be the real Clifford algebra with* p *positive and* q *negative squares and its usual* $\mathbb{Z}/2\mathbb{Z}$*-grading. The central division superalgebras*

$$(D_0, \dots, D_7) = (\mathbb{R}, \mathrm{Cl}_{0,1}, \mathrm{Cl}_{0,2}, \mathrm{Cl}_{0,3}, \mathbb{H}, \mathrm{Cl}_{3,0}, \mathrm{Cl}_{2,0}, \mathrm{Cl}_{1,0}), \tag{6.1}$$

where $\mathbb{R}$ *and* $\mathbb{H}$ *are concentrated in degree zero, represent the eight classes, with* $[D_r] + [D_s] = [D_{r+s}]$ *for* $r, s \in \mathbb{Z}/8\mathbb{Z}$, *and* $\epsilon(a) = a \bmod 2$.

(3) $\mathrm{BW}(\mathbb{F}_5) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. *Its nonzero elements are* $a = [\mathbb{F}_5[\xi]/(\xi^2 - 1)]$, $a' = [\mathbb{F}_5[\xi]/(\xi^2 - 2)]$, *with* ξ *odd, and* $a + a'$; *moreover* $\epsilon(a) = \epsilon(a') = 1$ *and* $\epsilon(a + a') = 0$. *For* $D = \mathbb{F}_5[\xi]/(\xi^2 - u)$ *with* $u \in \{1, 2\}$,

$$D \mathbin{\widehat{\otimes}} D \cong M_{1|1}(\mathbb{F}_5). \tag{6.2}$$

Proof. Let $\Gamma(k) = \mathbb{Z}/2\mathbb{Z} \times k^\times/(k^\times)^2$ with multiplication

$$(e, u)(e', u') = (e + e', (-1)^{ee'}uu'), \tag{6.3}$$

where the second components are square classes, and let $\theta : \mathrm{BW}(k) \to \Gamma(k)$ be the surjective homomorphism of [Wal64, Theorem 3] with kernel the Brauer group $\mathrm{Br}(k)$; it sends a class to its parity ϵ and, for $\epsilon = 1$, to the square of an odd central element modulo squares, or, for $\epsilon = 0$, to the discriminant of the center of the even part.

(1) For $k = \mathbb{C}$, $\mathrm{Br}(\mathbb{C}) = 0$ and every element of $\mathbb{C}^\times$ is a square, so $\mathrm{BW}(\mathbb{C}) \cong \mathbb{Z}/2\mathbb{Z}$ with ϵ an isomorphism; the superalgebra $\mathbb{C}[\xi]/(\xi^2 - 1)$ with ξ odd has center of dimension 2 and represents the nonzero class.

(2) For $k = \mathbb{R}$, Wall computes $\mathrm{BW}(\mathbb{R}) \cong \mathbb{Z}/8\mathbb{Z}$, generated by the class of $\mathrm{Cl}_{0,1}$, and the class of $\mathrm{Cl}_{p,q}$ corresponds to $q - p \bmod 8$ [Wal64, Theorem 3], [Lam05, Chapter IV, §§2–3]. The algebras in (6.1) are the division superalgebras among the Clifford algebras in the respective classes: $\mathrm{Cl}_{0,r}$ for $r \le 3$; the quaternion algebra $\mathbb{H}$, isomorphic to $\mathrm{Cl}_{0,2}$ as an ungraded algebra and taken with the trivial grading, in class 4; and $\mathrm{Cl}_{3,0}$, $\mathrm{Cl}_{2,0}$, $\mathrm{Cl}_{1,0}$ in classes $5, 6, 7$. For class 4: as an ungraded algebra $\mathrm{Cl}_{0,4} \cong M_2(\mathbb{H})$ [Lam05, Chapter V], and the grading of $\mathrm{Cl}_{0,4}$ is the conjugation by the even element $\omega = e_1e_2e_3e_4$, which satisfies $\omega^2 = 1$ and $\omega e_i = -e_i\omega$; in $M_2(\mathbb{H})$ the element ω is a non-central involution, hence conjugate to $\mathrm{diag}(1, -1)$, and conjugation by $\mathrm{diag}(1, -1)$ is the grading of $M_{1|1}(\mathbb{H})$ with $\mathbb{H}$ even. Thus $\mathrm{Cl}_{0,4} \cong M_{1|1}(\mathbb{H})$ as superalgebras, and $\mathbb{H}$ with the trivial grading is graded Morita equivalent to $\mathrm{Cl}_{0,4}$ by Proposition 2.6(2). Each is a division superalgebra: its even part is a division algebra and, when its odd part is nonzero, contains an odd unit, as every homogeneous element is a product of a unit of the even part and a generator. Since $[\mathrm{Cl}_{p,q}] + [\mathrm{Cl}_{p',q'}] = [\mathrm{Cl}_{p+p',q+q'}]$, we get $[D_r] + [D_s] = [D_{r+s}]$. The parity of $\mathrm{Cl}_{p,q}$ is $p + q \bmod 2$, because its ordinary center is $\mathbb{R}$ for $p + q$ even and $\mathbb{R} \oplus \mathbb{R}e_1 \cdots e_{p+q}$ for $p + q$ odd, where $e_1, \dots, e_{p+q}$ are the anticommuting generators of $\mathrm{Cl}_{p,q}$ with $e_i^2 = 1$ for $i \le p$ and $e_i^2 = -1$ for $i > p$; so $\epsilon(a) = a \bmod 2$.

(3) For $k = \mathbb{F}_5$, $\mathrm{Br}(\mathbb{F}_5) = 0$ by Wedderburn's theorem [Lam01, §13], and the squares in $\mathbb{F}_5^\times$ are 1 and 4, so $\mathbb{F}_5^\times$ modulo squares is $\mathbb{Z}/2\mathbb{Z}$ with nontrivial class 2. Since $-1 = 4 = 2^2$ is a square, the multiplication of $\Gamma(\mathbb{F}_5)$ is componentwise, and θ is an isomorphism because $\mathrm{Br}(\mathbb{F}_5) = 0$; so $\mathrm{BW}(\mathbb{F}_5) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. For $u \in \{1, 2\}$, $D = \mathbb{F}_5[\xi]/(\xi^2 - u)$ with ξ odd is a division superalgebra with center D, so $\epsilon([D]) = 1$ and $\theta([D]) = (1, u)$; thus $\theta(a) = (1, 1)$ and $\theta(a') = (1, 2)$ are distinct and $\theta(a + a') = (0, 2)$ has $\epsilon = 0$. Finally $D \mathbin{\widehat{\otimes}} D$ is generated by the odd elements $\xi_1 = \xi \mathbin{\widehat{\otimes}} 1$ and $\xi_2 = 1 \mathbin{\widehat{\otimes}} \xi$ with $\xi_i^2 = u$ and $\xi_1\xi_2 = -\xi_2\xi_1$; the odd element $\xi_1 + 2\xi_2$ satisfies $(\xi_1 + 2\xi_2)^2 = u + 4u = 0$, so $D \mathbin{\widehat{\otimes}} D$ is a four-dimensional central simple superalgebra with nonzero odd part that is not a division superalgebra, hence is isomorphic to $M_{1|1}(\mathbb{F}_5)$ by Proposition 2.6. This is (6.2), and it shows $2a = 2a' = 0$. □

Definition 6.2. For $a \in \mathbb{Z}/n\mathbb{Z}$ let $\tilde{a} \in \{0, \dots, n-1\}$ be its representative, and let $\gamma_n \in Z^2(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z})$ be the carry cocycle,

$$\tilde{a} + \tilde{b} = \widetilde{a + b} + n\,\gamma_n(a, b), \qquad \gamma_n(a, b) \in \{0, 1\}. \tag{6.4}$$

For $\mathrm{BW}(\mathbb{R}) = \mathbb{Z}/8\mathbb{Z}$ write $\gamma = \gamma_8$ and define $q : \mathbb{Z}/8\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ by

$$q(a) = \frac{\tilde{a}(\tilde{a} + 1)}{2} \bmod 2. \tag{6.5}$$

For a finite group G and $\rho \in \mathrm{Hom}(G, \mathbb{Z}/8\mathbb{Z})$, define $x_\rho, x'_\rho, x''_\rho : G \to \mathbb{Z}/2\mathbb{Z}$ and $\gamma_\rho \in Z^2(G, \mathbb{Z}/2\mathbb{Z})$ by

$$\widetilde{\rho(g)} = x_\rho(g) + 2x'_\rho(g) + 4x''_\rho(g), \qquad \gamma_\rho = \rho^*\gamma \bmod 2. \tag{6.6}$$

Lemma 6.3. *For $a, b \in \mathbb{Z}/8\mathbb{Z}$ and $\rho \in \mathrm{Hom}(G, \mathbb{Z}/8\mathbb{Z})$,*

$$q(a+b) = q(a)+q(b)+\epsilon(a)\epsilon(b), \qquad q(-a) = q(a)+\epsilon(a), \qquad q(a) = \epsilon(a)+\lfloor \tilde{a}/2 \rfloor \bmod 2, \tag{6.7}$$

$$x_\rho = \epsilon \circ \rho, \qquad \delta x'_\rho = x_\rho \smile x_\rho, \qquad q \circ \rho = x_\rho + x'_\rho. \tag{6.8}$$

Proof. The integer identity $(\tilde{a}+\tilde{b})(\tilde{a}+\tilde{b}+1)/2 = \tilde{a}(\tilde{a}+1)/2+\tilde{b}(\tilde{b}+1)/2+\tilde{a}\tilde{b}$ and $\tilde{a}\tilde{b} \equiv \epsilon(a)\epsilon(b) \pmod 2$ give the first identity in (6.7), once q is well defined: replacing $\tilde{a}$ by $\tilde{a}+8$ changes $\tilde{a}(\tilde{a}+1)/2$ by $8\tilde{a}+36$, an even integer. Taking $b=-a$ gives $q(-a) = q(a)+\epsilon(a)$, since $q(0)=0$ and $\epsilon(-a)=\epsilon(a)$. The third identity is checked on the eight values: $(q(a))_{a=0}^{7} = (0,1,1,0,0,1,1,0)$ and $(\epsilon(a) + \lfloor \tilde{a}/2 \rfloor)_{a=0}^{7} \equiv (0,1,1,0,0,1,1,0)$. For $\rho \in \mathrm{Hom}(G, \mathbb{Z}/8\mathbb{Z})$ write $x = x_\rho$ and $x' = x'_\rho$. By (6.6), $x = \epsilon \circ \rho$ is the reduction of $\widetilde{\rho(-)}$ modulo 2. For $g, h \in G$, reducing $\widetilde{\rho(g)} + \widetilde{\rho(h)} \equiv \widetilde{\rho(gh)} \pmod 8$ modulo 4 gives $x(g) + 2x'(g) + x(h) + 2x'(h) \equiv x(gh) + 2x'(gh) \pmod 4$; since $x(g)+x(h) = x(gh)+2x(g)x(h)$, this reads $2x(g)x(h)+2x'(g)+2x'(h) \equiv 2x'(gh) \pmod 4$, that is, $(\delta x')(g,h) = x(g)x(h)$. Finally $q \circ \rho = x_\rho + x'_\rho$ by the third identity in (6.7). □

Lemma 6.4. *Let $n \geq 2$ and let M be an abelian group, written multiplicatively, with trivial $\mathbb{Z}/n\mathbb{Z}$-action. Put*

$$M^n = \{t^n : t \in M\}, \qquad M[n] = \{t \in M : t^n = 1\},$$

let γ_n be the carry cocycle, and let Ψ_2, Ψ_3, and Ψ_4 be the maps on normalized cochains

$$\begin{aligned} \Psi_2(f) &= \prod_{j \in \mathbb{Z}/n\mathbb{Z}} f(1,j), \\ \Psi_3(\omega) &= \prod_{j \in \mathbb{Z}/n\mathbb{Z}} \omega(1,j,1), \\ \Psi_4(F) &= \prod_{i,j \in \mathbb{Z}/n\mathbb{Z}} F(i,1,j,1). \end{aligned} \tag{6.9}$$

(1) For $m \geq 1$, $H^{2m}(\mathbb{Z}/n\mathbb{Z}, M) \cong M/M^n$ and $H^{2m+1}(\mathbb{Z}/n\mathbb{Z}, M) \cong M[n]$.

(2) For $\eta \in C^1(\mathbb{Z}/n\mathbb{Z}, M)$, $\Psi_2(\delta\eta) = \eta(1)^n$. For $f \in Z^2(\mathbb{Z}/n\mathbb{Z}, M)$ there is $\eta \in C^1(\mathbb{Z}/n\mathbb{Z}, M)$ such that $f = \Psi_2(f)^{\gamma_n}\, \delta\eta$. For $t \in M$, the cocycle t^{γ_n} is a coboundary if and only if $t \in M^n$. Hence Ψ_2 induces an isomorphism $\overline{\Psi}_2 : H^2(\mathbb{Z}/n\mathbb{Z}, M) \to M/M^n$ such that $\overline{\Psi}_2^{-1}([t]) = [t^{\gamma_n}]$.

(3) For $t \in M[n]$, the normalized 3-cochain

$$\omega_t(a,b,c) = t^{\tilde{a}\gamma_n(b,c)} \tag{6.10}$$

is a cocycle such that $\Psi_3(\omega_t) = t$; for $t \in M$, the cocycle $t^{\gamma_n \smile \gamma_n}$ satisfies $\Psi_4(t^{\gamma_n \smile \gamma_n}) = t$. For $f \in C^2(\mathbb{Z}/n\mathbb{Z}, M)$ and $\omega \in C^3(\mathbb{Z}/n\mathbb{Z}, M)$,

$$\Psi_3(\delta f) = 1, \qquad \Psi_4(\delta\omega) = \Psi_3(\omega)^n.$$

Hence Ψ_3 and Ψ_4 induce surjective homomorphisms $\overline{\Psi}_3 : H^3(\mathbb{Z}/n\mathbb{Z}, M) \to M[n]$ and $\overline{\Psi}_4 : H^4(\mathbb{Z}/n\mathbb{Z}, M) \to M/M^n$ such that $\overline{\Psi}_3([\omega_t]) = t$ and $\overline{\Psi}_4([t^{\gamma_n \smile \gamma_n}]) = [t]$. If $M[n]$ is finite, then $\overline{\Psi}_3$ is an isomorphism; if M/M^n is finite, then $\overline{\Psi}_4$ is an isomorphism.

(4) For $m \in \{2,3,4\}$, the inclusion $\{\pm 1\} \subset \mathbb{R}^\times$ induces an isomorphism

$$H^m(\mathbb{Z}/n\mathbb{Z}, \{\pm 1\}) \longrightarrow H^m(\mathbb{Z}/n\mathbb{Z}, \mathbb{R}^\times);$$

moreover, for $m \geq 1$, $H^{2m}(\mathbb{Z}/n\mathbb{Z}, \mathbb{C}^\times) = 0$ and $H^{2m+1}(\mathbb{Z}/n\mathbb{Z}, \mathbb{C}^\times) \cong \mathbb{C}^\times[n]$.

Proof. Write $\Gamma = \mathbb{Z}/n\mathbb{Z}$, let $g \in \Gamma$ be the element 1, and let $N = \sum_{j \in \Gamma} g^j \in \mathbb{Z}[\Gamma]$.

(1) By [Wei94, Theorem 6.2.2], for $m \geq 1$, $H^{2m}(\Gamma, M) \cong M^\Gamma/NM$ and $H^{2m+1}(\Gamma, M) \cong \{t \in M : Nt = 1\}/(g-1)M$, where $\mathbb{Z}[\Gamma]$ acts on M through the trivial action of Γ, the group M being written multiplicatively. Thus N acts on M by $t \mapsto t^n$ and $g-1$ acts by $t \mapsto 1$, so that $M^\Gamma/NM = M/M^n$ and $\{t \in M : Nt = 1\}/(g-1)M = M[n]$.

Since $n \geq 2$, $\tilde{1} = 1$, and for $a \in \Gamma$, $\gamma_n(a,1) = \gamma_n(1,a) = 1$ if $\tilde{a} = n-1$ and $\gamma_n(a,1) = \gamma_n(1,a) = 0$ otherwise; hence

$$\sum_{j\in\Gamma} \gamma_n(1,j) = \sum_{j\in\Gamma} \gamma_n(j,1) = 1. \tag{6.11}$$

For $z \in M$ let $\zeta_z \in C^1(\Gamma, M)$ be the cochain $\zeta_z(b) = z^{\tilde{b}}$; by (A.1) and (6.4), $(\delta\zeta_z)(a,b) = z^{\tilde{a}+\tilde{b}-\widetilde{a+b}} = z^{n\gamma_n(a,b)}$ for $a, b \in \Gamma$. For $t \in M$, the cochains t^{γ_n} and $t^{\gamma_n \smile \gamma_n}$ are cocycles, because $\gamma_n \in Z^2(\Gamma,\mathbb{Z})$, $\gamma_n \smile \gamma_n \in Z^4(\Gamma,\mathbb{Z})$ by (A.6), and δ commutes with the homomorphism $\mathbb{Z} \to M$, $r \mapsto t^r$, applied to the values of a cochain, by Lemma A.2(2).

(2) Let $\eta \in C^1(\Gamma, M)$. By (A.1), $(\delta\eta)(a,b) = \eta(b)\eta(a+b)^{-1}\eta(a)$ for $a, b \in \Gamma$, so $\Psi_2(\delta\eta) = \prod_{j\in\Gamma} \eta(j)\eta(1+j)^{-1}\eta(1) = \eta(1)^n$, because $\prod_{j\in\Gamma} \eta(1+j) = \prod_{j\in\Gamma} \eta(j)$. Let $f \in Z^2(\Gamma, M)$ and $a, b \in \Gamma$. The cocycle identity $(\delta f)(a,b,1) = 1$ reads $f(a,b+1) = f(a,b)\, f(a+b,1)\, f(b,1)^{-1}$, and $(\delta f)(1,b,1) = 1$ reads $f(1,b)f(b,1)^{-1} = f(1,b+1)f(b+1,1)^{-1}$. By induction on $\tilde{b}$, starting from $f(a,0) = 1$ and $f(1,0)f(0,1)^{-1} = 1$,

$$f(a,b) = \prod_{r=0}^{\tilde{b}-1} f(a+r,1)\, f(r,1)^{-1}, \qquad f(1,b) = f(b,1), \tag{6.12}$$

where $a+r$ and r are read in Γ. Put $u = \prod_{j\in\Gamma} f(j,1)$, which equals $\Psi_2(f)$ by (6.12), and let $\eta_f \in C^1(\Gamma, M)$ be the cochain $\eta_f(a) = \prod_{r=0}^{\tilde{a}-1} f(r,1)^{-1}$. Then $\prod_{r=0}^{\tilde{b}-1} f(r,1) = \eta_f(b)^{-1}$, and $\prod_{r=0}^{\tilde{b}-1} f(a+r,1) = \prod_{i=\tilde{a}}^{\tilde{a}+\tilde{b}-1} f(i,1)$, with i read in Γ, equals $\eta_f(a+b)^{-1}\eta_f(a)$ if $\gamma_n(a,b) = 0$, since then $\tilde{a}+\tilde{b} = \widetilde{a+b}$, and equals $\prod_{i=\tilde{a}}^{n-1} f(i,1) \prod_{i=0}^{\widetilde{a+b}-1} f(i,1) = u\,\eta_f(a)\,\eta_f(a+b)^{-1}$ if $\gamma_n(a,b) = 1$, since then $\tilde{a}+\tilde{b} = \widetilde{a+b}+n$. Hence $f(a,b) = u^{\gamma_n(a,b)}\eta_f(a)\eta_f(b)\eta_f(a+b)^{-1} = u^{\gamma_n(a,b)}(\delta\eta_f)(a,b)$, that is, $f = \Psi_2(f)^{\gamma_n}\,\delta\eta_f$. For $t \in M$, $\Psi_2(t^{\gamma_n}) = t^{\sum_{j\in\Gamma}\gamma_n(1,j)} = t$ by (6.11). If $t^{\gamma_n} = \delta\eta$ with $\eta \in C^1(\Gamma, M)$, then $t = \Psi_2(\delta\eta) = \eta(1)^n \in M^n$; if $t = z^n$ with $z \in M$, then $t^{\gamma_n} = z^{n\gamma_n} = \delta\zeta_z$. The map Ψ_2 is a homomorphism $C^2(\Gamma, M) \to M$ such that $\Psi_2(B^2(\Gamma, M)) \subseteq M^n$, so it induces a homomorphism $\overline{\Psi}_2 : H^2(\Gamma, M) \to M/M^n$, which is surjective because $\overline{\Psi}_2([t^{\gamma_n}]) = [t]$ for $t \in M$, and injective because $f \in Z^2(\Gamma, M)$ with $\Psi_2(f) = z^n$, $z \in M$, satisfies $f = z^{n\gamma_n}\,\delta\eta_f = \delta(\zeta_z\eta_f)$. Thus $\overline{\Psi}_2^{-1}([t]) = [t^{\gamma_n}]$.

(3) Let $t \in M[n]$ and $a, b, c, d \in \Gamma$. The cochain ω_t is normalized, since $\tilde{0} = 0$ and $\gamma_n(0,c) = \gamma_n(b,0) = 0$. By (A.1), the cocycle identity $\gamma_n(c,d) - \gamma_n(b+c,d) + \gamma_n(b,c+d) - \gamma_n(b,c) = 0$, and (6.4),

$$\begin{aligned}(\delta\omega_t)(a,b,c,d) &= t^{\tilde{b}\gamma_n(c,d) - \widetilde{a+b}\gamma_n(c,d) + \tilde{a}(\gamma_n(b+c,d) - \gamma_n(b,c+d) + \gamma_n(b,c))} \\ &= t^{(\tilde{a}+\tilde{b}-\widetilde{a+b})\gamma_n(c,d)} = t^{n\gamma_n(a,b)\gamma_n(c,d)} = 1;\end{aligned}$$

so ω_t is a cocycle, and $\Psi_3(\omega_t) = t^{\sum_{j\in\Gamma}\gamma_n(j,1)} = t$ by (6.11). For $t \in M$, $\Psi_4(t^{\gamma_n\smile\gamma_n}) = t^{(\sum_{i\in\Gamma}\gamma_n(i,1))(\sum_{j\in\Gamma}\gamma_n(j,1))} = t$, again by (6.11). Let $f \in C^2(\Gamma, M)$ and $\omega \in C^3(\Gamma, M)$. By (A.1), for $i, j \in \Gamma$,

$$\begin{aligned}(\delta f)(1,j,1) &= f(j,1)\, f(1+j,1)^{-1}\, f(1,j+1)\, f(1,j)^{-1}, \\ (\delta\omega)(i,1,j,1) &= \omega(1,j,1)\,\omega(i+1,j,1)^{-1}\,\omega(i,j+1,1)\,\omega(i,1,j+1)^{-1}\,\omega(i,1,j).\end{aligned}$$

Taking the product over $j \in \Gamma$ in the first identity gives $\Psi_3(\delta f) = 1$, because $\prod_{j\in\Gamma} f(1+j,1) = \prod_{j\in\Gamma} f(j,1)$ and $\prod_{j\in\Gamma} f(1,j+1) = \prod_{j\in\Gamma} f(1,j)$. Taking the product over $i, j \in \Gamma$ in the second identity gives $\Psi_4(\delta\omega) = \Psi_3(\omega)^n$, because $\prod_{i,j\in\Gamma}\omega(i+1,j,1) = \prod_{i,j\in\Gamma}\omega(i,j+1,1)$ and $\prod_{i,j\in\Gamma}\omega(i,1,j+1) = \prod_{i,j\in\Gamma}\omega(i,1,j)$. The map Ψ_3 is a homomorphism $C^3(\Gamma, M) \to M$ such that $\Psi_3(B^3(\Gamma, M)) = 1$ and $\Psi_3(\omega)^n = \Psi_4(\delta\omega) = 1$ for $\omega \in Z^3(\Gamma, M)$; hence it induces a homomorphism $\overline{\Psi}_3 : H^3(\Gamma, M) \to M[n]$, which is surjective because $\overline{\Psi}_3([\omega_t]) = t$ for

$t \in M[n]$. The map Ψ_4 is a homomorphism $C^4(\Gamma, M) \to M$ such that $\Psi_4(B^4(\Gamma, M)) \subseteq M^n$; hence it induces a homomorphism $\overline{\Psi}_4 : H^4(\Gamma, M) \to M/M^n$, which is surjective because $\overline{\Psi}_4([t^{\gamma_n \smile \gamma_n}]) = [t]$ for $t \in M$. If $M[n]$ is finite, then $H^3(\Gamma, M)$ is finite of order $|M[n]|$ by (1), so the surjective homomorphism $\overline{\Psi}_3$ is bijective; if M/M^n is finite, then $H^4(\Gamma, M)$ is finite of order $|M/M^n|$ by (1), so $\overline{\Psi}_4$ is bijective.

(4) Let $\iota : \{\pm 1\} \to \mathbb{R}^\times$ be the inclusion and let $m \in \{2, 3, 4\}$. Composing cochains with ι is a homomorphism $C^m(\Gamma, \{\pm 1\}) \to C^m(\Gamma, \mathbb{R}^\times)$ that commutes with the coboundary (A.1); let $\iota_m : H^m(\Gamma, \{\pm 1\}) \to H^m(\Gamma, \mathbb{R}^\times)$ be the induced homomorphism. For $c \in C^m(\Gamma, \{\pm 1\})$, $\Psi_m(\iota \circ c) = \iota(\Psi_m(c))$ by (6.9). If n is even, then $\{\pm 1\}^n = \{1\}$, $(\mathbb{R}^\times)^n = \mathbb{R}_{>0}$, and $\{\pm 1\}[n] = \mathbb{R}^\times[n] = \{\pm 1\}$; if n is odd, then $\{\pm 1\}^n = \{\pm 1\}$, $(\mathbb{R}^\times)^n = \mathbb{R}^\times$, and $\{\pm 1\}[n] = \mathbb{R}^\times[n] = \{1\}$. In both cases ι induces bijections $\bar{\iota} : \{\pm 1\}/\{\pm 1\}^n \to \mathbb{R}^\times/(\mathbb{R}^\times)^n$ and $\bar{\iota} : \{\pm 1\}[n] \to \mathbb{R}^\times[n]$ between finite sets. For $M = \{\pm 1\}$ and $M = \mathbb{R}^\times$ write $\overline{\Psi}_m^M$ for the homomorphism $\overline{\Psi}_m$ of (2) and (3); it is an isomorphism, by (2) for $m = 2$ and by (3) for $m = 3, 4$, since $M[n]$ and M/M^n are finite. Then $\overline{\Psi}_m^{\mathbb{R}^\times} \circ \iota_m = \bar{\iota} \circ \overline{\Psi}_m^{\{\pm 1\}}$, so ι_m is an isomorphism. Since $\mathbb{C}^\times$ is divisible, (1) gives $H^{2m}(\Gamma, \mathbb{C}^\times) \cong \mathbb{C}^\times/\mathbb{C}^{\times n} = 0$ and $H^{2m+1}(\Gamma, \mathbb{C}^\times) \cong \mathbb{C}^\times[n]$ for $m \geq 1$. □

Proposition 6.5. *(1) Let $k = \mathbb{C}$. The torsor $\mathrm{Sec}(D)$ under $H^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ has two elements, and $o_4 = 1$ for both, because $H^4(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times) = 0$. Brauer–Wall reference categories over $\mathbb{C}$ exist. Up to monoidal superequivalence there are exactly four of them, and $\mathrm{SH}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$.*

(2) Let $k = \mathbb{R}$. The torsor $\mathrm{Sec}(D)$ under $H^2(\mathbb{Z}/8\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ has two elements, $H^4(\mathbb{Z}/8\mathbb{Z}, \mathbb{R}^\times) \cong \mathbb{Z}/2\mathbb{Z}$, and Q is an isomorphism; hence exactly one element of $\mathrm{Sec}(D)$ satisfies $o_4 = 1$. Brauer–Wall reference categories over $\mathbb{R}$ exist. Up to monoidal superequivalence there are exactly two of them, and $\mathrm{SH}_{\mathbb{R}}(\mathbb{Z}/8\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ is generated by the class of $(0, \omega_-)$, where

$$\omega_-(a, b, c) = (-1)^{\tilde{a}\gamma(b,c)}. \tag{6.13}$$

(3) Let $k = \mathbb{F}_5$ and let $a = [\mathbb{F}_5[\xi]/(\xi^2 - 1)]$ be as in Lemma 6.1(3). For every even system (M, A), the value $O(a, a, a, a)$ of the pentagon scalar is not a square in $\mathbb{F}_5$. In particular no reference system and no Brauer–Wall reference category over $\mathbb{F}_5$ exist, and $o_4(\mathbb{F}_5) \neq 1$.

Proof. In all three cases $\mathrm{Sec}(D)$ is a torsor under $H^2(\mathrm{BW}(k), \mathbb{Z}/2\mathbb{Z})$ and o_4 is defined (Lemma 5.9 and Proposition 5.13).

(1) By Lemma 6.4(1) and (4), $H^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ and $H^4(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times) = 0$, so $o_4(M) = 1$ for every even system (M, A) with respect to D, that is, for both classes of $\mathrm{Sec}(D)$, and Theorem 5.17 gives a reference category. In (4.98) for $G = \mathbb{Z}/2\mathbb{Z}$ and $k = \mathbb{C}$, $H^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})^+ = H^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ since $H^4(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times) = 0$, and $H^3(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times) \cong \mathbb{C}^\times[2]$; so $|\mathrm{SH}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z})| = 4$ and there are four reference categories by Proposition 5.18. Let x be the nontrivial character of $\mathbb{Z}/2\mathbb{Z}$ and let ω be the normalized 3-cochain with $\omega(1, 1, 1) = i$. Then $(\delta\omega)(1, 1, 1, 1) = \omega(1, 1, 1)^2 = -1$ and $\delta\omega = 1$ elsewhere, so $(x \smile x, \omega)$ is a twist with $\rho = 0$. Its square in $\mathrm{SH}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z})$ is $(0, \omega^2(-1)^{(x\smile x)\smile_1(x\smile x)}) = (0, \omega^2)$, because $((x \smile x) \smile_1 (x \smile x))(1, 1, 1) = x(0)x(1)x(1)x(1) + x(1)x(0)x(1)x(1) = 0$, and $\Psi_3(\omega^2) = -1$, so $[(0, \omega^2)] \neq 1$ in $H^3(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times)$ by Lemma 6.4(3). Hence $\mathrm{SH}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z})$ has an element of order 4 and is cyclic.

(2) By Lemma 6.4(1), (2), and (4), $H^2(\mathbb{Z}/8\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, generated by the class of γ reduced modulo 2, and $H^4(\mathbb{Z}/8\mathbb{Z}, \mathbb{R}^\times) \cong H^4(\mathbb{Z}/8\mathbb{Z}, \{\pm 1\}) \cong \mathbb{Z}/2\mathbb{Z}$. The cochain $(-1)^{\gamma\smile\gamma}$ has $\Psi_4((-1)^{\gamma\smile\gamma}) = (-1)^{(\sum_i \gamma(i,1))(\sum_j \gamma(j,1))} = -1 \notin (\mathbb{R}^\times)^8$, so it is not a coboundary by Lemma 6.4(3), and $Q([\gamma]) \neq 1$; thus Q is an isomorphism and $o_4([\gamma] \cdot M) = o_4(M)Q([\gamma]) \neq o_4(M)$ for every even system (M, A) with respect to D. Since $H^4(\mathbb{Z}/8\mathbb{Z}, \mathbb{R}^\times)$ has two elements, exactly one class in $\mathrm{Sec}(D)$ has $o_4 = 1$, and Theorem 5.17 gives a reference category. In (4.98) for $G = \mathbb{Z}/8\mathbb{Z}$, the class $[\gamma]$ does not lie in $H^2(\mathbb{Z}/8\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})^+$, since $(-1)^{\gamma\smile\gamma}$ is not a coboundary, so $H^2(\mathbb{Z}/8\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})^+ = 0$ and $\mathrm{SH}_{\mathbb{R}}(\mathbb{Z}/8\mathbb{Z}) \cong H^3(\mathbb{Z}/8\mathbb{Z}, \mathbb{R}^\times) \cong \{\pm 1\}$, generated

by the class of $(0,\omega_{-1})$ with $\omega_{-1}=\omega_-$ of (6.13), which is nontrivial by Lemma 6.4(3), since $\Psi_3(\omega_-)=-1$. Proposition 5.18 gives exactly two reference categories.

(3) By Lemma 5.12 and Proposition 5.13(4), we may take the representative D_a of Definition 2.7 to be $\mathbb{F}_5[\xi]/(\xi^2-1)$ with ξ odd, a division superalgebra of class a, with $2a=0$ by Lemma 6.1(3); and $D_0=\mathbb{F}_5$. Put $\varsigma=2\in\mathbb{F}_5$, so that $\varsigma^2=-1$. Write $H=\{0,a\}$. Definitions 5.8 and 5.10, Lemma 5.11, and Proposition 5.13 apply verbatim with $\mathrm{BW}(k)$ replaced by H, since their proofs use only the group structure of the index set and the representatives D_0 and D_a; we speak of Morita systems and even systems indexed by H. An even system (M,A) indexed by $\mathrm{BW}(\mathbb{F}_5)$ restricts to the even system $(M_{b,c},A_{b,c,d})_{b,c,d\in H}$ indexed by H, with the same value $O(a,a,a,a)$. It suffices to show that for every even system indexed by H the value $O(a,a,a,a)$ is not a square in $\mathbb{F}_5$.

An even system indexed by H. Let $\sigma:D_a\to D_a^{\mathrm{sop}}$ be the isomorphism of superalgebras with $\sigma(\xi)=\varsigma\xi$; it is one because $\sigma(\xi)\cdot_{\mathrm{sop}}\sigma(\xi)=-\varsigma^2\xi^2=1=\sigma(\xi^2)$. Let N be the super vector space D_a with the left action of $D_a\mathbin{\widehat{\otimes}}D_a$

$$(x\mathbin{\widehat{\otimes}}y)\cdot n=(-1)^{|y||n|}\,x\,n\,\sigma(y)\qquad(x,y,n\in D_a\text{ homogeneous}),\tag{6.14}$$

the product on the right taken in D_a; thus $(\xi\mathbin{\widehat{\otimes}}1)\cdot n=\xi n$ and $(1\mathbin{\widehat{\otimes}}\xi)\cdot n=(-1)^{|n|}\varsigma n\xi$. The composite of $x\mathbin{\widehat{\otimes}}y\mapsto x\mathbin{\widehat{\otimes}}\sigma(y)$ with (2.13) is an isomorphism of superalgebras $D_a\mathbin{\widehat{\otimes}}D_a\to\mathrm{End}_k(N)\cong M_{1|1}(\mathbb{F}_5)$ under which (6.14) is the action of $\mathrm{End}_k(N)$ on N; so N is the simple column module of $M_{1|1}(\mathbb{F}_5)$, with $D_0=\mathbb{F}_5$ acting on the right by scalars. Since $D_a\mathbin{\widehat{\otimes}}D_a$ and $\mathbb{F}_5$ are Morita equivalent by Proposition 2.6(2), Proposition 2.8 makes N a Morita equivalence $D_a\mathbin{\widehat{\otimes}}D_a\to\mathbb{F}_5$. Put $M_{0,0}=\mathbb{F}_5$, $M_{0,a}=M_{a,0}=D_a$, and $M_{a,a}=N$: a Morita system indexed by H. Its triple composites at (a,a,a) are

$$L=(N\mathbin{\widehat{\otimes}}D_a)\otimes_{k\mathbin{\widehat{\otimes}}D_a}D_a\cong N\mathbin{\widehat{\otimes}}D_a,\qquad U=(D_a\mathbin{\widehat{\otimes}}N)\otimes_{D_a\mathbin{\widehat{\otimes}}k}D_a\cong D_a\mathbin{\widehat{\otimes}}N,\tag{6.15}$$

through $(n\mathbin{\widehat{\otimes}}w)\otimes w'\mapsto n\mathbin{\widehat{\otimes}}ww'$ and $(w\mathbin{\widehat{\otimes}}n)\otimes w'\mapsto(-1)^{|n||w'|}ww'\mathbin{\widehat{\otimes}}n$ for homogeneous $n\in N$ and $w,w'\in D_a$. By (2.10), L is the $(D_a\mathbin{\widehat{\otimes}}D_a\mathbin{\widehat{\otimes}}D_a,D_a)$-bimodule $N\mathbin{\widehat{\otimes}}D_a$ with

$$(x\mathbin{\widehat{\otimes}}y\mathbin{\widehat{\otimes}}z)(n\mathbin{\widehat{\otimes}}w)=(-1)^{|z||n|}((x\mathbin{\widehat{\otimes}}y)\cdot n)\mathbin{\widehat{\otimes}}zw,\qquad(n\mathbin{\widehat{\otimes}}w)t=n\mathbin{\widehat{\otimes}}wt,\tag{6.16}$$

for homogeneous $x,y,z,w,t\in D_a$ and $n\in N$, and U is the $(D_a\mathbin{\widehat{\otimes}}D_a\mathbin{\widehat{\otimes}}D_a,D_a)$-bimodule $D_a\mathbin{\widehat{\otimes}}N$ with

$$(x\mathbin{\widehat{\otimes}}y\mathbin{\widehat{\otimes}}z)(w\mathbin{\widehat{\otimes}}n)=(-1)^{(|y|+|z|)|w|}\,xw\mathbin{\widehat{\otimes}}((y\mathbin{\widehat{\otimes}}z)\cdot n),\qquad(w\mathbin{\widehat{\otimes}}n)t=(-1)^{|n||t|}wt\mathbin{\widehat{\otimes}}n,\tag{6.17}$$

the left action on U being transported along the associativity isomorphism of $\mathbin{\widehat{\otimes}}$, which is the identity on pure tensors. The bimodule L is generated by $1\mathbin{\widehat{\otimes}}1$, since $(x\mathbin{\widehat{\otimes}}1\mathbin{\widehat{\otimes}}z)(1\mathbin{\widehat{\otimes}}1)t=x\mathbin{\widehat{\otimes}}zt$ by (6.16) and $(x\mathbin{\widehat{\otimes}}1)\cdot1=x$. Both L and U are Morita equivalences $D_a\mathbin{\widehat{\otimes}}D_a\mathbin{\widehat{\otimes}}D_a\to D_a$, so by Proposition 2.8 the homogeneous bimodule maps $L\to U$ form a line of invertible elements, and such a map is determined by the image v of $1\mathbin{\widehat{\otimes}}1$. An even $v=c_0(1\mathbin{\widehat{\otimes}}1)+c_1(\xi\mathbin{\widehat{\otimes}}\xi)\in U$, with $c_0,c_1\in\mathbb{F}_5$, must satisfy the relations of $1\mathbin{\widehat{\otimes}}1$ in L: the elements $(1\mathbin{\widehat{\otimes}}\xi\mathbin{\widehat{\otimes}}1)(1\mathbin{\widehat{\otimes}}1)=\sigma(\xi)\mathbin{\widehat{\otimes}}1=\varsigma(\xi\mathbin{\widehat{\otimes}}1\mathbin{\widehat{\otimes}}1)(1\mathbin{\widehat{\otimes}}1)$ and $(1\mathbin{\widehat{\otimes}}1\mathbin{\widehat{\otimes}}\xi)(1\mathbin{\widehat{\otimes}}1)=1\mathbin{\widehat{\otimes}}\xi=(1\mathbin{\widehat{\otimes}}1)\xi$ of L give the conditions $(1\mathbin{\widehat{\otimes}}\xi\mathbin{\widehat{\otimes}}1)v=\varsigma(\xi\mathbin{\widehat{\otimes}}1\mathbin{\widehat{\otimes}}1)v$ and $(1\mathbin{\widehat{\otimes}}1\mathbin{\widehat{\otimes}}\xi)v=v\xi$ in U. By (6.17) and (6.14),

$$\begin{aligned}(\xi\mathbin{\widehat{\otimes}}1\mathbin{\widehat{\otimes}}1)v&=c_0(\xi\mathbin{\widehat{\otimes}}1)+c_1(1\mathbin{\widehat{\otimes}}\xi),&\qquad(1\mathbin{\widehat{\otimes}}\xi\mathbin{\widehat{\otimes}}1)v&=c_0(1\mathbin{\widehat{\otimes}}\xi)-c_1(\xi\mathbin{\widehat{\otimes}}1),\\(1\mathbin{\widehat{\otimes}}1\mathbin{\widehat{\otimes}}\xi)v&=\varsigma c_0(1\mathbin{\widehat{\otimes}}\xi)+\varsigma c_1(\xi\mathbin{\widehat{\otimes}}1),&\qquad v\xi&=c_0(\xi\mathbin{\widehat{\otimes}}1)-c_1(1\mathbin{\widehat{\otimes}}\xi),\end{aligned}\tag{6.18}$$

using $(1\mathbin{\widehat{\otimes}}\xi)\cdot1=\varsigma\xi$, $(1\mathbin{\widehat{\otimes}}\xi)\cdot\xi=-\varsigma$, $(\xi\mathbin{\widehat{\otimes}}1)\cdot\xi=1$, and $(w\mathbin{\widehat{\otimes}}n)\xi=(-1)^{|n|}w\xi\mathbin{\widehat{\otimes}}n$. The first condition reads $c_0=\varsigma c_1$ and $-c_1=\varsigma c_0$, the second $\varsigma c_0=-c_1$ and $\varsigma c_1=c_0$; all four are the single relation $c_0=\varsigma c_1$, since $\varsigma^2=-1$. An odd $v=d_0(\xi\mathbin{\widehat{\otimes}}1)+d_1(1\mathbin{\widehat{\otimes}}\xi)$, with $d_0,d_1\in\mathbb{F}_5$,

must satisfy the same two relations with the sign of (2.5), $(1\,\widehat{\otimes}\,\xi\,\widehat{\otimes}\,1)v = \varsigma(\xi\,\widehat{\otimes}\,1\,\widehat{\otimes}\,1)v$ and $(1\,\widehat{\otimes}\,1\,\widehat{\otimes}\,\xi)v = -v\xi$; by (6.17) and (6.14) the first gives $d_1 = \varsigma d_0$ and the second $d_0 = \varsigma d_1$, so $d_0 = \varsigma^2 d_0 = -d_0$ and $d_0 = d_1 = 0$. So the line of homogeneous bimodule maps $L \to U$ is even, spanned by the map A_0 with $v = 1\,\widehat{\otimes}\,1 - \varsigma\,\xi\,\widehat{\otimes}\,\xi$, that is,

$$A_0(n\,\widehat{\otimes}\,w) = (n\,\widehat{\otimes}\,1\,\widehat{\otimes}\,1)\,v\,w = nw\,\widehat{\otimes}\,1 - (-1)^{|w|}\varsigma\,n\xi w\,\widehat{\otimes}\,\xi \qquad (n \in N,\ w \in D_a \text{ homogeneous}). \tag{6.19}$$

For a triple containing 0 the comparison is the unit identification, so M is even, and the even systems indexed by H with Morita system M are the $(M, \lambda A_0)$ with $\lambda \in \mathbb{F}_5^\times$.

The pentagon scalar of (M, A_0). When one of the four arguments is 0, $P_L = P_R$ by Proposition 5.13(1), applied to even systems indexed by H; the pentagon for $(0, a, a, a)$, for instance, still contains the comparison $A_{a,a,a}$, between the words $0((aa)a)$ and $0(a(aa))$, and the equality is the normalization proved there. For (a, a, a, a), write $E = D_a\,\widehat{\otimes}\,D_a\,\widehat{\otimes}\,D_a\,\widehat{\otimes}\,D_a$, let $\xi_1, \dots, \xi_4$ be the odd generators $\xi\,\widehat{\otimes}\,1\,\widehat{\otimes}\,1\,\widehat{\otimes}\,1, \dots, 1\,\widehat{\otimes}\,1\,\widehat{\otimes}\,1\,\widehat{\otimes}\,\xi$ of E, and let $W_1, \dots, W_5$ be the iterated bimodules of the words $((aa)a)a$, $(a(aa))a$, $a((aa)a)$, $a(a(aa))$, and $(aa)(aa)$; they are $(E, \mathbb{F}_5)$-bimodules of dimension 4. Using $M_{0,a} = M_{a,0} = D_a$, $M_{0,0} = \mathbb{F}_5$, and the unit isomorphisms, they are

$$\begin{aligned} &W_1 = (L\,\widehat{\otimes}\,D_a) \otimes_{D_a\widehat{\otimes}D_a} N, \quad W_2 = (U\,\widehat{\otimes}\,D_a) \otimes_{D_a\widehat{\otimes}D_a} N, \quad W_3 = (D_a\,\widehat{\otimes}\,L) \otimes_{D_a\widehat{\otimes}D_a} N, \\ &W_4 = (D_a\,\widehat{\otimes}\,U) \otimes_{D_a\widehat{\otimes}D_a} N, \quad W_5 = N\,\widehat{\otimes}\,N, \end{aligned} \tag{6.20}$$

with the generators $g_1 = ((1\,\widehat{\otimes}\,1)\,\widehat{\otimes}\,1) \otimes 1$, $g_2 = ((1\,\widehat{\otimes}\,1)\,\widehat{\otimes}\,1) \otimes 1$, $g_3 = (1\,\widehat{\otimes}\,(1\,\widehat{\otimes}\,1)) \otimes 1$, $g_4 = (1\,\widehat{\otimes}\,(1\,\widehat{\otimes}\,1)) \otimes 1$, and $g_5 = 1\,\widehat{\otimes}\,1$, where $1\,\widehat{\otimes}\,1$ denotes the generator of L or of U. Each W_i is generated by g_i as an E-module, because g_i is built from units and N is generated by 1. The five comparisons are $m_1 = (A_0\,\widehat{\otimes}\,\mathrm{id}_{D_a}) \otimes \mathrm{id}_N : W_1 \to W_2$, $m_3 = (\mathrm{id}_{D_a}\,\widehat{\otimes}\,A_0) \otimes \mathrm{id}_N : W_3 \to W_4$, and the unit identifications $m_2 : W_2 \to W_3$, $m_4 : W_1 \to W_5$, $m_5 : W_5 \to W_4$, which are E-linear and send $g_2 \mapsto g_3$, $g_1 \mapsto g_5$, $g_5 \mapsto g_4$, since they are composites of associativity isomorphisms of $\widehat{\otimes}$ and of balanced tensor products, interchange isomorphisms, and unit isomorphisms, all of which send pure tensors of units to pure tensors of units with sign $+1$. Thus $P_R(g_1) = m_5 m_4(g_1) = g_4$. For m_1, (6.19) gives $m_1(g_1) = ((1\,\widehat{\otimes}\,1 - \varsigma\,\xi\,\widehat{\otimes}\,\xi)\,\widehat{\otimes}\,1) \otimes 1$, where $\xi\,\widehat{\otimes}\,\xi \in U = D_a\,\widehat{\otimes}\,N$ is $\xi\,\widehat{\otimes}\,(\xi\,\widehat{\otimes}\,1) \cdot 1$, so that $(\xi\,\widehat{\otimes}\,\xi)\,\widehat{\otimes}\,1 = \xi_1\xi_2\,g_2$ in W_2, no sign arising because the factors passed are units; since $(\xi\,\widehat{\otimes}\,1) \cdot 1 = \varsigma^{-1}(1\,\widehat{\otimes}\,\xi) \cdot 1$ in N by (6.14), also $(\xi\,\widehat{\otimes}\,\xi)\,\widehat{\otimes}\,1 = \varsigma^{-1}\xi_1\xi_3\,g_2$, and

$$m_1(g_1) = (1 - \varsigma\varsigma^{-1}\xi_1\xi_3)\,g_2 = (1 - \xi_1\xi_3)\,g_2. \tag{6.21}$$

In the same way $m_3(g_3) = (1\,\widehat{\otimes}\,(1\,\widehat{\otimes}\,1 - \varsigma\,\xi\,\widehat{\otimes}\,\xi)) \otimes 1$ with $1\,\widehat{\otimes}\,(\xi\,\widehat{\otimes}\,\xi) = \xi_2\xi_3\,g_4 = \varsigma^{-1}\xi_2\xi_4\,g_4$, so

$$m_3(g_3) = (1 - \xi_2\xi_4)\,g_4, \qquad P_L(g_1) = m_3 m_2 m_1(g_1) = (1 - \xi_1\xi_3)(1 - \xi_2\xi_4)\,g_4, \tag{6.22}$$

by E-linearity of m_2 and m_3. It remains to evaluate the E-action on $g_4 \in W_4 = (D_a\,\widehat{\otimes}\,(D_a\,\widehat{\otimes}\,N)) \otimes_{D_a\widehat{\otimes}D_a} N$. Every element of W_4 is a combination of elements $(1\widehat{\otimes}(1\widehat{\otimes}n))\otimes n'$, because the balancing relations move the factors in D_a to the right: $(x\,\widehat{\otimes}\,(y\,\widehat{\otimes}\,n)) \otimes n' = (-1)^{|n|(|x|+|y|)}(1\,\widehat{\otimes}\,(1\widehat{\otimes}n)) \otimes (x\widehat{\otimes}y) \cdot n'$, by the right action $(w\widehat{\otimes}(w'\widehat{\otimes}n))(t\widehat{\otimes}t') = (-1)^{(|w'|+|n|)|t|+|n||t'|}wt\widehat{\otimes}(w't'\widehat{\otimes}n)$ of (2.10) and (6.17), applied with $w = w' = 1$. Writing $n \otimes n'$ for $(1\,\widehat{\otimes}\,(1\,\widehat{\otimes}\,n)) \otimes n'$, so that $W_4 = N \otimes_k N$ with $g_4 = 1 \otimes 1$, the generators act by

$$\begin{aligned} &\xi_1(n \otimes n') = (-1)^{|n|}\,n \otimes \xi n', \quad \xi_2(n \otimes n') = (-1)^{|n|+|n'|}\varsigma\,n \otimes n'\xi, \\ &\xi_3(n \otimes n') = \xi n \otimes n', \qquad\qquad \xi_4(n \otimes n') = (-1)^{|n|}\varsigma\,n\xi \otimes n' : \end{aligned} \tag{6.23}$$

ξ_3 and ξ_4 act on the inner N through $\xi\,\widehat{\otimes}\,1$ and $1\,\widehat{\otimes}\,\xi$ by (6.14), while ξ_1 and ξ_2 act on the two factors D_a and are moved to the right N by the balancing relations with the

signs $(-1)^{|n|}$ and $(-1)^{|n|}(-1)^{|n'|}\varsigma$. Hence $\xi_2\xi_4(1\otimes 1) = \xi_2(\varsigma\,\xi\otimes 1) = -\varsigma^2\,\xi\otimes\xi = \xi\otimes\xi$, $\xi_1\xi_3(1\otimes 1) = \xi_1(\xi\otimes 1) = -\xi\otimes\xi$, and $\xi_1\xi_3(\xi\otimes\xi) = \xi_1(1\otimes\xi) = 1\otimes 1$, so

$$P_L(g_1) = (1-\xi_1\xi_3)(1\otimes 1 - \xi\otimes\xi) = 1\otimes 1 - \xi\otimes\xi + \xi\otimes\xi + 1\otimes 1 = 2\,g_4. \tag{6.24}$$

Since P_L and P_R are E-linear and W_1 is generated by g_1, $P_L = 2P_R$, that is, $O(a,a,a,a) = 2$.

All even systems indexed by H. Let (M', A') be an even system indexed by H. Then $M'_{0,0} = \mathbb{F}_5$ and $M'_{0,a} = M'_{a,0} = D_a$ by normalization, and $M'_{a,a}$ and N are Morita equivalences $D_a \mathbin{\widehat{\otimes}} D_a \to \mathbb{F}_5$, so $M'_{a,a}$ is evenly isomorphic to N or to ΠN by Proposition 2.8. Write $O' = O_{M',A'}$. In the first case, let $\phi : M' \to M$ be an isomorphism of Morita systems (with $\phi_{0,0}$, $\phi_{0,a}$, $\phi_{a,0}$ identities), and put $\phi^L = (\phi_{a,a} \mathbin{\widehat{\otimes}} \mathrm{id}_{D_a}) \otimes \mathrm{id}_{D_a} : L' \to L$ and $\phi^U = (\mathrm{id}_{D_a} \mathbin{\widehat{\otimes}} \phi_{a,a}) \otimes \mathrm{id}_{D_a} : U' \to U$, where L' and U' are the triple composites of M' at (a,a,a), using $2a = 0$ and $M_{0,a} = M'_{0,a} = D_a$. By Lemma 5.11(3), the comparisons $\phi^U A'_{a,a,a}(\phi^L)^{-1}$ at (a,a,a), together with the unit identifications at the other triples, form an even system with Morita system M and pentagon scalar O'; so $\phi^U A'_{a,a,a}(\phi^L)^{-1} = \lambda_0 A_0$ for some $\lambda_0 \in \mathbb{F}_5^\times$, and $O'(a,a,a,a) = 2\lambda_0^2$. In the second case $M' \cong \Pi^n M$ for the normalized $n \in Z^2(H, \mathbb{Z}/2\mathbb{Z})$ with $n(a,a) = 1$, and (5.18) gives $O' = O\,(-1)^{n\smile n}\,\delta\lambda$ for a normalized $\lambda \in C^3(H, \mathbb{F}_5^\times)$; at (a,a,a,a) we have $(-1)^{n(a,a)n(a,a)} = -1$ and $(\delta\lambda)(a,a,a,a) = \lambda(a,a,a)^2$, so $O'(a,a,a,a) = -2\lambda(a,a,a)^2$. The nonzero squares in $\mathbb{F}_5$ are 1 and 4, so $\lambda(a,a,a)^2 \in \{1,4\}$ and $\pm 2\lambda(a,a,a)^2 \in \{2,3\}$; hence in both cases $O'(a,a,a,a)$ is not a square in $\mathbb{F}_5$. This proves the first assertion. A reference system has $O(a,a,a,a) = 1$, a square, so none exists; by Theorem 5.17 no Brauer–Wall reference category over $\mathbb{F}_5$ exists, and $o_4(\mathbb{F}_5) \neq 1$. □

7 Stacking over $\mathbb{C}$ and $\mathbb{R}$

Definition 7.1. Identify $\mathrm{BW}(\mathbb{C}) = \mathbb{Z}/2\mathbb{Z}$ with $\{0,1\} \subset \mathbb{Z}$, let $\mathrm{BW}(\mathbb{R}) = \mathbb{Z}/8\mathbb{Z}$ with $\epsilon(a) = a \bmod 2$ (Lemma 6.1), and let $q : \mathbb{Z}/8\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ be the function (6.5). The *standard hexagon scalars* over $\mathbb{C}$ and over $\mathbb{R}$ are the pairs $\kappa^{\mathbb{C}} = (\kappa_1^{\mathbb{C}}, \kappa_2^{\mathbb{C}})$ of maps $\mathrm{BW}(\mathbb{C})^3 \to \mathbb{C}^\times$ and $\kappa^{\mathbb{R}} = (\kappa_1^{\mathbb{R}}, \kappa_2^{\mathbb{R}})$ of maps $\mathrm{BW}(\mathbb{R})^3 \to \mathbb{R}^\times$ given by

$$\begin{aligned} \kappa_1^{\mathbb{C}}(a;b,c) &= 1, \qquad & \kappa_2^{\mathbb{C}}(a,b;c) &= i^{abc}, \\ \kappa_1^{\mathbb{R}}(a;b,c) &= 1, \qquad & \kappa_2^{\mathbb{R}}(a,b;c) &= (-1)^{\epsilon(a)\epsilon(b)q(c)}. \end{aligned} \tag{7.1}$$

Lemma 7.2. *Let $k = \mathbb{R}$, let (R, β_1, β_2) be an exchange datum of a Brauer–Wall reference category $\mathcal{C}$, and put $\tau = (\mathbb{Z}/8\mathbb{Z}, \mathrm{id}, 0, \omega_-)$ with ω_- of* (6.13)*. Then Definition 3.22 defines a Brauer–Wall reference category $\mathcal{C}^\tau = \mathcal{P}_{\mathcal{C}}(\tau)$ with components $\mathcal{C}^\tau_a$ the families supported at a, and*

$$R^\tau_{X,Y} = r^{-1}_{Y\otimes X} R_{X,Y} r_{X\otimes Y} : (X\otimes Y)\otimes \mathbf{1} \longrightarrow (Y\otimes X)\otimes \mathbf{1}, \tag{7.2}$$

on families supported at single elements, is part of an exchange datum $(R^\tau, \beta_1^-, \beta_2^-)$ of $\mathcal{C}^\tau$ such that

$$\beta_1^-(a;b,c) = \beta_1(a;b,c)\,(-1)^{\tilde{a}\gamma(b,c)}, \qquad \beta_2^-(a,b;c) = \beta_2(a,b;c)\,(-1)^{\gamma(a,b)\tilde{c}}. \tag{7.3}$$

Proof. Write $\theta = \omega_-$. By Lemma 6.1, $\mathrm{BW}(\mathbb{R}) = \mathbb{Z}/8\mathbb{Z}$, and θ is a normalized 3-cocycle by Lemma 6.4(3); so τ is a twist. By Proposition 3.24, $\mathcal{C}^\tau \in \mathcal{W}$, every simple object of $\mathcal{C}^\tau$ is isomorphic to a unique X_a, $a \in \mathbb{Z}/8\mathbb{Z}$, and X_a has type a. Hence $\mathcal{C}^\tau$ is a Brauer–Wall reference category, and its component $\mathcal{C}^\tau_a$ consists of the families supported at a.

Let X, Y, Z be families supported at $a, b, c \in \mathbb{Z}/8\mathbb{Z}$ with values X_a, Y_b, Z_c. Then $(X \otimes_\tau Y)_{a+b} = (X_a \otimes Y_b)\otimes \mathbf{1}$, and the associator $\alpha^\tau_{X,Y,Z}$ of (3.59) is $\theta(a,b,c)A^0_{a,b,c}(X_a, Y_b, Z_c)$, where A^0 of (3.55) is built from unitors alone. Let $u_L = e_{0,(X_a\otimes Y_b)\otimes Z_c}((e_{0,X_a\otimes Y_b} \otimes \mathrm{id}_{Z_c}) \otimes \mathrm{id}_{\mathbf{1}})$ and

$u_R = D_R^{X_a,Y_b,Z_c}(0,0)$ in the notation of (3.108), so that $D_L^{X_a,Y_b,Z_c}(0,0) = \alpha_{X_a,Y_b,Z_c} u_L$; both are composites of right unitors. Equation (3.109) with all four indices 0 gives $u_R\,\alpha^{\tau}_{X,Y,Z}\,u_L^{-1} = \theta(a,b,c)\,\alpha_{X_a,Y_b,Z_c}$, and by (7.2), $r_{Y_b\otimes X_a} R^{\tau}_{X,Y} r^{-1}_{X_a\otimes Y_b} = R_{X_a,Y_b}$. Conjugating by the right unitors in this way, both hexagons for R^τ in $\mathcal{C}^\tau$ become the hexagons (4.54) and (4.55) for R in $\mathcal{C}$ with each associator $\alpha_{U,V,W}$ replaced by $\theta(u,v,w)\alpha_{U,V,W}$, u,v,w the types of U,V,W. In the left hexagon the left side acquires $\theta(b,c,a)\theta(a,b,c)$ and the right side $\theta(b,a,c)$; in the right hexagon the right side acquires $\theta(c,a,b)\theta(a,c,b)^{-1}\theta(a,b,c)$. Let $(\beta_1^\theta,\beta_2^\theta)$ be the hexagon scalars of R^τ; solving for them gives

$$\beta_1^\theta(a;b,c) = \beta_1(a;b,c)\frac{\theta(a,b,c)\theta(b,c,a)}{\theta(b,a,c)}, \qquad \beta_2^\theta(a,b;c) = \beta_2(a,b;c)\frac{\theta(a,c,b)}{\theta(c,a,b)\theta(a,b,c)}. \quad (7.4)$$

The degree and supernaturality conditions hold because the unitors are even and natural, so R^τ is a family of nonzero elements of the lines of Lemma 4.21 for the reference category $\mathcal{C}^\tau$, with the unit values, and $\mathcal{C}^\tau$ has the same exchange degree as $\mathcal{C}$, since R^τ has the degree of R; its hexagon scalars are (7.4). Since $\theta = \omega_-$ and γ is symmetric, $\theta(a,b,c)\theta(b,c,a)/\theta(b,a,c) = (-1)^{\tilde{a}\gamma(b,c)}$ and $\theta(a,c,b)/(\theta(c,a,b)\theta(a,b,c)) = (-1)^{\tilde{c}\gamma(a,b)}$, which gives (7.3). □

Proposition 7.3. *(1) Let $k = \mathbb{C}$. Every Brauer–Wall reference category $\mathcal{C}$ over $\mathbb{C}$ has exchange degree $b_{\mathcal{C}}(a,b) = ab$, and there is a Brauer–Wall reference category over $\mathbb{C}$ with an exchange datum (R,β_1,β_2) such that $(\beta_1,\beta_2) = \kappa^{\mathbb{C}}$.*

(2) Let $k = \mathbb{R}$. Every Brauer–Wall reference category $\mathcal{C}$ over $\mathbb{R}$ has exchange degree $b_{\mathcal{C}}(a,b) = \epsilon(a)\epsilon(b)$, and there is a Brauer–Wall reference category over $\mathbb{R}$ with an exchange datum (R,β_1,β_2) such that $(\beta_1,\beta_2) = \kappa^{\mathbb{R}}$.

Proof. (1) By Lemma 6.1, $\mathrm{BW}(\mathbb{C}) = \mathbb{Z}/2\mathbb{Z}$ with $\epsilon = \mathrm{id}$; since $b_{\mathcal{C}}$ is biadditive with $b_{\mathcal{C}}(1,1) = \epsilon(1) = 1$ by (4.56), $b_{\mathcal{C}}(a,b) = ab$. Brauer–Wall reference categories over $\mathbb{C}$ exist by Proposition 6.5(1). Let $\mathcal{C}$ be one and let (R,β_1,β_2) be an exchange datum of $\mathcal{C}$ (Proposition 4.23). The hexagon scalars equal 1 whenever an argument is 0, so they are determined by $A = \beta_1(1;1,1)$ and $B = \beta_2(1,1;1)$. At $(a,b,c,d) = (1,1,1,1)$ the third identity of (4.58) has the sign $(-1)^{b_{\mathcal{C}}(1,1)b_{\mathcal{C}}(1,1)} = -1$ and reads $A^2 = -B^2$; hence $\varsigma = B/A$ satisfies $\varsigma^2 = -1$, that is, $\varsigma \in \{i,-i\}$. A rescaling by λ with $\ell = \lambda(1,1)$ replaces (A,B) by $(A\lambda(1,0)\lambda(1,1)^{-2}, B\lambda(0,1)\lambda(1,1)^{-2}) = (A\ell^{-2}, B\ell^{-2})$ by (4.57); choosing ℓ with $\ell^2 = A$ gives an exchange datum $(\lambda R,\beta_1^\lambda,\beta_2^\lambda)$ of $\mathcal{C}$ (Proposition 4.25) with $\beta_1^\lambda(a;b,c) = 1$ and $\beta_2^\lambda(a,b;c) = \varsigma^{abc}$. If $\varsigma = i$, this is $\kappa^{\mathbb{C}}$. If $\varsigma = -i$, let $\mathcal{C}^{\mathrm{conj}}$ be the complex-conjugate category: it has the same objects, morphisms, composition, and tensor product as $\mathcal{C}$, and the scalar action $c\cdot f = \bar{c}f$ for $c\in\mathbb{C}$ and a morphism f. All structure maps are additive and $\mathbb{C}$-bilinear for this scalar action, so $\mathcal{C}^{\mathrm{conj}}$ is a monoidal Π-supercategory over $\mathbb{C}$ whose underlying category is again essentially small, semisimple, rigid, with finite-dimensional Hom spaces. The endomorphism superalgebra of an object in $\mathcal{C}^{\mathrm{conj}}$ is the complex-conjugate algebra of its endomorphism superalgebra in $\mathcal{C}$; for $D_0 = \mathbb{C}$ and $D_1 = \mathbb{C}[\xi]/(\xi^2-1)$ the conjugate algebra is isomorphic to the algebra itself via $c+c'\xi \mapsto \bar{c}+\bar{c}'\xi$. Hence simple objects keep their types, and $\mathcal{C}^{\mathrm{conj}}$ is a Brauer–Wall reference category. The exchange isomorphisms of $\mathcal{C}$ remain exchange isomorphisms of $\mathcal{C}^{\mathrm{conj}}$, and the hexagon identities, which read (left side) $=\beta\cdot$(right side) with $\beta\in\mathbb{C}^\times$ acting by the scalar multiplication of $\mathcal{C}$, read (left side) $=\bar{\beta}\cdot$(right side) for the scalar multiplication of $\mathcal{C}^{\mathrm{conj}}$. Thus the hexagon scalars of $\mathcal{C}^{\mathrm{conj}}$ for the exchange isomorphisms λR are the complex conjugates of $\beta_1^\lambda,\beta_2^\lambda$, namely 1 and $\overline{(-i)^{abc}} = i^{abc}$; so $(\lambda R,\kappa^{\mathbb{C}})$ is an exchange datum of the Brauer–Wall reference category $\mathcal{C}^{\mathrm{conj}}$.

(2) By Lemma 6.1, $\mathrm{BW}(\mathbb{R}) = \mathbb{Z}/8\mathbb{Z}$ with $\epsilon(a) = a \bmod 2$; since $b_{\mathcal{C}}$ is biadditive with $b_{\mathcal{C}}(1,1) = \epsilon(1) = 1$, $b_{\mathcal{C}}(a,b) = \tilde{a}\tilde{b}\,b_{\mathcal{C}}(1,1) \bmod 2 = \epsilon(a)\epsilon(b)$. Hence in (4.58) the sign in the third identity is $(-1)^{\epsilon(a)\epsilon(b)\epsilon(c)\epsilon(d)}$, and the exponent in the fourth is $(\epsilon(a)+\epsilon(b))\epsilon(c)\epsilon(a)\epsilon(b)+$

$\epsilon(a)(\epsilon(b)+\epsilon(c))\epsilon(b)\epsilon(c) = 4\epsilon(a)\epsilon(b)\epsilon(c) = 0$ in $\mathbb{Z}/2\mathbb{Z}$, since $\epsilon^2 = \epsilon$. Thus every exchange datum (R, β_1, β_2) of a Brauer–Wall reference category over $\mathbb{R}$ satisfies, for $a, b, c, d \in \mathbb{Z}/8\mathbb{Z}$,

$$\begin{aligned}
\beta_1(a;b,c)\beta_1(a;b+c,d) &= \beta_1(a;b,c+d)\beta_1(a;c,d),\\
\beta_2(a+b,c;d)\beta_2(a,b;d) &= \beta_2(b,c;d)\beta_2(a,b+c;d),\\
\frac{\beta_1(b;c,d)\beta_1(a;c,d)}{\beta_1(a+b;c,d)} &= (-1)^{\epsilon(a)\epsilon(b)\epsilon(c)\epsilon(d)}\frac{\beta_2(a,b;c)\beta_2(a,b;d)}{\beta_2(a,b;c+d)},\\
\frac{\beta_1(a;b,c)}{\beta_1(a;c,b)} &= \frac{\beta_2(b,a;c)}{\beta_2(a,b;c)}.
\end{aligned} \tag{7.5}$$

Brauer–Wall reference categories over $\mathbb{R}$ exist by Proposition 6.5(2); let $\mathcal{C}$ be one, with an exchange datum (R, β_1, β_2).

Normal forms. In this proof, the rescaling of a pair κ by λ means the pair given by (4.57), so that rescaling the exchange datum by λ rescales (β_1, β_2) by λ. For $e = (e_1, e_2) \in \{0,1\}^2$ put

$$\kappa_1^e(a;b,c) = (-1)^{e_1\tilde{a}\,\gamma(b,c)}, \qquad \kappa_2^e(a,b;c) = (-1)^{\epsilon(a)\epsilon(b)q(c)+e_2\,\gamma(a,b)\,\tilde{c}}, \tag{7.6}$$

so that $\kappa^{(0,0)} = \kappa^{\mathbb{R}}$. We show that (β_1, β_2) is a rescaling of κ^e for some e. We use the following facts about normalized cochains on $\mathbb{Z}/n\mathbb{Z}$ with values in an abelian group K_0, written multiplicatively; the second follows from Lemma 6.4(2). (a) For $\eta \in C^1(\mathbb{Z}/n\mathbb{Z}, K_0)$, $(\delta\eta)(b,c) = \eta(b)\eta(c)\eta(b+c)^{-1}$; the 1-cocycles are the homomorphisms $b \mapsto z^{\tilde{b}}$ with $z \in K_0$, $z^n = 1$; and for $z \in K_0$ the cochain $\eta(b) = z^{\tilde{b}}$ has $\delta\eta = z^{n\gamma_n}$ by (6.4). (b) Every $f \in Z^2(\mathbb{Z}/n\mathbb{Z}, K_0)$ is of the form $f = u^{\gamma_n}\delta\eta$ with $u = \Psi_2(f)$ and $\eta \in C^1(\mathbb{Z}/n\mathbb{Z}, K_0)$, and u^{γ_n} is a coboundary if and only if $u \in K_0^n$.

The identities (7.5) are multiplicative in κ except for the sign in the third one. Hence a pair $\kappa^{\mathbb{R}}\mu$ satisfies (7.5) if and only if $\mu = (\mu_1, \mu_2)$ satisfies the four identities with the sign omitted; we call such a pair μ *sign-free*. Rescalings act on μ as on κ. That $\kappa^{\mathbb{R}}$ satisfies (7.5) is the case q of Lemma 6.3: the first identity is trivial, the second is $(\epsilon(a+b)\epsilon(c)+\epsilon(a)\epsilon(b))q(d) = (\epsilon(b)\epsilon(c)+\epsilon(a)\epsilon(b+c))q(d)$, the third is $\epsilon(a)\epsilon(b)(q(c)+q(d)+q(c+d)) = \epsilon(a)\epsilon(b)\epsilon(c)\epsilon(d)$, and the fourth is the symmetry of $\epsilon(a)\epsilon(b)$. For $\mu^e = \kappa^e/\kappa^{\mathbb{R}}$, that is $\mu_1^e(a;b,c) = (-1)^{e_1\tilde{a}\gamma(b,c)}$ and $\mu_2^e(a,b;c) = (-1)^{e_2\gamma(a,b)\tilde{c}}$, the first two identities are the cocycle identity of γ, in the third both sides equal $(-1)^{8e_i\gamma(a,b)\gamma(c,d)} = 1$ because $\tilde{a} + \tilde{b} - \widetilde{a+b} = 8\gamma(a,b)$, and the fourth holds because γ is symmetric. Thus each κ^e satisfies (7.5) with values in $\{\pm 1\}$.

Let $\kappa = (\beta_1, \beta_2)$, which satisfies (7.5) with values in $K_0 = \mathbb{R}^\times$, so that $K_0^8 = \mathbb{R}_{>0}$ and $\{z \in K_0 : z^8 = 1\} = \{\pm 1\}$, and put $\mu = \kappa/\kappa^{\mathbb{R}}$, a sign-free pair. By the first identity, $\mu_1(a;-,-) \in Z^2(\mathbb{Z}/8\mathbb{Z}, K_0)$ for each a, so by (b) there are $u(a) = \Psi_2(\mu_1(a;-,-)) \in K_0$, with $u(0) = 1$, and $\eta_a \in C^1(\mathbb{Z}/8\mathbb{Z}, K_0)$, with $\eta_0 = 1$, such that $\mu_1(a;b,c) = u(a)^{\gamma(b,c)}(\delta\eta_a)(b,c)$, where $\eta_0 = 1$ is possible because $\mu_1(0;b,c) = 1 = u(0)$; the rescaling $\lambda(a,b) = \eta_a(b)$ replaces $\mu_1(a;b,c)$ by $u(a)^{\gamma(b,c)}$. The third identity now reads $\big(u(a)u(b)u(a+b)^{-1}\big)^{\gamma(c,d)} = \delta(\mu_2(a,b;-))(c,d)$; applying Ψ_2 in (c,d) to both sides gives $u(a)u(b)u(a+b)^{-1} = \mu_2(a,b;1)^8 \in \mathbb{R}_{>0}$. Writing $u(a) = s(a)|u(a)|$ with $s(a) \in \{\pm 1\}$, the function s is a homomorphism $\mathbb{Z}/8\mathbb{Z} \to \{\pm 1\}$, so $s(a) = (-1)^{e_1\tilde{a}}$ for some $e_1 \in \{0,1\}$, and the rescaling $\lambda(a,b) = |u(a)|^{\tilde{b}/8}$, with $\delta\lambda(a,-) = |u(a)|^\gamma$ by (a), replaces u by s. Now $\mu_1 = \mu_1^e$, and the third identity gives $\delta(\mu_2(a,b;-)) = 1$. By (a), $\mu_2(a,b;c) = v(a,b)^{\tilde{c}}$ with $v(a,b) = \mu_2(a,b;1) \in \{\pm 1\}$, and the second identity at $d = 1$ says $v \in Z^2(\mathbb{Z}/8\mathbb{Z}, \{\pm 1\})$; by (b), $v = (-1)^{e_2\gamma}\delta\eta'$ with $e_2 \in \{0,1\}$ and $\eta' : \mathbb{Z}/8\mathbb{Z} \to \{\pm 1\}$. The rescaling $\lambda(a,b) = \eta'(a)^{\tilde{b}}$ leaves μ_1 unchanged, since $\delta\lambda(a,-) = \eta'(a)^{8\gamma} = 1$, and replaces $\mu_2(a,b;c)$ by $\big(v(a,b)\eta'(a+b)\eta'(a)^{-1}\eta'(b)^{-1}\big)^{\tilde{c}} = (-1)^{e_2\gamma(a,b)\tilde{c}}$. Hence μ is a rescaling of μ^e, and κ is a rescaling of κ^e.

The cases $e_1 \neq e_2$ do not occur. We show $e_1 = e_2$. Let G be a finite group and $s, t \in \mathrm{Tw}(G)$, and for a pair κ of hexagon scalars of $\mathcal{C}$ write $K_{s,t}^\kappa$ for the cochain $K_{s,t}$ of (4.62)

computed with κ. If $\kappa' = (\beta_1^\lambda, \beta_2^\lambda)$ is obtained from $\kappa = (\beta_1, \beta_2)$ by (4.57), then (4.62) and (4.57) give $K_{s,t}^{\kappa'} = K_{s,t}^{\kappa}(\delta\lambda_{s,t})^{-1}$ with $\lambda_{s,t}(g,h) = \lambda(\rho_t(g), \rho_s(h))$, so the products defined by κ' and by κ differ by the gauge $(\mathrm{id}_G, 0, \lambda_{s,t})$ and induce the same operation on $\mathcal{T}_{\mathbb{R}}(G)$. Since (R, β_1, β_2) is an exchange datum of $\mathcal{C}$, Theorem 4.30 shows that the operation induced by κ^e on $\mathcal{T}_{\mathbb{R}}(G)$ is commutative for every finite group G. Take $G = \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ with generators g of $\mathbb{Z}/8\mathbb{Z}$ and h of $\mathbb{Z}/2\mathbb{Z}$, let $\rho : G \to \mathbb{Z}/8\mathbb{Z}$ be the first projection, let $\pi' : G \to \mathbb{Z}/2\mathbb{Z}$ be the second projection, and put $\rho' = 4\pi'$, $x_1 = \epsilon \circ \rho$, and $y_2 = \pi'$. Then $\epsilon \circ \rho' = 0$, $q \circ \rho' = 0$, and $\gamma_{\rho'} = y_2 \smile y_2$, so (4.60) gives

$$p_\rho \star p_{\rho'} = (\rho + \rho', 0, (-1)^{e_2\, y_2 \smile y_2 \smile x_1}), \qquad p_{\rho'} \star p_\rho = (\rho + \rho', 0, (-1)^{e_1\, x_1 \smile y_2 \smile y_2}), \tag{7.7}$$

because in both products the parity cochains vanish, the last term of Ξ vanishes, and K^{κ^e} reduces to the sign in (7.7), as $\epsilon \circ \rho' = 0$, $q \circ \rho' = 0$, and $\widetilde{\rho'}$ takes even values. A gauge $(\mathrm{id}_G, \lambda', \beta')$ between two twists with the same ρ and $\nu = 0$ has $\delta\lambda' = 0$ and multiplies ω by the coboundary $(\delta\beta')^{-1}$ with $\beta' \in C^2(G, \mathbb{R}^\times)$; comparing signs, the two twists in (7.7) are gauge equivalent if and only if $e_1\, x_1 \smile y_2 \smile y_2 + e_2\, y_2 \smile y_2 \smile x_1 \in B^3(G, \mathbb{Z}/2\mathbb{Z})$. The linear form $\Psi(f) = f(g,h,h) + f(h,g,h) + f(h,h,g)$ on $C^3(G, \mathbb{Z}/2\mathbb{Z})$ vanishes on coboundaries, because for $b' \in C^2(G, \mathbb{Z}/2\mathbb{Z})$, in the expansion of $\delta b'(g,h,h) + \delta b'(h,g,h) + \delta b'(h,h,g)$ with $h^2 = 1$ and $gh = hg$ every term occurs twice, while $\Psi(x_1 \smile y_2 \smile y_2) = \Psi(y_2 \smile y_2 \smile x_1) = 1$. Hence the two twists are gauge equivalent only if $e_1 = e_2$.

Conclusion. Let $(\lambda R, \kappa^e)$ be the rescaled exchange datum of $\mathcal{C}$, with $e \in \{(0,0), (1,1)\}$. If $e = (0,0)$, then $\kappa^e = \kappa^{\mathbb{R}}$. If $e = (1,1)$, Lemma 7.2 applied to $\mathcal{C}$ with this exchange datum gives a Brauer–Wall reference category $\mathcal{C}^\tau$ with an exchange datum whose hexagon scalars are $\kappa_1^{(1,1)}(a;b,c)(-1)^{\tilde{a}\gamma(b,c)} = 1$ and $\kappa_2^{(1,1)}(a,b;c)(-1)^{\gamma(a,b)\tilde{c}} = \kappa_2^{\mathbb{R}}(a,b;c)$, that is, $\kappa^{\mathbb{R}}$. □

Definition 7.4. By Proposition 7.3, we may fix a Brauer–Wall reference category $\mathcal{C}_{\mathbb{C}}$ over $\mathbb{C}$ with an exchange datum (R, β_1, β_2) such that $(\beta_1, \beta_2) = \kappa^{\mathbb{C}}$, and a Brauer–Wall reference category $\mathcal{C}_{\mathbb{R}}$ over $\mathbb{R}$ with an exchange datum (R, β_1, β_2) such that $(\beta_1, \beta_2) = \kappa^{\mathbb{R}}$. We call them the *standard reference categories* and the fixed exchange data their *standard exchange data.* In the rest of this section, for $k = \mathbb{C}$ and for $k = \mathbb{R}$, the product $\star$ of Definition 4.27 and the group structure on $\mathcal{T}_k(G)$ of Theorem 4.30 are those defined by the standard reference category over k with its standard exchange datum.

Proposition 7.5. *Let G be a finite group and let $s, t \in \mathrm{Tw}(G)$.*

(1) Let $k = \mathbb{C}$, and let $x = \rho_s$ and $y = \rho_t$, regarded as elements of $Z^1(G, \mathbb{Z}/2\mathbb{Z})$. Then

$$\begin{aligned} s \star t &= \big(G,\ \rho_s + \rho_t,\ \nu_s + \nu_t + y \smile x,\ \omega_s\omega_t\, i^{\,y \smile y \smile x}\, (-1)^{\Xi^{\mathbb{C}}_{s,t}}\big), \\ \Xi^{\mathbb{C}}_{s,t} &= \nu_s \smile_1 \nu_t + (y \smile x) \smile_1 (\nu_s + \nu_t) + y \smile (yx) \smile x, \end{aligned} \tag{7.8}$$

and

$$\begin{aligned} s^\vee &= \big(G,\ -\rho_s,\ \nu_s + x \smile x,\ \omega_s^{-1}\, i^{\,x \smile x \smile x}\, (-1)^{\Xi^{\mathbb{C}\vee}_s}\big), \\ \Xi^{\mathbb{C}\vee}_s &= (\nu_s + x \smile x) \smile_1 \nu_s, \end{aligned} \tag{7.9}$$

where $i^{\,y \smile y \smile x}(g,h,\ell) = i^{y(g)y(h)x(\ell)}$ and $i^{\,x \smile x \smile x}(g,h,\ell) = i^{x(g)x(h)x(\ell)}$.

(2) Let $k = \mathbb{R}$, and let $x = \epsilon \circ \rho_s$ and $y = \epsilon \circ \rho_t$. Then

$$\begin{aligned} s \star t &= \big(G,\ \rho_s + \rho_t,\ \nu_s + \nu_t + y \smile x,\ \omega_s\omega_t(-1)^{\Xi^{\mathbb{R}}_{s,t}}\big), \\ \Xi^{\mathbb{R}}_{s,t}(g,h,\ell) &= (\nu_s \smile_1 \nu_t)(g,h,\ell) + ((y \smile x) \smile_1 (\nu_s + \nu_t))(g,h,\ell) \\ &\quad + y(g)y(h)x(h)x(\ell) + y(g)y(h)\, q(\rho_s(\ell)), \end{aligned} \tag{7.10}$$

and

$$\begin{aligned} s^\vee &= \big(G,\ -\rho_s,\ \nu_s + x \smile x,\ \omega_s^{-1}(-1)^{\Xi^{\mathbb{R}\vee}_s}\big), \\ \Xi^{\mathbb{R}\vee}_s(g,h,\ell) &= ((\nu_s + x \smile x) \smile_1 \nu_s)(g,h,\ell) + x(g)x(h)\, q(\rho_s(\ell)). \end{aligned} \tag{7.11}$$

Proof. (1) By Definition 7.4 and Proposition 7.3, $b_{\mathcal{C}_{\mathbb{C}}}(a,b) = ab$ and $(\beta_1,\beta_2) = \kappa^{\mathbb{C}}$, so (4.62) with $\rho_s = x$ and $\rho_t = y$ gives $c_{s,t} = y \smile x$ and

$$K_{s,t}(g,h,\ell) = \frac{\kappa_2^{\mathbb{C}}(y(g),y(h);x(\ell))}{\kappa_1^{\mathbb{C}}(y(g);x(h),x(\ell))} = i^{y(g)y(h)x(\ell)}, \tag{7.12}$$

so that the last term of $\Xi_{s,t}$ is $y(g)y(h)x(h)x(\ell) = (y \smile (yx) \smile x)(g,h,\ell)$; substituting into (4.60) gives (7.8). For $s^\vee$, we have $c_{s,s} = x \smile x$ and, since $-a = a$ in $\mathbb{Z}/2\mathbb{Z}$, $K_s^\vee(g,h,\ell) = i^{x(g)x(h)x(\ell)}$; in $\Xi_s^\vee$ the last term is $x(g)x(h)x(h)x(\ell) = (x \smile x \smile x)(g,h,\ell)$, and $(x \smile x) \smile_1 (\nu_s + x \smile x) = (x \smile x) \smile_1 \nu_s$ because

$$\big((x \smile x) \smile_1 (x \smile x)\big)(g,h,\ell) = \big(x(g)+x(h)\big)x(g)x(h)x(\ell) + x(g)x(h)\big(x(h)+x(\ell)\big)x(\ell) = 0. \tag{7.13}$$

Hence $\Xi_s^\vee = (\nu_s + x \smile x) \smile_1 \nu_s + x \smile x \smile x$, and since $(-1)^{x\smile x\smile x}\, i^{-x\smile x\smile x} = i^{x\smile x\smile x}$, substituting into (4.61) gives (7.9).

(2) By Definition 7.4 and Proposition 7.3, $b_{\mathcal{C}_{\mathbb{R}}}(a,b) = \epsilon(a)\epsilon(b)$ and $(\beta_1,\beta_2) = \kappa^{\mathbb{R}}$, so (4.62) gives $c_{s,t} = y \smile x$ and $K_{s,t}(g,h,\ell) = (-1)^{y(g)y(h)q(\rho_s(\ell))}$, and (4.60) is (7.10). For $s^\vee$, we have $c_{s,s} = x \smile x$ and $K_s^\vee(g,h,\ell) = (-1)^{x(g)x(h)q(-\rho_s(\ell))} = (-1)^{x(g)x(h)(q(\rho_s(\ell))+x(\ell))}$ by (6.7), and $\Xi_s^\vee = (\nu_s + x \smile x) \smile_1 \nu_s + x \smile x \smile x$ as in (1); the two terms $x(g)x(h)x(\ell)$ cancel, and substituting into (4.61) gives (7.11). □

Theorem 7.6. *Let $k = \mathbb{C}$ and let G be a finite group. If some $\rho \in \mathrm{Hom}(G,\mathbb{Z}/2\mathbb{Z})$ is not the reduction modulo 2 of a homomorphism $G \to \mathbb{Z}/4\mathbb{Z}$, then the exact sequence*

$$0 \longrightarrow \mathrm{SH}_{\mathbb{C}}(G) \longrightarrow \mathcal{T}_{\mathbb{C}}(G) \xrightarrow{d} \mathrm{Hom}(G,\mathbb{Z}/2\mathbb{Z}) \longrightarrow 0 \tag{7.14}$$

of Theorem 4.32 does not split. For $G = \mathbb{Z}/2\mathbb{Z}$ and $\rho = \mathrm{id}$,

$$\mathrm{SH}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}, \qquad \mathcal{T}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/8\mathbb{Z} \text{ generated by } [p_\rho], \tag{7.15}$$

and (7.14) *does not split.*

Proof. The sequence (7.14) is exact by Theorem 4.32. For $\rho \in \mathrm{Hom}(G,\mathbb{Z}/2\mathbb{Z})$, (7.8) with $s = t = p_\rho$ and $x = y = \rho$ gives $p_\rho \star p_\rho = (G, 0, \rho \smile \rho, i^{\rho\smile\rho\smile\rho}(-1)^{\rho\smile(\rho\rho)\smile\rho})$, and $\rho\rho = \rho$ since ρ takes values in $\{0,1\}$; hence

$$p_\rho \star p_\rho = (G, 0,\ \rho \smile \rho,\ (-i)^{\rho\smile\rho\smile\rho}). \tag{7.16}$$

Suppose that $\sigma : \mathrm{Hom}(G,\mathbb{Z}/2\mathbb{Z}) \to \mathcal{T}_{\mathbb{C}}(G)$ is a homomorphism with $d \circ \sigma = \mathrm{id}$, and write $\mathcal{T}_{\mathbb{C}}(G)$ multiplicatively. For $\rho \in \mathrm{Hom}(G,\mathbb{Z}/2\mathbb{Z})$ put $f(\rho) = [p_\rho]\sigma(\rho)^{-1} \in \mathrm{SH}_{\mathbb{C}}(G)$. Since $\mathcal{T}_{\mathbb{C}}(G)$ is abelian and $\sigma(\rho)^2 = \sigma(2\rho) = \sigma(0) = 1$,

$$[p_\rho]^2 = f(\rho)^2\sigma(\rho)^2 = f(\rho)^2. \tag{7.17}$$

The map $[(\nu,\omega)] \mapsto [\nu]$ of (4.98) sends $[p_\rho]^2 = [p_\rho \star p_\rho]$ to $[\rho \smile \rho]$ by (7.16), and sends $f(\rho)^2$ to $2\bar{f}(\rho) = 0$ in the $\mathbb{F}_2$-vector space $H^2(G,\mathbb{Z}/2\mathbb{Z})$, where $\bar{f}(\rho)$ is the image of $f(\rho)$. Hence $[\rho \smile \rho] = 0$ for every ρ. Let $\rho \in \mathrm{Hom}(G,\mathbb{Z}/2\mathbb{Z})$. Since $[\rho \smile \rho] = 0$, there is $\lambda \in C^1(G,\mathbb{Z}/2\mathbb{Z})$ such that $\rho \smile \rho = \delta\lambda$. Let $\tilde{\rho}, \tilde{\lambda} : G \to \{0,1\} \subset \mathbb{Z}$ be the lifts of ρ and λ; then $\tilde{\rho}(1) = \tilde{\lambda}(1) = 0$, and $\tilde{\rho}(gh) \equiv \tilde{\rho}(g) + \tilde{\rho}(h) \pmod 2$ for $g,h \in G$. For $g,h \in G$, the integer $(\delta\tilde{\rho})(g,h) = \tilde{\rho}(h) - \tilde{\rho}(gh) + \tilde{\rho}(g)$ is 2 if $\tilde{\rho}(g) = \tilde{\rho}(h) = 1$ and 0 otherwise, so $\delta\tilde{\rho} = 2\,\tilde{\rho} \smile \tilde{\rho}$ in $C^2(G,\mathbb{Z})$; and $\tilde{\rho} \smile \tilde{\rho} - \delta\tilde{\lambda}$ has values in $2\mathbb{Z}$, since its reduction modulo 2 is $\rho \smile \rho - \delta\lambda = 0$. Define $\hat{\rho} : G \to \mathbb{Z}$ by $\hat{\rho} = \tilde{\rho} - 2\tilde{\lambda}$. Then $\delta\hat{\rho} = 2(\tilde{\rho} \smile \tilde{\rho} - \delta\tilde{\lambda})$ has values in $4\mathbb{Z}$, so the reduction ρ' of $\hat{\rho}$ modulo 4 satisfies $\rho'(gh) = \rho'(g) + \rho'(h)$ for $g,h \in G$, that is, $\rho' \in \mathrm{Hom}(G,\mathbb{Z}/4\mathbb{Z})$, and

the reduction of ρ' modulo 2 is ρ. Hence every $\rho \in \mathrm{Hom}(G, \mathbb{Z}/2\mathbb{Z})$ is the reduction modulo 2 of a homomorphism $G \to \mathbb{Z}/4\mathbb{Z}$, which proves the non-splitting criterion.

Let $G = \mathbb{Z}/2\mathbb{Z}$ and $\rho = \mathrm{id}$. Then $\mathrm{SH}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z}$ by Proposition 6.5(1), so $|\mathcal{T}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z})| = 8$ by exactness. By (7.16), $p_\rho \star p_\rho$ is the twist $(\mathbb{Z}/2\mathbb{Z}, 0, \rho \smile \rho, \omega')$ with $\omega'(1,1,1) = -i$. As in the proof of Proposition 6.5(1), the square of this element in $\mathrm{SH}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z})$ is $(0, \omega)$ with $\omega(1,1,1) = i^2 = -1$, which is nontrivial in $H^3(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times)$ since $\Psi_3(\omega) = -1$. Hence $[p_\rho]^2$ has order 4, $[p_\rho]$ has order 8, and $\mathcal{T}_{\mathbb{C}}(\mathbb{Z}/2\mathbb{Z})$ is cyclic of order 8; it is not isomorphic to $\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, so (7.14) does not split. This also follows from the criterion, since ρ does not lift to $\mathbb{Z}/4\mathbb{Z}$: $[\rho \smile \rho] \neq 0$ in $H^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z})$ because $\Psi_2(\rho \smile \rho) = \rho(1)\rho(1) = 1$. □

Theorem 7.7. *Let $k = \mathbb{R}$ and let G be a finite group. The exact sequence*

$$0 \longrightarrow \mathrm{SH}_{\mathbb{R}}(G) \longrightarrow \mathcal{T}_{\mathbb{R}}(G) \xrightarrow{d} \mathrm{Hom}(G, \mathbb{Z}/8\mathbb{Z}) \longrightarrow 0 \tag{7.18}$$

of Theorem 4.32 splits.

Proof. By Proposition 7.5(2) the product on $\mathrm{Tw}(G)$ is given by (7.10), and the sequence (7.18) is exact by Theorem 4.32. We use the notation of Definition 6.2.

Powers of p_ρ. Write $p_\rho^{\star 1} = p_\rho$ and $p_\rho^{\star m} = p_\rho^{\star(m-1)} \star p_\rho$. Fix ρ and write $x = x_\rho$, $x' = x'_\rho$; by Lemma 6.3, $q \circ \rho = x + x'$ and $\delta x' = x \smile x$. If $\epsilon \circ \rho = 0$, then in $p_{(m-1)\rho} \star p_\rho$ both parity characters vanish, so $\nu = 0$ and all terms of $\Xi^{\mathbb{R}}$ vanish; hence $p_{(m-1)\rho} \star p_\rho = p_{m\rho}$, and $p_\rho^{\star m} = p_{m\rho}$ by induction. For general ρ we compute successively, using $q(2a) = \epsilon(a)$ and $q(3a) = q(a) + \epsilon(a)$ from (6.7). In $p_\rho \star p_\rho$, both characters are x, $\nu = x \smile x$, and the exponent is $x \smile (xx) \smile x + x \smile x \smile (q \circ \rho) = x \smile x \smile x + x \smile x \smile (x + x') = x \smile x \smile x'$, since $xx = x$; so

$$p_\rho \star p_\rho = \big(G,\ 2\rho,\ x \smile x,\ (-1)^{x \smile x \smile x'}\big). \tag{7.19}$$

The gauge $(\mathrm{id}_G, x', (-1)^{x' \smile x'})$ transforms it into $p_{2\rho}$: by (3.48) the parity cochain of the transformed twist is $x \smile x + \delta x' = 0$, and by (3.49) its scalar cochain is

$$(-1)^{x \smile x \smile x'}\,(-1)^{\delta(x' \smile x')}\,(-1)^{0 \smile x' + x' \smile (x \smile x)} = 1,$$

because $\delta(x' \smile x') = x \smile x \smile x' + x' \smile x \smile x$ by (A.6). Thus $[p_\rho]^2 = [p_{2\rho}]$. In $p_\rho^{\star 2} \star p_\rho$, the characters $\epsilon \circ \rho_s$ and $\epsilon \circ \rho_t$ of (7.10) are $\epsilon \circ 2\rho = 0$ and x, so the term $y \smile x$ of (7.10) vanishes, $\nu_s \smile_1 \nu_t = (x \smile x) \smile_1 0 = 0$, the term $y(g)y(h)x(h)x(\ell)$ vanishes, and $y(g)y(h)q(\rho_s(\ell)) = x(g)x(h)q(2\rho(\ell))$, which is $(x \smile x \smile x)(g,h,\ell)$; hence $p_\rho^{\star 3} = (3\rho, x \smile x, (-1)^{x \smile x \smile (x+x')})$, and the gauge $(\mathrm{id}_G, x', (-1)^{x' \smile x' + x \smile x'})$ transforms it into $p_{3\rho}$, since $\delta(x \smile x') = x \smile x \smile x$. In $p_\rho^{\star 3} \star p_\rho$ both characters $\epsilon \circ \rho_s$ and $\epsilon \circ \rho_t$ are x, so $\nu = x \smile x + x \smile x = 0$, $\nu_s \smile_1 \nu_t = 0$, $(x \smile x) \smile_1 (x \smile x) = 0$ by (7.13), the term $y(g)y(h)x(h)x(\ell)$ is $(x \smile x \smile x)(g,h,\ell)$, and $y(g)y(h)q(\rho_s(\ell)) = x(g)x(h)q(3\rho(\ell))$, which is $(x \smile x \smile x')(g,h,\ell)$ because $q(3\rho(\ell)) = q(\rho(\ell)) + x(\ell) = x'(\ell)$; the total exponent $x \smile x \smile x + x \smile x \smile x'$ cancels the exponent of ω_s. Hence $p_\rho^{\star 4} = (4\rho, 0, 1) = p_{4\rho}$. Thus $[p_\rho]^m = [p_{m\rho}]$ for $1 \le m \le 4$. For $m = 4 + b$ with $0 \le b \le 4$, $[p_\rho]^m = [p_{4\rho}][p_{b\rho}] = [p_{4\rho} \star p_{b\rho}] = [p_{m\rho}]$, because $\epsilon \circ 4\rho = 0$ and $q \circ 4\rho = 0$ make every exponent in (7.10) vanish in $p_{4\rho} \star p_{b\rho}$; in particular $[p_\rho]^8 = [p_{8\rho}] = [p_0] = 1$, and negative exponents follow by taking inverses. Since $d[p_\rho] = \rho$, the order of ρ divides the order of $[p_\rho]$, and $[p_\rho]^{\mathrm{ord}\,\rho} = [p_0] = 1$; so the two orders agree.

Splitting. Choose a decomposition $\mathrm{Hom}(G, \mathbb{Z}/8\mathbb{Z}) = \bigoplus_{i \in I} \langle e_i \rangle$ into cyclic groups and define

$$h : \mathrm{Hom}(G, \mathbb{Z}/8\mathbb{Z}) \longrightarrow \mathcal{T}_{\mathbb{R}}(G), \qquad h\Big(\sum_i a_i e_i\Big) = \prod_i [p_{e_i}]^{a_i} \qquad (0 \le a_i < \mathrm{ord}\, e_i). \tag{7.20}$$

The group $\mathcal{T}_{\mathbb{R}}(G)$ is abelian (Theorem 4.30), and $[p_{e_i}]^{\mathrm{ord}\, e_i} = [p_{(\mathrm{ord}\, e_i) e_i}] = [p_0] = 1$, so h is a well-defined homomorphism; and $d \circ h = \mathrm{id}$ since d is a homomorphism with $d[p_{e_i}] = e_i$. Hence (7.18) splits. □

A Cochains

Definition A.1. Let G be a group and M an abelian group. For $n \geq 1$, a *normalized inhomogeneous n-cochain* is a map $c : G^n \to M$ such that $c(g_1, \ldots, g_n) = 0$ if some $g_i = 1$. We write $C^n(G, M)$ for the abelian group of these cochains, and we put $C^0(G, M) = M$. When M is written multiplicatively, normalization means value 1 and the formulas below are read multiplicatively; all coefficient groups carry the trivial G-action. The *coboundary map* $\delta : C^n(G, M) \to C^{n+1}(G, M)$ is

$$\begin{aligned}(\delta c)(g_1, \ldots, g_{n+1}) = {} & c(g_2, \ldots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i c(g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) \\ & + (-1)^{n+1} c(g_1, \ldots, g_n).\end{aligned} \tag{A.1}$$

On $C^0(G, M) = M$ the coboundary map is 0, the constant 1 in multiplicative notation, in accordance with the trivial action; thus $B^1(G, M) = 0$. A cochain is a *cocycle* if its coboundary is 0. We write $Z^n(G, M)$ for the group of cocycles, $B^n(G, M) = \delta C^{n-1}(G, M)$ for the coboundaries, and $H^n(G, M) = Z^n(G, M)/B^n(G, M)$. A group homomorphism $\phi : G \to H$ induces the *pullback* $\phi^* : C^n(H, M) \to C^n(G, M)$ given by

$$(\phi^* c)(g_1, \ldots, g_n) = c(\phi(g_1), \ldots, \phi(g_n)). \tag{A.2}$$

A homomorphism of abelian groups $\psi : M \to M'$ induces $\psi_* : C^n(G, M) \to C^n(G, M')$, $c \mapsto \psi \circ c$; for $c \in C^n(G, \mathbb{Z}/2\mathbb{Z})$ we write $(-1)^c \in C^n(G, k^\times)$ for $\psi_* c$ with $\psi : \mathbb{Z}/2\mathbb{Z} \to k^\times$, $r \mapsto (-1)^r$. If M is a ring, the *cup product* of $a \in C^p(G, M)$ and $b \in C^q(G, M)$ is the cochain $a \smile b \in C^{p+q}(G, M)$ given by

$$(a \smile b)(g_1, \ldots, g_{p+q}) = a(g_1, \ldots, g_p) b(g_{p+1}, \ldots, g_{p+q}). \tag{A.3}$$

For $v, w \in C^2(G, \mathbb{Z}/2\mathbb{Z})$, put

$$(v \smile_1 w)(g, h, \ell) = v(gh, \ell) w(g, h) + v(g, h\ell) w(h, \ell). \tag{A.4}$$

Lemma A.2. *Let G and H be groups, let M and M' be abelian groups, let $\phi : G \to H$ be a group homomorphism, and let $\psi : M \to M'$ be a homomorphism of abelian groups.*

(1) $\delta \circ \delta = 0$, and $H^n(G, M)$ is the cohomology of G with coefficients in M regarded as a trivial G-module [Bro82, Chapter I, §5 and Chapter III, §1].

(2) $\delta(\phi^* c) = \phi^*(\delta c)$ for $c \in C^n(H, M)$, and $\delta(\psi_* c) = \psi_*(\delta c)$ for $c \in C^n(G, M)$. In particular, for $c \in C^n(G, \mathbb{Z}/2\mathbb{Z})$,

$$\delta((-1)^c) = (-1)^{\delta c}. \tag{A.5}$$

(3) If M is a commutative ring, then for $a \in C^p(G, M)$ and $b \in C^q(G, M)$,

$$\delta(a \smile b) = \delta a \smile b + (-1)^p a \smile \delta b. \tag{A.6}$$

Proof. (1) In $\delta(\delta c)$, the term of (A.1) obtained by first multiplying the entries at positions $i, i+1$ and then those at positions $j, j+1$ occurs twice with opposite signs, once in each order, and the terms obtained from the first and last terms of (A.1) cancel in the same way; so $\delta \circ \delta = 0$. The complex $(C^n(G, M), \delta)_{n \geq 0}$ is the complex of G-equivariant maps from the normalized bar resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ to M, and the normalized bar resolution is a free resolution [Bro82, Chapter I, §5]; hence the cohomology of $(C^n(G, M), \delta)_{n \geq 0}$ is the cohomology of G with coefficients in M [Bro82, Chapter III, §1].

(2) The right side of (A.1) is a sum with integer coefficients of values of c at arguments obtained from $(g_1, \dots, g_{n+1})$ by multiplying adjacent entries or deleting the first or last entry. Both operations commute with ϕ^*, since ϕ is a homomorphism, and with ψ_*, since ψ is additive. Equation (A.5) is the case $\psi : \mathbb{Z}/2\mathbb{Z} \to k^\times$, $r \mapsto (-1)^r$.

(3) Let $g_1, \dots, g_{p+q+1} \in G$ and expand the three terms of (A.6) at $(g_1, \dots, g_{p+q+1})$ by (A.1) and (A.3). The first term of $\delta a \smile b$ and the last term of $(-1)^p a \smile \delta b$,

$$a(g_2, \dots, g_{p+1})b(g_{p+2}, \dots, g_{p+q+1}), \qquad (-1)^{p+q+1}a(g_1, \dots, g_p)b(g_{p+1}, \dots, g_{p+q}),$$

are the first and the last term of $\delta(a \smile b)$; the last term of $\delta a \smile b$ and the first term of $(-1)^p a \smile \delta b$,

$$(-1)^{p+1}a(g_1, \dots, g_p)b(g_{p+2}, \dots, g_{p+q+1}), \qquad (-1)^p a(g_1, \dots, g_p)b(g_{p+2}, \dots, g_{p+q+1}),$$

cancel; and the terms of $\delta a \smile b$ with $1 \le i \le p$ together with the terms of $(-1)^p a \smile \delta b$ with $1 \le j \le q$ are the terms of $\delta(a \smile b)$ with $1 \le i \le p+q$, the index j corresponding to $i = p + j$. □

Lemma A.3. *Let $v, w, \nu \in Z^2(G, \mathbb{Z}/2\mathbb{Z})$ and $\lambda \in C^1(G, \mathbb{Z}/2\mathbb{Z})$.*

(1)

$$\delta(v \smile_1 w) = v \smile w + w \smile v. \tag{A.7}$$

(2)

$$\begin{aligned}(\nu + \delta\lambda) \smile (\nu + \delta\lambda) + \nu \smile \nu &= \delta\big((\nu + \delta\lambda) \smile \lambda + \lambda \smile \nu\big) \\ &= \delta(\lambda \smile \nu + \nu \smile \lambda + \lambda \smile \delta\lambda).\end{aligned} \tag{A.8}$$

Proof. All calculations take place over $\mathbb{Z}/2\mathbb{Z}$, where (A.6) reads $\delta(a \smile b) = \delta a \smile b + a \smile \delta b$.

(1) Expand the coboundary of $v \smile_1 w$ by (A.1) and use $v(gh, \ell) = v(g, h\ell) + v(h, \ell) + v(g, h)$ and $w(gh, \ell) = w(g, h\ell) + w(h, \ell) + w(g, h)$. Cancellation gives

$$\delta(v \smile_1 w)(g, h, \ell, m) = v(g, h)w(\ell, m) + w(g, h)v(\ell, m),$$

which is (A.7).

(2) Put $\bar\nu = \nu + \delta\lambda$, so that $\delta\lambda = \bar\nu + \nu$ and $\delta\bar\nu = \delta\nu = 0$ by Lemma A.2(1). By (A.6), $\delta(\bar\nu \smile \lambda) = \bar\nu \smile \delta\lambda = \bar\nu \smile \bar\nu + \bar\nu \smile \nu$ and $\delta(\lambda \smile \nu) = \delta\lambda \smile \nu = \bar\nu \smile \nu + \nu \smile \nu$, whose sum is $\bar\nu \smile \bar\nu + \nu \smile \nu$; and $\delta(\lambda \smile \nu + \nu \smile \lambda + \lambda \smile \delta\lambda) = \delta\lambda \smile \nu + \nu \smile \delta\lambda + \delta\lambda \smile \delta\lambda = (\bar\nu + \nu) \smile \nu + \nu \smile (\bar\nu + \nu) + (\bar\nu + \nu) \smile (\bar\nu + \nu) = \bar\nu \smile \bar\nu + \nu \smile \nu$. □

Acknowledgements

The author is deeply grateful to Yuji Tachikawa for his generous support and encouragement throughout this work. This research was supported by FoPM, WINGS Program, the University of Tokyo. This work was supported by JST SPRING, Japan Grant Number JPMJSP2108.

Use of AI tools

The author used the coding assistants Codex (OpenAI) and Claude Code (Anthropic) in preparing this paper. They were used to draft proofs from the statements, definitions, and outlines written by the author, to carry out and check sign computations and cochain identities, including brute-force verification of the formulas in Sections 4 and 7, to reorganize the manuscript, and to check references. The mathematical content, the choice of statements, and the overall structure are the author's, and the author has verified every argument and takes full responsibility for the paper.

References

[Bro82] K. S. Brown, *Cohomology of groups*, Graduate Texts in Mathematics **87**, Springer, 1982.

[BM18] G. Brumfiel and J. Morgan, *The Pontrjagin dual of 3-dimensional spin bordism*, arXiv:1612.02860v2, 2018. https://arxiv.org/abs/1612.02860v2.

[BE17] J. Brundan and A. P. Ellis, *Monoidal supercategories*, Commun. Math. Phys. **351** (2017), 1045–1089. https://arxiv.org/abs/1603.05928v3.

[CGLW12] X. Chen, Z.-C. Gu, Z.-X. Liu, and X.-G. Wen, *Symmetry-protected topological orders in interacting bosonic systems*, Science **338** (2012), no. 6114, 1604–1606. https://arxiv.org/abs/1301.0861.

[CGLW13] X. Chen, Z.-C. Gu, Z.-X. Liu, and X.-G. Wen, *Symmetry protected topological orders and the group cohomology of their symmetry group*, Phys. Rev. B **87** (2013), 155114. https://arxiv.org/abs/1106.4772.

[DN21] A. Davydov and D. Nikshych, *Braided Picard groups and graded extensions of braided tensor categories*, Selecta Math. (N.S.) **27** (2021), article 65. https://doi.org/10.1007/s00029-021-00670-1.

[EV07] A. Elduque and O. Villa, *The existence of superinvolutions*, arXiv:0704.3127v1, 2007. https://arxiv.org/abs/0704.3127v1.

[ENO10] P. Etingof, D. Nikshych, and V. Ostrik, *Fusion categories and homotopy theory*, with an appendix by E. Meir, Quantum Topol. **1** (2010), no. 3, 209–273. https://doi.org/10.4171/QT/6.

[FH21] D. S. Freed and M. J. Hopkins, *Reflection positivity and invertible topological phases*, Geom. Topol. **25** (2021), 1165–1330. https://arxiv.org/abs/1604.06527.

[GJF19] D. Gaiotto and T. Johnson-Freyd, *Symmetry protected topological phases and generalized cohomology*, J. High Energy Phys. **2019**, no. 5 (2019), article 007. https://doi.org/10.1007/JHEP05(2019)007.

[GW14] Z.-C. Gu and X.-G. Wen, *Symmetry-protected topological orders for interacting fermions—Fermionic topological nonlinear σ models and a special group supercohomology theory*, Phys. Rev. B **90** (2014), 115141. https://arxiv.org/abs/1201.2648v4.

[Haz16] R. Hazrat, *Graded rings and graded Grothendieck groups*, arXiv:1405.5071v4, 2016. https://arxiv.org/abs/1405.5071v4.

[JMPP21] C. Jones, S. Morrison, D. Penneys, and J. Plavnik, *Extension theory for braided-enriched fusion categories*, arXiv:1910.03178v2, 2021. https://arxiv.org/abs/1910.03178v2.

[JS93] A. Joyal and R. Street, *Braided tensor categories*, Adv. Math. **102** (1993), 20–78.

[Kap14] A. Kapustin, *Symmetry protected topological phases, anomalies, and cobordisms: beyond group cohomology*, arXiv:1403.1467v3, 2014. https://arxiv.org/abs/1403.1467v3.

[Kaw26] K. Kawasumi, *Real RMod classification and stacking*, Lean 4 source code and verification records, 2026. Commit `7f74b4d`. `https://github.com/KotaroKawasumi/realrmod-lean/tree/7f74b4d288c748e285be46fd439d53880b805beb`.

[Kel05] G. M. Kelly, *Basic concepts of enriched category theory*, Repr. Theory Appl. Categ. **10** (2005). `https://tac.mta.ca/tac/reprints/articles/10/tr10.pdf`.

[Kwo13] S. Kwok, *Super Morita theory*, arXiv:1301.5246v2, 2013. `https://arxiv.org/abs/1301.5246v2`.

[Lam01] T. Y. Lam, *A first course in noncommutative rings*, 2nd ed., Graduate Texts in Mathematics **131**, Springer, 2001.

[Lam05] T. Y. Lam, *Introduction to quadratic forms over fields*, Graduate Studies in Mathematics **67**, American Mathematical Society, 2005.

[ML98] S. Mac Lane, *Categories for the working mathematician*, 2nd ed., Graduate Texts in Mathematics **5**, Springer, 1998.

[MP19] S. Morrison and D. Penneys, *Monoidal categories enriched in braided monoidal categories*, Int. Math. Res. Not. IMRN **2019**, no. 11, 3527–3579. `https://doi.org/10.1093/imrn/rnx217`.

[MPP18] S. Morrison, D. Penneys, and J. Plavnik, *Completion for braided enriched monoidal categories*, arXiv:1809.09782, 2018. `https://arxiv.org/abs/1809.09782`.

[PSS23] J. Plavnik, S. Sanford, and D. Sconce, *Tambara–Yamagami categories over the reals: the split case*, arXiv:2303.17843, 2023. `https://arxiv.org/abs/2303.17843`.

[San24] S. Sanford, *Fusion categories over non-algebraically closed fields*, arXiv:2401.02354v3, 2024. `https://arxiv.org/abs/2401.02354v3`.

[San26] S. Sanford, *Putting the Brauer back in Brauer-Picard*, arXiv:2604.10869, 2026. `https://arxiv.org/abs/2604.10869`.

[SP11] C. J. Schommer-Pries, *The classification of two-dimensional extended topological field theories*, arXiv:1112.1000v2, 2011. `https://arxiv.org/abs/1112.1000v2`.

[SS21] S. V. Sam and A. Snowden, *Supersymmetric monoidal categories*, arXiv:2011.12501v2, 2021. `https://arxiv.org/abs/2011.12501v2`.

[TY98] D. Tambara and S. Yamagami, *Tensor categories with fusion rules of self-duality for finite abelian groups*, J. Algebra **209** (1998), 692–707.

[Wal64] C. T. C. Wall, *Graded Brauer groups*, J. Reine Angew. Math. **213** (1964), 187–199. `https://doi.org/10.1515/crll.1964.213.187`.

[Wei94] C. A. Weibel, *An introduction to homological algebra*, Cambridge Studies in Advanced Mathematics **38**, Cambridge University Press, 1994.